GE VERNOVA

ADMS

16.25.1

Display Reference Manual

Change History

Date	Change Description
Jun 13, 2023	ADMS 3.16: • Updated section Convergence Summary for P, Q, Capacitor, and Generator. • Updated section Load Shed Sort Values. • Added section DNA Scheduler Monitor. • Updated section Fault Section Determination Stage 2. • Added section Transformer Bank. • Added section Transformer Bank. • Added section Transformer Bank Toolbar. • Removed section Multi-Interval Power Flow Job Status. • Editorial and formatting updates.
Jul 21, 2023	ADMS 3.16.0.1: • Updated section Fault Location Results.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Oct 25, 2023	ADMS 16.2.0: • Updated section Station Commissioning/Decommissioning. • Updated Capacitor Toolbar. • Updated Planned Outage Toolbar. • Updated Power Transformer Toolbar.
Apr 05, 2024	ADMS 16.7.0 • Updated document format and styling.

Contents

1. ADMS DMS Tabular Displays

This chapter contains descriptions of the tabular displays associated with the Distribution Management System (DMS) part of the ADMS.

The displays are grouped by type and arranged in the order that they are accessed from the RT DMS menu or from other displays.

1.1. ADMS Auto-Refreshable Displays

In most cases, displays present static figures that are not refreshed automatically.

However, some displays do have an optional display that automatically refreshes the presented information on a periodic basis.

The name of an optional auto-refreshable display is formed from the name of the original display plus the suffix "RF".

This is the list of displays that have auto-refreshable options:

- Station Abnormal Summary.
- Abnormal Components per Feeder.
- Feeder Exception Summary.
- Station Branch Flows.
- Station-Feeder Branch Flows.
- Station Exception.
- De-Energization.
- De-Energization Customers per Feeder.
- High Voltages.
- High Voltages per Feeder.
- Voltage Imbalance Information.
- Voltage Imbalance Information per Feeder.
- Low Voltages.
- Low Voltages per Feeder.
- Distributed Energy Resource - Power Transformer.
- Distributed Energy Resource - Feeder.

1.2. Meter Measurement Data

The Meter Measurement Data display provides the data on customer meters. This display can be accessed by selecting the Go to Station Meter Measurement Tabular option from the RT DMS menu (note that substation name must be entered in the Browser toolbar's text box).

Display call-up URL:

`netview://<Mode/gridtabular/metermeasurements_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the Meter Measurement Data display:

- **Load ID:** ID of the load.
- **Meter ID:** ID of the meter.
- **Device Type:** Type of the device.
- **Energizing Station:** Name of the energizing substation.
- **Energizing Feeder:** Name of the energizing feeder.
- **Last Communication Time:** Time of the last communication with the meter.
- **Measured Volt Phase A/B/C:** Phase A, B, and C measured voltages (in V and per-unit).
- **Calculated Volt Phase A/B/C:** Phase A, B, and C DPF-calculated voltages (in per-unit).
- **Difference Volt Phase A/B/C:** The difference between the measured and DPF-calculated voltages at phase A, B, and C (in percent).
- **Real Power Phase A/B/C:** Real power in phases A, B, and C in kW.
- **Reactive Power Phase A/B/C:** Reactive power in phases A, B, and C in kVAR.
- **Voltage Exception:** Voltage exception state.

Navigation:

- "Locate Transformer" navigates to the distribution transformer on the geographic view.
- "Transformer Loads" opens the Distribution Transformer Load display.

1.3. System Secondary Measurement Data

The System Secondary Measurement Data display provides a list of secondary voltage measurements known by ADMS. This display can be accessed by selecting the Go to System Secondary Measurement Tabular option from the RT DMS menu.

Display call-up URL:

`netview://<Mode/gridtabular/Secondarymeasurements_GridDisplay`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the System Secondary Measurement Data display:

- **ID:** SCADA composite identifier of the measurement.
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- **Device Name:** Name of the measurement component or node.
- **Device ID:** ID of the measurement component or node.
- **Station Name:** Nominal station name.
- **Nominal Feeder:** Nominal feeder name.
- **Energizing Feeder:** Energizing feeder name.
- **Measurement Value:** Measurement value.
- **SCADA Quality:** SCADA quality, such as good, suspect, test mode, or replaced.
- **Measurement Validation (DNAF):** Measurement validation status, such as valid or invalid.
- **Phasing:** Phase designation (for example, ABC) for three phases. A for only phase A.
- **Measured Voltage (User Voltage Base):** Measured voltage, expressed against the configurable voltage base value.
- **Calculated Voltage (User Voltage Base):** Calculated voltage, expressed against the configurable voltage base value.
- **Mismatch:** Absolute difference between the calculated and measured voltage values expressed against the configurable voltage base value.

Navigation:

- "Locate" navigates to the measurement or the substation circle on the geographic view.

1.4. Station Static Schedules

The Station Static Schedules display provides details for static load schedules for substations. The display can be accessed by selecting the Go to Station Static Schedule Tabular option from the RT, ST, or SIM menu. (Note that the substation name must be entered in the Browser toolbar's text box.)

Display call-up URL:

`netview://<Mode>/gridtabular/StationSchedules_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Station Static Schedules display:

- **ID:** ID of the schedule.
- **Number of Points:** Number of schedule points.
- **Name:** Name of the schedule.
- **Schedule Type:** Schedule type (for example, Load-to-Voltage Dependency).
- **X Axis Type:** Type of the schedule's X axis (for example, Temp, Energy, Time, Loading).
- **X Axis Units:** Units of the schedule's X axis (for example, Degree, kWh, Hours, % Rating).
- **Y Axis Type:** Type of the schedule's Y axis (for example, Limit, Billing Rate, Multiplier, Voltage,

Coefficient, Real Power).

- **Y Axis Unit 1:** The first unit of the schedule's Y axis (for example, Amp, Dollars, per unit, % V, No Units, % kW).
- **Y Axis Unit 2:** The second unit of the schedule's Y axis (for example, None, per unit, No Units, PF).
- **Peak Load:** Flag indicating the peak load (Yes or No).
- **Day Type:** Day type (for example, Weekday, Weekend, Holiday).
- **Season:** Season (Summer, Fall, Winter, or Spring).
- **Linear:** Flag indicating whether the schedule is linear (Yes or No).
- **Normalized:** Flag indicating whether the schedule is normalized (Yes or No).

Navigation:

- "Schedule Points" opens the Schedule Points (Static) display.
- "Chart" opens a dialog box with the schedule chart.

1.4.1. Schedule

The Schedule display provides details for static load schedules. The display can be accessed by clicking the Schedules button on the Stations List or Distribution Transformer Load display.

Display call-up URL:

`netview://<Mode>/gridtabular/schedules_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Schedule display:

- **Name:** Name of the schedule.
- **Number of Points:** Number of schedule points.
- **Schedule Type:** The schedule type (for example, Load-to-Voltage Dependency).
- **X Axis Type:** Type of the schedule's X axis (for example, Temp, Energy, Time, Loading).
- **X Axis Units:** Units of the schedule's X axis (for example, Degree, kWh, Hours, % Rating).
- **Y Axis Type:** Type of the schedule's Y axis (for example, Limit, Billing Rate, Multiplier, Voltage, Coefficient, Real Power).
- **Y Axis Unit 1:** The first unit of the schedule's Y axis (for example, Amp, Dollars, per unit, % V, No Units, % kW).
- **Y Axis Unit 2:** The second unit of the schedule's Y axis (for example, None, per unit, No Units, PF).
- **Peak Load:** Flag indicating the peak load (Yes or No).
- **Day Type:** Day type (for example, Weekday, Weekend, Holiday).

- **Season:** Season (Summer, Fall, Winter, or Spring).
- **Linear:** Flag indicating whether the schedule is linear (Yes or No).
- **P Average:** Average real power. This column is hidden by default.
- **Q Average:** Average reactive power. This column is hidden by default.
- **Normalized:** Flag indicating whether the schedule is normalized (Yes or No).
- **ID:** ID of the schedule.

Navigation:

- "Schedule Points" opens the Schedule Points (Static) display.
- "Chart" opens a dialog box with the schedule chart.

1.4.2. Schedule Points (Static)

The Schedule Points display shows the details about static schedule points. The display can be accessed by clicking the Schedule Points button on the Schedule display.

Display call-up URL:

`netview://<Mode>/gridtabular/schedulepointsstatic_GridDisplay/<Substation Name>/<Schedule ID>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Schedule Points display:

- **Schedule ID:** ID of the schedule.
- **X Value:** X value of the schedule point.
- **Y Value 0:** Y0 value of the schedule point.
- **Y Value 1:** Y1 value of the schedule point.

1.5. Station Dynamic Schedules

The Dynamic Schedule for Station display provides details for dynamic load schedules for substations. The display can be accessed by selecting the Go to Station Dynamic Schedule Tabular option from the RT, ST, or SIM menu. (Note that the substation name must be entered in the Browser toolbar’s text box.)

Display call-up URL:

`netview://<Mode>/gridtabular/StationSchedulesDynamic_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Station:** Station name.

- **Device ID:** Device ID.
- **Schedule Type:** Enumeration of the schedule type (for example, 2 = Load-to-Voltage Dependency).
- **X Value:** Value of the schedule's X axis.
- **X Unit:** Unit of the schedule's X axis.
- **Y Value:** Value of the schedule's Y axis.
- **Y Unit:** Unit of the schedule's Y axis.
- **Y2 Value:** Value of the schedule's Y2 axis.
- **Y2 Unit:** Unit of the schedule's Y2 axis.
- **Y3 Value:** Value of the schedule's Y3 axis.
- **Y3 Unit:** Unit of the schedule's Y3 axis.
- **Linear:** Flag indicating whether the schedule is linear (Yes or No).
- **Normalized:** Flag indicating whether the schedule is normalized (Yes or No).

Navigation:

- "Schedule Points" opens the Station Dynamic Schedules display.

1.5.1. Schedule Points (Dynamic)

The Schedule Points display shows details about dynamic schedule points for a device. The display can be accessed from the Station Dynamic Schedules display.

Display call-up URL:

`netview://<Mode>/gridtabular/schedulesdynamic_griddisplay/<Substation>/<LoadID>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Schedule Points display:

- **Station:** Substation name.
- **Device ID:** ID of the device.
- **Schedule Type:** The schedule type (for example, Dynamic Schedule Real Power).
- **X Value:** X-value of the schedule point.
- **X Unit:** Unit of the schedule's X-axis (for example, DateTime).
- **Y Value:** Y-value of the schedule point.
- **Y Unit:** Unit of the schedule's Y-axis (for example, kW).
- **Y2 Value:** Y2-value of the schedule point.
- **Y2 Unit:** Unit of the schedule's Y2-axis (for example, kW).
- **Y3 Value:** Y3-value of the schedule point.

- **Y3 Unit:** Unit of the schedule's Y3-axis (for example, kW).

Navigation:

- "Schedule Points" opens the Schedule Points (Dynamic) display.

1.6. DMS Site Overview

The DMS Site Overview display shows the detailed status of connected servers (Standby, DNAF server, Read-Only Server, and so on), critical server tasks, and model synchronization data. This display allows the user to check the state of the overall DMS-only system without looking through various log files. The user can also determine if the servers are ready for a failover upgrade from a client.

The display shows data relevant to a specific system configuration, such as DMS-only, Read-Only, and Customer Only. In addition, the display can be configured to show Browser clients connected to the server.

> **Note**
>
> Clients and servers (for example, Switch Order server) must connect to the OMS server (NetServer) managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.

The DMS Site Overview display can be accessed from the ADMS menu bar > RT DMS menu > DMS Site Overview.

Display call-up URL:

`netview://RT/gridtabular/DMSSiteOverview_GridDisplay`

The following information is shown on the DMS Site Overview display:

- **Time Stamp:** Connection time stamp.
- **IP Address:** IP address of the connection host.
- **Host Name:** Name of the connection host. This field does not appear by default. You can add it to the display using the %HOSTNAME query field.
- **Application Type:** Application that is connected to the server (Call server, Crew server, DNAF, NetClient, or OMS Main).
- **DMS Dynamic Sequence:** The sequence number acknowledged by the client (main and all connected clients). In a well-behaving system, the sequence number of all clients will be the same and will match the sequence if the dynamics file (dynamics.sav) is viewed in the Snapshot Viewer application.
- **Sync Age:** Number of seconds since last synchronization for all connected clients.
- **Confirmed:** Shows 1 if role confirmation was sent.
- **Role:** Server role (Enabled, Standby, or Read-Only).

- **Switch Order Files Count:** The number of files stored both in the %IDMS_ROOT%\dynamics\server\SwitchOrderServer\DailyLogs and %IDMS_ROOT%\dynamics\server\SwitchOrderServer\SwitchOrders folders.

1.6.1. ADMS Site Overview

The ADMS Site Overview display shows the detailed status of connected servers (Standby, Read-Only Server, Call Server, Crew Server, and so on), critical server tasks, and model synchronization data. This display allows the user to check the state of the overall ADMS system without looking through various log files. The user can also determine if the servers are ready for a failover upgrade from a client.

The display shows data relevant to a specific system configuration, such as DMS+OMS, Read-Only, and Customer Only. In addition, the display can be configured to show Browser clients connected to the server.

> **Note**
>
> Clients and servers (for example, Switch Order server) must connect to the OMS server (NetServer) managed connection port for the OMS server to forward data to the DMS server for Site Overview display queries.

The ADMS Site Overview display can be accessed from the ADMS menu bar > OMS menu > OMS Site Overview.

Display call-up URL:

`netview://RT/gridtabular/IDMSSiteOverview_GridDisplay`

The following information is shown on the ADMS Site Overview display:

- **Time Stamp:** Connection time stamp.
- **IP Address:** IP address of the connection host.
- **Host Name:** Name of the connection host. This field does not appear by default. It can be added to the display using the %HOSTNAME query field.
- **Application Type:** Application that is connected to the server (Call server, Crew server, DNAF, NetClient, or OMS Main).
- **DMS Dynamic Sequence:** The sequence number acknowledged by the client (main and all connected clients). In a well-behaving system, the sequence number of all clients will be the same and will match the sequence if the dynamics file (dynamics.sav) is viewed in the Snapshot Viewer application.
- **Sync Age:** Number of seconds since the last synchronization for all connected clients.
- **Confirmed:** Can be 0 or 1. Shows 1 if role confirmation was sent.
- **Role:** Server role (Enabled, Standby, or Read-Only).
- **Last Message Time:** Last message time stamp.

- **Incident Journal Time:** Incident journal creation time stamp.
- **Call Journal Time:** Call journal creation time stamp.
- **Crew Journal Time:** Crew journal creation time stamp.
- **Call Server Journal Time:** Call Server journal creation time stamp.
- **Alive:** Shows 1 for active connections and 0 for inactive connections.
- **Switch Order Files Count:** The number of files stored both in the %IDMS_ROOT%\dynamics\server\SwitchOrderServer\DailyLogs and %IDMS_ROOT%\dynamics\server\SwitchOrderServer\SwitchOrders folders.

1.7. Advanced Application Plan Status System View

On the Advanced Application Plan Status System View display, the dispatcher can view the execution summary for advanced application plans and also cancel and remove the plans. This display can be accessed from the RT DMS menu by clicking Advanced Application Plan Status View.

For LVM, this display shows the main plan, bank-level sub-plans, and gateway dispatch sub-plans, if any.

Display call-up URL:

`netview://<Mode>/gridtabular/PlanStatusView_GridDisplay`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the Advanced Application Plan Status System View display:

- **Application:** The advanced application plan type, such as FISR Fault, LVM, or AFR Manual.
- **Station:** Name of the substation.
- **Plan ID:** ID of the plan.
- **Rank:** Rank of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Execution Status:** Plan execution status (Not Executed, Queued, Completed, Failed, Executing, or Canceled). For the main LVM plan, the lowest execution status among the sub-plans is shown, as appropriate.
- **Reason Plan Not Executed:** The reason the plan is terminated or canceled (Timed Out, Rejected – LVM Plan Execution in Progress, Third-Party Owned Equipment – Restrictive Execution Mode Applied, New Fault Event).
- **Execution Mode:** Plan execution mode, such as Closed Loop.
- **Start Time:** Plan execution start time.
- **Completion Time:** Plan completion time.
- **Time Elapsed:** Time spent for the plan execution in seconds.
- **Solution Time:** Solution evaluation request time.

- **Execution Time:** Solution execution time.
- **Dispatch Start Time:** The start date and time for a plan dispatch. Applies to LVM DER dispatch plans.
- **Dispatch Duration:** Number of minutes that the plan is dispatched. Applies to LVM DER dispatch plans.

Navigation and controls:

- "Plan Details" navigates to the FISR Plan Details, AFR Plan Details, or LVM Plan Statistics display. Applies to FISR, AFR, and LVM only.
- "Show Control Moves" opens the Plan Control Moves display with plan steps in the lower pane.
- "Prevent Plan Start" cancels the plan. Cancelled plans cannot be implemented on the corresponding plan display. This action only prevents implementation of plans where implementation has not yet started. If a plan is sent for implementation and has already started at least one step, the plan will complete to prevent partial implementation.
- "Remove" removes the plan from the Advanced Application Plan Status System View display and from the corresponding plan display.
- "Mark Devices" marks optimization devices included in the plan.
- "Feeders in Plan" navigates to a geographic view filtered to show only feeders included in the plan.

1.7.1. Plan Control Moves

The Plan Control Moves display can be accessed from the Advanced Application Plan Status System View display. On the Plan Control Moves display, you can view the switching steps for the selected plan.

Display call-up URL:

`netview://<Mode>/gridtabular/PlanStepsView_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Step Number:** Step number of the move in the plan.
- **Subplan ID:** Sub-plan identifier. Applies to FISR plans when the parallel switching sequence functionality is enabled, and plans are split into sub-plans to be implemented in parallel.
- **Device Type:** The device-type of the component (for example, SWITCH, JUMPER).
- **Device Name:** The name/number of the device as derived from the Asset Management System (the source of the Station Model Files).
- **Device ID:** Unique identifier of the device.
- **Station:** Dynamic substation name.

- **Controllable:** Indicates if the device can be controlled (yes or no).
- **Initial State:** Initial position of the device.
- **Target State:** Final position of the device.
- **Status:** Step implementation status (pending, not executed, or completed).
- **Completion Time:** Step completion time.
- **Control Type:** The control method used for the device (for example, Raise/Lower).
- **Settings Group Template:** The name of the associated settings group template, if modeled.
- **Number of State Change Devices:** Shows the number of switching devices that changed their state during the solution evaluation and impact the associated plan.
- **Restoration Section Information:** Description of the involved restoration section information.

Navigation and controls:

- "Locate" locates the device on the geographic view in the current viewport.
- "Settings Group Template" shows the Settings Group Templates and Protection Settings Template Information displays with the devices' settings group template information.
- "Number of State Change Devices" shows the State Change Switching Devices display in the new viewport.

1.8. Station Exception

The Station Exception display summarizes abnormalities and violations by station and is used as a basis for navigating to more-detailed information about the exceptions. A separate Station Exception is provided for real-time, study, and simulator modes.

The Station Exception provides a high-level perspective of the activities within the dispatchers' Area of Responsibilities (AOR). It only displays stations where the dispatcher has been granted write access.

This display has the following features:

- By default, substations are sorted and presented to the user to show the worst conditions in the following priority order: de-energization (number of de-energized customers), abnormals, high-voltage violations, low-voltage violations, voltage imbalance violations, overload violations, flow imbalance violations, and ground return current rate.
- Exceptions and violations are highlighted in red on the display. These red values are linked to displays giving more details.
- Clicking the substation name navigates to the exception summary for each feeder of that substation.

The Station Exception display can be accessed from the RT, ST, or SIM menu. It can also be accessed from the geographic view by pressing Ctrl + S.

Display call-up URL:

`netview://<Mode>/gridtabular/stationException_GridDisplay/<Substation Name>`

Or:

`netview://<Mode>/gridtabular/stationExceptionRF_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

Display call-up URL for SIM mode:

`netview://SIM/gridtabular/stationExceptionSIM_GridDisplay/<Substation Name>`

The following information is shown on the StationException display:

- **Station:** The list of Substations for which the dispatcher currently has operational control. Clicking the Substation Name navigates to the Feeder Exception Summary display for that station.
- **Area:** The area or division in which the substation is located.
- **De-Energized Customers:** The number of customers (loads) that are currently de-energized. Clicking the de-energization value navigates to the De-Energization display for the substation.
- **Abnormal Topology Conditions:** The total number of switches in their abnormal state and temporary modifications belonging to the substation. Clicking the value navigates to the Station Abnormal Summary display.
- **Highest Load or Bus Voltage (%):** The highest voltage that exists on a load or a bus in the substation as calculated by the Limit Monitor functions and expressed as a percentage. If there is a high voltage violation inside or outside the station, the value is highlighted in red. Clicking the value navigates to the High Voltages display. Right-clicking the value navigates to the device with highest voltage inside or outside the station on the geographic display.
- **Lowest Load or Bus Voltage (%):** The lowest voltage that exists on a load or a bus in the substation as calculated by the Limit Monitor functions and expressed as a percentage. If there is a low voltage violation inside or outside the station, the value is highlighted in red. Clicking the value navigates to the Low Voltages display. Right-clicking the value navigates to the device with lowest voltage inside or outside the station on the geographic display.
- **Largest Voltage Imbalance:** The largest voltage imbalance value that exists at the substation as calculated by the Network Analysis functions, expressed as a percentage of nominal voltage. If the cell background is red, this indicates that the value exceeds the voltage imbalance limit. Clicking the value navigates to the Voltage Imbalance Information display for the substation.
- **Largest Flow Imbalance:** The largest flow imbalance that exists at the substation as calculated by the Network Analysis functions. If the cell background is red, this indicates that flow imbalance exceeds the flow imbalance limit. Clicking the red value navigates to the Flow Imbalance Violation display for the substation.
- **Maximum Loading:** The highest loading that exists for a network element connected to the substation as calculated by the Network Analysis functions, expressed as a percentage of the branch flow limits (flow/limit * 100). This is the greatest overload of the highest overload level. For example, if the highest violation limit of Level 1 is 300% (for example, due to a manual entry)

and the highest violation limit of Level 3 is 200%, the highest violation limit of Level 3 (200%) is reported.

If the loading exceeds the limit, then the cell background is red. If a network element connected to this station has capacity margin violations without flow violations, then the cell background is blue.

Clicking the cell navigates to the Branch Flows display for the station.

- **Ground Return Current:** The highest level of imbalance flow that exists for a breaker or recloser, expressed as a percentage of the limit value. If the cell background is red, then it indicates that the imbalance flow exceeds the limit. Clicking the value navigates to the Ground Return Current Information display for the substation.
- **Neutral Current:** The neutral current percentage, which is calculated as the neutral current divided by the product of the neutral relay current trip setting and the user-configured safety factor. If the calculated percentage is greater than 100, the cell color is shown in red to indicate that there is a neutral current violation. Clicking the value navigates to the Neutral Current Information display, which shows neutral relay setting and neutral current for the breakers and reclosers.
- **DPF Status:** The current status of the Distribution Power Flow solution for the substation. The possible states are Unsolved, Solving, Solved, Suspect, Rejected, or Failed.
- **PRV Status:** The current status of the Protection Validation solution for the substation. The possible states are Unsolved, Solving, Solved – No Violations, Solved – Interruption Violation, Solved – Pick-up Violation, Solved – Violations, Rejected or Failed.
- **Reserved Capacity Impact**: Navigates to the Reserved Capacity Impact for Station display.

Navigation:

- "Station" opens the Feeder Exception Summary display for the selected station.
- "De-Energized Customers" navigates to the De-Energization display for the substation.
- Abnormal Device Count: If the cell background is red, opens the Station Abnormal Summary display.
- Highest Load or Bus Voltage: Clicking the value opens the High Voltages display. Right-clicking the value navigates to the device with highest voltage inside or outside the station on the geographic display.
- Lowest Load or Bus Voltage: Clicking the value opens the Low Voltages display. Right-clicking the value navigates to the device with lowest voltage inside or outside the station on the geographic display.
- Largest Voltage Imbalance: Navigates to the Voltage Imbalance Information display.
- Largest Flow Imbalance: If the cell background is red, then it opens the Flow Imbalance Violation display.
- Maximum Loading: Navigates to the Branch Flows display.
- Ground Return Current: Navigates to the Ground Return Current Information display.

- Neutral Current: Navigates to the Neutral Current Information display.
- DPF status: Opens the Distribution Power Flow Summary for Station and DPF Feeder Results in the same window.
- PRV Status Impact: Opens the Protection Validation Results display.
- Reserved Capacity Impact: Opens the Reserved Capacity Impact for Station display.

1.8.1. Feeder Exception Summary

The Feeder Exception Summary display has the same capability as the Station Exception display, with the difference that it shows the exceptions by feeder. When you select values in a column, a detailed summary is presented, showing the exceptions for that particular feeder only. This display can be accessed by clicking a station name on the Station Exception display.

Display call-up URL:

`netview://<Mode>/gridtabular/FeederException_griddisplay/<Substation Name>`

Or:

`netview://<Mode>/gridtabular/feederExceptionRF_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the FeederException display:

- **Feeder:** List of feeders at the substation.
- **Area:** The area or region in which the feeder is located.
- **De-Energized Customers:** The number of customers that are currently de-energized. Clicking the de-energization value navigates to the De-Energization Customers per Feeder display for the feeder.
- **Abnormal Topology:** The number of switches in their abnormal state and temporary modifications belonging to the feeder. Clicking the value navigates to the Abnormal Components per Feeder display.
- **Highest Load or Bus Voltage (%):** The highest voltage that exists on the feeder as calculated by the Network Analysis functions, expressed as a percentage above nominal voltage. If there is a high voltage violation on the feeder, then the value is highlighted in red. Clicking the value navigates to the High Voltages per Feeder display for the feeder. Right-clicking the value navigates to the device with highest voltage on the geographic display.
- **Lowest Load or Bus Voltage (%):** The lowest voltage that exists at the feeder as calculated by the Network Analysis functions, expressed as a percentage below nominal voltage. If there is a low voltage violation on the feeder, then the value is highlighted in red. Clicking the value navigates to the Low Voltages per Feeder display for the feeder. Right-clicking the value navigates to the device with lowest voltage on the geographic display.
- **Largest Voltage Imbalance:** The largest voltage imbalance value that exists at the feeder as calculated by the Network Analysis functions, expressed as a percentage of nominal voltage. If

the cell background is red, then this indicates that voltage imbalance violations exist for the feeder. Clicking the value navigates to the Voltage Imbalance Information per Feeder display for the feeder.

- **Largest Flow Imbalance:** The largest flow imbalance that exists at the feeder as calculated by the Network Analysis functions. If the cell background is red, then this indicates that flow imbalance violations exist for the feeder. Clicking the red value navigates to the Flow Imbalance Violation per Feeder display for the feeder.
- **Maximum Loading:** The highest loading that exists for a network element connected to the substation as calculated by the Network Analysis functions, expressed as a percentage of the branch flow limits (flow/limit * 100). This is the greatest overload of the highest overload level. For example, if the highest violation limit of Level 1 is 300% (for example, due to a manual entry) and the highest violation limit of Level 3 is 200%, then the highest violation limit of Level 3 (200%) is reported.

  If the loading exceeds the limit, then the cell background is red. If a network element connected to this station has capacity margin violations without flow violations, then the cell background is blue.

  Clicking the cell navigates to the Branch Flows per Feeder display for the station.
- **Ground Return Current:** Ground return current. If the cell background is red, then it indicates that there is a violation. Clicking a cell value navigates to the Ground Return Current Information per Feeder display.
- **Neutral Current:** The neutral current percentage, which is calculated as the neutral current divided by the product of the neutral relay current trip setting and the user-configured safety factor. If the calculated percentage is greater than 100, the content color is changed to red to indicate that there is a neutral current violation. Clicking the value navigates to the Neutral Current Information per Feeder display, which displays the shows neutral relay setting and neutral current for the breakers and reclosers.
- **DPF Status:** The current status of the Distribution Power Flow (DPF) solution for the Feeder. The possible states are Unsolved (indicating that the DPF has not been run), Solved (indicating that the DPF has been run successfully), Suspect (indicating that the solution has unrealistically low voltages) or Not Converged (indicating that the DPF was unable to converge on a solution).
- **Reserved Capacity Impact:** Navigates to the Reserved Capacity Impact for Feeder display.
- **Distributed Energy Resource - Power Transformer:** Navigates to the Distributed Energy Resource - Power Transformer display.
- **Distributed Energy Resource - Feeder:** Navigates to the Distributed Energy Resource - Feeder display.

Navigation:

- De-Energized Customers: If there are de-energized customers, opens the De-Energization Customers per Feeder display.
- Abnormal Topology: If cell is red, opens the Abnormal Components per Feeder display.
-

Feeder Breaker: Navigates to the feeder breaker on the station internals display.

- Highest Load or Bus Voltage: Clicking the value opens the High Voltages per Feeder display. Right-clicking navigates to the device with highest voltage inside or outside the station on the geographic display.
- Lowest Load or Bus Voltage: Clicking the value opens the Low Voltages per Feeder display. Right-clicking the value navigates to the device with lowest voltage inside or outside the station on the geographic display.
- Largest Voltage Imbalance: Navigates to the Voltage Imbalance Information per Feeder display.
- Largest Flow Imbalance: If cell is red, then it opens the Flow Imbalance Violation per Feeder display.
- Maximum Loading: Navigates to the Branch Flows per Feeder display.
- Ground Return Current: Navigates to the Ground Return Current Information per Feeder display.
- Neutral Current: Navigates to the Neutral Current Information per Feeder display.
- Reserved Capacity Impact: Navigates to the Reserved Capacity Impact for Feeder display.
- Distributed Energy Resource - Power Transformer: Navigates to the Distributed Energy Resource - Power Transformer display.
- Distributed Energy Resource - Feeder: Navigates to the Distributed Energy Resource - Feeder display.

1.8.2. Station Abnormal Summary

The Station Abnormal Summary display shows switches and permanent jumpers in abnormal position, temporary grounds, and cuts and jumpers. The Station Abnormal Summary display can be accessed by clicking the Abnormal Topology on the Station Exception display.

The date and time the abnormal condition was detected are provided for switches, permanent jumpers in abnormal position, temporary grounds, cuts, and jumpers. Each abnormal condition also references the area, substation, and type of change that occurred.

Switches and temporary modifications are listed by switch name in chronological order of the abnormality event. Clicking the Locate button navigates the user to the specified device on the geographic display. Clicking the Tag button opens the tagging operations pop-up.

Display call-up URL:

`netview://<Mode>/gridtabular/Abnormal_griddisplay/<Substation Name>`

Or:

`netview://<Mode>/gridtabular/AbnormalRF_griddisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Abnormal display:

- **Date Time:** Date and time when the change of state occurred. This is the time that the model was changed, for non-telemetered devices it does not necessarily match the time that the switching took place in the field.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (for example, SWITCH, JUMPER).
- **Name:** The name of the network component.
- **ID:** The unique identifier of the network component.
- **Description:** Description of the network component.
- **Tag:** Indicates whether the device is currently tagged. If clicked, opens the Tagging Operations pop-up.
- **Status:** The current status of the abnormal component, such as Open or Disconnected.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Associated Transformer:** ID of the associated transformer if the component is an elbow switch.
- **Is Orphan:** Indicator (Yes/No) of whether the device is orphaned.
- **Op - Center:** The name of the service center.

Navigation:

- "Locate" locates the component on the geographic (or schematic) network view in the same viewport.
- "Attributes" opens the Operator Attributes pop-up.
- "Tag" opens the Tagging Operations pop-up.

1.8.3. Abnormal Components per Feeder

The Abnormal Components per Feeder display shows network elements in abnormal position with respect to the selected feeder. It informs about the time when the abnormal position of the network element was detected. Each abnormal condition also references the area, substation, and type of change that occurred. This display can be accessed from the Feeder Exception Summary.

Display call-up URL:

`netview://<Mode>/gridtabular/AbnormalFdr_griddisplay/<Substation Name>/<Feeder Name>`

Or:

`netview://<Mode>/gridtabular/AbnormalFdrRF_griddisplay/<Substation Name>/<Feeder Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the AbnormalFdr display:

- **Date Time:** Date and time when the change of state occurred. This is the time that the model was changed, for non-telemetered devices it does not necessarily match the time that the switching took place in the field.

- **Name:** The name of the network component.
- **Component Type:** The device-type of the component (for example, SWITCH, JUMPER).
- **Attributes:** Displays the operator attributes of the component.
- **Locate:** Clicking this button navigates to a geographic display of the network, centered on the component.
- **Feeder Name:** Name of the associated feeder.
- **Tag:** Indicates whether the device is currently tagged. If clicked, opens the Tag Operations pop-up.
- **Status:** The current status of the device (Open, Closed, or Inserted).
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Associated TF:** ID of the associated transformer if the component is an elbow switch.

Navigation:

- "Locate" locates the component on the geographic (or schematic) network view in the same viewport.
- "Attributes" opens the Operator Attributes pop-up.

1.8.4. De-Energization

The De-Energization display shows de-energized loads and transformers along with the associated customer count and the date and time of de-energization. This display can be accessed via the Station Exception or Feeder Exception Summary displays. Clicking the Locate button navigates to the specified device on the geographic display.

Display call-up URL:

`netview://<Mode>/gridtabular/deenergization_GridDisplay/<Substation Name>`

Or:

`netview://<Mode>/gridtabular/deenergizationRF_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the De-Energization display:

- **Time De-Energized:** Date and time when the de-energization occurred.
- **Name:** Name of the network component.
- **Load ID:** Identifier of the network component.
- **Feeder:** Feeder Name.
- **Component Type:** The device-type of the component (for example, Load).
- **Customer Count:** The number of associated customers that were de-energized.

Navigation:

- "Locate" locates the load in the current viewport.

1.8.5. De-Energization Customers per Feeder

The De-Energization Customers per Feeder display is similar to the De-Energization display with the difference that it focuses on a specific feeder.

Display call-up URL:

netview://<Mode>/gridtabular/deenergizationFdr_GridDisplay/<Substation Name>/<Feeder Name>

Or:

netview://<Mode>/gridtabular/deenergizationFdrRF_GridDisplay/<Substation Name>/<Feeder Name>

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

1.8.6. High Voltages

The High Voltages display shows violations calculated by the Limit Monitor function that is run as part of the Distribution Power Flow (DPF) solution. This display can be accessed from the Station Exception display by right-clicking the Highest Load or Bus Voltage cell.

Voltage violations are evaluated for all loads. The violations are ordered such that the most severe violations are presented first.

Dispatchers are alerted to feeder voltage violations through the issuing of an alarm. Navigation is provided from the feeder voltage alarm on the Alarm Summary to this Voltage Violation Detail.

Display call-up URL:

`netview://<Mode>/gridtabular/voltageViolationHigh_GridDisplay/<Substation Name>/`

Or:

`netview://<Mode>/gridtabular/voltageViolationHighRF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the voltageViolationHigh display:

- **Name:** The name of the component where the violation exists.
- **ID:** The unique identifier of the component. If the component is a load, the name of the load is displayed. If the component is a bus, the ID of the bus is displayed.
- **Identifier:** The identifier (name) of the component.
- **Feeder:** Feeder name.

- **Component Type:** The device-type of the component (for example, Load, Line).
- **Connection Type**: The connection type of the component (for example, Delta, Wye).
- **Limit Name:** The name of the Limit that has been violated (for example, NORMAL, EMERGENCY).
- **Limit Level:** Level of the highest violated limit.
- **Limit Value:** The voltage limit value expressed in terms of the 120 V base.
- **Voltage:** The actual voltage calculated by DPF, expressed in terms of the 120 V base.
- **Voltage:** The amount that the actual voltage deviates from the nominal value, expressed as a percentage.

Navigation:

- "Locate" locates the device in the current viewport.

1.8.7. High Voltages per Feeder

The High Voltages per Feeder display shows violations calculated by the Limit Monitor function that is run as part of the DPF solution. This display can be accessed via the Feeder Exception Summary display by right-clicking the Highest Load or Bus Voltage cell.

Voltage violations are evaluated for all loads. The violations are ordered such that the most severe violations are presented first.

Dispatchers are alerted to feeder voltage violations through the issuing of an alarm. Navigation is provided from the feeder voltage alarm on the Alarm Summary to this Voltage Violation Detail.

The High Voltages per Feeder display is similar to the High Voltages display, except that it only shows data for the selected feeder.

Display call-up URL:

`netview://<Mode>/gridtabular/voltageViolationHighFdr_GridDisplay/<Substation Name>/<Feeder Name>`

Or:

`netview://<Mode>/gridtabular/voltageViolationHighFdrRF_GridDisplay/<Substation Name>/<Feeder Name>`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on this display:

- **ID:** The Name of the component where the violation exists.
- **Identifier:** The unique identifier of the component. If the component is a load, the name of the load is displayed. If the component is a bus, the ID of the bus is displayed.
- **Feeder:** Feeder Name.
-

Component Type: The device-type of the component (for example, Load, Line).

- **Connection Type**: The connection type of the component (for example, Delta, Wye).
- **Limit Name:** The name of the Limit that has been violated (for example, NORMAL, EMERGENCY).
- **Limit Level:** Level of the highest violated limit.
- **Limit Value:** The voltage limit value expressed in terms of the 120 V base.
- **Voltage:** The actual voltage calculated by DPF, expressed in terms of the 120 V base.
- **Voltage:** The amount that the actual voltage deviates from the nominal value, expressed as a percentage.

Navigation:

- "Locate" locates the device in the current viewport.

1.8.8. Low Voltages

The Low Voltages display shows violations calculated by the Limit Monitor function that is run as part of the DPF solution. This display can be accessed from the Station Exception display by right-clicking the Lowest Load or Bus Voltage cell.

Voltage violations are evaluated for all loads. The violations will be ordered such that the most severe violations are presented first.

Dispatchers will be alerted to feeder voltage violations through the issuing of an alarm. Navigation will be provided from the feeder voltage alarm on the Alarm Summary to this Voltage Violation Detail.

Display call-up URL:

```
netview://<Mode>/gridtabular/voltageViolationLow_GridDisplay/<Substation Name>/
```

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the Low Voltages display:

- **Name:** The name of the component where the violation exists.
- **ID:** The unique identifier of the component. If the component is a load, then the name of the load is displayed. If the component is a bus, the ID of the bus is displayed.
- **Identifier:** The identifier (name) of the component.
- **Feeder:** Feeder Name.
- **Component Type:** The device-type of the component (for example, Load, Line).
- **Connection Type**: The connection type of the component (for example, Delta, Wye).
- **Limit Name:** The name of the Limit that has been violated (for example, NORMAL, EMERGENCY).

- **Limit Level:** Level of the highest violated limit.
- **Limit Value:** The voltage limit value expressed in terms of the 120 V base.
- **Voltage:** The actual voltage calculated by DPF, expressed in terms of the 120 V base.
- **Voltage:** The amount that the actual voltage deviates from the nominal value, expressed as a percentage.

Navigation:

- "Locate" locates the device in the current viewport

1.8.9. Low Voltages per Feeder

The Low Voltages per Feeder display shows violations calculated by the Limit Monitor function that is run as part of the DPF solution. This display can be accessed from the Feeder Exception Summary display by right-clicking the Lowest Load or Bus Voltage cell.

Voltage violations are evaluated for all loads. The violations will be ordered such that the most severe violations are presented first.

Dispatchers will be alerted to feeder voltage violations through the issuing of an alarm. Navigation will be provided from the feeder voltage alarm on the Alarm Summary to this Voltage Violation Detail.

The Low Voltages per Feeder display is similar to the Low Voltages display, except it only shows information for the selected feeder.

Display call-up URL:

`netview://<Mode>/gridtabular/voltageViolationLowFdr_GridDisplay/<Substation Name>/<Feeder Name>`

Or:

`netview://<Mode>/gridtabular/voltageViolationLowFdrRF_GridDisplay/<Substation Name>/<Feeder Name>`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the Low Voltages per Feeder display:

- **ID:** The Name of the component where the violation exists.
- **Identifier:** The unique identifier of the component. If the component is a load, the name of the load is displayed. If the component is a bus, the ID of the bus is displayed.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (for example, Load, Line).
- **Connection Type**: The connection type of the component (for example, Delta, Wye).
- **Limit Name:** The name of the Limit that has been violated (for example, NORMAL, EMERGENCY).

- **Limit Level:** Level of the highest violated limit.
- **Limit Value:** The voltage limit value expressed in terms of the 120 V base.
- **Voltage:** The actual voltage calculated by DPF (Distribution Power Flow), expressed in terms of the 120 V base.
- **Voltage:** The amount that the actual voltage deviates from the nominal value, expressed as a percentage.

Navigation:

- "Locate" locates the device in the current viewport.

1.8.10. Voltage Imbalance Information

The Voltage Imbalance Information display shows the voltage imbalance value that evaluated for all three-phase loads and are ordered such that the loads with the highest imbalance value are presented first. You can access the Voltage Imbalance Information display by clicking the Voltage Imbalance field on the Station Exception display.

Dispatchers are alerted to feeder voltage violations through the issuing of an alarm. Navigation is provided from the feeder voltage alarm on the Alarm Summary to these Voltage Imbalance Information details.

Display call-up URL:

`netview://<Mode>/gridtabular/voltageViolationImbalance_GridDisplay/<Substation Name>/`

Or:

`netview://<Mode>/gridtabular/voltageViolationImbalanceRF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the voltageViolationImbalance display:

- **Name:** The name of the network component.
- **ID:** The unique identifier of the component where the violation exists.
- **Energizing Feeder:** Energizing feeder name.
- **Component Type:** The device-type of the component (for example, LOAD, LINE).
- **Limit Value:** The violation limit level in percentage.
- **Voltage Imbalance Percent:** The actual level of voltage imbalance calculated by DPF (Distribution Power Flow) in percentage.

Navigation:

- "Locate" locates the device in the current viewport.

1.8.11. Voltage Imbalance Information per Feeder

The Voltage Imbalance Information per Feeder display shows the voltage imbalance value that evaluated for all three-phase loads and are ordered such that the loads with the highest imbalance value are presented first. You can access the Voltage Imbalance Information display by clicking the Voltage Imbalance field on the Feeder Exception Summary display.

Dispatchers are alerted to feeder voltage violations through the issuing of an alarm. Navigation is provided from the feeder voltage alarm on the Alarm Summary to this Voltage Imbalance Information Detail.

The Voltage Imbalance Information per Feeder display is similar to the Voltage Imbalance Information display except that it shows information for the given feeder.

Display call-up URL:

```
netview://<Mode>/gridtabular/voltageViolationImbalanceFdr_GridDisplay/<Substation
Name>/<Feeder Name>
```

Or:

```
netview://<Mode>/gridtabular/voltageViolationImbalanceFdrRF_GridDisplay/<Substation
Name>/<Feeder Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the voltageViolationImbalanceFdr display:

- **Name:** The name of the network component.
- **ID:** The unique identifier of the component where the violation exists.
- **Energizing Feeder:** Energizing feeder name.
- **Component Type:** The device-type of the component (for example, LOAD, LINE).
- **Limit Value:** The violation limit level in percentage.
- **Voltage Imbalance Percent:** The actual level of voltage imbalance calculated by DPF (Distribution Power Flow) in percentage.

Navigation:

- "Locate" locates the device in the current viewport.

1.8.12. Flow Imbalance Violation

The Flow Imbalance Violation display show violations calculated by the Limit Monitor function that is run as part of the Power Flow solution. You can access the Flow Imbalance Violation display by clicking the Flow Violation field on the Station Exception display.

Flow violations are evaluated for all switches. The violations are ordered such that the most severe violations are presented first.

Display call-up URL:

`netview://<Mode>/gridtabular/flowViolationImbalance_GridDisplay/<Substation Name>/`

Or:

`netview://<Mode>/gridtabular/flowViolationImbalanceRF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the flowViolationImbalance display:

- **Name:** The name of the network component.
- **ID:** The unique identifier of the component where the violation exists.
- **Feeder:** Feeder Name.
- **Component Type:** The device-type of the component (for example, Switch, Breaker).
- **Amps A/B/C:** Phase A/B/C current in amps.
- **Limit Value:** The violation limit level in percentage.
- **Flow Imbalance:** The actual level of flow imbalance calculated by DPF in percentage.

Navigation:

- "Locate" locates the device in the current viewport

1.8.13. Flow Imbalance Violation per Feeder

The Flow Imbalance Violation per Feeder display shows violations calculated by the Limit Monitor function that is run as part of the Power Flow solution. You can access the Flow Imbalance Violation per Feeder display by clicking the Flow Violation field on the Feeder Exception Summary display.

Flow violations are evaluated for all switches. The violations are ordered such that the most severe violations are presented first.

The Flow Imbalance Violation per Feeder display is similar to the Flow Imbalance Violation display except it only shows the flow violations for the selected feeder.

Display call-up URL:

`netview://<Mode>/gridtabular/flowViolationImbalanceFdr_GridDisplay/<Substation Name>/<Feeder Name>`

Or:

`netview://<Mode>/gridtabular/flowViolationImbalanceFdrRF_GridDisplay/<Substation Name>/<Feeder Name>`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the voltageViolationImbalanceFdr display:

- **ID:** The unique identifier of the component where the violation exists.

- **Name:** The name of the network component.
- **Feeder:** Feeder Name.
- **Component Type:** The device-type of the component (for example, Switch, Breaker).
- **Amps A/B/C:** Phase A/B/C current in amps.
- **Limit Value:** The violation limit level in percentage.
- **Flow Imbalance:** The actual level of flow imbalance calculated by DPF in percentage.

Navigation:

- "Locate" locates the device in the current viewport

1.8.14. Branch Flows

The Branch Flows display shows the flow limit details calculated by the Limit Monitor function run as part of the Distribution Power Flow solution. This display can be accessed by: Station Exception > Maximum Loading (%).

Branch loading is determined for lines, switching devices, transformers, and reactors. The branch flows are ordered such that the highest flows are presented first.

Notice the Locate button on these displays. Clicking this button navigates the user to the geographic or feeder schematic display with the component in question centered on the display and highlighted.

Dispatchers are alerted to feeder overloads through the issuing of an alarm.

Display call-up URL:

`netview://<Mode>/gridtabular/BranchFlows_GridDisplay/<Substation Name>`

Or:

`netview://<Mode>/gridtabular/BranchFlowsRF_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Branch Flows per Feederdisplay:

- **Name:** The name of the network component.
- **ID:** The unique identifier of the component.
- **Feeder:** Feeder Name.
- **Component Type:** The device-type of the component (for example, Load, Line).
- **Limit Name:** The type of limit that has been violated (that is, Long Term, Medium Term, Short Term). For components without violations, shows No Violation.
- **Limit Level:** The highest violated limit value. The values are expressed as numbers from 1 (least severe) to 3 (most severe). Note that an overload caused by a manual entry is always considered a Level 1 overload.

- **Limit Value (per Phase):** The flow limit level per phase (amps).
- **Derated Limit Value (per Phase)**: The derated flow limit level per phase (amps).
- **Flow Value:** The actual current level calculation by DPF (Distribution Power Flow) in amperes.
- **% Limit:** The actual flow value expressed as a percentage of the limit value.

Navigation:

- "Locate" locates component in the same viewport on the geographic display.

1.8.15. Branch Flows per Feeder

This display is similar to the Branch Flows display with the difference that it focuses on the currently selected feeder.

Display call-up URL:

`netview://<Mode>/gridtabular/BranchFlowsfdr_GridDisplay/<Substation Name>/<Feeder Name>`

Or:

`netview://<Mode>/gridtabular/BranchFlowsfdrRF_GridDisplay/<Substation Name>/<Feeder Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Branch Flows per Feeder display:

- **Name:** The name of the network component.
- **ID:** The unique identifier of the component.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (for example, Load, Line).
- **Limit Name:** The type of limit that has been violated (that is, Long Term, Medium Term, Short Term). For components without violations, shows No Violation.
- **Limit Level:** The highest violated limit value. The values are expressed as numbers from 1 (least severe) to 3 (most severe). Note that an overload caused by a manual entry is always considered a Level 1 overload.
- **Flow Limit Phase A/B/C:** The current flow limit level for phases A, B, and C (amps).
- **Flow Value Phase A/B/C:** The individual phase flow value for phases A, B, and C (amps).
- **Flow Value:** The actual current level calculation by DPF (Distribution Power Flow) in amperes.
- **% Limit:** The actual flow value expressed as a percentage of the limit value.

Navigation:

- "Locate" locates a component in the same viewport on the geographic display.

1.8.16. Ground Return Current Information

This display lists switches limit amperage and imbalance flow values. The display header also shows the neutral current safety factor. This display can be opened from the Station Exception display.

Display call-up URL:

`netview://<Mode>/gridtabular/groundreturn_GridDisplay/`

Or:

`netview://<Mode>/gridtabular/groundreturnrf_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the switch.
- **ID:** Switch unique identifier.
- **Alternate ID:** Switch second unique identifier.
- **Energizing Feeder:** Name of the energizing feeder.
- **Switch Type:** Switch type.
- **Minimum Ground Pick-up Limit (Amps):** Minimum ground pick-up limit in Amps.
- **Ground Imbalance Flow (Amps):** Ground imbalance flow value in Amps.
- **Minimum Ground Pick-up Limit (%):** Minimum ground pick-up limit in percentage.

Navigation:

- "Locate" navigates to the switch on the geographic view.

1.8.17. Ground Return Current Information per Feeder

This display lists feeders with limit and imbalance flow values in amperes. The display header also shows the neutral current safety factor.

This display is similar to the Ground Return Current Information display with the exception that it has a feeder view. This display can be opened from the Feeder Exception Summary display.

Display call-up URL:

`netview://<Mode>/gridtabular/groundreturnfdr_GridDisplay/<StationName>/<FeederName>`

Or:

`netview://<Mode>/gridtabular/groundreturnfdrRF_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

-

Name: Name of the switch.

- **ID:** Switch unique identifier.
- **Alternate ID:** Switch second unique identifier.
- **Energizing Feeder:** Name of the energizing feeder.
- **Switch Type:** Type of switch.
- **Minimum Ground Pick-up Limit (Amps):** Minimum ground pick-up limit in Amps.
- **Ground Imbalance Flow (Amps):** Ground imbalance flow value in Amps.
- **Minimum Ground Pick-up Limit (%):** Minimum ground pick-up limit in percentage.

Navigation:

- "Locate" navigates to the network element on the geographic view.

1.8.18. Neutral Current Information

The Neutral Current Information display shows neutral relay setting and neutral current for the breakers and reclosers for a station. The display header also shows the neutral current safety factor. This display can be opened from the Station Exception display.

Display call-up URL:

`netview://<Mode>/gridtabular/NeutralRelayPRV_GridDisplay/<StationName>`

Or:

`netview://<Mode>/gridtabular/NeutralRelayPRVRefreshable_GridDisplay/<StationName>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** The name of the switching device.
- **ID:** The unique identifier of the switching device.
- **Alternate ID:** The second or alternate identifier of the switching device.
- **Energizing Station:** The energizing station for the switching device.
- **Energizing Feeder:** The energizing feeder for the switching device.
- **Device Type:** The type of the switching device (for example, breaker or recloser).
- **Switch Status:** The current state of the switching device (for example: open or closed).
- **Neutral Relay Setting:** The maximum allowed neutral trip setting (in Amps) for a switching device.
- **Neutral Current:** The calculated neutral current (in Amps) of the switching device.
- **Neutral Current Violation (%):** Shows whether the calculated neutral current exceeds the product of the neutral relay settings of the switch and the user configured safety factor. The

field background is shown in red if the Neutral Current Violation value is above 100%.

Navigation:

- "Locate" navigates to the switching device on the geographic view.

1.8.19. Neutral Current Information per Feeder

The Neutral Current Information per Feeder display shows neutral relay setting and neutral current for the breakers and reclosers for a given feeder. The display header also shows the neutral current safety factor. This display can be opened from the Feeder Exception Summary display.

Display call-up URL:

```
netview://<Mode>/gridtabular/NeutralRelayPRVFeeder_GridDisplay/<StationName>/<FeederName>
```

Or:

```
netview://<Mode>/gridtabular/NeutralRelayPRVFeederRefreshable_GridDisplay/<StationName>/<FeederName>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** The name of the switching device.
- **ID:** The unique identifier of the switching device.
- **Alternate ID:** The second or alternate identifier of the switching device.
- **Energizing Station:** The energizing station for the switching device.
- **Energizing Feeder:** The energizing feeder for the switching device.
- **Device Type:** The type of the switching device (for example, breaker or recloser).
- **Switch Status:** The current state of the switching device (for example: open or closed).
- **Neutral Relay Setting:** The maximum allowed neutral trip setting (in Amps) for a switching device.
- **Neutral Current:** The calculated neutral current (in Amps) of the switching device.
- **Neutral Current Violation (%):** Shows whether the calculated neutral current exceeds the product of the neutral relay settings of the switch and the user configured safety factor. The field background is shown in red if the Neutral Current Violation value is above 100%.

Navigation:

- "Locate" navigates to the switching device on the geographic view.

1.8.20. Reserved Capacity Impact for Station

The Reserved Capacity Impact display is accessed from the Station Exception and Station Exception

RF displays. It provides detailed information about the components affected by reserved capacity objects in the station.

Display call-up URL:

`netview://<Mode>/gridtabular/ReservedCapacityImpact_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on Reserved Capacity Impact display:

- **Name:** Name of the component.
- **ID:** ID of the component.
- **Locate:** Navigates to the component on the network view.
- **Feeder:** The nominal feeder of the component.
- **Component Type:** The component type string for the component.
- **Flow Value:** The DPF calculated flow value with units.
- **RC Impact:** The reserved capacity derating value for the component.
- **Limit Value:** The value for the maximum violated limit level or limit level 1 if no limit is violated.
- **% Limit:** The loading of the component calculated by ((Flow Value + RC Impact) / Limit Value).
- **Capacity Available:** The capacity available excluding the reserved capacity impact value.
- **RC Limit Value:** The derated Limit Value.
- **RC % Limit:** The loading of the component calculated by (Flow Value / (Limit Value – RC Impact)).
- **RC Capacity Available:** The capacity available including the reserved capacity impact value.

1.8.21. Reserved Capacity Impact for Feeder

The Reserved Capacity Impact display is accessed from the Feeder Exception and Feeder Exception RF displays. The display provides the same information as the Reserved Capacity Impact for Station display using components impacted by reserved capacity on the chosen feeder.

Display call-up URL:

`netview://<Mode>/gridtabular/ReservedCapacityImpactfdr_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

For the description of the display fields, refer to Reserved Capacity Impact for Station.

1.8.22. Distributed Energy Resource - Power Transformer

The Distributed Energy Resource- Power Transformer display shows the details about distributed energy resources that are associated with the specified power transformer.

To open this display, select Distributed Energy Resource - Power Transformer from the Feeder Exception display.

Display call-up URL:

netview://<Mode>/gridtabular/DERPowerTF_GridDisplay/<Substation Name>/<Power Transformer Name>/<Power Transformer ID>

Or:

netview://<Mode>/gridtabular/DERPowerTFRF_GridDisplay/<Substation Name>/<Power Transformer Name>

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

For the description of the display fields, refer to Distributed Energy Resource.

1.8.23. Distributed Energy Resource - Feeder

The Distributed Energy Resource- Feeder display shows the details about distributed energy resources that are associated with the specified feeder.

To open this display, select Distributed Energy Resource - Feeder from the Feeder Exception display.

Display call-up URL:

netview://<Mode>/gridtabular/DERFeeder_GridDisplay/<Substation Name>/<Feeder Name>

or:

netview://<Mode>/gridtabular/DERFeederRF_GridDisplay/<Substation Name>/<Feeder Name>

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

For the description of the display fields, refer to Distributed Energy Resource.

1.9. Stations List

The Stations List display can be accessed from the real-time and study menu from the main toolbar. This display also can be accessed from the geographic view by pressing Ctrl + U.

It lists available substations and provides links to navigate to geographic view, station internals, and SCADA One-Line displays.

Display call-up URL:

`netview://<Mode>/gridtabular/stationslist_GridDisplay/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Station:** Name of the substation

Navigation:

- "Geographic View" navigates to the geographic view of the related substation.
- "SCADA Oneline" navigates to the SCADA one-line display in the same viewport.
- "Station Internals" navigates to substation internal display of the related substation in the same viewport.
- "Conversion History" navigates to the Conversion History display in the same viewport.
- "Schedules" navigates to the Schedule display.

1.9.1. Conversion History

The Conversion History display can be accessed from the Stations List display by clicking a button in the Conversion History column, or by clicking the Conversion History button on the substation toolbar. It displays the conversion history of the selected substation by Converter tool.

Display call-up URL:

```
netview://<Mode>/gridtabular/StationPedigree_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationPedigree_GridDisplay display:

- **File Name:** Name of the file used to generate the substation model file. The file may be substation external file, internal file, placement files, Global.csv file, global_internals.xml file, DMS_Defaults.csv, Substation model file, Converter.exe file.
- **File Path:** Directory name.
- **Date Created:** Date when the file was created.
- **Date Modified:** Date when the file was modified for the last time.
- **Version:** Version information of Converter.exe. This column is not applicable for other files.
- **Size:** File size in kB.
- **Total Fatals:** Total number of fatal errors during the model file’s conversion.
- **Total Errors:** Total number of errors during the model file’s conversion.
- **Total Warnings:** Total number of warnings during the model file’s conversion.

1.9.2. Shunts for Stations

This display lists feeders (shunts) and allows navigating to the feeders by clicking the Go to Geographic button.

Display call-up URL:

`netview://<Mode>/gridtabular/Shimp_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Shunt:** Name (number) of feeder.
- **ON Status:** Clickable check box, by default it is unchecked

Navigation:

- "Go to Geographic" locates feeder device in current viewport.

1.9.3. Transfer Capacity

The available transfer capacity is defined as the reserve loading capacity on a feeder at any given time as viewed from any of its feeder backbone tie switches (open switches). The Transfer Capacity is evaluated with contingencies on two feeders and so it is bi-directional. Results are presented for transfer in each direction.

Bi-directional transfer capacity values for each designated tie-switch in the station, along with the associated normally closed switch to achieve these values, will be presented.

Display call-up URL:

`netview://<Mode>/gridtabular/TransferCapacity_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Tie Switch Name:** Name of the switch.
- **Second ID:** Identifier of the switch.
- **Current Status:** Current status of the switch.
- **From Feeder:** The name of Feeder which the load is being transferred from.
- **To Feeder:** The name of the Feeder which the load is being transferred to.
- **Transfer Capacity Phase A/B/C:** The maximum phase A, B, and C loads that can be transferred by closing the switch in Amps.
- **Limiting Switch:** The switch that needs to be opened to maintain the transferred load within the limits.
- **Mark Devices:** Marks devices on the geographic view.
- **From Feeder:** The name of Feeder which the load is being transferred from.
- **To Feeder:** The name of the Feeder which the load is being transferred to.
- **Transfer Capacity Phase A/B/C:** The maximum phase A, B, and C loads that can be transferred by closing the switch in Amps.
- **Limiting Switch:** The switch that needs to be opened to maintain the transferred load within

the limits.

1.9.4. Station Internal Component Results

The Station Internal Component Results display gives the power flow results for each of the line sections, switches, and power transformers/regulators within the station model.

Display call-up URL:

`netview://<Mode>/gridtabular/stationInternals_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the StationInternals display:

- **ID:** ID of component.
- **Name:** Name of component.
- **Component Type:** Type of component.
- **Phase A/B/C Voltage Side 1 (LL):** Voltage magnitude and phase angle at the side 1 of the component for phases A, B, and C.
- **Phase A/B/C Voltage Side 2 (LL):** Voltage magnitude and phase angle at the side 2 of the component for phases A, B, and C.
- **Amps A/B/C:** Current magnitude in the component for phases A, B, and C.

Navigation:

- "Analyst Attributes" opens the Analyst Attributes pop-up with the attribute data for the selected component.
- "Analyst Outputs" opens the Analyst Output pop-up with the full DPF output data for the selected component.

1.10. Area List

This display lists the areas the current user has permissions on. The Area List display can be called up from the RT DMS menu.

Display call-up URL:

`netview://<Mode>/gridtabular/AreaList_griddisplay`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the AreaList display:

- **Areas:** Area name.

Navigation:

- "Go to the Geographic Overview for Area" calls up the Navigation Overview focused on the selected area.
- "Go to a list of Stations for Area" opens the Stations for Area display.

1.10.1. Stations for Area

The Stations for Area display shows the substations of a specific area.

Display call-up URL:

netview://<Mode>/gridtabular/AreaStations_GridDisplay/<Area Name>

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the AreaStations display:

- **Station:** Substation name.

Navigation:

- "Go to List of Areas": Opens the Area List display.
- "Go to the Geographic": Opens the geographic view centered on the selected substation.
- "Go to Station OneLine": Opens the SCADA One Line display for the selected substation.
- "Go to the List of Feeders": Opens the Station Feeder Summary display listing all feeders associated with the selected substation.

1.10.2. Station Feeder Summary

The Station Feeder Summary display provides general feeder details for the substation. The display can be accessed by clicking the Go to the List of Feeders button on the Stations for Area display.

Display call-up URL:

`netview://<Mode>/gridtabular/FeederSummary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the FeederSummary display:

- **Feeder Name:** Feeder name.
- **Feeder ID:** Feeder ID.
- **Alternate Name:** Feeder alternate name.
- **Address:** Feeder address.
- **Service Center:** Service center the feeder belongs to.
- **Nominal Voltage (kV):** Feeder’s nominal voltage in kV.
- **Feeder Color:** Feeder color as defined in the Viewer Configuration Editor.

- **Station Name:** Name of the substation the feeder belongs to.

1.11. Station Output Summary

The Station Output Summary provides navigation to DNAF results for a particular station. For example, Bus Load Allocation results for a station can be accessed from this display. This display can be easily tailored to remove or add the desired navigation points for your dispatcher. Preferable access may be through the analyst pull-down menus.

The display can be accessed from the RT, ST, or SIM menu. It can also be accessed from the geographic view by pressing Ctrl + O.

Display call-up URL:

`netview://<Mode>/gridtabular/stnOutput_GridDisplay/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the stnOutput display:

- **Station:** The list of stations for which the user currently has operational responsibility. In the case of study mode, it is all the stations in the network model.
- **LVM Solution:** This indicates whether Load and Voltage/VAR Management switching plans have been generated for this station. Clicking the value navigates to the details of the latest plan.
- **FISR Solutions:** This indicates whether Fault Isolation and Service Restoration switching plans have been generated for this station. Clicking the value navigates to the details of the plan.
- **AFR Solutions:** This indicates whether Automatic Feeder Reconfiguration switching plans have been generated for this station. Clicking the value navigates to the details of the plan.
- **PLOS Solutions:** This indicates whether Planned Outage Switching plans have been generated for this station. Clicking the value navigates to the details of the plan.
- **FISRCA Solutions:** Indicates whether Fault Isolation and Service Restoration Contingency Analysis plans have been generated. Clicking the value navigates to FISRCA Solution Overview display. Available in Study mode only.

Navigation:

- "DPF Results" opens the following displays in one window: Distribution Power Flow Summary for Station and DPF Feeder Results.
- "SCC Results" opens the Short Circuit Calculation Summary display.
- "PRV Results" opens the Protection Validation Results display.
- "FL Results" opens the Fault Location display.
- "LVM Solution" navigates to the Load and Volt/VAR Management display.
- "FISR Solutions" navigates to the FISR Results Summary display.
- "FISRCA Solutions" navigates to FISRCA Solution Overview display. Available in Study mode only.

- "AFR Solutions" navigates to the AFR Results Summary display.
- "PLOS Solutions" navigates to the Planned Outage Service display.
- "Application Diagnostic Summary" navigates to the DNAF Application Diagnostic Information display.

1.12. SCADA/DER Devices

The SCADA/DER Devices display lists SCADA devices and distributed energy resource objects that exist in the system. You can access this display by clicking the SCADA/DER Devices Tree View Display option in the RT DMS menu.

Display call-up URL:

`netview://<Mode>/gridtabular/DeviceTreeView_GridDisplay`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Element tree:** Hierarchical element tree, which includes service centers, substations, feeders, and devices. The elements are listed in alphabetical order.
- **Name:** Device name.
- **Service Center:** Service center that the device belongs to.
- **Station:** Substation that is associated with the device.
- **Feeder:** Feeder that is associated with the device.
- **Is SCADA:** Specifies whether this is a SCADA device.
- **Is DER:** Specifies whether this energy resource is controlled by the 2030.5 Gateway.
- **Control ID:** Control measurement ID.
- **Control Type:** Control type.
- **Status:** Control status.
- **Available Status:** The list of available statuses.
- **Last Execution Status:** Status of last executed control.
- **Last Execution Time:** The control’s last execution time.

Navigation:

- "Locate" navigates to the device on the network view.

1.12.1. Distributed Energy Resource

The Distributed Energy Resource display shows the details about distributed energy resources in the system.

To open this display, select the DNAF Display Plugin option from the RT DMS drop-down menu, and then select Distributed Energy Resource.

Display call-up URL:

`netview://<Mode>/gridtabular/GENALLDPF_GridDisplay`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Optimization Group:** Name of the optimization group.
- **Substation:** Name of the substation.
- **Real Time Power Transformer:** Energizing power transformer name.
- **Real Time Feeder:** Energizing feeder name.
- **Name:** Name of the energy resource.
- **Connected:** Connection status (Connected or Disconnected).
- **Aggregator:** Aggregator name.
- **Type:** DER device type, such as Wind or Solar.
- **Enrolled Program:** Enrolled programs of the DER.
- **Capacity (kVA):** Capacity in kVA.
- **Real Power Phase A/B/C:** The real power value at phases A, B, and C in kW.
- **Total Real Power (kW):** The total real power in kW.
- **Reactive Power Phase A/B/C:** The reactive power value at phases A, B, and C in kVAR.
- **Reactive Power Phase A/B/C:** The reactive power value at phases A, B, and C in kVAR.
- **Total Reactive Power (kVAR):** The total reactive power in kVAR.
- **Real Power Target (kW):** Real power target in kW.
- **Reactive Power Target (kVAR):** Reactive power target in kVAR.

Navigation:

- "Locate" navigates to the energy resource on the network view.
- "Analyst Attributes" displays the Analyst Attributes pop-up.
- "Analyst Outputs" displays the Analyst Output pop-up.
- "Operator Attributes" displays the Operator Attributes pop-up.
- "Operator Outputs" displays the Operator Output pop-up.

1.12.2. DER Measurement Data

The DER Measurement Data display shows distributed energy resources measurement data for a substation. To open this display for a station, enter the station name in the text entry field and select the Go to Station DER Measurement Tabular option from the RT DMS display.

This display shows the following DER statuses and measurements: real and reactive power, frequency, phase voltage, connection status, operational energy storage capacity, energy storage state of charge, operational status (for 2030.5 Gateway communicated DERs only), and alarm state (for 2030.5 Gateway communicated DERs only).

Display call-up URL:

`netview://<Mode>/gridtabular/DerMeasurements_GridDisplay/<Station Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **ID:** Energy resource unique identifier.
- **Name:** Name of the energy resource.
- **Station:** Name of the substation.
- **Nominal Feeder:** Nominal feeder name.
- **Energizing Feeder:** Energizing feeder name.
- **Device Type:** Energy resource device type.
- **Program Enrollment:** Enrolled programs of the energy resource.
- **Data Source:** Measurement data source (SCADA or 2030.5 Gateway).
- **Measurement Type:** Measurement type (for example, kW).
- **Value:** Measurement value.
- **Measurement Quality:** Measurement quality. The supported measurement quality bits are Valid, Manually Edited, Estimated Using Reference Day, Estimated Using Linear Interpolation, Questionable, Derived, and Projected (Forecast). The measurement quality value is set to Suspect if the Questionable bit is set.
- **Phasing:** Measurement phasing.
- **Reading Time:** Measurement reading time.
- **Time of Last Update:** Measurement last update time.

Navigation:

- "Locate" navigates to the energy resource on the network view.

1.13. Distribution Power Flow Summary for Station

The Station Distribution Power Flow Summary display provides high-level DPF output for the substation and can be accessed by clicking the DPF Results button on the Station Output Summary display. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Distribution Power Flow option. The display contains the following information:

- Information about DPF convergence for the station.
- Value of supply bus voltage in KV, per unit, and equivalent 120 V value.
- Power transformer load tap changer (LTC) positions and loadings.
- Substation capacitor steps.
- Voltage regulator tap positions and loading.
- Feeder capacitor status (ON/OFF).

Display call-up URL:

`netview://<Mode>/gridtabular/subDPF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the subDPF display:

- **Station:** The substation for which the power flow analysis was run.
- **DPF Status:** The status of the Distribution Power Flow results – Solved (DPF solved successfully), Unsolved (DPF has not run), Suspect (DPF solution may have caused violations), or Diverged (DPF solution has not converged).

Navigation:

- "Convergence Data" opens the DPF Station Results display.
- "Load Category Data" opens the BLA Load Category Data display.
- "BLA Results Summary" opens the BLA Results Summary display.
- "Measurement Comparison" opens the Measurement Comparison Data display.
- "Transformer and Regulator Summary" opens the Transformer and Regulator DPF Results display.
- "Capacitor Summary" opens the Capacitor DPF Results display.
- "Generator / Distributed Energy Resource Summary" opens the Generator/Distributed Energy Resource DPF Results display.
- "Bus Voltage Summary" opens the Bus Voltage Summary display.
- "Interface Summary" opens the DPF Interface Summary display.
- "Load Summary" opens the Load DPF Results display for the selected substation.
- "Meter Data" opens the Meter Data Results display.
- "Suspect Source Voltage" opens the DPF Interface Summary display.
- "Suspect Load" opens the Suspect Load Results display.
- "Suspect Control" opens the Suspect Controls display.
- "Suspect Generator / Distributed Energy Resource" opens the Suspect Generator/Distributed Energy Resource Results display.
- "Loss/PQI Summary" opens the Station Losses Summary, Station Power Transformer and

Regulator Losses, and Feeder Losses and Power Quality Indices displays for the selected substation.

- "Distribution Transformer Summary" opens the Distribution Transformer Summary display.
- "Feeder Summary" opens the Feeder Summary display.
- "Reserved Capacity Output" opens the Reserved Capacity Output display.
- "Sensor Location Summary" opens the Sensor Location Summary display.

1.13.1. DPF Station Results

The DPF Station Results display provides information about DPF convergence for the substation. It can be accessed by clicking the Convergence Data button on the Distribution Power Flow Summary for Station display.

Display call-up URL:

`netview://<Mode>/gridtabular/subDPFConvergeData_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the subDPFConvergeData display:

- **Station:** The substation for which the power flow was run.
- **DPF Status:** The status of the Distribution Power Flow results – Solved (DPF solved successfully), Unsolved (DPF has not run), Suspect (DPF solution may have caused violations), or Diverged (DPF solution has not converged).
- **DPF Evaluation Time:** Date and time DPF (Distribution Power Flow) last evaluated.
- **DPF Iterations:** The number of iterations executed by DPF to reach a solution.
- **P Mismatch:** Real power mismatch for the final iteration of the DPF calculation in kW.
- **Q Mismatch:** Reactive power mismatch for the final iteration of the DPF calculation in kVAR.
- **V Mismatch:** Voltage magnitude mismatch for the final iteration of the DPF calculation in per unit.
- **Largest Voltage Mismatch Connected Device ID:** ID of the device connected to the largest voltage mismatch node from the final iteration of the latest DPF solution.
- **Largest Voltage Mismatch Connected Device Name:** Name of the device connected to the largest voltage mismatch node from the final iteration of the latest DPF solution.
- **BLA Solution Iterations:** Number of BLA (Bus Load Allocation) DPF (Distribution Power Flow) iterations.
- **Converged Real Power:** Indication that the BLA Analysis has converged for the real power value (Yes/No/Unsolved).
- **Converged Reactive Power:** Indication that the BLA Analysis has converged for the reactive power value (Yes/No/Unsolved).
-

Converged Apparent Power: Indication that the BLA Analysis has converged for the apparent power value (Yes/No/Unsolved).

- **Suspect Source Voltage:** The number of suspect source voltages.
- **Suspect Load:** The number of suspect loads.
- **Suspect Control:** The number of suspect controls.

Navigation:

- "Locate Largest Voltage Mismatch Connected Device" navigates to the device connected to the largest voltage mismatch node on the network view.

1.13.2. BLA Load Category Data

The Bus Load Allocation Load Category Data display provides Bus Load Allocation calculation related to load categories. It can be accessed by clicking the Load Category Data button on the Distribution Power Flow Summary for Station display.

Display call-up URL:

`netview://<Mode>/gridtabular/BLALoadCategory_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the BLALoadCategory display:

- **Load Category Name:** Name of load category.
- **Current Load Schedule:** Name of current load schedule.
- **Percent Accuracy:** Accuracy of load schedule data in percentage.
- **Hour 1:** Hour of first schedule point.
- **Schedule Point 1 – Real Power:** Real power value for the current time point in the named schedule.
- **Schedule Point 1 – Reactive Power:** Reactive power value for the current time point in the named schedule.
- **Schedule Point 1 – Power Factor:** Power factor value for the current time point in the named schedule.
- **Hour 2:** Hour of second schedule point.
- **Schedule Point 2 – Real Power:** Real power value for the current time point in the named schedule.
- **Schedule Point 2 – Reactive Power:** Reactive power value for the current time point in the named schedule.
- **Schedule Point 2 – Power Factor:** Power factor value for the current time point in the named schedule.
- **P – Interpolated Final or Actual:** Interpolated real power multiplier or actual value if sub hour

schedule is provided.

- **Q – Interpolated Final or Actual:** Interpolated reactive power multiplier or actual value if sub hour schedule is provided.
- **PF – Interpolated Final or Actual:** Interpolated power factor multiplier or actual value if sub hour schedule is provided.
- **Load Multiplier Schedule:** Load multiplier schedule name (for example, Residential Multiplier Schedule).
- **Schedule Multiplier - Real Power:** Multiplier for the real power schedule.
- **Schedule Multiplier - Reactive Power:** Multiplier for the reactive power schedule.
- **Secondary Voltage Schedule:** Name of the current secondary voltage schedule (for example, VoltDrop).
- **Cold Load Pickup Schedule:** Name of the current cold load pickup schedule.

1.13.3. Measurement Comparison Data

The Measurement Comparison Data display shows a comparison between measured and calculated values for a station. This gives an indication of how closely the values that were calculated by BLA/DPF match the values from SCADA. This display can be accessed from the Distribution Power Flow Summary for Station display by clicking Measurement Comparison.

Display call-up URL:

`netview://<Mode/gridtabular/meascomparison_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the Measurements display:

- **Device ID:** ID of measurement component or node.
- **Device Name:** Name of the device that the measurement is associated with.
- **Device Type:** Type of the device that the measurement is associated with.
- **Energizing Feeder:** Energizing feeder name.
- **Energizing Station:** Energizing station name.
- **Measurement Type:** Type of the measurement (for example, Amp or Volt).
- **Measurement Phase:** Phase designation (for example, ABC for three phases, A for only phase A).
- **Measurement Value:** Measurement value from SCADA.
- **Calculated Value:** Calculated value from DPF.
- **Difference:** Difference between the measured and calculated values, in percentage.
- **BLA Scaling Factor:** Bus Load Allocation scaling factor for the measurement. This is the value to which the scheduled nominal loads for a measurement island were scaled by BLA to match the

target measured flow values from SCADA.

- **Terminal Identifier:** Terminal identifier (from side, or to side).

Navigation:

- "Locate" navigates to the measurement or the substation circle on the geographic view.

1.13.4. Transformer and Regulator DPF Results

The Transformer and Regulator DPF Results display is accessed by clicking the Transformer and Regulator Summary on the Distribution Power Flow Summary for Station display. It provides a summary of the power flow results for each of the tap-change transformers and regulators on each station.

Display call-up URL:

`netview://<Mode>/gridtabular/tfDPF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the tfDPF display:

- **Name:** Transformer Name.
- **ID:** Unique identifier.
- **Nominal Feeder:** The nominal (owning) feeder for the device.
- **Energizing Feeder:** The energizing feeder for the device.
- **Type:** Transformer type (Power TF, Regulator, and so on).
- **Source Phasing:** The phase connections of the transformer source-side winding.
- **Rating:** The VA capacity of the transformer.
- **Regulator Effective Size:** Regulator effective size in kVA.

Navigation:

- "Locate" locates the transformer on the geographic view.
- "Primary Winding Results" opens the Transformer Primary Winding Results display.
- "Secondary Winding Results" opens the Transformer Secondary Winding Results display.
- "Tertiary Winding Results" opens the Transformer Tertiary Winding Results display.
- "Analyst Attributes" displays the Analyst Attributes pop-up.
- "Analyst Outputs" displays the Analyst Output pop-up.
- "Operator Attributes" displays the Operator Attributes pop-up.
- "Operator Outputs" displays the Operator Output pop-up.

1.13.4.1. Transformer Primary Winding Results

The Transformer Primary Winding Results display provides information about type, voltage, and power flows on the primary side of the transformer or regulator.

This display can be accessed from the Transformer and Regulator DPF Results display.

Display call-up URL:

`netview://<Mode>/gridtabular/tfDPFSource_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the tfDPFSource display:

- **Name:** Name of the transformer or regulator.
- **ID:** ID of the transformer or regulator.
- **Type:** Type of the transformer.
- **Tap Changer Type:** Type of the tap changer (for example, Regulating Tap).
- **Tap position:** Position of the regulating tap.
- **Regulation Status:** Regulation status. Can be Regulating, Off, or none.
- **Blocked Status:** Blocked status. Can be Blocked or Unblocked.
- **Voltage Reduction Status:** Voltage reduction status. Can be Enabled, Disabled, or none.
- **Average Voltage:** The Line-to-Line average voltage in the primary side of the transformer. Expressed as a 120 V base value.
- **Total Real Power:** Total real power in MW.
- **Total Reactive Power:** Total reactive power in MVAR.
- **Total Apparent Power:** Total apparent power in MVA.
- **Three Phase Power Factor:** Three Phase Power Factor.

Navigation:

- "Locate" locates the device on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.

1.13.4.2. Transformer Secondary Winding Results

The Transformer Secondary Winding Results display provides information about type, voltage, and power flows on the load side of the transformer or regulator.

This display can be accessed from the Transformer and Regulator DPF Results display.

Display call-up URL:

`netview://<Mode>/gridtabular/tfDPFLoad_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the tfDPFLoad display:

- **Name:** Name of the transformer or regulator.
- **ID:** ID of the transformer or regulator.
- **Type:** Type of the transformer.
- **Tap Changer Type:** Type of the tap changer (for example, Regulating Tap).
- **Tap position:** Position of the regulating tap.
- **Regulation Status:** Regulation status. Can be Regulating, Off, or none.
- **Blocked Status:** Blocked status. Can be Blocked or Unblocked.
- **Voltage Reduction Status:** Voltage reduction status. Can be Enabled, Disabled, or none.
- **Average Voltage:** The Line-to-Line average voltage in the primary side of the transformer. Expressed as a 120 V base value.
- **Total Real Power:** Total real power in MW.
- **Total Reactive Power:** Total reactive power in MVAR.
- **Total Apparent Power:** Total apparent power in MVA.
- **Three Phase Power Factor:** Three Phase Power Factor.

Navigation:

- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.
- "Locate" locates the device on the geographic view.

1.13.4.3. Transformer Tertiary Winding Results

The Transformer Tertiary Winding Results display provides information about type, voltage, and power flows on the load side of the transformer or regulator.

This display can be accessed from the Transformer and Regulator DPF Results display.

Display call-up URL:

`netview://<Mode>/gridtabular/tfDPFTertiary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the tfDPFTertiary display:

- **Name:** Name of the transformer or regulator.
- **ID:** ID of the transformer or regulator.
- **Type:** Type of the transformer.
- **Tap Changer Type:** Type of the tap changer (for example, Regulating Tap).
- **Tap position:** Position of the regulating tap.
- **Regulation Status:** Regulation status. Can be Regulating, Off, or none.
- **Blocked Status:** Blocked status. Can be Blocked or Unblocked.
- **Voltage Reduction Status:** Voltage reduction status. Can be Enabled, Disabled, or none.
- **Average Voltage:** The Line-to-Line average voltage in the primary side of the transformer. Expressed as a 120 V base value.
- **Total Real Power:** Total real power in MW.
- **Total Reactive Power:** Total reactive power in MVAR.
- **Total Apparent Power:** Total apparent power in MVA.
- **Three Phase Power Factor:** Three Phase Power Factor.

Navigation:

- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.
- "Locate" locates the device on the geographic view.

1.13.5. Capacitor DPF Results

The Capacitor DPF Results display is accessed by clicking the Capacitor Summary button on the Distribution Power Flow Summary for Station display. It provides a summary of the power flow results for each capacitor in the selected station.

Display call-up URL:

`netview://<Mode>/gridtabular/capDPF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the capDPF display:

- **Name:** Name of capacitor.
- **ID:** ID of capacitor.

- **Nominal Feeder:** Nominal (owning) feeder name.
- **Energizing Feeder:** Energizing feeder name.
- **Control Type:** The control method used to control the capacitor.
- **Current Operation Mode:** Operation mode of the capacitor (Remote/Fixed).
- **Status:** Last solved capacitor status.
- **Reactive Power Phase A/B/C:** The reactive power flow in phases A, B, and C in kVAR.
- **Maximum Daily Operations:** Number of capacitor maximum daily operations.
- **Current Number of Daily Switching Operations:** Current number of daily switching operations.
- **Minimum Time Between Switching Operations:** Minimum time between switching operations. Expressed in hours.
- **Time of Last Switching Operation:** Time of last switching operation.
- **Position Validation:** Is the current step position of the capacitor bank valid (Yes/No).

Navigation:

- "Locate" locates the device on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output op-up.

1.13.6. Generator/Distributed Energy Resource DPF Results

The Generator/Distributed Energy Resource DPF Results display can be accessed by clicking the Generator/Distributed Energy Resource Summary button on the Distribution Power Flow Summary for Station display.

This display gives the power flow results for each of the generators/energy resources within the station model.

Display call-up URL:

`netview://<Mode>/gridtabular/genDPF_griddisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of generator.
- **ID:** ID of generator
- **Nominal Feeder:** The name of the nominal (owning) feeder.

- **Energizing Feeder:** The name of the energizing feeder.
- **Capacity (kVA):** DER capacity in kVA.
- **Rated Power Factor:** DER rated power factor.
- **Generation Capacity Multiplier (%):** User-entered generation capacity multiplier (%) from the DNAF pop-up display in study mode.
- **Connection Status:** Connection status (Connected or Disconnected).
- **Real Power Phase A/B/C:** Phase A, B, and C real power at generator terminal in kW.
- **Total Real Power:** Total real power value in kW.
- **Reactive Power Phase A/B/C:** Phase A, B, and C reactive power at generator terminal in kVAR.
- **Total Reactive Power:** Total reactive power value in kVAR.
- **Voltage Phase A/B/C:** Phase A, B, and C voltage magnitude and phase angle at generator terminal.
- **Phase A/B/C:** Phase A, B, and C current magnitude (in Amps) and phase angle at generator terminal.

Navigation:

- "Locate" locates the device on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.
- "DER Associated Inverters" opens the Associated Inverter Details display.
- "DER Autonomous Modes" opens the Autonomous Modes Details display.

Controls:

- "Disconnect" disconnects the DER from the network to prevent backfeed in the transmission system.

1.13.6.1. Associated Inverter Details

The Associated Inverter Details display shows details for inverters that are associated with an energy resource. This display can be accessed from the Generator/Distributed Energy Resource DPF Results display.

Display call-up URL:

```
netview://<Mode>/gridtabular/DER_Assoc_Inverter_GridDisplay/<Substation Name>/<Energy Resource Node ID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the inverter.
- **ID:** Unique identifier of the inverter.
- **Model Name:** Name of the inverter model.
- **Description:** Description of the inverter.
- **Phase:** Device phasing.
- **Node:** ID of the node.
- **Station:** The name of the nominal substation the inverter belongs to.
- **Feeder:** The name of the nominal substation the inverter belongs to.
- **Disconnect Capability:** The disconnect capability (Physical, Virtual, Physical-Virtual, or None).
- **Power Factor Convention:** The power factor type (IEC Convention or IEEE Convention).
- **Maximum Real Power Delivered (kW):** The maximum real power the DER can deliver to the grid (in kW).
- **Maximum Apparent Power Delivered (kVA)**: The maximum apparent power the DER can deliver to the grid (in kVA).
- **Maximum Reactive Power Delivered (kVAR):** The maximum reactive power the DER can deliver to the grid (in kVAR).
- **Maximum Real Power Absorbed (kW):** The maximum real power the DER can absorb from the grid (in kW). For energy storage charging.
- **Maximum Apparent Power Absorbed (kVA):** The maximum apparent power the DER can absorb from the grid (in kVA). For energy storage charging.
- **Maximum Reactive Power Absorbed (kVAR):** The maximum reactive power the DER can absorb from the grid (in kVAR). For energy storage charging.
- **Peak Power Limit (kW):** The peak power target power level limit (in kW).
- **Rating (Amps):** The maximum nameplate AC current level of the DER (in Amps).
- **Normal Operating Voltage (V):** The normal operating voltage for the DER site/service connection (in V).
- **Voltage Reference Offset (V):** An offset voltage adjustment for the DER, relative to the normal operating voltage (in V).
- **Short Circuit Rating Multiplier:** The short circuit multiplier of the nameplate AC rating.
- **Disconnect Expiration Timeout (sec):** The time after which a command to disconnect expires and the device can reconnect (in seconds).
- **Power Factor Reversion Timeout (sec):** The time after which a DER returns to its default power factor setting (in seconds).

Navigation:

- "Locate" locates the inverter on the network view.

1.13.6.2. Autonomous Modes Details

The Autonomous Modes Details display shows details for modes that are associated with an energy resource or an inverter. This display can be accessed from the Generator/Distributed Energy Resource DPF Results display, energy resource toolbar, or inverter toolbar.

Display call-up URL:

```
netview://<Mode>/gridtabular/DER_Autonomous_modes_GridDisplay/<Substation Name>/<Energy
Resource ID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

Or

```
netview://<Mode>/gridtabular/Inverter_Autonomous_modes_GridDisplay/<Substation
Name>/<Inverter ID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the inverter mode (for example, VOLTVAR1).
- **Mode Type:** For example, Volt-VAR.
- **Active Inverter Mode:** Specifies whether the active inverter mode is enabled, disabled, or not applicable.
- **Active Inverter Curve Index:** Shows the active inverter curve index.
- **Active Inverter Curve Used:** Specifies whether the active inverter curve is used.
- **Season:** Summer, Fall, Winter, or Spring.
- **Day Type:** For example, Weekday, Weekend, Holiday.
- **Number of Points:** The number of schedule points.
- **X Axis Type:** Type of the schedule's X axis (for example, Voltage).
- **X Axis Units:** The unit of the schedule's X axis (for example, % V).
- **Y Axis Type:** Type of the schedule's Y axis (for example, Reactive Power).
- **Y1 Units:** The first unit of the schedule's Y axis (for example, % kW).
- **Linear or Block Type:** Shows the schedule type (Linear or Block).

Navigation:

- "Schedule Points" opens the DER Autonomous Modes X and Y Points Summary display.
- "Chart" shows the energy resource chart based on schedule points.

1.13.6.3. DER Autonomous Modes X and Y Points Summary

The DER Autonomous Modes X and Y Points Summary display shows the details about DER autonomous mode points. The display can be accessed by clicking the Number of Points button on the Autonomous Modes Details display.

Display call-up URL:

```
netview://<Mode>/gridtabular/DER_Autonomous_modes_XYPoints_GridDisplay/<LINEAR or
BLOCK>/<Mode Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **X Points:** X value of the mode point.
- **Y Points:** Y value of the mode point.

1.13.7. Bus Voltage Summary

The Bus Voltage Summary display can be accessed by clicking the Bus Voltage Summary button on the Distribution Power Flow Summary for Station display.

Display call-up URL:

```
netview://<Mode>/gridtabular/BusVoltage_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Bus Voltage Summary display:

- **Name:** Name of the bus.
- **ID:** ID of the bus.
- **Nominal Feeder:** Nominal (owning) feeder name.
- **Real Time Feeder:** Energizing feeder name.
- **Phase A/B/C Measured Voltage (SCADA):** SCADA voltage measurement values for phases A, B, and C.
- **Phase A/B/C Measured Voltage (User Voltage Base):** The voltage measurement values for phases A, B, and C expressed as 120 V base values.
- **Phase A/B/C Calculated Voltage:** The calculated values of phase A, B, and C voltages.
- **Phase A/B/C Voltage Mismatch:** The value of a difference between the measured and calculated voltages in phases A, B, and C.

Navigation:

- "Locate" navigates to the equipment on the geographic display in the same viewport.

1.13.8. DPF Interface Summary

The Interface Summary display provides information about amperage, voltage, and angles at each phase of the interface. It also contains real and reactive power values at each phase.

This display can be accessed by clicking the Interface Summary button on the Distribution Power Flow Summary for Station display.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/intfcused_GridDisplay/<Substation Name>`

Second Pane:

`netview://<Mode>/gridtabular/intcdpfflow_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Intfcused display:

- **Name:** Name of the Interface.
- **ID:** Unique identifier of the interface.
- **Station:** The substation of the interface.
- **Study Interface:** Study interface indicator; can be Yes or No.
- **Connected:** Connection indicator; can be Yes or No.
- **Connected Branch ID:** Unique identifier of the connected branch.
- **Voltage Phase A/B/C:** Voltage values and angles at phases A, B, and C expressed as 120 V base values.
- **Suspect:** Interface suspect indicator; can be Yes or No.

The following information is shown on the intcdpfflow display:

- **Name:** Name of the Interface.
- **ID:** Unique identifier of the interface.
- **Total Real Power:** Total real power value in kW.
- **Total Reactive Power:** Total reactive power value in kVAR.
- **Total Current Injection:** The total current injected in Amps.
- **Real Power Phase A/B/C:** The real power value at phases A, B, and C in kW.
- **Reactive Power Phase A/B/C:** The reactive power value at phases A, B, and C in kVAR.
- **Phase A/B/C:** Current values and angles in phases A, B, and C in Amps.

1.13.9. Load DPF Results

The Load DPF Results display shows Distribution Power Flow output related to loads associated with the substation. The display can be accessed by clicking the Load Summary button on the Distribution Power Flow Summary for Station display.

Display call-up URL:

`netview://<Mode>/gridtabular/LoadDpf_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the device.
- **ID:** Unique identifier of the device.
- **Nominal Feeder:** Nominal (owning) feeder name.
- **Energizing Feeder:** Energizing feeder name.
- **Load Has Embedded Generation:** Flag indicating that a load contains embedded generation.
- **Phase:** Device phasing.
- **Voltage Phase A/B/C:** Phase A, B, and C voltages.
- **Real Power Phase A/B/C:** Real power values at phases A, B, and C in kW.
- **Total Real Power:** Total real power value in kW.
- **Reactive Power Phase A/B/C:** Reactive power values at phases A, B, and C in kVAR.
- **Total Reactive Power:** Total reactive power value in kVAR.
- **Three Phase Power Factor:** Three-phase power factor.

Navigation:

- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.
- "Locate" locates the device on the geographic view.

1.13.10. Meter Data Results

The Meter Data Results display can be accessed from the Distribution Power Flow Summary for Station display. It provides the data for loads where SCADA voltage measurements are available.

Display call-up URL:

`netview://<Mode>/gridtabular/LoadDPFMeterData_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Meter Data Results display:

- **Name:** Name of the load.
- **ID:** ID of the load.

- **Nominal Feeder:** The name of the nominal (owning) feeder.
- **Energizing Feeder:** The name of the feeder the load is connected to.
- **Meter Data:** The SCADA voltage measurement value.

Navigation:

- "Locate" navigates to the load on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.

1.13.11. Suspect Load Results

The Suspect Load Results display provides information for suspect load data for a substation. This display can be accessed from the Distribution Power Flow Summary for Station display.

Display call-up URL:

`netview://<Mode>/gridtabular/suspectLoad_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following columns are shown on the Suspect Load Results display:

- **Name:** Name of the network element.
- **ID:** Unique identifier of the network element.
- **Nominal Feeder:** The name of the nominal (owning) feeder.
- **Energizing Feeder:** The name of the energizing feeder.
- **Suspect Load Phase A/B/C:** Suspect load of phases A, B, and C.

Navigation:

- "Locate" locates the device in the current viewport.

1.13.12. Suspect Controls

The Suspect Controls display provides information about suspect device LVM control results for a substation. This display can be accessed from the Distribution Power Flow Summary for Station display.

Display call-up URL:

`netview://<Mode>/gridtabular/suspectControl_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the network element.
- **ID:** Unique identifier of the network element.
- **Nominal Feeder:** The name of the nominal (owning) feeder.
- **Energizing Feeder:** The name of the energizing feeder.
- **Type:** Type of the device. Can be Regulator, LTC, or Capacitor.
- **Control Type:** The control method used for the device.
- **Current Operation Mode:** Current operation mode.

Navigation:

- "Locate" locates the device on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.

1.13.13. Suspect Generator/Distributed Energy Resource Results

The Suspect Generator/Distributed Energy Resource Results display provides information for suspect generator/distributed energy resource data for a substation. This display can be accessed from the Distribution Power Flow Summary for Station display.

Display call-up URL:

```
netview://<Mode>/gridtabular/SuspectEnergySource_GridDisplay/<Substation Name>/
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following columns are shown on the Suspect Load Results display:

- **Name:** Name of the network element.
- **ID:** Unique identifier of the network element.
- **Nominal Feeder:** The name of the nominal (owning) feeder.
- **Energizing Feeder:** The name of the energizing feeder.
- **Maximum Voltage Before Separation From the Network:** Maximum voltage before separation from the network (per unit).
- **Minimum Voltage Before Separation From the Network:** Minimum voltage before separation from the network (per unit).
- **Connection Status:** Connection status.

- **Voltage Phase A/B/C:** Voltage of phases A, B, and C.

Navigation:

- "Locate" locates the device in the current viewport.
- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.

1.13.14. Station Losses Summary

This display can be accessed by clicking Loss/PQI Summary button on the Distribution Power Flow Summary for Station display.

The Station Losses Summary display informs about total feeder and substation losses as well as total losses of substation elements such as power transformers, station regulators, and station series reactors (current-limiting reactors).

Display call-up URL:

`netview://<Mode>/gridtabular/StationLosses_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationLosses display:

- **Name:** Name of the substation.
- **Total Loss kW:** Station real power loss in kilowatts.
- **Total Loss kVAR:** Station reactive power loss in kilovolt-amperes reactive.
- **Power Transformer Loss kW:** Total power transformer real power losses in kW.
- **Power Transformer Loss kVAR:** Total power transformer reactive power losses in kVAR.
- **Station Regulator Loss kW:** Total station regulator real power loss in kilowatts.
- **Station Regulator Loss kVAR:** Total station regulator reactive power loss in kilovolt amperes reactive.
- **Station Series Reactor Loss kW:** Total station series reactor real power loss in kilowatts.
- **Station Series Reactor Loss kVAR:** Total station series reactor reactive power loss in kilovolt amperes reactive.
- **Feeder Loss kW:** Total feeder real power loss in kilowatts. The losses include losses in lines, distribution transformers, and other equipment.
- **Feeder Loss kVAR:** Total feeder reactive power loss in kilovolt amperes reactive. The losses include losses in lines, distribution transformers, and other equipment.

1.13.15. Station Power Transformer and Regulator Losses

This display can be accessed by clicking Loss/PQI Summary button on the Distribution Power Flow Summary for Station display.

This display informs about transformer losses. Transformer losses named "core loss" are mainly related to heat and do not include losses due to resistance in the conductors of the windings, which is called "copper loss" (proportional to the resistance of the wire and the square of the current).

Display call-up URL:

`netview://<Mode>/gridtabular/StationTfLosses_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationTfLosses display:

- **Name:** Name of the transformer.
- **ID:** ID of the transformer.
- **Type:** Type of transformer. Can be Power TF, or Regulator.
- **Source Phasing:** Phasing of the transformer.
- **Feeder Name:** Nominal feeder name.
- **Energizing Feeder:** Energizing feeder name.
- **Rating:** Rating of the transformer.
- **Real Copper Loss:** Real copper loss of the transformer.
- **Reactive Copper Loss:** Reactive copper loss of the transformer.
- **Real Core Loss:** Real core loss of the transformer.
- **Reactive Core Loss:** Reactive core loss of the transformer.

1.13.16. Feeder Losses and Power Quality Indices

The Feeder Losses and Power Quality Indices display informs about losses related to elements along the feeder. This display can be accessed by clicking Loss/PQI Summary button on the Distribution Power Flow Summary for Station display.

Information on this display is recalculated each time the Distribution Power Flow (DPF) solves for a station. Instantaneous losses and Power Quality Index (PQI) values are presented. Loss and PQI results for individual components can be accessed from the geographic display (operator output display).

Display call-up URL:

`netview://<Mode>/gridtabular/FeederLossesPQI_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Losses and Power Quality Indices display:

- **Name:** Name of the associated feeder.
- **Total Feeder Loss KW:** Total feeder real power losses in kW. The losses include losses in lines, distribution transformers, and other equipment.
- **Total Feeder Loss KVAR:** Total feeder reactive power losses in kVAR. The losses include losses in lines, distribution transformers, and other equipment.
- **Total Distribution Transformer Loss KW:** Total real power losses of all the distribution transformers along the feeder in kW.
- **Total Distribution Transformer Loss KVAR:** Total reactive power losses of all the distribution transformers along the feeder in kVAR.
- **Total Regulator Loss KW:** Total real power losses of all the regulators along the feeder in kW.
- **Total Regulator Loss KVAR:** Total reactive power losses of all the regulators along the feeder in kVAR.
- **High Voltage Power Quality Index:** Feeder high power quality index in % kW.
- **Low Voltage Power Quality Index:** Feeder low power quality index in % kW.
- **Imbalance Voltage Power Quality Index:** Feeder imbalance power quality index in % kW.

1.13.17. Distribution Transformer Summary

The Distribution Transformer Summary display is accessed from the Distribution Power Flow Summary for Station display. It provides detailed information about the distribution transformers of a station.

Display call-up URL:

`netview://<Mode>/gridtabular/DistTfSummary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on Distribution Transformer Summary display:

- **Name:** Name of the distribution transformer.
- **ID:** ID of the distribution transformer.
- **Nominal Feeder:** Name of the nominal (owning) feeder.
- **Energizing Feeder:** Name of the energizing feeder.
- **Transformer Type:** Type of the distribution transformer.
- **Banked Transformer:** Indicates whether the distribution transformer is banked. Can be Yes or No.
- **Level 1 Limit:** The level 1 limit value of the distribution transformer.
- **Rating:** Transformer rating.
-

Total kVA Demand: Transformer total kVA demand.

- **Loading Level Against Level 1 Limit:** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.
- **Secondary Real Power Phase A/B/C:** Secondary real power in phases A, B, and C in kW.
- **Secondary Reactive Power Phase A/B/C:** Secondary reactive power in phases A, B, and C in kVAR.

Navigation:

- "Locate" navigates to the distribution transformer on the geographic view.
- "Transformer Analyst Attributes" opens the Analyst Attributes pop-up for the transformer.
- "Transformer Analyst Output" opens the Analyst Output pop-up for the transformer.
- "Downstream Loads" opens the Distribution Transformer Load display.

1.13.18. Feeder Summary

The Feeder Summary display is accessed from the Distribution Power Flow Summary for Station display. It provides information about measured and calculated current for the feeders of a station.

Display call-up URL:

`netview://<Mode>/gridtabular/BLAFeederAmpSummary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Amp Summary display:

- **Name:** Name of the feeder.
- **Feeder Average Calculated Voltage (kV LL):** Average line-to-line voltage calculated for feeder in kV.
- **Measured Flow Phase A/B/C:** Flow measured at phases A, B, and C in Amps.
- **Calculated Flow Phase A/B/C:** Flow calculated at phases A, B, and C in Amps.
- **Amp Flow Imbalance:** Flow imbalance in percentage.
- **Measurement Error Count (per DPF):** Count of measurement errors.
- **Cumulative Measurement Error Count:** Count of cumulative measurement error.
- **Voltage Imbalance:** Voltage imbalance in percentage.
- **Customers De-Energized:** The number of de-energized customers.
- **Three-Phase Connected kVA:** Total three-phase connected power in kVA.
- **Three-Phase Connected Customers:** Total three-phase connected customers.
- **Three-Phase Nominal Customers:** The nominal number of three-phase connected customers.
- **Feeder Connected Customers A/B/C:** Number of customers connected to phases A, B, and C of

the feeder.

- **Feeder Nominal Customers A/B/C:** Nominal number of customers connected to phases A, B, and C of the feeder.

Navigation:

- "Voltage Summary" opens the Feeder Voltage Summary display.
- "Loading Summary" opens the Feeder Loading Summary display.
- "Loss Summary" opens the Feeder Loss Summary display.

1.13.18.1. Feeder Voltage Summary

The Feeder Voltage Summary display can be accessed from the Feeder Summary display. It provides the feeder voltage details.

Display call-up URL:

`netview://<Mode>/gridtabular/FeederVoltSummary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Voltage Summary display:

- **Feeder:** The name of the feeder.
- **Feeder Highest Voltage Device ID:** ID of the device with maximum voltage.
- **Feeder Maximum Voltage Value:** Maximum voltage value per unit.
- **Feeder Maximum Voltage Device A/B/C ID:** ID of maximum voltage device for phases A, B, and C.
- **Feeder Maximum Voltage Value A/B/C:** Maximum voltage value for phases A, B, and C, per unit.
- **Feeder Lowest Voltage Device ID:** ID of the device with minimum voltage.
- **Feeder Minimum Voltage Value:** Minimum voltage value per unit.
- **Feeder Lowest Voltage Device A/B/C ID:** ID of minimum voltage device for phases A, B, and C.
- **Feeder Minimum Voltage Value A/B/C:** Minimum voltage value for phases A, B, and C, per unit.
- **Feeder Maximum Voltage Drop Device ID:** ID of the device with maximum voltage drop.
- **Feeder Maximum Voltage Drop Value:** Maximum voltage drop value per unit.
- **Feeder Maximum Voltage Drop Device A/B/C ID:** ID of maximum voltage drop device for phases A, B, and C.
- **Feeder Maximum Voltage Drop Value A/B/C:** Maximum voltage drop value for phases A, B, and C, per unit.

Navigation:

- "Locate Highest Voltage Device" navigates to the device with maximum voltage on the geographic view.
- "Locate Highest Voltage Device A/B/C" navigates to the phase A/B/C device with maximum voltage on the geographic view.
- "Locate Lowest Voltage Device" navigates to the device with minimum voltage on the geographic view.
- "Locate Lowest Voltage Device A/B/C" navigates to the phase A/B/C device with minimum voltage on the geographic view.
- "Locate Maximum Voltage Drop Device" navigates to the device with maximum voltage drop on the geographic view.
- "Locate Maximum Voltage Drop Device A/B/C" navigates to the phase A/B/C device with maximum voltage drop on the geographic view.

1.13.18.2. Feeder Loading Summary

The Feeder Loading Summary display can be accessed from the Feeder Summary display. It provides the feeder voltage details.

Display call-up URL:

`netview://<Mode>/gridtabular/FeederLoadSummary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Loading Summary display:

- **Feeder:** The name of the feeder.
- **Feeder Percent Loading:** Feeder loading in percentage.
- **Feeder Percent Loading A/B/C:** Feeder loading in phases A, B, and C in percentage.
- **Feeder Maximum Percent Loaded Device ID:** ID of the maximum loaded device.
- **Feeder Maximum Percent Loaded Value:** The value of maximum feeder loading in percentage.
- **Feeder Maximum Percent Loaded Device A/B/C ID:** ID of the maximum loaded device for phases A, B, and C.

Navigation:

- "Locate Maximum Percent Loaded Device" navigates to the device with maximum loading on the geographic view.
- "Locate Maximum Percent Loaded Device A/B/C" navigates to the phase A/B/C device with maximum loading on the geographic view.

1.13.18.3. Feeder Loss Summary

The Feeder Loss Summary display can be accessed from the Feeder Summary display. It provides the feeder voltage details.

Display call-up URL:

`netview://<Mode>/gridtabular/FeederLossSummary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Loss Summary display:

- **Feeder:** The name of the feeder.
- **Feeder Line Loss:** Feeder line loss in kW.
- **Feeder Loss Percentage:** Feeder line loss in percentage.
- **Feeder Load:** Feeder load in kW.
- **Feeder Load:** Feeder load in kVAR.
- **Feeder Maximum Loss Device ID:** ID of the device with maximum losses.
- **Feeder Maximum Loss Value:** The value of maximum feeder loss in kW.
- **Feeder Maximum Loss Device A/B/C ID:** ID of the with maximum loss device for phases A, B, and C.
- **Feeder Maximum Loss Value A/B/C:** The value of maximum feeder loss in phases A, B, and C in kW.

Navigation:

- "Locate Maximum Loss Device" navigates to the device with maximum loss on the geographic view.
- "Locate Maximum Loss Device A/B/C" navigates to the A/B/C phase device with maximum loss on the geographic view.

1.13.18.4. Feeder Min Max Summary

The Feeder Min Max Summary display provides detailed information about feeder measurements for a station.

Display call-up URL:

`netview://<Mode>/gridtabular/FeederMinMax_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Min Max Summary display:

- **Feeder:** Name of the feeder.
- **Feeder kV LL Average:** Average line-to-line voltage in kV.

- **Num Deenergized:** The number of de-energized customers.
- **Feeder Loss - Line:** Real power loss in kW.
- **Feeder Loss Percentage:** Real power loss in percentage.
- **Feeder Load (kW):** Feeder load in kW.
- **Feeder Load (kVAR):** Feeder load in kVAR.
- **Percentage Imbalance on Amps:** Current imbalance in percentage.
- **Percentage Imbalance on Volts:** Voltage imbalance in percentage.
- **Feeder Percent Loading:** Feeder loading in percentage.
- **Feeder Percent Loading A/B/C:** Feeder loading in percentage for phases A, B, and C.
- **Feeder Max Volt Device:** ID of the device that has the highest voltage, which exceeds the high voltage limit value.
- **Feeder Max Volt Value:** Maximum voltage limit value.
- **Feeder Max Volt Device A/B/C:** ID of the device that has the highest voltage, which exceeds the high voltage limit value for phase A, B, or C.
- **Feeder Max Volt Value A/B/C:** Maximum voltage limit value for phases A, B, and C.
- **Feeder Max Percent Loaded Device:** ID of the device that has the maximum loading percentage.
- **Max Percent Loaded Value:** Maximum loading percentage.
- **Feeder Max Percent Loaded Device A/B/C:** ID of the device that has the maximum loading percentage for phases A, B, and C.
- **Max Percent Loaded Value A/B/C:** Maximum loading percentage for phases A, B, and C.
- **Feeder Three Phase Conn:** Total three-phase load connected to the feeder in kVA.
- **Feeder Three Phase Conn Customers:** Total number of customers that have the three-phase power supply.
- **Feeder Conn Customers A/B/C:** Total number of customers that are supplied from phases A, B, and C.
- **Feeder Min Volt Device:** ID of the device that has the minimum voltage, which is below the low voltage limit value.
- **Feeder Min Volt Value:** Minimum voltage limit value.
- **Feeder Min Volt Device A/B/C:** ID of the device that has the highest voltage, which exceeds the low voltage limit value for phases A, B, and C.
- **Feeder Min Volt Value A/B/C:** Minimum voltage limit value for phases A, B, and C.
- **Feeder Max Voltage Drop Device:** ID of the device with the maximum voltage drop value.
- **Feeder Max Voltage Drop Value:** Voltage drop limit value.
- **Feeder Max Voltage Drop Device A/B/C:** ID of the device with the maximum voltage drop value for phases A, B, and C.

- **Feeder Max Voltage Drop Value A/B/C:** Voltage drop limit value for phases A, B, and C.
- **Feeder kW Loss Device:** ID of the device with the maximum kW loss value.
- **Feeder Max kW Loss Value:** Maximum kW loss value.
- **Feeder kW Loss Device A/B/C:** ID of the device with the maximum kW loss value for phases A, B, and C.
- **Feeder Max kW Loss Value A/B/C:** Maximum kW loss value for phases A, B, and C.

1.13.19. Reserved Capacity DPF Results

The Reserved Capacity DPF Results display is accessed from the Distribution Power Flow Summary for Station display. It provides detailed information about the reserved capacity objects of a station.

Display call-up URL:

`netview://<Mode>/gridtabular/ReservedCapacityDPF_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on Reserved Capacity DPF Results display:

- **Name:** Name of the reserved capacity.
- **ID:** ID of the reserved capacity.
- **Nominal Feeder:** Name of the nominal (owning) feeder.
- **Real Time Feeder:** Name of the energizing feeder.
- **Phase:** Reserved capacity phasing.
- **Units:** The measurement units of the reserved capacity (1 = kW and 2 = kVA).
- **Real Power A/B/C:** Real power in phases A, B, and C in kW.
- **Reactive Power A/B/C:** Reactive power in phases A, B, and C in kVAR.
- **Firm Commitment:** The firm commitment flag (can be Yes or No).
- **Feeder Node ID 1/2:** The ID of the network nodes 1 and 2.
- **Station Reserved Capacity Value 1/2:** The value of the reserved capacity for the station.
- **Feeder Reserved Capacity Value 1/2:** The value of the reserved capacity for the station.
- **Load ID 1/2:** The ID of the associated loads 1 and 2.

Navigation:

- "Locate" navigates to the reserved capacity on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up for the device.
- "Operator Attributes" opens the Operator Attributes pop-up for the device.
- "Loads" opens the Reserved Capacity Associated Loads display.

1.13.19.1. Reserved Capacity Associated Loads

The Reserved Capacity Associated Loads display shows the information for the loads that are associated with the selected reserved capacity. The display can be accessed by clicking the Loads button on the Reserved Capacity DPF Results display.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReservedCapacityAssociatedLoads_GridDisplay/<Reserved Capacity ID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the load.
- **ID:** Unique identifier of the load.
- **Nominal Feeder:** Nominal (owning) feeder name.
- **Real Time Feeder:** Energizing feeder name.
- **Phase:** Device phasing.
- **Voltage Phase A/B/C:** Phase A, B, and C voltages.
- **Real Power Phase A/B/C:** Real power values at phases A, B, and C in kW.
- **Total Real Power:** Total real power value in kW.
- **Reactive Power Phase A/B/C:** Reactive power values at phases A, B, and C in kVAR.
- **Total Reactive Power:** Total reactive power value in kVAR.
- **Three Phase Power Factor:** Three-phase power factor.

Navigation:

- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.
- "Operator Outputs" opens the Operator Output pop-up.
- "Locate" locates the device on the geographic view.

1.13.19.2. Reserved Capacity Output

The Reserved Capacity Output display is accessed from the Distribution Power Flow Summary for Station display. It provides detailed information about the reserved capacity objects in the station.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReservedCapacityOutput_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Reserved Capacity Output display:

- **Name:** Name of the reserved capacity.
- **ID:** ID of the reserved capacity.
- **Primary Energization:** Energizing station and feeder for the primary reserved capacity.
- **Primary Line Commitment:** Line commitment value for the primary reserved capacity.
- **Primary Station Commitment:** Station commitment value for the primary reserved capacity.
- **Primary Load P:** Real power load value for the primary reserved capacity.
- **Primary Load Q:** Reactive power load value for the primary reserved capacity.
- **Alternate Energization:** Energizing station and feeder for the alternate reserved capacity.
- **Alternate Line Commitment:** Line commitment value for the alternate reserved capacity.
- **Alternate Station Commitment:** Station commitment value for the alternate reserved capacity.
- **Alternate Load P:** Real power load value for the alternate reserved capacity.
- **Alternate Load Q:** Reactive power load value for the alternate reserved capacity.

1.13.20. Sensor Location Summary

The Sensor Location Summary display is accessed from the Distribution Power Flow Summary for Station display. It provides a comparison of the measured and calculated values for each sensor location object.

Display call-up URL:

```
netview://<Mode>/gridtabular/DpfSensorLocation_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on Sensor Location Summary display:

- **Name:** Name of the sensor location.
- **ID:** ID of the sensor location.
- **Node ID:** ID of the node connected to the sensor location object.
- **Flow Branch ID:** ID of the flow branch connected to the sensor location object.
- **Locate:** Navigates to the sensor location in the geographic view.
- **Measured Amps (Phase A):** Amp measurement value for phase A.
- **Calculated Amps (Phase A):** DPF calculated amp value for phase A.
- **Measured Amps (Phase B):** Amp measurement value for phase B.
- **Calculated Amps (Phase B):** DPF calculated amp value for phase B.
- **Measured Amps (Phase C):** Amp measurement value for phase C.

- **Calculated Amps (Phase C):** DPF calculated amp value for phase C.
- **Measured Volts (Phase A):** Volt measurement value for phase A.
- **Calculated Volts (Phase A):** DPF calculated volt value for phase A.
- **Measured Volts (Phase B):** Volt measurement value for phase B.
- **Calculated Volts (Phase B):** DPF calculated volt value for phase B.
- **Measured Volts (Phase C):** Volt measurement value for phase C.
- **Calculated Volts (Phase C):** DPF calculated volt value for phase C.
- **Fault Status (Phase A):** Fault status measurement for phase A.
- **Fault Status (Phase B):** Fault status measurement for phase B.
- **Fault Status (Phase C):** Fault status measurement for phase C.

1.13.21. DPF Feeder Results

The DPF Feeder Results display shows Distribution Power Flow output related to feeders associated with the substation. It shows current and real/reactive power with respect to each phase of the feeder. High and Low voltage at some point of the feeder is also shown.

This display can be accessed by clicking the DPF Output button on the Station Output Summary display. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Distribution Power Flow option.

Display call-up URL:

`netview://<Mode>/gridtabular/feederDPF_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the feederDPF display:

- **Feeder:** Feeder name.
- **Operating Voltage:** The nominal operating voltage for the feeder in kV.
- **Highest Load or Bus Voltage:** The highest voltage calculated at some point on the feeder, expressed in percentage.
- **Lowest Load or Bus Voltage:** The lowest voltage calculated at some point on the feeder, expressed in percentage.
- **Current Phase A/B/C:** The A, B, and C phase currents at the feeder head in Amps.
- **Real Power Phase A/B/C:** The real power calculated at the feeder head for phases A, B, and C in kW.
- **Total Real Power:** The total real power calculated at the feeder head in kW.
- **Reactive Power Phase A/B/C:** The reactive power calculated at the feeder head for phases A, B, and C in kVAR.
-

Total Reactive Power: The total reactive power calculated at the feeder head in kVAR.

- **Three-Phase Power Factor:** The three-phase power factor calculated at the head of the feeder.

Navigation:

- "Feeder Head Branch" navigates to the feeder head branch.

1.13.22. DPF Analyst/Debug Display

The DPF Analyst/Debug display provides voltage drop check details. The display can be accessed by clicking DPF Analyst Results from the real-time and study menus on the main toolbar.

Display call-up URL:

`netview://<Mode>/gridtabular/DPFAnalystDebug_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the DPF Analyst/Debug display:

- **Station:** Station name.
- **Area:** Area name.
- **Voltage Change – Conductors:** The total number of lines with voltage changes that exceed the limit. If clicked, opens the Regulators Voltage Drop Details display.
- **Voltage Change – Distribution Transformers:** The total number of distribution transformers with voltage changes that exceed the limit. If clicked, opens the Distribution Transformers Voltage Drop Details display.
- **Voltage Change – Step Transformers:** The total number of step transformers with voltage changes that exceed the limit. If clicked, opens the Step Transformers Voltage Drop Details display.
- **Voltage Change – Auto Transformers:** The total number of auto transformers with voltage changes that exceed the limit. If clicked, opens the Auto Transformers Voltage Drop Details display.
- **Voltage Change – Power Transformers:** The total number of power transformers with voltage changes that exceed the limit. If clicked, opens the Power Transformers Voltage Drop Details display.
- **Voltage Change – Regulators:** The total number of regulators with voltage changes that exceed the limit. If clicked, opens the Regulators Voltage Drop Details display.
- **Feeder Main Open Switching Device Maximum Phase Angle Mismatch:** Shows the largest angle difference (in red if any one open switch exceeds the limit set in the DNAF Configuration Editor). If clicked, opens the Feeder Main Switch Summary display.

Navigation:

- "Locate" navigates to the station on the geographic view.

1.13.22.1. Lines Voltage Drop Details

The Line Voltage Drop Details display provides voltage drop check details for all the lines that violate the voltage drop limit in the selected station. The display can be accessed from the DPF Analyst/Debug Display when the column Voltage Change – Conductors is red.

Display call-up URL:

`netview://<Mode>/gridtabular/LineVoltDropDetails_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Lines Voltage Drop Details display:

- **Device ID:** Device ID of the line.
- **Device Name:** Device name.
- **Feeder:** Name of the feeder the line belongs to.
- **Side 1 (From) Phase A/B/C Voltage:** The voltage in the from-node of the line for phases A, B, and C.
- **Side 2 (To) Phase A/B/C Voltage:** The voltage in the to-node of the line for phases A, B, and C.
- **Phase A/B/C Voltage Difference:** The per unit voltage drop between the from-node and the to-node for phases A, B, and C, in percentage.

Navigation:

- "Locate" navigates to the line on the geographic view.

1.13.22.2. Distribution Transformers Voltage Drop Details

The Distribution Transformers Voltage Drop Details display provides voltage drop check details for all the distribution transformers that violate the voltage drop limit in the selected station.

The display can be accessed from the DPF Analyst/Debug Display when the column Voltage Change – Distribution Transformers is red.

Display call-up URL:

`netview://<Mode>/gridtabular/DistTfVoltDropDetails_griddisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Distribution Transformers Voltage Drop Details display:

- **Device ID:** ID of the distribution transformer.
- **Device Name:** The name of the distribution transformer.
- **Feeder:** Name of the feeder the distribution transformer belongs to.
- **Side 1 (Primary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the primary side of the distribution transformer.

- **Side 1 (Primary) Average Voltage:** The line-to-line average voltage in the primary side of the distribution transformer.
- **Side 2 (Secondary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the secondary side of the distribution transformer.
- **Side 2 (Secondary) Average Voltage:** The Line-to-Line average voltage in the secondary side of the distribution transformer.
- **Average Voltage Difference:** The average voltage drop difference between the primary and secondary side of the distribution transformer, in percentage.

Navigation:

- "Locate" navigates to the distribution transformer on the geographic view.

1.13.22.3. Power Transformers Voltage Drop Details

The Power Transformers Voltage Drop Details display provides voltage drop check details for all the power transformers that violate the voltage drop limit in the selected station.

The display can be accessed from the DPF Analyst/Debug Display when the column Voltage Change – Power Transformers is red.

Display call-up URL:

```
netview://<Mode>/gridtabular/PowerTfVoltDropDetails_griddisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Power Transformers Voltage Drop Details display:

- **Device ID:** ID of the power transformer.
- **Device Name:** Name of the power transformer.
- **Feeder:** Name of the feeder the transformer belongs to.
- **Side 1 (Primary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the primary side of the power transformer.
- **Side 1 (Primary) Average Voltage:** The line-to-line average voltage in the primary side of the power transformer.
- **Side 2 (Secondary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the secondary side of the power transformer.
- **Side 2 (Secondary) Average Voltage:** The Line-to-Line average voltage in the secondary side of the power transformer.
- **Average Voltage Difference:** The average voltage drop difference between the primary and secondary side of the power transformer, in percentage.

Navigation:

- "Locate" navigates to the power transformer on the geographic view.

1.13.22.4. Auto Transformers Voltage Drop Details

The Auto Transformers Voltage Drop Details display provides voltage drop check details for all the auto transformers that violate the voltage drop limit in the selected station.

The display can be accessed from the DPF Analyst/Debug Display when the column Voltage Change – Auto Transformers is red.

Display call-up URL:

```
netview://<Mode>/gridtabular/AutoTfVoltDropDetails_griddisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Auto Transformers Voltage Drop Details display:

- **Device ID:** ID of the auto transformer.
- **Device Name:** Name of the auto transformer.
- **Feeder:** Name of the feeder the transformer belongs to.
- **Side 1 (Primary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the primary side of the auto transformer.
- **Side 1 (Primary) Average Voltage:** The line-to-line average voltage in the primary side of the auto transformer.
- **Side 2 (Secondary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the secondary side of the auto transformer.
- **Side 2 (Secondary) Average Voltage:** The Line-to-Line average voltage in the secondary side of the auto transformer.
- **Average Voltage Difference:** The average voltage drop difference between the primary and secondary side of the auto transformer, in percentage.

Navigation:

- "Locate" navigates to the transformer on the geographic view.

1.13.22.5. Step Transformers Voltage Drop Details

The Step Transformers Voltage Drop Details display provides voltage drop check details for all the step-up-down transformers that violate the voltage drop limit in the selected station.

The display can be accessed from the DPF Analyst/Debug Display when the column Voltage Change – Step Transformers is red.

Display call-up URL:

```
netview://<Mode>/gridtabular/StepTfVoltDropDetails_griddisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Step Transformers Voltage Drop Details display:

- **Device ID:** ID of the step-up-down transformer.
- **Device Name:** Name of the step-up-down transformer.
- **Feeder:** Name of the feeder the transformer belongs to.
- **Side 1 (Primary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the primary side of the transformer.
- **Side 1 (Primary) Average Voltage:** The line-to-line average voltage in the primary side of the transformer.
- **Side 2 (Secondary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the secondary side of the transformer.
- **Side 2 (Secondary) Average Voltage:** The Line-to-Line average voltage in the secondary side of the transformer.
- **Average Voltage Difference:** The average voltage drop difference between the primary and secondary side of the transformer, in percentage.

Navigation:

- "Locate" navigates to the transformer on the geographic view.

1.13.22.6. Regulators Voltage Drop Details

The Regulators Voltage Drop Details display provides voltage drop check details for all the regulators that violate the voltage drop limit in the selected station.

The display can be accessed from the DPF Analyst/Debug Display when the column Voltage Change – Regulators is red.

Display call-up URL:

`netview://<Mode>/gridtabular/RegulatorVoltDropDetails_griddisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Regulators Voltage Drop Details display:

- **Device ID:** ID of the regulator.
- **Device Name:** Name of the regulator.
- **Feeder:** Name of the feeder the regulator belongs to.
- **Side 1 (Primary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the primary side of the regulator.
- **Side 1 (Primary) Average Voltage:** The line-to-line average voltage in the primary side of the regulator.
- **Side 2 (Secondary) Phase A/B/C Voltage:** The voltage of phases A, B, and C in the secondary side of the regulator.
- **Side 2 (Secondary) Average Voltage:** The Line-to-Line average voltage in the secondary side of

the regulator.

- **Average Voltage Difference:** The average voltage drop difference between the primary and secondary side of the regulator, in percentage.

Navigation:

- "Locate" navigates to the regulator on the geographic view.

1.13.22.7. Feeder Main Switch Summary

The Feeder Main Switch Summary display provides the feeder main boundary switches whose state is currently open in the station. The Angle Difference cell is red when the value is greater than Max Phase Angle Value set in the system.

The display can be accessed from the DPF Analyst/Debug Display via the column "Feeder Main Switch".

Display call-up URL:

`netview://<Mode>/gridtabular/StationFdrMainBoundarySwt_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on Feeder Main Switch Summary display:

- **ID:** ID of the switch.
- **Name:** Name of the switch.
- **Station:** Name of the station the switch belongs to.
- **Feeder:** Name of the feeder the switch belongs to.
- **Side 1 Phase A/B/C Angle:** The phase angle values of the From node for phases A, B, and C.
- **Side 2 Phase A/B/C Angle:** The phase angle values of the To node for phases A, B, and C.
- **Phase A/B/C Angle Difference:** The difference of phase angle values between from node and to node for phases A, B, and C, in degrees.

Navigation:

- "Locate" navigates to the switch on the geographic view.

1.14. Short Circuit Calculation Summary

The Short Circuit Calculation (SCC) Summary display can be accessed from the Station Output Summary and Base Application Status displays. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Short Circuit Calculation option. Short Circuit Calculation Results are shown for each simulated fault type in a substation.

Display call-up URL:

`netview://<Mode>/gridtabular/shortcircuitcalculation_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the Short Circuit Calculation Summary display:

- **Station:** The name of the substation.
- **Device ID:** The unique identifier of the selected device.
- **Type of Fault:** The simulated fault type (AG, BG, CG, ABG, ACG, BCG, AB, BC, AC, ABC, ABCG).
- **Fault Impedance:** Fault impedance evaluation in R + jX ohms.
- **Phase A/B/C Fault Current:** Phase A, B, and C currents in Amps.
- **Phase A/B/C Post-Fault Voltage:** Phase A, B, and C voltage after the fault (expressed as a p.u. value).
- **Node ID:** The unique identifier of the simulated fault node of the selected device.

1.15. Protection Validation Results

The Protection Validation Results display can be accessed from the Station Output Summary display. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Protection Validation option. Protection validation and short circuit results are shown for all of the breakers and overcurrent relays within a substation.

Display call-up URL:

`netview://<Mode>/gridtabular/faultanalysis_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the Protection Validation Results display:

- **Energizing Station:** The dynamic (energizing) station for the device.
- **Energizing Feeder:** The dynamic (energizing) feeder for the device.
- **Fault Impedance Evaluation:** The fault impedance evaluation setting for the dynamic (energizing) feeder.
- **Fault Impedance - Bolted Fault:** The bolted fault impedance (zero-fault impedance) in R + jX ohms.
- **Fault Impedance - User Configured Fault Impedance:** The user-configured fault impedance in R + jX ohms.
- **Device Name:** The name of the associated switch.
- **Device Type:** The type of switch (for example, breaker, recloser) and the relay/recloser type (for example mechanical, electronic).
- **Device Status:** The current state of the switch (Open or Closed).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or

Sectionalizer).

- **Application Protection Zone Self-Protected Power Transformer:** Shows the name of the bounding self-protected power transformer. If the device name is not provided, this field shows the device ID.
- **Maximum Interruption Violation:** Indicator that the calculated maximum fault current exceeds the capacity of the switch.
- **Minimum Phase Pick-Up Violation:** Indicator that the minimum calculated fault current is less than the minimum pick-up current of the protection relay.
- **Minimum Ground Pick-Up Violation:** Indicator that the minimum ground fault current is less than the minimum pick-up current of the ground fault relay.

Navigation:

- "Locate" opens the Network View in the current viewport and locates the switch.
- "Application Protection Zone Self-Protected Power Transformer" navigates to the bounding self-protected power transformer on the geographic view.
- "Maximum Interruption Results" opens the Protection Validation (Maximum Interruption) Results display.
- "Minimum Phase Pick-up Results" opens the Protection Validation (Minimum Phase Pick-Up) Results display.
- "Minimum Ground Pick-up Results" opens the Protection Validation (Minimum Ground Pick-Up) Results display.
- "Validation and Coordination Results" opens the Protection Validation and Coordination display.

1.15.1. Protection Validation and Coordination

This display can be accessed from Protection Validation Results display.

The Protection Validation and Coordination display informs about Short Circuit Analysis (Fault Analysis) and shows the information related to phase and ground protecting devices.

Display call-up URL:

```
netview://<Mode>/gridtabular/FAValidCoord_GridDisplay/<Substation Name>/
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the FAValidCoord display:

- **Switch Name:** Name of the switch.
- **Alternate ID:** Feeder second unique identifier.
- **Energizing Station:** The name of the associated substation to which the selected element is connected.
- **Energizing Feeder:** The name of the associated feeder.

- **Switch Type:** Type of the element.
- **Switch Status:** Status, can be closed or open.
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Protection Validation:** Short Circuit Validation.
- **Minimum Phase Protective Device Name:** The name of phase protective device name.
- **Minimum Phase Protective Device ID:** Unique identifier of minimum phase protective device.
- **Minimum Phase Pick-Up Current:** Minimum phase pick-up current limit in Amps.
- **Minimum Phase Fault Current:** The lowest short circuit current calculated at some point in the network downstream of the switch in Amps.
- **Minimum Ground Protective Device Name:** The name of minimum ground protective device.
- **Minimum Ground Protective Device ID:** Unique identifier of minimum ground protective device.
- **Minimum Ground Pick-Up Current:** Minimum ground pick-up current limit in Amps.
- **Minimum Ground Fault Current:** Minimum ground fault current in Amps.
- **Protection Coordination:** Coordination of short circuit.
- **Upstream Protection Device:** Upstream protection device name.

Navigation:

- "Locate" locates the switch in the current viewport.

1.15.2. Protection Validation Results for Plan

The Protection Validation Results for Plan display can be accessed from FISR Plan, FISRCA Plan, PLOS Plan, and AFR Plan displays by clicking the Protection Validation Results button. Protection validation and short circuit results are shown for all of the breakers and reclosed in the plan.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysis_GridDisplay/<Substation Name>/
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Switch Alternate ID:** The second unique identifier of the switch.
- **Fault Impedance Evaluation:** Fault impedance evaluation type (Bolted Fault or User Configured Impedance).
- **Fault Impedance - Bolted Fault:** The bolted fault impedance (zero-fault impedance) in R + jX

ohms.

- **Fault Impedance - User Configured Fault Impedance:** The user-configured fault impedance in R + jX ohms.
- **Device Type:** The type of switching device (for example, breaker, recloser).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Neutral Relay Setting:** Neutral relay current trip setting in Amps.
- **Neutral Current:** The neutral current in Amps.
- **Neutral Current Percentage:** The neutral current percentage, which is calculated as the neutral current divided by the product of the neutral relay current trip setting and the user-configured safety factor. If the calculated percentage is greater than 100, the cell color is shown in red to indicate that there is a neutral current violation.
- **Switch Status:** Status of the switch.
- **Application Protection Zone Self-Protected Power Transformer:** Shows the name of the bounding self-protected power transformer. If the device name is not provided, this field shows the device ID.
- **Maximum Interruption Violation:** Flag indicating the maximum interruption violation.
- **Minimum Phase Pick-Up Violation:** Flag indicating minimum pick up violation.
- **Minimum Ground Pick-Up Violation:** Indicator that the minimum ground fault current is less than the minimum pick-up current of the ground fault relay.
- **Peak Load Violation:** Flag indicating peak load violation.

Navigation:

- "Switch Analyst Attributes" opens the Analyst Attributes display for the switch.
- "Switch Operator Attributes" opens the Operator Attributes display for the switch.
- "Locate" locates the device in the current viewport.
- "Application Protection Zone Self-Protected Power Transformer" navigates to the bounding self-protected power transformer on the geographic view.
- "Maximum Interruption Results" opens the Protection Validation (Maximum Interruption) Results display.
- "Minimum Phase Pick-up Results" opens the Protection Validation (Minimum Phase Pick-Up) Results display.
- "Minimum Ground Pick-up Results" opens the Protection Validation (Minimum Ground Pick-Up) Results display.

1.15.3. Protection Validation (Maximum Interruption) Results

This display can be accessed from the Protection Validation Results displays.

The Protection Validation (Maximum Interruption) Results display provides information about maximum interruption fault current, maximum interruption current, and maximum fault current. The results are obtained by simulating a fault on the load side of the breakers and reclosers in a substation that are involved in the reconfiguration plan.

Display call-up URL:

```
netview://<Mode>/gridtabular/FaultAnalysisMax_GridDisplay/<Substation Name>
```

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysisMax_GridDisplay/<Substation Name>/<PlanID>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Alternate ID:** The second unique identifier of the switch.
- **Energizing Station:** The dynamic (energizing) station for the breaker or recloser. Appears for substation application protection validation results only.
- **Energizing Feeder:** The dynamic (energizing) feeder for the breaker or recloser. Appears for substation application protection validation results only.
- **Device Type:** The type of switching device (for example, breaker, recloser).
- **Switch Status:** The current state of the switch (Open or Closed).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Relay Type:** The overcurrent protection type. Applies only to breakers.
- **Maximum Interruption Current:** The maximum interruption capacity of the switch in Amps.
- **Maximum Fault Current with Bolted Fault (Amps):** The maximum short circuit current (in Amps) calculated for a location immediately downstream of the protective device evaluated with a zero-fault impedance.
- **Maximum Fault Current with Configured Fault Impedance (Amps):** The maximum short circuit current (in Amps) calculated for a location immediately downstream of the protective device evaluated with a user configured fault impedance.
- **Maximum Capacity Ratio with Bolted Fault (%):** The ratio of the maximum calculated fault current evaluated with a zero-fault impedance to the maximum interruption current of the switch, in percentage.
- **Maximum Capacity Ratio with Configured Fault Impedance (%):** The ratio of the maximum calculated fault current evaluated with a user configured fault impedance to the maximum interruption current of the switch, in percentage.
- **Maximum Interruption Violation:** Indicator that the calculated maximum fault current exceeds the capacity of the switch.

Navigation:

- "Locate" locates the device on the geographic view.
- "Switch Analyst Attributes" opens the Analyst Attributes pop-up. Appears for reconfiguration application plan protection validation results only.
- "Switch Analyst Outputs" opens the Analyst Output pop-up. Appears for reconfiguration application plan protection validation results only.

1.15.4. Protection Validation (Minimum Phase Pick-Up) Results

This display can be accessed from the Protection Validation Results displays.

The Protection Validation (Minimum Phase Pick-up) Results display provides information about minimum phase pick-up current, minimum phase fault current, and minimum phase pick-up ratio. The results are obtained by simulating a phase-to-phase fault at the highest impedance distance from the protection device.

Display call-up URL:

`netview://<Mode>/gridtabular/FaultAnalysisMinPhase_GridDisplay/<Substation Name>`

`netview://<Mode>/gridtabular/ReconfigFaultAnalysisMinPhase_GridDisplay/<Substation Name>/<PlanID>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the FaultAnalysisMinPhase display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Alternate ID:** The second unique identifier of the switch.
- **Energizing Station:** The dynamic (energizing) station for the breaker or recloser. Appears for substation application protection validation results only.
- **Energizing Feeder:** The dynamic (energizing) feeder for the breaker or recloser. Appears for substation application protection validation results only.
- **Device Type:** The type of switching device (for example, breaker, recloser).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Switch Status:** The current state of the switch (Open or Closed).
- **Relay Type:** The overcurrent protection type. This field applies only to breakers.
- **Minimum Phase Pick-up Current with Safety Factor (Amps):** The minimum phase pick-up current (multiplied by a phase trip safety factor) setting of the phase fault relay in Amps.
- **Minimum Phase Fault Current with Bolted Fault (Amps):** The minimum phase short circuit current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance.
- **Minimum Fault Current (Amps):** The minimum of the minimum phase short circuit current (in

Amps) and minimum phase ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance.

- **Minimum Phase Pick-up Ratio with Bolted Fault (%):** The ratio of the minimum phase pick-up current (multiplied by a phase trip safety factor) to the minimum phase fault current evaluated with a zero-fault impedance in percentage.
- **Minimum Phase Fault Current with Configured Fault Impedance (Amps):** The minimum phase short circuit current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a user configured fault impedance.
- **Minimum Phase Pick-up Ratio with Configured Fault Impedance (%):** The ratio of the minimum phase pick-up current (multiplied by a phase trip safety factor) to the minimum phase fault current evaluated with a user configured fault impedance in percentage.
- **Minimum Phase Pick-up Violation:** Indicator that the minimum calculated fault current is less than the minimum pick-up current of the protection relay.
- **Phase Fault Sensitivity with Bolted Fault (%):** The ratio of the minimum phase fault current evaluated with a zero-fault impedance to the minimum phase pick-up current (multiplied by a phase trip safety factor).
- **Phase Fault Sensitivity with Configured Fault Impedance (%):** The ratio of the minimum phase fault current evaluated with a user configured fault impedance to the minimum phase pick-up current (multiplied by a phase trip safety factor).

Navigation:

- "Locate" locates the switch on the geographic view.
- "Minimum Phase Fault Location" locates the device on the geographic view.
- "Switch Analyst Attributes" opens the Analyst Attributes pop-up. Appears for reconfiguration application plan protection validation results only.
- "Switch Analyst Outputs" opens the Analyst Output pop-up. Appears for reconfiguration application plan protection validation results only.

1.15.5. Protection Validation (Minimum Ground Pick-Up) Results

This display can be accessed from the Protection Validation Results displays.

The Protection Validation (Minimum Ground Pick-up) Results display provides information about minimum ground pick-up current, minimum ground fault current, and minimum ground pick-up ratio. The results are obtained by simulating a phase-to-ground fault at the highest impedance distance from the protection device.

Display call-up URL:

```
netview://<Mode>/gridtabular/FaultAnalysisMinGround_GridDisplay/<Substation Name>
```

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysisMinGround_GridDisplay/<Substation
Name>/<PlanID>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the FaultAnalysisMinGround display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Alternate ID:** The second unique identifier of the switch.
- **Energizing Station:** The dynamic (energizing) station for the breaker or recloser. Appears for substation application protection validation results only.
- **Energizing Feeder:** The dynamic (energizing) feeder for the breaker or recloser. Appears for substation application protection validation results only.
- **Device Type:** The type of switching device (for example, breaker, recloser).
- **Switch Status:** The current state of the switch (Open or Closed).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Relay Type:** The overcurrent protection type. Applies only to breakers.
- **Minimum Ground Pick-up Current with Safety Factor (Amps):** The minimum ground pick-up current (multiplied by a ground trip safety factor) setting of the ground fault relay in Amps.
- **Minimum Ground Fault Current with Bolted Fault (Amps):** The minimum ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance.
- **Minimum Ground Pick-up Ratio with Bolted Fault (%):** The ratio of the minimum ground pick-up current (multiplied by a ground trip safety factor) to the minimum ground fault current evaluated with a zero-fault impedance in percentage.
- **Minimum Ground Fault Current with Configured Fault Impedance (Amps):** The minimum ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a user configured fault impedance.
- **Minimum Ground Pick-up Ratio with Configured Fault Impedance (%):** The ratio of the minimum ground pick-up current (multiplied by a ground trip safety factor) to the minimum ground fault current evaluated with a user configured fault impedance, in percentage.
- **Minimum Ground Pick-up Violation:** Indicator that the minimum ground fault current is less than the minimum pick-up current of the ground fault relay.
- **Minimum Ground Fault Type:** Minimum ground fault type.
- **Ground Fault Sensitivity with Bolted Fault (%):** The ratio of the minimum ground fault current evaluated with a zero-fault impedance to the minimum ground pick-up current (multiplied by a ground trip safety factor).

- **Ground Fault Sensitivity with Configured Fault Impedance (%):** The ratio of the minimum ground fault current evaluated with a user configured fault impedance to the minimum ground pick-up current (multiplied by a ground trip safety factor).

Navigation:

- "Locate" locates the switch on the geographic view.
- "Switch Analyst Attributes" opens the Analyst Attributes pop-up. Appears for reconfiguration application plan protection validation results only.
- "Switch Analyst Outputs" opens the Analyst Output pop-up. Appears for reconfiguration application plan protection validation results only.
- "Minimum Ground Fault Location" locates the fault location on the geographic view.

1.16. Fault Location

The Fault Location display shows a list of potential trigger devices and the number of fault locations provided by Fault Analysis functions with respect to a specific breaker.

This display can be accessed from the Station Output Summary display. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Fault Location option.

Display call-up URL:

`netview://<Mode>/gridtabular/faultlocationmain_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Trigger Branch Name:** The name of the device.
- **Trigger Branch ID:** Unique identifier of the device.
- **Solution ID:** Identifier of solution.
- **Solution Time:** Time of solution.
- **Number of Potential Fault Locations:** Number of potential fault locations.

Navigation:

- "Locate" navigates to the trigger branch devices on the geographic view.
- "View Potential Fault Locations" navigates to the Fault Location Results display.

1.16.1. Fault Location Results

The Fault Location Results display provides details on potential fault locations for the substation branches. The display can be accessed from the Fault Location display.

The FaultLocationresult display lists potential fault locations.

Display call-up URL:

`netview://<Mode>/gridtabular/faultlocationresult_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Band:** Error band for the fault location. (0 = Fault is on this branch. 1-3 = Fault is progressively further away).
- **Station Name:** Name of the substation.
- **Branch Device ID:** Unique identifier of branch device.
- **Phase A/B/C Calculated Fault Current:** Calculated fault current at fault location for phases A, B, and C, in Amps.
- **Difference A/B/C:** Difference between the measured fault current (or maximum manually entered fault current) and the calculated fault current at the fault location for phases A, B, and C, in percentage.
- **Phase A/B/C Calculated Fault Voltage:** Calculated fault voltage at fault location for phases A, B, and C (expressed as a p.u. value).

Navigation:

- "Fault Location" navigates to fault location on the geographical view
- "Locate in URD" navigates to fault location on the loop schematic view.

1.17. LVM Station Summary

The LVM Station Summary display provides the summary of LVM application results by station. This display can be accessed from RT and ST menus.

Display call-up URL:

`Netview://<Mode>/gridtabular/LVMStationSummary_GRIDDISPLAY`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the LVMStationSummary display:

- **Station:** Name of substation.
- **LVM Enabled**: Indicates whether the substation has some feeders with the DNAF LVM application enabled (Yes or No).
- **Active Problem Formulation(s)**: The substation and its feeder’s active problem formulations.
- **LVM Solution Type**: Type of the LVM solution (for example, optimization).
- **Time Last Solved (with a Plan):** The LVM application’s last execution time.

- **Time Since Last Plan was Created (Hours:Minutes)**: The time elapsed since the last time LVM was executed.
- **Maximum Bus Voltage Mismatch**: The maximum difference between measured and calculated voltages at station internal and external buses.
- **Capacitor Operations in Monitored Interval**: The maximum number of operations over all the capacitors in the substation during the monitoring interval.
- **LTC and Regulator Operations in Monitored Interval**: The maximum number of operations over all the LTCs and regulators in the substation during the monitoring interval.
- **Consecutive Infeasible Solutions**: The number of consecutively generated infeasible LVM solutions. The cell is red if this counter exceeds the maximum value set in the DNAF Configuration Editor.
- **Daily Infeasible Solutions**: The number of all infeasible solutions generated in one day. The cell is red if this counter exceeds the maximum value set in the DNAF Configuration Editor.
- **Consecutively Discarded Solutions**: The number of consecutively discarded optimization LVM solutions. The cell is red if this counter exceeds the maximum value set in the DNAF Configuration Editor.
- **Daily Discarded Solutions**: The number of all discarded optimization LVM solutions on a given day. The cell is red if this counter exceeds the maximum value set in the DNAF Configuration Editor.

  The maximum number of daily discarded solutions is counted from midnight. At the beginning of each day, the discarded solutions count is reset to zero. If the interval duration is not an integral divisor of 24 hours, the last interval is truncated at the next midnight.
- **Optimization Group Name**: Name of the optimization group the station belongs to.
- **LVM Throttling Status**: Indicates whether real/reactive power throttling is enabled (Yes/No).
- **LVM Throttling Delay**: Shows the current LVM throttling delay, in seconds.

Navigation:

- "Locate" locates the substation on the geographic view.
- "Maximum Voltage Mismatch (Station Internal Buses)" navigates to the Bus Voltage Summary display to view the measured and calculated voltages and the voltage mismatch for every station internal device.
- "Capacitor Operations in Monitored Interval" navigates to the Capacitor Control display.
- "LTC and Regulator Operations in Monitored Interval" navigates to the LTC/Regulator Control display.
- "LVM Solution Details" navigates to the Load and Volt/VAR Management display to view the plan details.
- "Capacitor Summary" navigates to the Capacitor Control display.
- "LTC and Regulator Summary" navigates to the LTC/Regulator Control display.

1.17.1. Capacitor Control

The Capacitor Control display provides a summary of all remotely controllable capacitors for the selected substation. This display can be accessed by clicking the Capacitor Summary or Capacitor Operations in Monitored Interval field on the LVM Station Summary display. You can also access this display from the substation toolbar by clicking Network Analysis Inputs and selecting the Capacitor option.

Display call-up URL:

`netview://<Mode>/gridtabular/StationCapCtrls_griddisplay/<Sub Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationCapCtrls display:

- **Name:** Name of capacitor.
- **ID:** ID of capacitor.
- **Locate:** Clicking the Locate button navigates to the equipment on the geographic display.
- **Nominal Feeder:** The nominal feeder of the equipment.
- **Control Type:** The control type of this capacitor. It may be "local control only", "remote control only", or "remote and local control".
- **Current Operation Mode:** The capacitor's current operation mode. It may be "Local" or "Remote".
- **Current Status:** The status of capacitor. It may be "On" or "Off".
- **LVM Not Eligible (Daily Count):** The daily number of times a capacitor is not available for use by LVM based on the regular LVM eligibility criteria.
- **Maximum Number of Daily Operations:** The maximum number of operations in one day. This is set by the DNAF Configuration Editor.
- **Daily Operation Count:** The accumulated operation numbers in one day.
- **Monitor Duration Period:** The duration of monitoring (in hours).
- **Monitor Starting Time:** The time when monitoring starts.
- **Monitor Ending Time:** The time when monitoring will end.
- **Operation Count in Monitored Interval:** The number of capacitor operations within the time interval specified by the "Monitor Starting Time" and "Monitor Ending Time" columns.

1.17.2. LTC/Regulator Control

The LTC/Regulator Control display provides the summary of all remotely controllable LTC and regulators for the selected substation. This display can be accessed by clicking the LTC and Regulator Summary or LTC and Regulator Operations in Monitored Interval field on the LVM Station Summary display.

Display call-up URL:

`netview://<Mode>/gridtabular/StationLTCRegCtrls_griddisplay/<Sub Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationLTCRegCtrls display:

- **Device Name:** Name of device.
- **Device ID:** ID of device.
- **Device Type:** The type (LTC or Regulator) of device.
- **Locate:** Clicking the Locate button navigates to the equipment on the geographic display.
- **Nominal Feeder:** The nominal feeder of the equipment.
- **Control Type:** The control type of this capacitor. It may be "local control only", "remote control only", or "remote and local control".
- **Current Operation Mode:** The capacitor's current operation mode. It may be "Local" or "Remote".
- **Tap Position:** Position of the regulating tap.
- **LVM Not Eligible (Daily Count):** The daily number of times a capacitor is not available for use by LVM based on the regular LVM eligibility criteria.
- **Maximum Number of Daily Operations:** The maximum number of operations in one day. This is set by the DNAF Configuration Editor.
- **Daily Operation Count:** The accumulated operation numbers in one day.
- **Monitor Duration Period:** The duration of monitoring (in hours).
- **Monitor Starting Time:** The time when monitoring started.
- **Monitor Ending Time:** The time when monitoring will end.
- **Operation Count in Monitored Interval:** The number of capacitor's operations within the time interval specified by column "Monitor Starting Time" and "Monitor Ending Time".

1.17.3. Heuristic LVM Summary

The Heuristic LVM Summary display show LVM execution mode and plan execution status.

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_Summary_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Execution Mode:** Plan execution mode (1 – Advisory, 2 – Semi-Automatic, 3 – Closed Loop).
- **Status:** Plan execution status.

Navigation:

- "Details" opens the displays as follows (based on the execution mode):
- **Advisory (1):** Capacitors Evaluated and Tap Changers Evaluated.
- **Semi-Automatic (2):** Problem Formulation Details and Problem Formulation Input Parameters.
- **Closed Loop (3):** LTC/Regulators Evaluated and Delayed Tap Moves.

1.17.3.1. Capacitors Evaluated

The Capacitors Evaluated display can be accessed from the Heuristic LVM Summary display by clicking the Details button, and from the Load and Volt/VAR Management display by clicking the Plan Statistics button.

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_CapacitorsEvaluated_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of regulating device.
- **Device ID:** Unique identifier of the regulating device
- **Nominal Feeder:** Feeder of the regulating device.
- **Selected for Plan:** A flag indicating that the device is selected for the LVM plan.
- **Total Voltage Violation:** Total voltage violation expressed as a 120 V base value.
- **Total Voltage Change:** Total voltage violation expressed as a 120 V base value.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.2. Tap Changers Evaluated

The Tap Changers Evaluated display can be accessed from the Heuristic LVM Summary display by clicking the Details button, and from the Load and Volt/VAR Management display.

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_TapChangersEvaluated_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of the regulating device.
- **Device ID:** Unique identifier of the regulating device.

- **Device Feeder:** Feeder of the regulating device.
- **Selected for Plan:** Flag indicating that the device is selected for the LVM plan.
- **Total Voltage Violation:** Total voltage violation expressed as a 120 V base value.
- **Total Voltage Change:** Total voltage violation expressed as a 120 V base value.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.3. LTC/Regulators Evaluated

The LTC/Regulators Evaluated display can be accessed from the Heuristic LVM Summary display by clicking the Details button, and from the Load and Volt/VAR Management.

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_LTC_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of regulating device.
- **Device ID:** Unique identifier of the regulating device
- **Device Station:** Substation of the regulating device.
- **Device Feeder:** Feeder of the regulating device.
- **LVM Eligibility:** LVM eligibility status of the regulating device.
- **LVM Evaluation Order:** LVM evaluation order.
- **Regulator Tap Step Size:** The size of the tap step expressed in percentage.
- **Regulation Margin:** Regulation margin expressed in percentage.
- **Regulation Margin Limit:** Regulation margin limit expressed in percentage.
- **Minimum Available Voltage Margin:** Available voltage margin expressed in percentage.
- **Voltage Imbalance:** Voltage imbalance expressed in percentage.
- **Voltage Imbalance Limit:** Voltage imbalance limit expressed in percentage.
- **Estimated Voltage after Tap Move:** Estimated voltage after the tap move expressed as a 120 V base value.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.4. Problem Formulation Details

The Problem Formulation Details display can be accessed from the Heuristic LVM Summary display by clicking the Details button, and from the Load and Volt/VAR Management.

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_ProbFormDetail_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Capacitor Evaluated:** Name of the device.
- **Selected for Plan:** Flag indicating that the device is selected for the LVM plan.
- **Evaluated:** Flag indicating that the device is evaluated.
- **Total Objective:** Total objective.
- **Lowest Average Voltage Component:** The weight for the lowest average voltage component in the Heuristic LVM capacitor control algorithm.
- **Voltage Reduction Component:** The weight for the voltage reduction component in the Heuristic LVM capacitor control algorithm.
- **Power Factor Component:** The weight for the power factor component in the Heuristic LVM capacitor control algorithm.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.5. Problem Formulation Input Parameters

The Problem Formulation Input Parameters display can be accessed from the Heuristic LVM Summary display by clicking the Details button, and from the Load and Volt/VAR Management.

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_ProbFormParam_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Lowest Average Voltage Exponential Weight (LV1):** The first lowest average voltage coefficient used in the Heuristic LVM capacitor control algorithm.
- **Lowest Average Voltage Exponential Weight (LV2):** The second lowest average voltage coefficient used in the Heuristic LVM capacitor control algorithm.
- **Power Factor Exponential Weight (LV1):** The first power factor coefficient used in the Heuristic LVM capacitor control algorithm.
-

Power Factor Voltage Exponential Weight (LV2): The second power factor coefficient used in the Heuristic LVM capacitor control algorithm.

1.17.3.6. Delayed Tap Moves

The Delayed Tap Moves display can be accessed from the Heuristic LVM Summary display by clicking the Details button, and from the Load and Volt/VAR Management.

Display call-up URL:

```
netview://<Mode>/gridtabular/HeuLVM_DelayedTapMove_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of regulating device.
- **Device ID:** Unique identifier of the regulating device
- **Device Station:** Substation of the regulating device.
- **Device Feeder:** Feeder of the regulating device.
- **Initial Position:** Initial position of the device.
- **Recommended Position:** Plan recommended final position of the device.
- **Move Creation Time:** Time when the move was created.
- **Time Delay Schedule:** Name of the time delay schedule.
- **Scheduled Time Delay:** Schedule time delay in minutes and seconds.
- **Remaining Time to Execution:** Time that remains for the execution.
- **Move Execution Time:** Time of the move execution.
- **Elapsed Time Since Move Creation:** Time since the move was created in minutes and seconds.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.7. Field Regulators Evaluated

Display call-up URL:

```
netview://<Mode>/gridtabular/HeuLVM_FieldReg_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of regulating device.
- **Device ID:** Unique identifier of the regulating device

- **Device Station:** Substation of the regulating device.
- **Device Feeder:** Feeder of the regulating device.
- **LVM Eligibility:** LVM eligibility status of the regulating device.
- **LVM Evaluation Order:** LVM evaluation order.
- **Regulator Tap Step Size:** The size of the tap step expressed in percentage.
- **Regulation Margin:** Regulation margin expressed in percentage.
- **Regulation Margin Limit:** Regulation margin limit expressed in percentage.
- **Available Voltage Margin:** Available voltage margin expressed in percentage.
- **Voltage Imbalance:** Voltage imbalance expressed in percentage.
- **Voltage Imbalance Limit:** Voltage imbalance limit expressed in percentage.
- **Estimated Voltage after Tap Move:** Estimated voltage after the tap move expressed as a 120 V base value.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.8. Station Regulators Evaluated

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_StationReg_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of regulating device.
- **Device ID:** Unique identifier of the regulating device
- **Device Station:** Substation of the regulating device.
- **Device Feeder:** Feeder of the regulating device.
- **LVM Eligibility:** LVM eligibility status of the regulating device.
- **LVM Evaluation Order:** LVM evaluation order.
- **Regulator Tap Step Size:** The size of the tap step expressed in percentage.
- **Regulation Margin:** Regulation margin expressed in percentage.
- **Regulation Margin Limit:** Regulation margin limit expressed in percentage.
- **Available Voltage Margin:** Available voltage margin expressed in percentage.
- **Voltage Imbalance:** Voltage imbalance expressed in percentage.
- **Voltage Imbalance Limit:** Voltage imbalance limit expressed in percentage.
- **Estimated Voltage after Tap Move:** Estimated voltage after the tap move expressed as a 120 V

base value.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.3.9. Best Capacitor

Display call-up URL:

`netview://<Mode>/gridtabular/HeuLVM_BestCap_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Name:** Name of regulating device.
- **Device ID:** Unique identifier of the regulating device
- **Device Station:** Substation of the regulating device.
- **Nominal Feeder:** Feeder of the regulating device.

Navigation:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.4. LVM Plans Summary

The Load and Voltage/VAR Management Plans Summary display shows a summary of the selected LVM plan. This display can be accessed from the Load and Volt/VAR Management display.

Display call-up URL:

`netview://<Mode>/gridtabular/vvcplansmain_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Station:** Name of substation.
- **LVM Station Group:** Name of the optimization group.
- **Problem Formulation:** Name of problem formulation.

Navigation:

- "Plan Statistics" opens the LVM Plan Statistics display.
- "Plan Performance Indices" opens the LVM Plan Weighted Performance Indexes display.

1.17.4.1. LVM Plan Statistics

The Load and Voltage/VAR Management Plan Statistics can be accessed from the LVM Plans Summary display.

LVM Plan Statistics are shown in two panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/vvcplanstat1_GridDisplay/<Station Name>/<Plan ID>`

Or, for mixed problem formulations at feeder, transformer, or station levels:

`netview://<Mode>/gridtabular/vvcplanstatmixed1_GridDisplay/<Station Name>/<Plan ID>`

Second Pane:

`netview://<Mode>/gridtabular/vvcplanstat2_GridDisplay/<Station Name>/<Plan ID>`

Or, for mixed problem formulations at feeder, transformer, or station levels:

`netview://<Mode>/gridtabular/vvcplanstatmixed2_GridDisplay/<Station Name>/<Plan ID>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the vvcplanstat1 display:

- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Post-Plan Total Real Demand:** Total area real power demand after switching plan in kW.
- **Post-Plan Total Reactive Demand:** Total area reactive power demand after switching plan in kVAR.
- **Real Power Demand Reduction:** Changes of area real power demand after the switching plan in kW.
- **Reactive Power Demand Reduction:** Changes of area reactive power demand after the switching plan in kVAR.
- **Pre-Plan Real Power Loss:** Losses before the switching plan in kW.
- **Pre-Plan Reactive Power Loss:** Losses before the switching plan in kVAR.
- **Post-Plan Real Power Loss:** Losses after the switching plan in kW.
- **Post-Plan Reactive Power Loss:** Losses after the switching plan in kVAR.
- **Minimum Target Power Factor:** Minimum target power factor if the problem formulation is Area Reactive Support.
- **Maximum Target Power Factor:** Maximum target power factor if the problem formulation is Area Reactive Support.
- **Pre-Plan Area Power Factor:** Power factor before switching plan.
- **Post-Plan Area Power Factor:** Power factor after switching plan.

- **Pre-Plan Power Transformer Power Factor(s):** Pre-plan power transformer power factor.
- **Post-Plan Power Transformer Power Factor(s):** Post-plan power transformer power factor.

The following information is shown on the vvcplanstat2 display:

- **Maximum Segment Loading:** Maximum segment loading after the LVM plan in percentage.
- **Maximum Loading Segment ID:** Unique identifier of the maximum loaded segment.
- **Maximum Device Voltage:** Maximum load voltage after the LVM plan in percentage.
- **Maximum Voltage Device ID:** Unique identifier of the max voltage load.
- **Minimum Device Voltage:** Minimum load voltage after the LVM plan in percentage.
- **Minimum Voltage Device ID:** Unique identifier of the load minimum voltage.
- **Maximum Regulator Group Voltage Imbalance:** Maximum regulator group voltage imbalance after the LVM plan in percentage.
- **Maximum Regulator Group Voltage Imbalance Device ID:** Unique identifier of the device with the maximum regulator group voltage imbalance after the LVM plan.

Navigation:

- "Locate Maximum Loading" navigates to the Geographic Network view centered on the segment with the maximum loading.
- "Locate Maximum Voltage" navigates to the Geographic Network view centered on the load with the maximum voltage.
- "Locate Minimum Voltage" navigates to the Geographic Network view centered on the load with the minimum voltage.
- "Locate Regulator Group" navigates to the Geographic Network view centered on the device with the maximum regulator group voltage imbalance.
- "Bus Voltages" navigates to the LVM Bus Voltages display.

1.17.4.2. LVM Bus Voltages

The LVM Bus Voltages display provides a summary of all telemetered bus voltages in the station (measured and calculated values). The display can be accessed from the LVM Plan Statistics display.

Display call-up URL:

```
netview://<Mode>/gridtabular/vvcBusVoltage_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Name:** Name of the regulating device.
- **Feeder:** Feeder of the regulating device.
- **Phase A/B/C Pre-Plan Voltage:** Phase A/B/C pre-plan voltage for the regulating device.

- **Phase A/B/C Post-Plan Voltage:** Phase A/B/C post-plan voltage for the regulating device.
- **High Voltage Limit:** High voltage limit for the regulating device.
- **Original High Voltage Limit:** Original high voltage limit for the regulating device.
- **Manual High Voltage Limit:** Manual high voltage limit for the regulating device.
- **High Voltage Limit Source:** High voltage limit source for the regulating device.
- **Low Voltage Limit:** Low voltage limit for the regulating device.
- **Original Low Voltage Limit:** Original low voltage limit for the regulating device.
- **Manual Low Voltage Limit:** Manual low voltage limit for the regulating device.
- **Low Voltage Limit Source:** Low voltage limit source for the regulating device.

1.17.4.3. LVM Plan Weighted Performance Indexes

The Load and Voltage/VAR Management Plan Weighted Performance Indexes display can be accessed from the LVM Plans Summary display.

Weighted performance indexes of the selected Load and Voltage/VAR Management (LVM) plan are shown in this display in two panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/vvcperfindex1_GridDisplay/`

Or, for mixed problem formulations at feeder, transformer, or station levels:

`netview://<Mode>/gridtabular/vvcperfindexmixed1_GridDisplay/`

Second Pane:

`netview://<Mode>/gridtabular/vvcperfindex2_GridDisplay/`

Or, for mixed problem formulations at feeder, transformer, or station levels:

`netview://<Mode>/gridtabular/vvcperfindexmixed2_GridDisplay/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on vvcperfindex1 (vvcperfindexmixed1) display:

- **Plan ID:** The plan ID.
- **Total Weighted PI:** Total weighted performance index.
- **Control Moves:** Number of moves in the plan performance index.
- **Total Cost:** Control cost performance index.
- **Total Intermediate Cost:** Control cost performance index during temporary moves.
- **Flow Violation:** Flow violation performance index.
-

Intermediate Flow Violation: Flow violation performance index during temporary moves.

- **Bus Voltage Violation:** Bus voltage violation performance index.
- **Intermediate Bus Voltage Violation:** Bus voltage violation performance index during temporary moves.
- **Load Voltage Violation:** Load voltage violation performance index.
- **Intermediate Bus Voltage Violation:** Load voltage violation performance index during temporary moves.
- **Losses Weighted PI:** Losses performance index.
- **Real and Reactive Flow:** The sum of the real and reactive flow performance index.
- **Bus Voltage Imbalance Weighted PI:** Bus voltage imbalance performance index.

The following information is shown on the vvcperfindex2 (vvcperfindexmixed2) display:

- **Losses:** Losses performance index.
- **Margin:** Margin to limit performance index.
- **Maximum Segment Margin PI**: Maximum segment margin performance index.
- **Feeder Voltage Drop:** Feeder voltage drop performance index.
- **TCUL Range:** Tap changer under load (LTC) range performance index.
- **Bus Voltage Imbalance:** Bus voltage imbalance performance index.

1.17.5. LVM Plans

The LVM Plans display shows a summary of LVM plans for a substation's LVM solution areas. This display can be accessed from the Station Output Summary display.

Display call-up URL:

`netview://<Mode>/gridtabular/LVMPlans_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Plan Type:** The plan type (station level plan, bank-level sub-plan, or gateway dispatch sub-plan).
- **Number of Moves:** Number of control operations needed for the plan.
- **Plan ID:** The plan ID.

Navigation:

- "Show Steps" opens the Load and Volt/VAR Management display.
- "Plan Statistics" opens the LVM Plan Statistics display.
- "Plan Performance Indices" opens the LVM Plan Weighted Performance Indexes display.
- "Implement Plan" clicking this button implements the plan (applies to study mode only).

- "Restore Plan" clicking this button restores the network to its previous state (applies to study mode only).

1.17.5.1. LVM Plan Steps

The Load and Voltage/VAR Management Plan Steps display lists the recommended control actions, including capacitor switching, regulator/LTC tap positions, and power factor and VAR output recommendations for generators and DERs. This display can be accessed from the LVM Plans display.

Display call-up URL:

`netview://<Mode>/gridtabular/vvcplansteps2_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Step Number:** Step number.
- **Device Name:** Name of the regulating device.
- **Device ID:** Unique identifier of the regulating device
- **Energizing Station:** Energizing substation of the regulating device.
- **Energizing Feeder:** Energizing feeder of the regulating device.
- **Device Type:** The type of regulating device, such as a capacitor, tap changer, or DER.
- **Control Type:** The type of regulating device control, such as open/close a device or raise/lower a step.
- **Phase:** Device phasing.
- **Initial Position or Setting:** Initial position or setting of the regulating device.
- **Recommended Final Position or Setting:** Plan recommended final position or setting of the regulating device.

Navigation and controls:

- "Locate" navigates to the Geographic Network view centered on the selected device.
- "Mark Device" marks the plan devices on the Geographic Network view.

1.17.6. Load and Volt/VAR Management

The Load and Voltage/VAR Management display lists the recommended control actions, including capacitor switching, regulator/LTC tap positions, and power factor and VAR output recommendations for generators and DERs. This display can be accessed from the Station Output Summary and Advanced Application Status displays. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Load and Volt/VAR Management option. Each of the recommended switching steps of the best LVM plan is shown in this display.

Display call-up URL:

```
netview://<Mode>/gridtabular/vvcplansteps_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Step Number:** Step number.
- **Energizing Station:** Energizing substation of the regulating device.
- **Energizing Feeder:** Energizing feeder of the regulating device.
- **Device Name:** Name of the regulating device.
- **Device Type:** The type of regulating device, such as a capacitor, tap changer, or DER.
- **Control Type:** The type of regulating device control, such as open/close a device or raise/lower a step.
- **Phase:** Device phasing.
- **Initial Position or Setting:** Initial position or setting of the regulating device.
- **Recommended Final Position or Setting:** Plan recommended final position or setting of the regulating device.

Navigation and controls:

- "Locate" navigates to the Geographic Network view centered on the selected device.
- "Plan Summary" opens the LVM Plans Summary display.
- "Plan Statistics" opens the LVM Plan Statistics display.
- "Plan Performance Indices" opens the LVM Plan Weighted Performance Indexes display.
- "Plan Violations" opens the LVM Violations display.
- "Implement Plan" clicking this button implements the plan (applies to study mode only).
- "Restore Plan" clicking this button restores the network to its previous state (applies to study mode only).
- "Mark Device" marks the plan devices on the Geographic Network view.

1.17.6.1. LVM Violations

The Load and Voltage/VAR Management Violations display shows a summary of pre-plan and post-plan voltage and flow violations for the selected LVM plan. This display can be accessed from the Load and Volt/VAR Management display.

Display call-up URL:

```
netview://<Mode>/gridtabular/LVMViolations_GridDisplay/<StationName>/<PlanID>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Pre Or Post:** Specifies, whether this is a pre-plan or post-plan violation.
- **Type:** Violation type (Violation or Constraint).
- **Substation:** Name of the substation.
- **Feeder:** Name of the feeder.
- **Plan ID:** ID of the plan.
- **Device ID:** Unique identifier of the object that the violation limit is associated with.
- **Device Type:** Type of the object that the violation limit is associated with (for example, a load).
- **Overload Percent:** The violation value in percent.
    - For overload or high voltage violations, this value is calculated as 100*(Actual Value)/(Limit Value).
    - For low voltage violations, this value is calculated as 100*(Limit Value)/(Actual Value).
- **Actual Value:** The actual violation value. For kVAR violations, this is the actual flow level calculation by DPF (Distribution Power Flow), in kVAR. For voltage violations, this is the violation value in percent.
- **Limit Value:** The violation limit level. For kVAR violations, the value is shown in kVAR; for voltage violations, the value is shown in percent.
- **Limit Type:** The type of limit that has been violated (for example, kVAR, Voltage Imbalance, or Low Voltage LL).
- **Violation Phasing:** Phasing information.
- **Maximum Sensitivity Control Type:** LVM control type with the greatest sensitivity on the violation (for example, Capacitor or Generator Power Factor).
- **Maximum Sensitivity Device ID:** ID of the LVM plan device with the greatest sensitivity on the violation.
- **Maximum Sensitivity Value:** The difference between the actual value and the limit value for the LVM plan device with the greatest sensitivity on the violation (in 120 V base).

Navigation and controls:

- "Locate" navigates to the Geographic Network view centered on the selected device.

1.17.7. LVM Device Exclusions List (For System)

The LVM Device Exclusions List – Unprocessed and LVM Device Exclusions List – Processed displays provide a summary of all devices in the system that have at least one exclusion reason. These displays can be accessed by selecting the LVM Device Exclusions List option from the RT DMS menu. On these displays, you can also remove the LVM Control Failure flag and toggle the Control Processed flag for devices.

Display call-up URL:

```
netview://RT/gridtabular//?Frame1=LVMDeviceExclusionsListGlobalUnprocessed_GridDisplay&Frame2
=LVMDeviceExclusionsListGlobalprocessed_GridDisplay
```

The following information is shown on these displays:

- **Name:** Name of the device.
- **ID:** ID of the device.
- **Device Type:** Type of the device.
- **Energizing Station:** Name of the energizing station.
- **Energizing Area:** Name of the energizing area.
- **Energizing Optimization Group:** Name of the energizing optimization group.
- **Energizing Feeder:** Name of the energizing feeder.
- **Energizing Transformer:** Name of the energizing transformer.
- **Exclusion Reasons:** The number of exclusion reasons for the device. Hovering over the cell shows a tooltip that lists the exclusion reasons and confirmation type (for control failure devices only).
- **Exclusion Time:** The timestamp of the earliest exclusion (only for devices with a Control Failure or Manual Exclusion).
- **Tag Types:** The type of each tag placed on the device.
- **Control Process Status:** The processing status for the device (Processed or Unprocessed).
- **Control Type:** The type of control the device failed with (Status Control or Analog Setpoint Control). Applies to control failure devices only.
- **Control Failed Time:** Date and time of the control failure. Applies to control failure devices only.
- **Indirect Checked Measurement Name:** The name of the measurement that was used for indirect confirmation. Applies to control failure devices only.
- **LVM Eligibility Status:** Current LVM eligibility status for the device.
- **LVM Eligibility Reason:** Current reason for LVM ineligibility for the device.

Navigation and controls:

- Energizing Station: Opens the LVM Device Exclusions List for Station display.
- Locate: Navigates to the Geographic Network view centered on the selected device.
- Control Process Status: Right-clicking this button toggles the "processed" flag on the device; left-clicking refreshes the display to show changes.
- Reset Control Failure: Right-clicking this button toggles the "control failure" flag on the device; left-clicking refreshes the display to show changes.

1.17.7.1. LVM Device Exclusions List (For Station)

The LVM Device Exclusions List – Unprocessed and LVM Device Exclusions List – Processed displays provide a summary of all station devices that have at least one exclusion reason. These displays can be accessed by clicking the Energizing Station button on the system-wide LVM Device Exclusions List displays. On these displays, you can also remove the LVM Control Failure flag and toggle the Control Processed flag for devices.

Display call-up URL:

```
netview://RT/gridtabular//<StationName>/?Frame1=LVMDeviceExclusionsListUnprocessed_GridDispla
y&Frame2=LVMDeviceExclusionsListprocessed_GridDisplay
```

The following information is shown on these displays:

- **Name:** Name of the device.
- **ID:** ID of the device.
- **Device Type:** Type of the device.
- **Energizing Station:** Name of the energizing station.
- **Energizing Area:** Name of the energizing area.
- **Energizing Feeder:** Name of the energizing feeder.
- **Energizing Optimization Group:** Name of the energizing optimization group.
- **Energizing Transformer:** Name of the energizing transformer.
- **Exclusion Reasons:** The number of exclusion reasons for the device. Hovering over the cell shows a tooltip that lists the exclusion reasons and confirmation type (for control failure devices only).
- **Exclusion Time:** The timestamp of the earliest exclusion (only for devices with a Control Failure or Manual Exclusion).
- **Tag Types:** The type of each tag placed on the device.
- **Control Process Status:** The processing status for the device (Processed or Unprocessed).
- **Control Type:** The type of control the device failed with (Status Control or Analog Setpoint Control). Applies to control failure devices only.
- **Control Failed Time:** Date and time of the control failure. Applies to control failure devices only.
- **Indirect Checked Measurement Name:** The name of the measurement that was used for indirect confirmation. Applies to control failure devices only.
- **LVM Eligibility Status:** Current LVM eligibility status for the device.
- **LVM Eligibility Reason:** Current reason for LVM ineligibility for the device.

Navigation and controls:

- Locate: Navigates to the Geographic Network view centered on the selected device.
- Control Process Status: Right-clicking this button toggles the "processed" flag on the device; left-clicking refreshes the display to show changes.

- Reset Control Failure: Right-clicking this button toggles the "control failure" flag on the device. Left-clicking this button refreshes the display to show changes.

1.18. FISR Results Summary

The Fault Isolation and Service Restoration Results Summary display can be accessed from the Station Output Summary display by clicking the FISR Solution button. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Fault Isolation and Service Restoration option. This display shows list of Fault Isolation and Service Restoration plans available. Clicking Get FISR Plans navigates to FISR Plan display with detailed information about the plan.

Display call-up URL:

`netview://<Mode>/gridtabular/FISRPlansMain_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Number of Plans:** Number of FISR plans.
- **Solution Time:** Time when the solution was completed.
- **Peak Solution Time:** Last FISR peak solution time.
- **Execution Time:** Time when the plan was executed.
- **Problem Formulation:** Problem formulation name.
- **Plan Available:** Specifies whether the plan is available.
- **Status:** FISR solution status.
- **Tripped Device Station:** For FISR triggered by permanent fault, the station name for the first tripped device.
- **Tripped Device Name:** For FISR triggered by permanent fault, the name of the first tripped device.
- **Tripped Device Id:** For FISR triggered by permanent fault, the id of the first tripped device.
- **Tripped Device Type:** For FISR triggered by permanent fault, the device type for the first tripped device.
- **Tripped Device Operation Time:** For FISR triggered by permanent fault, the operation time for the first tripped device.

Navigation:

- "Get FISR Plans" opens the FISR Plan display.
- "Plan Statistics" opens the FISR Plan Statistics display.
- "Plan Performance Indexes" opens the FISR Plan Weighted Performance Indexes display.

1.18.1. FISR Plan

The FISR Plan display shows the list of FISR plans with basic details. This display can be accessed from the FISR Results Summary display.

From the FISR Plan display, you can navigate to the FISR Plan Details display to show detailed plan information.

Display call-up URL for RT mode:

```
netview://RT/gridtabular/FISRPlans_RTGridDisplay/<Substation Name>/<Solution ID>
```

Display call-up URL for ST mode:

```
netview://ST/gridtabular/FISRPlans_STGridDisplay/<Substation Name>/<Solution ID>
```

The following information is shown on the FISRPlans display:

- **Plan Type:** Type of plan (FISR Fault or FISR Loss of Voltage).
- **Rank:** Rank of the plan.
- **State:** State of the plan, such as "Feasible" or "Invalid - Topology Change".
- **Full Restoration:** Indicates whether this is a full restoration plan (Yes or No). This field shows No for partial restoration plans (plans that were solved for a part of the faulted sections) and reduced restoration plans (plans that do not restore sections where violations appear or would be made worse by the plan).
- **Third Party Owned Equipment - Revert Plan Execution Mode:** Shows the new the plan execution mode if the plan execution mode changed due to the presence of third party-owned equipment in a switching plan (for example, Revert to Semi Automatic).
- **Switching Validation Check State:** State of the switching validation check.
- **Number of Moves:** Number of moves in the plan.
- **Number of Customers De-Energized:** Number of customers de-energized after the plan.
- **Number of Customers Restored:** Number of customers restored after the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan.
- **Unserved kW:** Total real power of all loads that remain de-energized after the plan, in kW.
- **Maximum Segment Loading:** Maximum segment loading after reconfiguration, in percentage. Maximum segment loading can be in neighboring stations in the FISR study area (violations in neighboring stations do not prevent FISR from running).
    - It is against derated limit if adjusting limits based on reserved capacity is enabled.
    - It is against regular limit if adjusting limits based on reserved capacity is not enabled.
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration in percentage.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration in percentage.

- **Highest Device Voltage (%):** Maximum load voltages after reconfiguration in percentage.
- **Lowest Device Voltage (%):** Minimum load voltages after reconfiguration in percentage.
- **Maximum Backfeed:** Maximum power transformer backfeed, kW per phase.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Initial Branch Flow Violation:** Indicates whether branch flow violations existed in the initial loading.
- **Final Branch Flow Violation:** Indicates whether branch flow violations existed in the final loading (considering the DER re-energization delay).
- **Protection Validation:** The protection validation application solution status for the plan.
- **Total PI:** Total performance index.
- **Problem Formulation:** Problem formulation name.
- **Number of Analyzed Tap Changing Transformers:** Number of tap changing transformers in the network selection that were analyzed in scope of the solution.
- **Implement ST Plan:** Click this button to implement the plan (applies to study mode only).
- **Restore ST Plan:** Click this button to reverse (undo) the plan (applies to study mode only).
- **Implement RT Plan:** Click this button to implement the plan (applies to real-time mode only). This button is available for semi-automatic plan execution mode only.
- **Reject RT Plan:** Click this button to reject the plan with comments (applies to real-time mode only). This button is available for semi-automatic plan execution mode only.
- **Comments (RT):** Write the comments when rejecting the plan (applies to real-time mode only).
- **Implemented if Closed Loop:** If Yes, the FISR execution mode is "Closed Loop" or "Closed Loop – Calculated Voltage Limit Violation not Worse" and the plan is implemented automatically (applies to real-time mode only).

Navigation:

- "Switching Validation Check State" opens the Switching Validation Check State display.
- "Show Steps" opens the Switching Plan Control Moves display.
- "Mark Devices" opens the Geographic Network view in the new viewport and marks the devices included in the switching plan.
- "Protection Settings Group Updates" opens the Protection Settings Group Updates display.
- "Number of Analyzed Tap Changing Transformers" navigates to the Plan Estimated Tap Position display in the new viewport.
- "Protection Validation Results" opens the Protection Validation Results for Plan display.
- "Plan Details" opens the FISR Plan Details display.
- "Display Feeders in Plan" opens the Geographic Network view in the new viewport, which displays the scope of feeders in the plan.

1.18.1.1. FISR Plan Details

The Fault Isolation and Service Restoration Plan Details display can be accessed from FISR Plan display. The FISR Plan Details display contains the detailed information related to the selected FISR plan and is shown in three panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/FISRPlanDetail1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/FISRPlanDetail2_GridDisplay/<Substation Name>/`

Third Pane:

`netview://<Mode>/gridtabular/FISRPlanDetail3_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on FISRPlanDetail1:

- **Station Name:** Name of the substation.
- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Number of Customers De-Energized:** Number of de-energized customers.
- **Number of Customers Restored:** Number of customers restored after the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan.
- **Number of Higher Priority Customers Not Restored:** Number of higher priority customers not restored after the plan.
- **Number of Transformers De-Energized:** Number of transformers de-energized.
- **Number of Transformers Restored:** Number of transformers restored after the plan.
- **Number of Transformers Not Restored:** Number of transformers not restored after the plan.
- **Restored kW:** Total real power of all energized loads after the plan minus the total real power of all energized loads before the plan, in kW.
- **Unserved kW:** Total real power of all loads that remain de-energized after the plan, in kW.
- **True kW De-Energized:** Total real power of the de-energized loads at the peak evaluation time, without CLPU, in kW.
- **kW with CLPU De-Energized:** Total real power of the de-energized loads at the peak evaluation time, with CLPU, in kW. Applicable only when CLPU is considered for reconfiguration applications.
- **Outage Time:** Outage time calculated by the system.

- **Duration of Plan Validity:** The estimated time period after the outage, after which overloads caused by CLPU occur, in hours. Applicable only when CLPU is considered for reconfiguration applications.
- **Plan No Longer Valid Time:** Time when the plan is considered no longer valid.
- **Number of Measurement Islands:** Number of measurement islands in the network selection.
- **Number of Non-Converged BLA Measurement Islands:** Number of BLA not converged measurement islands in the reconfiguration path of the plan.
- **Number of Scheduled Loading Measurement Islands:** Number of scheduled loading measurement islands in the reconfiguration path of the plan.
- **Number of Analyzed Tap Changing Transformers:** Number of tap changing transformers in the network selection that were analyzed in scope of the solution.
- **Reenergized Real Generation:** Real power of generators and DERs re-energized by the switching plan, in kW. These objects will not run or contribute power during intermediate steps.
- **Reenergized Reactive Generation:** Reactive power of generators and DERs re-energized by the switching plan, in kVAR. These objects will not run or contribute power during intermediate steps.
- **Reenergized Real Embedded Generation:** Real power of embedded generation re-energized by the switching plan, in kW. These objects will not run or contribute power during intermediate steps.
- **Reenergized Reactive Embedded Generation:** Reactive power of embedded generation re-energized by the switching plan, in kVAR. These objects will not run or contribute power during intermediate steps.

Navigation:

- "Number of Non-Converged BLA Measurement Islands" navigates to the Non-Converged BLA and Scheduled Loading Measurement Islands display in the new viewport.
- "Number of Scheduled Loading Measurement Islands" navigates to the Non-Converged BLA and Scheduled Loading Measurement Islands display in the new viewport.
- "Number of Analyzed Tap Changing Transformers" navigates to the Plan Estimated Tap Position display in the new viewport.

The following information is shown on FISRPlanDetail2:

- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with the maximum loading against regular flow limit.
- **Maximum Segment Flow Value (Amps):** The maximum segment flow value after reconfiguration, in Amp.
- **Maximum Segment Flow Value Before DERs are Energized (Amps):** The initial flow with DERs off for the maximum loaded segment in the final loading, in Amp.

- **Maximum Segment Loading Before DERs are Energized (%):** The initial loading with DERs off for the maximum loaded segment in the final loading, in percentage.
- **Highest Device Voltage (%):** Maximum load voltage after reconfiguration, in percentage.
- **Highest Voltage Device ID:** The name of the device with the highest voltage (for the network selection/study area).
- **Lowest Device Voltage (%):** Minimum load voltage after reconfiguration, in percentage.
- **Lowest Voltage Device ID:** The name of the device with the lowest voltage (for the network selection/study area).
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration in percentage.
- **Highest Violated Voltage Limit:** Highest violated voltage limit after reconfiguration in percentage.
- **Highest Violated Voltage Device ID:** ID of the load with highest voltage violation.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration in percentage.
- **Lowest Violated Voltage Limit (%):** Lowest violated voltage limit after reconfiguration in percentage.
- **Lowest Violated Voltage Device ID:** ID of the load with lowest voltage violation.
- **Maximum Transformer Backfeed (kW/ph):** Maximum transformer backfeed, kW per phase.
- **Maximum Backfeed Power Transformer ID:** ID of the transformer with maximum backfeed.
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
- **Lost Revenue:** Estimated revenue-cost of production.
- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading (%):** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.
- **Type of Flow Limit Used:** Type of flow limit used by FISR to calculate the loading of the segment with maximum loading, either Regular or Derated.
- **Regular Limit Maximum Segment Loading (%) at Plan No Longer Valid Time:** Maximum segment loading against regular flow limit after reconfiguration in percentage at Plan No Longer Valid Time when cold load pickup is considered.
- **Regular Limit Maximum Loading Segment ID at Plan No Longer Valid Time:** ID of the segment with the maximum loading against regular flow limit at Plan No Longer Valid Time when cold load pickup is considered.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Maximum Phase Trip Loading Device ID:** ID of the recloser/breaker with maximum loading

against phase trip rating.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit.
- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the load with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the load with the minimum voltage.
- "Locate Highest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the highest voltage violation.
- "Locate Lowest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the lowest violated voltage.
- "Locate Maximum Backfeed Power Transformer" navigates to the Geographic Network view centered on the transformer with the maximum backfeed flow.
- "Protection Validation Results" opens the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.
- "Locate Regular Limit Maximum Overloading Segment at Plan No Longer Valid Time" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit at Plan No Longer Valid Time when cold load pickup is considered.
- "Locate Maximum Phase Trip Loading Device" navigates to the Geographic Network view centered on the recloser/breaker with the maximum loading against phase trip rating.

The following information is shown on FISRPlanDetail3. The plans are ranked based on their total weighted PI, sorted from smallest to largest value. The total weighted PI is the sum of the weighted individual PIs (weight × PI). The weight of each individual PI is configured in the DNAF Configuration Editor (DCE). If the weight is 0, it means this PI is not considered. A PI with a larger weight has greater importance.

- **Total Weighted PI:** The total weighted PI is the sum of the weighted individual PIs.
- **Third-Party Owned Equipment Control Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of steps involving third-party owned equipment in the plan.
- **Number of Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of moves in the plan.
- **Cost Weighted PI:** This individual weighted PI indicates the cost of control moves. It is the sum of the cost of all control moves in the plan. The cost of the control is determined by the adjacency to the reconfigured area, whether it is an automatic or manual control, and the operation cost defined in the switch model.
- **Intermediate Cost Weighted PI:** This individual weighted PI indicates the cost of control

moves during the intermediate switching state/solution.

- **Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations. It is calculated using the equation $\sum f_i^2$ to sum all branch violations, where f_i is zero if the branch i is not overloaded, and f_i is the difference between the flow value and the flow limit if branch i is overloaded.
- **Intermediate Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations during the intermediate switching state/solution.
- **Voltage Violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations. It is calculated using the equation $\sum v_i^2$ to sum all load voltage violations, where v_i is zero if the load voltage is within limit for load i, and v_i is the difference between the voltage value and the limit for load i if the load voltage upper or lower limit is violated.
- **Intermediate Voltage violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations during the intermediate switching state/solution.
- **Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads.
- **Intermediate Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads during the intermediate switching state/solution.
- **Losses Weighted PI:** This individual weighted PI indicates the cost for the real power losses (in kW).
- **Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers.
- **Intermediate Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers during the intermediate switching state/solution.
- **Non-Converged or Scheduled Loading BLA Islands Weighted PI:** This individual weighted PI indicates the cost for the number of non-converged or scheduled loading BLA measurement islands involved in the plan.
- **Reverse Flow Through Non-Reversible Device Weighted PI:** This individual weighted PI indicates the cost for the number of reverse flow occurrences through non-reversible protective reclosers or breakers.
- **Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **Intermediate Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority during the intermediate switching state/solution.
- **CMI Weighted PI:** This individual weighted PI indicates the cost for the Customer Minutes Interrupted (CMI) for the plan. It is the sum of the interrupted minutes for all de-energized customers, which is calculated based on the automatic and manual switching device operation times that are specified in the problem formulation.
- **Priority CMI Weighted PI:** This individual weighted PI indicates the cost for the priority

Customer Minutes Interrupted (CMI) for the plan using the equation $\sum(P_i \times t_i)$, where P_i is the priority of that customer and t_i is the duration of the customer outage. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.

- **LMI Weighted PI:** This individual weighted PI indicates the cost for the Load Minutes Interrupted (LMI) in kW for the plan. It is the sum of the de-energized load in kW multiplied by the duration of the load outage (in minutes).
- **Margin Weighted PI:** This individual weighted PI indicates the cost for the maximum flow margin of all branches after the final switching step (final network configuration). The margin is 0 if a branch is overloaded, and it is the difference between the flow value and the limit if a branch is not overloaded.
- **Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops.
- **Intermediate Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops during the intermediate switching state/solution.
- **Maximum Segment Margin Weighted PI**: This individual weighted PI indicates the cost for the maximum segment loading in the plan. This PI is useful for ranking plans that do not have flow violations, allowing a plan with the least heavily loaded branch device to be ranked higher.
- **MAIFI Weighted PI:** This individual weighted PI indicates the cost for Momentary Average Interruption Frequency Index (MAIFI), which tracks the average frequency of momentary interruptions per customer over a predefined area.
- **SAIDI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Duration Index (SAIDI), which tracks the average duration the customers are interrupted per customer over a predefined area.
- **SAIFI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Frequency Index (SAIFI), which tracks the average frequency of sustained interruptions per customer over a predefined area.
- **Feeder Voltage Drop Weighted PI:** This individual weighted PI indicates the cost for the largest voltage drop among all feeders in the plan.
- **TCUL Range Weighted PI:** This individual weighted PI indicates the cost for the deviation of the voltage from all feeders connected to a primary bus with a TCUL.
- **Protection Validation Weighted PI:** This individual weighted PI indicates the cost for the final plan step (final network configuration) execution of the Protection Validation function, which is the summation of the maximum interruption, maximum phase pickup, or minimum ground pickup violation using a pre-defined equation (found in the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*). This PI is only calculated for the top-ranked plans.
- **Protection Margin Weighted PI:** This individual weighted PI indicates the cost for the distance between the current/load flow (multiplied by a safety factor) to the flow limit.
- **Protection Reach Weighted PI:** This individual weighted PI indicates the cost for the protection reach. It checks whether a protection settings group can satisfy the minimum fault current.
- **Paralleling Circuits Switching Validation Check Weighted PI:** This individual weighted PI

indicates the cost for the number of switching steps with a "Paralleling Circuits with Reclosing Capable Protective Devices" switching check violation.

- **Ferroresonance Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Ferroresonance Switching Conditions" switching check violation.
- **3-Wire/4-Wire Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Feeding Single Phase-Ground Loads from 3 or 4 Wire Feeder Sections" switching check violation.

1.18.1.2. FISR Plan Protection Validation Results

The Protection Validation Results for Plan display can be accessed from FISR Plan, FISRCA Plan, PLOS Plan, and AFR Plan displays by clicking the Protection Validation Results button. Protection validation and short circuit results are shown for all of the breakers and reclosers in the plan.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysis_GridDisplay/<Substation Name>/
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Switch Alternate ID:** The second unique identifier of the switch.
- **Fault Impedance Evaluation:** Fault impedance evaluation type (Bolted Fault or User Configured Impedance).
- **Fault Impedance - Bolted Fault:** The bolted fault impedance (zero-fault impedance) in R + j X ohms.
- **Fault Impedance - User Configured Fault Impedance:** The user-configured fault impedance in R + j X ohms.
- **Device Type:** The type of switching device (for example, fuse, breaker, recloser – hydraulic, or recloser - electronic).
- **Operation Mode:** The current operation mode of the recloser (switch, recloser, or sectionalizer).
- **Neutral Relay Setting:** Neutral relay current trip setting in Amps.
- **Neutral Current:** The neutral current in Amps.
- **Neutral Current Percentage:** The neutral current percentage, which is calculated as the neutral current divided by the product of the neutral relay current trip setting and the user-configured safety factor. If the calculated percentage is greater than 100, the cell color is shown in red to indicate that there is a neutral current violation.
- **Switch Status:** Status of the switch.

- **Application Protection Zone Self-Protected Power Transformer:** Shows the name of the bounding self-protected power transformer. If the device name is not provided, this field shows the device ID.
- **Maximum Interruption Violation:** Indicator that the calculated maximum fault current exceeds the capacity of the switch. If the text is "Yes", then the cell color is shown in red to indicate that there is a maximum interruption violation.
- **Minimum Phase Pick-up Violation:** Indicator that the minimum calculated fault current is less than the minimum pick-up current of the protection relay. If the text is "Yes", then the cell color is shown in red to indicate that there is a minimum phase pick-up violation.
- **Minimum Ground Pick-up Violation:** Indicator that the minimum ground fault current is less than the minimum pick-up current of the ground fault relay. If the text is "Yes", then the cell color is shown in red to indicate that there is a minimum ground pick-up violation.
- **Peak Load Violation:** Indicator that the maximum of the three phase load currents through the protective device exceeds the phase trip current. If the text is "Yes", then the cell color is shown in red to indicate that there is a peak load violation.

Navigation:

- "Switch Analyst Attributes" opens the Analyst Attributes display for the switch.
- "Switch Operator Attributes" opens the Operator Attributes display for the switch.
- "Locate" locates the device in the current viewport.
- "Application Protection Zone Self-Protected Power Transformer" navigates to the bounding self-protected power transformer on the geographic view.
- "Maximum Interruption Results" opens the Protection Validation (Maximum Interruption) Results display.
- "Minimum Phase Pick-up Results" opens the Protection Validation (Minimum Phase Pick-Up) Results display.
- "Minimum Ground Pick-up Results" opens the Protection Validation (Minimum Ground Pick-Up) Results display.

1.18.1.3. FISR Protection Validation Maximum Interruption Results

This display can be accessed from the Protection Validation Results display.

The Protection Validation (Maximum Interruption) Results display provides information about the maximum interruption fault current, the maximum interruption current, and the maximum fault current for each protective device in a FISR plan. The results are obtained by simulating a fault on the load side of the breakers and reclosers that are involved in the reconfiguration plan.

Display call-up URL:

`netview://<Mode>/gridtabular/ReconfigFaultAnalysisMax_GridDisplay/<Substation Name>/<PlanID>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Alternate ID:** The second unique identifier of the switch.
- **Device Type:** The type of switching device (for example, fuse, breaker, recloser – hydraulic, or recloser - electronic).
- **Switch Status:** The current state of the switch (Open or Closed).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Relay Type:** The overcurrent protection type. Applies only to breakers (for example, fuse, mechanical relay, or electronic relay).
- **Maximum Interruption Current:** The maximum interruption capacity of the switch in Amps.
- **Maximum Fault Current with Bolted Fault (Amps):** The maximum short circuit current (in Amps) calculated for a location immediately downstream of the protective device evaluated with a zero-fault impedance.
- **Maximum Fault Current with Configured Fault Impedance (Amps):** The maximum short circuit current (in Amps) calculated for a location immediately downstream of the protective device evaluated with a user configured fault impedance.
- **Maximum Capacity Ratio with Bolted Fault (%):** The ratio of the maximum calculated fault current evaluated with a zero-fault impedance to the maximum interruption current of the switch, in percentage.
- **Maximum Capacity Ratio with Configured Fault Impedance (%):** The ratio of the maximum calculated fault current evaluated with a user configured fault impedance to the maximum interruption current of the switch, in percentage.
- **Maximum Interruption Violation:** Indicator that the calculated maximum fault current exceeds the capacity of the switch. If the text is "Yes", then the cell color is shown in red to indicate that there is a maximum interruption violation.

Navigation:

- "Locate" locates the device on the geographic view.
- "Switch Analyst Attributes" opens the Analyst Attributes pop-up. This pop-up is displayed only for the reconfiguration application plan protection validation results.
- "Switch Analyst Outputs" opens the Analyst Output pop-up. This pop-up is displayed only for the reconfiguration application plan protection validation results.

1.18.1.4. FISR Protection Validation Minimum Phase Pick-Up Results

This display can be accessed from the Protection Validation Results display.

The Protection Validation (Minimum Phase Pick-up) Results display provides information about the minimum phase pick-up current, the minimum phase fault current, and the minimum phase pick-up ratio. The results are obtained by simulating a phase-to-phase fault at the highest impedance distance from the breakers and reclosers that are involved in the reconfiguration plan.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysisMinPhase_GridDisplay/<Substation Name>/<PlanID>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the FaultAnalysisMinPhase display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Alternate ID:** The second unique identifier of the switch.
- **Device Type:** The type of switching device (for example, fuse, breaker, recloser – hydraulic, or recloser - electronic).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Switch Status:** The current state of the switch (Open or Closed).
- **Relay Type:** The overcurrent protection type. This field applies only to breakers (for example, fuse, mechanical relay, or electronic relay).
- **Minimum Phase Pick-up Current with Safety Factor (Amps):** The minimum phase pick-up current (multiplied by a safety factor) setting of the phase fault relay in Amps.
- **Minimum Phase Fault Current with Bolted Fault (Amps):** The minimum phase short circuit current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance.
- **Minimum Phase Pick-up Ratio with Bolted Fault (%):** The ratio of the minimum phase pick-up current (multiplied by a safety factor) to the minimum phase fault current evaluated with a zero-fault impedance in percentage.
- **Minimum Phase Fault Current with Configured Fault Impedance (Amps):** The minimum phase short circuit current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a user configured fault impedance.
- **Minimum Phase Pick-up Ratio with Configured Fault Impedance (%):** The ratio of the minimum phase pick-up current (multiplied by a safety factor) to the minimum phase fault current evaluated with a user configured fault impedance in percentage.
- **Minimum Phase Pick-up Violation:** Indicator that the minimum calculated fault current is less than the minimum pick-up current of the protection relay. If the text is "Yes", then the cell color is shown in red to indicate that there is a minimum phase pick-up violation.
- **Phase Fault Sensitivity with Bolted Fault (%):** The ratio of the minimum phase fault current evaluated with a zero-fault impedance to the minimum phase pick-up current (multiplied by a

safety factor).

- **Phase Fault Sensitivity with Configured Fault Impedance (%):** The ratio of the minimum phase fault current evaluated with a user configured fault impedance to the minimum phase pick-up current (multiplied by a safety factor).

Navigation:

- "Locate" locates the switch on the geographic view.
- "Minimum Phase Fault Location" locates the device on the geographic view.
- "Switch Analyst Attributes" opens the Analyst Attributes pop-up. This pop-up is displayed only for the reconfiguration application plan protection validation results.
- "Switch Analyst Outputs" opens the Analyst Output pop-up. This pop-up is displayed only for the reconfiguration application plan protection validation results.

1.18.1.5. FISR Protection Validation Minimum Ground Pick-Up Results

This display can be accessed from the Protection Validation Results display.

The Protection Validation (Minimum Ground Pick-up) Results display provides information about the minimum ground pick-up current, the minimum ground fault current, and the minimum ground pick-up ratio. The results are obtained by simulating a phase-to-ground fault at the highest impedance distance from the breakers and reclosers that are involved in the reconfiguration plan.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysisMinGround_GridDisplay/<Substation
Name>/<PlanID>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the FaultAnalysisMinGround display:

- **Switch Name:** The name of the associated switch.
- **Switch ID:** The unique identifier of the switch.
- **Alternate ID:** The second unique identifier of the switch.
- **Device Type:** The type of switching device (for example, fuse, breaker, recloser – hydraulic, or recloser - electronic).
- **Switch Status:** The current state of the switch (Open or Closed).
- **Operation Mode:** The current operation mode of the recloser (Switch, Recloser, or Sectionalizer).
- **Relay Type:** The overcurrent protection type. Applies only to breakers (for example, fuse, mechanical relay, or electronic relay).
- **Minimum Ground Pick-up Current with Safety Factor (Amps):** The minimum ground pick-up current (multiplied by a safety factor) setting of the ground fault relay in Amps.
-

- **Minimum Ground Fault Current with Bolted Fault (Amps):** The minimum ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance.
- **Minimum Ground Pick-up Ratio with Bolted Fault (%):** The ratio of the minimum ground pick-up current (multiplied by a safety factor) to the minimum ground fault current evaluated with a zero-fault impedance in percentage.
- **Minimum Ground Fault Current with Configured Fault Impedance (Amps):** The minimum ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a user configured fault impedance.
- **Minimum Ground Pick-up Ratio with Configured Fault Impedance (%):** The ratio of the minimum ground pick-up current (multiplied by a safety factor) to the minimum ground fault current evaluated with a user configured fault impedance, in percentage.
- **Minimum Ground Pick-up Violation:** Indicator that the minimum ground fault current is less than the minimum pick-up current of the ground fault relay. If the text is "Yes", then the cell color is shown in red to indicate that there is a minimum ground pick-up violation.
- **Minimum Ground Fault Type:** Minimum ground fault type.
- **Ground Fault Sensitivity with Bolted Fault (%):** The ratio of the minimum ground fault current evaluated with a zero-fault impedance to the minimum ground pick-up current (multiplied by a safety factor).
- **Ground Fault Sensitivity with Configured Fault Impedance (%):** The ratio of the minimum ground fault current evaluated with a user configured fault impedance to the minimum ground pick-up current (multiplied by a safety factor).

Navigation:

- "Locate" locates the switch on the geographic view.
- "Switch Analyst Attributes" opens the Analyst Attributes pop-up. This pop-up is displayed only for the reconfiguration application plan protection validation results.
- "Switch Analyst Outputs" opens the Analyst Output pop-up. This pop-up is displayed only for the reconfiguration application plan protection validation results.
- "Minimum Ground Fault Location" locates the fault location on the geographic view.

1.18.2. FISR Plan Statistics

The Fault Isolation and Service Restoration Plan Statistics display shows the detailed statistics associated with each plan. This display can be accessed from the FISR Results Summary display.

Each advanced application function has a different set of plan statistics, which are represented on this display. The statistics for FISR is shown in two panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/FISRPlanStat1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/FISRPlanStat2_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on FISRPlanStat1:

- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Number of Customers De-Energized:** Number of customers de-energized.
- **Number of Customers Restored:** Number of customers restored after the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan.
- **Number of Higher Priority Customers Not Restored:** Number of higher priority customers not restored after the plan.
- **Number of Transformers De-Energized:** Number of transformers de-energized.
- **Number of Transformers Restored:** Number of transformers restored after the plan.
- **Number of Transformers Not Restored:** Number of transformers not restored after the plan.
- **Restored kW:** Total real power of all energized loads after the plan minus the total real power of all energized loads before the plan, in kW.
- **Unserved kW:** Total real power of all loads that remain de-energized after the plan, in kW.
- **Restored DER Load Phase A/B/C:** Total real power of all DER loads after the plan minus the total real power of all energized DER loads before the plan per phase, in kW.
- **Restored Native Load Phase A/B/C:** Total real power of all native loads after the plan minus the total real power of all energized native loads before the plan per phase, in kW.

The following information is shown on FISRPlanStat2:

- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against regular flow limit.
- **Highest Device Voltage:** Maximum load voltages after reconfiguration in percentage.
- **Highest Voltage Device ID:** ID of the load with maximum voltage.
- **Lowest Device Voltage:** Minimum load voltages after reconfiguration in percentage.
- **Lowest Voltage Device ID:** ID of the load with minimum voltage.
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
-

Lost Revenue: Lost revenue.

- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading:** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit.
- "Locate Maximum Load Voltage" navigates to the Geographic Network view centered on the load with the maximum voltage.
- "Locate Minimum Load Voltage" navigates to the Geographic Network view centered on the load with the minimum voltage.
- "Protection Validation Details" opens the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.

1.18.3. FISR Plan Weighted Performance Indexes

The Fault Isolation and Service Restoration Plan Weighted Performance Indexes display shows the detailed Weighted Performance Indexes associated with each plan. This display can be accessed from the FISR Results Summary display by pressing the Plan Performance Indexes button.

The Weighted Performance Indexes for FISR are shown in two panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/ReconfigPerfIndex1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/ ReconfigPerfIndex2_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the top pane:

- **Total Weighted PI:** The total weighted PI is the sum of the weighted individual PI’s.
- **Third-Party Owned Equipment Control Moves:** This individual weighted PI indicates the cost for the total number of steps involving third-party owned equipment in the plan.
- **Control Moves:** This individual weighted PI indicates the cost for the total number of moves in the plan.

- **Total Cost:** This individual weighted PI indicates the cost of control moves. It is the sum of the cost of all control moves in the plan. The cost of the control is determined by the adjacency to the reconfigured area, whether it is an automatic or manual control, and the operation cost defined in the switch model.
- **Total Intermediate Cost:** This individual weighted PI indicates the cost of control moves during the intermediate switching state/solution.
- **Flow Violations:** This individual weighted PI indicates the cost for flow violations. It is calculated using the equation $\sum f_i^2$ to sum all branch violations, where f_i is zero if the branch i is not overloaded, and f_i is the difference between the flow value and the flow limit if branch i is overloaded.
- **Intermediate Flow Violations:** This individual weighted PI indicates the cost for flow violations during the intermediate switching state/solution.
- **Bus Voltage Violation:** This individual weighted PI indicates the cost for bus voltage violations. It is calculated using the equation $\sum v_i^2$ to sum all bus voltage violations, where v_i is zero if the bus voltage is within limit for bus i, and v_i is the difference between the voltage value and the limit for bus i if the bus voltage upper or lower limit is violated.
- **Intermediate Bus Voltage violations:** This individual weighted PI indicates the cost for bus voltage violations during the intermediate switching state/solution.
- **Load Voltage Violations:** This individual weighted PI indicates the cost for load voltage violations. It is calculated using the equation $\sum v_i^2$ to sum all load voltage violations, where v_i is zero if the load voltage is within limit for load i, and v_i is the difference between the voltage value and the limit for load i if the load voltage upper or lower limit is violated.
- **Intermediate Load Voltage Violations:** This individual weighted PI indicates the cost for load voltage violations during the intermediate switching state/solution.
- **Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads.
- **Intermediate Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads during the intermediate switching state/solution.
- **Losses Weighted PI:** This individual weighted PI indicates the cost for the real power losses (in kW).
- **Non-Converged or Scheduled Loading BLA Islands Weighted PI:** This individual weighted PI indicates the cost for the number of non-converged or scheduled loading BLA measurement islands involved in the plan.
- **Reverse Flow for Non-Reversible Reclosure/Breaker:** This individual weighted PI indicates the cost for the number of reverse flow occurrences through non-reversible protective reclosers or breakers.

The following information is shown on the bottom pane:

- **Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers.
- **Intermediate Dropped Customer Count Weighted PI:** This individual weighted PI indicates

the cost for the number of de-energized customers during the intermediate switching state/solution.

- **Dropped Priority Customer Weighted PI:** This individual weighted PI indicates the cost for the dropped priority customer. The priority customer is assigned based on the costs of uninterrupted service that are given for each load category.
- **Intermediate Dropped Priority Customer Weighted PI:** This individual weighted PI indicates the cost for the dropped priority customer during the intermediate switching state/solution.
- **Customer Minutes Interrupted:** This individual weighted PI indicates the cost for the Customer Minutes Interrupted (CMI) for the plan. It is the sum of the interrupted minutes for all de-energized customers, which is calculated based on the automatic and manual switching device operation times that are specified in the problem formulation.
- **Priority Customer Minutes Interrupted:** This individual weighted PI indicates the cost for the priority Customer Minutes Interrupted (CMI) for the plan using the equation $\sum(P_i \times t_i)$, where P_i is the priority of that customer and t_i is the duration of the customer outage. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **Load Minutes Interrupted:** This individual weighted PI indicates the cost for the Load Minutes Interrupted (LMI) in kW for the plan. It is the sum of the de-energized load in kW multiplied by the duration of the load outage (in minutes).
- **Maximum Margin:** This individual weighted PI indicates the cost for the maximum flow margin of all branches after the final switching step (final network configuration). The margin is 0 if a branch is not overloaded, and it is the difference between the flow value and the limit if a branch is overloaded.
- **Radial Condition:** This individual weighted PI indicates the cost for the total number of loops.
- **Intermediate Radial Condition:** This individual weighted PI indicates the cost for the total number of loops during the intermediate switching state/solution.
- **Momentary Frequency - MAIFI:** This individual weighted PI indicates the cost for Momentary Average Interruption Frequency Index (MAIFI), which tracks the average frequency of momentary interruptions per customer over a predefined area.
- **Interruption Duration - SAIDI:** This individual weighted PI indicates the cost for System Average Interruption Duration Index (SAIDI), which tracks the average duration the customers are interrupted per customer over a predefined area.
- **Interruption Frequency - SAIFI:** This individual weighted PI indicates the cost for System Average Interruption Frequency Index (SAIFI), which tracks the average frequency of sustained interruptions per customer over a predefined area.
- **Feeder Voltage Drop Weighted PI:** This individual weighted PI indicates the cost for the largest voltage drop among all feeders in the plan.
- **Voltage Equalization:** This individual weighted PI indicates the cost for the deviation of the voltage from all feeders connected to a primary bus with a Load Tap Changing (LTC) transformer.
- **Protection Validation Weighted PI:** This individual weighted PI indicates the cost for the final

plan step (final network configuration) execution of the Protection Validation function, which is the summation of the maximum interruption, maximum phase pickup, and minimum ground pickup violation using a pre-defined equation. Refer to the *section 10.8 (Appendix B)* in the "*Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*". This PI is only calculated for the top-ranked plans.

- **Protection Margin:** This individual weighted PI indicates the cost for the distance between the current/load flow (multiplied by a safety factor) to the flow limit.
- **Protection Reach:** This individual weighted PI indicates the cost for the protection reach. It checks whether a protection settings group can satisfy the minimum fault current.
- **Paralleling Circuits Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Paralleling Circuits with Reclosing Capable Protective Devices" switching check violation.
- **Ferroresonance Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Ferroresonance Switching Conditions" switching check violation.
- **3-Wire/4-Wire Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Feeding Single Phase-Ground Loads from 3 or 4 Wire Feeder Sections" switching check violation.
- **Maximum Number of Switching Steps:** This individual weighted PI indicates the cost for the maximum number of switching steps in the plan.
- **Maximum Number of Operations:** This individual weighted PI indicates the cost for the maximum number of operations.
- **Maximum Segment Margin**: This individual weighted PI indicates the cost for the maximum segment loading in the plan. This PI is useful for ranking plans that do not have flow violations, allowing a plan with the least heavily loaded branch device to be ranked higher.

1.19. FISRCA Solution Overview

The Fault Isolation and Service Restoration Contingency Analysis Solution Overview display can be accessed from the Station Output Summary display by clicking FISRCA Solutions. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the FISR Contingency Analysis option.

This display contains a list of links to FISRCA solutions and the values of accumulated number of customers not restored respectively.

Display call-up URL:

`netview://ST/gridtabular/FISRCASummary_GridDisplay/<Substation Name>`

The following information is shown on this display:

- **Accumulated Number of Customers Not Restored:** Accumulated number of customers not restored after the plan has been executed.

- **Execution Time:** FISRCA (Fault Isolation and Service Restoration Contingency Analysis) execution time.
- **Solution Time:** Evaluation time of the FISRCA solution.
- **Peak Solution Time:** FISRCA peak solution time.

Navigation:

- "View Solution Results Summary" navigates to the FISRCA Results Summary display.

1.19.1. FISRCA Results Summary

The Fault Isolation and Service Restoration Contingency Analysis Results Summary display provides a list of FISRCA plans and basic information related to the plans.

This display allows navigating to displays containing plan statistics and detailed plan information, including performance index information.

This display can be accessed by clicking View Solution Results Summary on the FISRCA summary display.

Display call-up URL:

`netview://<Mode>/gridtabular/FISRCAsolutionSummary_GridDisplay/`

Where, <Mode> is only ST (Study mode).

The following information is shown on this display:

- **Name:** Switchable section name.
- **Name of Boundary Switches:** Number of boundary switches.
- **Fault Location:** Navigates to the fault location in the geographic view
- **Plan Available:** Plan available indicator.
- **Number of Customers Not Restored Outside of the Faulted Switchable Section:** Number of customers not restored outside of the faulted switchable section.
- **Total Number of Customers Not Restored:** Total number of customers not restored after reconfiguration.

Navigation:

- "Get FISRCA Plans" navigates to the FISRCA Plan display.
- "Plan Statistics" navigates to the FISRCA Plan Statistics display.
- "Plan Performance Indexes" navigates to the FISRCA Plan Weighted Performance Indexes display.

1.19.2. FISRCA Plan

The Fault Isolation and Service Restoration Contingency Analysis Plan display can be accessed from FISRCA Results Summary display.

This display contains the list of FISRCA plans with basic information related to the plans. It allows navigating to FISRCA Plan Detail display with detailed information related to FISRCA plans.

This display allows navigating to the Switching Plan Control Moves display and marking devices in the FISRCA plan in geographic view.

Display call-up URL:

`netview://<Mode>/gridtabular/FISRCAPlans_GridDisplay/<Solution ID>`

Where, <Mode> is only ST (Study mode).

The following information is shown on this display:

- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Number of Customers De-Energized:** Number of customers de-energized after the plan.
- **Number of Customers Restored:** Number of customers restored after the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan.
- **Unserved kW:** Total real power of all loads that remain de-energized after the plan, in kW.
- **Maximum Segment Loading:** Maximum segment loading after reconfiguration, in percentage. The maximum segment loading is calculated against the regular limit if adjusting limits without a reserved capacity, or against the derated limit if adjusting limits with a reserved capacity.
- **Highest Violated Voltage:** Highest load voltage violation after reconfiguration, in percentage.
- **Lowest Violated Voltage:** Lowest load voltage violation after reconfiguration, in percentage.
- **Highest Device Voltage** Maximum load voltages after reconfiguration in percentage.
- **Lowest Device Voltage:** Minimum load voltages after reconfiguration in percentage.
- **Maximum Phase Trip Loading:** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Initial Branch Flow Violation:** Indicates whether branch flow violations existed in the initial loading.
- **Final Branch Flow Violation:** Indicates whether branch flow violations existed in the final loading (considering the DER re-energization delay).
- **Protection Validation:** The protection validation application solution status for the plan.
- **Total PI:** Total performance index.
- **Problem Formulation:** Problem formulation name.

Controls:

- **Implement Plan:** Click this button to implement the plan (applies to study mode only).
- **Restore Plan:** Click this button to reverse (undo) the plan steps that were implemented and close the upstream protective device (applies to study mode only).

Navigation:

- "Show Steps" opens the FISRCA Plan and Switching Plan Control Moves displays in the same window.
- "Mark Devices" opens the geographic view with marked switches included in the switching plan.
- "Protection Validation Results" navigates to the Protection Validation Results display.
- "Plan Details" opens the FISR Plan Details display.
- "Display Feeders in Plan" opens the geographic view filtered by the feeders included in the switching plan.

1.19.2.1. FISRCA Plan Detail

The Fault Isolation and Service Restoration Contingency Analysis Plan Detail can be accessed from the FISR Plan and FISRCA Results Summary displays.

This display contains detailed information related to FISRCA plans and is shown in four panes.

Display call-up URL:

First pane:

`netview://<Mode>//gridtabular/FISRCAPlanDetail1_GridDisplay/`

Second pane:

`netview://<Mode>/gridtabular/FISRCAPlanDetail2_GridDisplay/`

Third pane:

`netview://<Mode>/gridtabular/FISRCAPlanDetail3_GridDisplay/`

Fourth pane:

`netview://<Mode>/gridtabular/FISRCAPlanDetail4_GridDisplay/`

Where, <Mode> is only ST (Study mode).

The following information is shown on FISRCAPlanDetail1_GridDisplay:

- **Station Name:** Name of the substation.
- **Rank:** Rank of the plan.
- **State:** State of the plan provided by the solution (Good / Good with neighboring violation / Good – Violation / Bad).
- **Number of Moves:** Number of moves in the plan.

- **Number of Customers De-Energized:** Number of customers de-energized after the plan.
- **Number of Customers Restored:** Number of customers restored after the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan.
- **Number of Higher Priority Customers Not Restored:** Number of higher priority customers not restored after the plan.
- **Number of Transformers De-Energized:** Number of transformers de-energized.
- **Number of Transformers Restored:** Number of transformers restored after the plan.
- **Number of Transformers Not Restored:** Number of transformers not restored after the plan.
- **Restored kW:** Total real power of all energized loads after the plan minus the total real power of all energized loads before the plan, in kW.
- **Unserved kW:** Total real power of all loads that remain de-energized after the plan, in kW.
- **Number of Measurement Islands:** Number of measurement islands in the network selection.
- **Number of Non-Converged BLA Measurement Islands:** Number of BLA not converged measurement islands in the reconfiguration path of the plan.
- **Number of Scheduled Loading Measurement Islands:** Number of scheduled loading measurement islands in the reconfiguration path of the plan.

The following information is shown on FISRCAPlanDetail2_GridDisplay:

- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** The ID of the segment with the maximum loading against regular flow limit.
- **Maximum Segment Flow Value (Amps):** The maximum segment flow value after reconfiguration, in Amps.
- **Highest Device Voltage:** Maximum load voltages after reconfiguration in percentage.
- **Highest Voltage Device ID:** ID of the load with maximum voltage.
- **Lowest Device Voltage:** Minimum load voltages after reconfiguration in percentage.
- **Lowest Voltage Device ID:** ID of the load with minimum voltage.
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration, in percentage.
- **Highest Violated Voltage Limit:** Highest violated voltage limit after reconfiguration, in percentage.
- **Highest Violated Voltage Device ID:** ID of the load with the highest voltage violation.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration, in percentage.
- **Lowest Violated Voltage Limit (%):** Lowest violated voltage limit after reconfiguration, in percentage.

- **Lowest Violated Voltage Device ID:** ID of the load with the lowest voltage violation.
- **Customer Value of Uninterrupted Services:** Customer value of uninterrupted service.
- **Lost Revenue:** Utilities value of lost revenue (adjusted for saving in supply costs).
- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading (%):** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.
- **Type of Flow Limit Used:** Type of flow limit used by FISRCA to calculate the loading of the segment with maximum loading, either Regular or Derated.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Maximum Phase Trip Loading Device ID:** ID of the recloser/breaker with maximum loading against phase trip rating.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the load with the maximum loading.
- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the load with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the load with the minimum voltage.
- "Locate Highest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the highest voltage violation.
- "Locate Lowest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the lowest violated voltage.
- "Protection Validation Results" opens the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.
- "Locate Maximum Phase Trip Loading Device" navigates to the Geographic Network view centered on the recloser/breaker with the maximum loading against phase trip rating.

The weighted performance index (PI) information is shown on the FISRCAPlanDetail3_GridDisplay and the FISRCAPlanDetail4_GridDisplay. The plans are ranked based on their total weighted PI, sorted from smallest to largest value. The total weighted PI is the sum of the weighted individual PIs (weight × PI). The weight of each individual PI is configured in the DNAF Configuration Editor (DCE). If the weight is 0, it means this PI is not considered. A PI with a larger weight has greater importance.

The following information is shown on FISRCAPlanDetail3_GridDisplay:

- **Total Weighted PI:** The total weighted PI is the sum of the weighted individual PIs.
- **Third-Party Owned Equipment Control Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of steps involving third-party owned equipment in the plan.
- **Number of Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of moves in the plan.
- **Cost Weighted PI:** This individual weighted PI indicates the cost of control moves. It is the sum of the cost of all control moves in the plan. The cost of the control is determined by the adjacency to the reconfigured area, whether it is an automatic or manual control, and the operation cost defined in the switch model.
- **Intermediate Cost Weighted PI:** This individual weighted PI indicates the cost of control moves during the intermediate switching state/solution.
- **Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations. It is calculated using the equation $\sum f_i^2$ to sum all branch violations, where f_i is zero if the branch i is not overloaded, and f_i is the difference between the flow value and the flow limit if branch i is overloaded.
- **Intermediate Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations during the intermediate switching state/solution.
- **Voltage violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations. It is calculated using the equation $\sum v_i^2$ to sum all load voltage violations, where v_i is zero if the load voltage is within limit for load i, and v_i is the difference between the voltage value and the limit for load i if the load voltage upper or lower limit is violated.
- **Intermediate Voltage violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations during the intermediate switching state/solution.
- **Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads.
- **Intermediate Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads during the intermediate switching state/solution.
- **Losses Weighted PI:** This individual weighted PI indicates the cost for the real power losses (in kW).
- **Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers.
- **Intermediate Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers during the intermediate switching state/solution.
- **Non-Converged or Scheduled Loading BLA Islands Weighted PI:** This individual weighted PI indicates the cost for the number of non-converged or scheduled loading BLA measurement islands involved in the plan.
- **Reverse Flow Through Non-Reversible Device Weighted PI:** This individual weighted PI indicates the cost for the number of reverse flow occurrences through non-reversible protective

reclosers or breakers.

The following information is shown on FISRCAPlanDetail4_GridDisplay:

- **Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **Intermediate Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority during the intermediate switching state/solution.
- **CMI Weighted PI:** This individual weighted PI indicates the cost for the Customer Minutes Interrupted (CMI) for the plan. It is the sum of the interrupted minutes for all de-energized customers, which is calculated based on the automatic and manual switching device operation times that are specified in the problem formulation.
- **Priority CMI Weighted PI:** This individual weighted PI indicates the cost for the priority Customer Minutes Interrupted (CMI) for the plan using the equation $\sum(P_i \times t_i)$, where P_i is the priority of that customer, and t_i is the duration of the customer outage. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **LMI Weighted PI:** This individual weighted PI indicates the cost for the Load Minutes Interrupted (LMI) in kW for the plan. It is the sum of the de-energized load in kW multiplied by the duration of the load outage (in minutes).
- **Margin Weighted PI:** This individual weighted PI indicates the cost for the maximum flow margin of all branches after the final switching step (final network configuration). The margin is 0 if a branch is overloaded, and it is the difference between the flow value and the limit if a branch is not overloaded.
- **Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops.
- **Intermediate Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops during the intermediate switching state/solution.
- **Maximum Segment Margin Weighted PI**: This individual weighted PI indicates the cost for the maximum segment loading in the plan. This PI is useful for ranking plans that do not have flow violations, allowing a plan with the least heavily loaded branch device to be ranked higher.
- **MAIFI Weighted PI:** This individual weighted PI indicates the cost for Momentary Average Interruption Frequency Index (MAIFI), which tracks the average frequency of momentary interruptions per customer over a predefined area.
- **SAIDI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Duration Index (SAIDI), which tracks the average duration the customers are interrupted per customer over a predefined area.
- **SAIFI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Frequency Index (SAIFI), which tracks the average frequency of sustained interruptions per customer over a predefined area.
- **Feeder Voltage Drop Weighted PI:** This individual weighted PI indicates the cost for the largest voltage drop among all feeders in the plan.

- **TCUL Range Weighted PI:** This individual weighted PI indicates the cost for the deviation of the voltage from all feeders connected to a primary bus with a TCUL.
- **Protection Validation Weighted PI:** This individual weighted PI indicates the cost for the final plan step (final network configuration) execution of the Protection Validation function, which is the summation of the maximum interruption, maximum phase pickup, or minimum ground pickup violation using a pre-defined equation (found in the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*). This PI is only calculated for the top-ranked plans.
- **Protection Margin Weighted PI:** This individual weighted PI indicates the cost for the distance between the current/load flow (multiplied by a safety factor) to the flow limit.
- **Protection Reach Weighted PI:** This individual weighted PI indicates the cost for the protection reach. It checks whether a protection settings group can satisfy the minimum fault current.
- **Paralleling Circuits Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Paralleling Circuits with Reclosing Capable Protective Devices" switching check violation.
- **Ferroresonance Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Ferroresonance Switching Conditions" switching check violation.
- **3-Wire/4-Wire Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Feeding Single Phase-Ground Loads from 3 or 4 Wire Feeder Sections" switching check violation.

1.19.2.2. FISRCA Protection Validation Results for Plan

The Protection Validation Results for the Plan display can be accessed from FISR Plan, FISRCA Plan, PLOS Plan, and AFR Plan displays by clicking the Protection Validation Results button. Protection validation and short circuit results are shown for all of the breakers and reclosers in the plan.

Display call-up URL:

```
netview://<Mode>/gridtabular/ReconfigFaultAnalysis_GridDisplay/<Substation Name>/
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

> **Note**
>
> This display is identical to the FISR results plan display. For detailed information, refer to the FISR Plan Protection Validation Results section.

1.19.3. FISRCA Plan Statistics

The Fault Isolation and Service Restoration Contingency Analysis Plan Statistics display can be accessed from the FISRCA Results Summary display.

This display contains the list of FISRCA plans and information related to the result after plans have been performed. This display can be accessed from the FISRCA solution summary display and it is displayed in two panes.

Display call-up URL:

First pane:

`netview://<Mode>/gridtabular/FISRCAplanstat1_GridDisplay/`

Second pane:

`netview://<Mode>/gridtabular/FISRCAplanstat2_GridDisplay/`

Where, <Mode> is only ST (Study mode).

The following information is shown on FISRCAplanstat1_GridDisplay:

- **Rank:** Rank of the plan.
- **State:** State of the plan provided by the solution (Good / Good with neighboring violation / Good – Violation / Bad).
- **Number of Moves:** Number of moves in the plan.
- **Number of Customers De-Energized:** Number of customers de-energized after the plan.
- **Number of Customers Restored:** Number of customers restored after the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan.
- **Number of Higher Priority Customers Not Restored:** Number of higher priority customers not restored after the plan.
- **Number of Transformers De-Energized:** Number of transformers de-energized.
- **Number of Transformers Restored:** Number of transformers restored after the plan.
- **Number of Transformers Not Restored:** Number of transformers not restored after the plan.
- **Restored kW:** Total real power of all energized loads after the plan minus the total real power of all energized loads before the plan, in kW.
- **Unserved kW:** Total real power of all loads that remain de-energized after the plan, in kW.
- **Restored DER Load Phase A/B/C:** Total real power of all DER loads after the plan minus the total real power of all energized DER loads before the plan per phase, in kW.
- **Restored Native Load Phase A/B/C:** Total real power of all native loads after the plan minus the total real power of all energized native loads before the plan per phase, in kW.

The following information is shown on FISRCAplanstat2_GridDisplay:

- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against regular flow limit.
- **Highest Device Voltage:** Maximum load voltages after reconfiguration.

- **Highest Voltage Device ID:** ID of the load with maximum voltage.
- **Lowest Device Voltage:** Minimum load voltages after reconfiguration.
- **Lowest Voltage Device ID:** ID of the load with minimum voltage.
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
- **Lost Revenue:** Lost revenue.
- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading:** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.

Navigation:

- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the load with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the load with the minimum voltage.
- "Protection Validation Details" navigates to the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against the derated flow limit because of reserved capacity.

1.19.4. FISRCA Plan Weighted Performance Indexes

Fault Isolation and Service Restoration Contingency Analysis Weighted Performance Indexes display shows the detailed Weighted Performance Indexes associated with each plan. This display can be accessed from the FISRCA Results Summary display by pressing the Plan Performance Indexes button.

The Weighted Performance Indexes for FISRCA is shown in two panes.

First Pane:

`netview://<Mode>/gridtabular/ReconfigPerfIndex1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/ ReconfigPerfIndex2_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

> **Note**
>
> This display is identical to the FISR results plan display. For detailed information, refer to the FISR Plan Weighted Performance Indexes section.

1.20. AFR Results Summary

The Automatic Feeder Reconfiguration Results Summary display can be accessed from the Station Output Summary display. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Automatic Feeder Reconfiguration option. Summary of AFR results is shown in this display.

Display call-up URL:

`netview://<Mode>/gridtabular/afrplansmain_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on the display:

- **Number of Plans:** Number of AFR plans.
- **Solution Time:** Solution time of the plan.
- **Peak Solution Time:** Peak solution time of the plan.
- **Execution Time:** Execution time of the plan.
- **Problem Formulation:** Problem formulation name.

Navigation:

- "Get AFR Plans" opens the AFR Plan display in the current viewport.
- "Plan Statistics" opens the AFR Plan Statistics display in the current viewport.
- "Plan Performance Indexes" opens the AFR Plan Weighted Performance Indexes display in the current viewport.

1.20.1. AFR Plan

The Automatic Feeder Reconfiguration Plan display shows a summary of AFR plans. The AFR Plan display can be accessed from the AFR Results Summary display.

Display call-up URL for RT mode:

`netview://RT/gridtabular/afrplans_RTGridDisplay/<Substation Name>/<Solution ID>`

Display call-up URL for ST mode:

`netview://ST/gridtabular/afrplans_STGridDisplay/<Substation Name>/<Solution ID>`

The following information is shown on the display:

- **Plan Type:** The plan type, such as AFR Manual or AFR Overload.
- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Third Party Owned Equipment - Revert Plan Execution Mode:** Shows the new plan execution

mode if the plan execution mode changed due to the presence of third-party-owned equipment in a switching plan (for example, Revert to Semi Automatic).

- **Switching Validation Check State:** State of the switching validation check.
- **Number of Moves:** Number of moves in the plan.
- **Post-Plan Number of Customers De-Energized:** Number of customers de-energized after the plan.
- **Lost kW:** Total real power of all the energized loads before the plan minus the total real power of all the energized load after the plan, in kW.
- **Maximum Segment Loading:** Maximum segment loading after reconfiguration, in percentage.
    - It is against derated limit if adjusting limits based on reserved capacity is enabled.
    - It is against regular limit if adjusting limits based on reserved capacity is not enabled.
- **Highest Device Voltage (%):** Maximum load voltages after reconfiguration in percentage.
- **Lowest Device Voltage (%):** Minimum load voltages after reconfiguration in percentage.
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration in percentage.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration in percentage.
- **Maximum Backfeed:** Maximum backfeed, kW per phase.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Total PI:** Total performance index.
- **Problem Formulation:** Problem formulation name.
- **Number of Analyzed Tap Changing Transformers:** Number of tap changing transformers in the network selection that were analyzed in scope of the solution.
- **Protection Validation:** The protection validation application solution status for the plan.
- **Implement ST Plan:** Click this button to implement the plan (applies to study mode only).
- **Restore ST Plan:** Click this button to reverse (undo) the plan (applies to study mode only).
- **Implement RT Plan:** Click this button to implement the plan (applies to real-time mode only). This button is available for semi-automatic plan execution mode only.
- **Reject RT Plan:** Click this button to reject the plan with comments (applies to real-time mode only). This button is available for semi-automatic plan execution mode only.
- **Comments (RT):** Write the comments when rejecting the plan (applies to real-time mode only).
- **Implemented if Closed Loop:** If Yes, the AFR execution mode is "Closed Loop" or "Closed Loop – Calculated Voltage Limit Violation not Worse" and the plan is implemented automatically (applies to real-time mode only).

Navigation:

- "Switching Validation Check State" opens the Switching Validation Check State display.
- "Show Steps" opens the Switching Plan Control Moves display in the bottom part of current viewport.
- "Protection Settings Group Updates" opens the Protection Settings Group Updates display.
- "Mark Devices" opens the geographic view in the new viewport and marks the devices included in the plan.
- "Number of Analyzed Tap Changing Transformers" navigates to the Plan Estimated Tap Position display in the new viewport.
- "Protection Validation Results" opens the Protection Validation Results display in the new viewport.
- "Plan Details" opens the AFR Plan Details display in new viewport.

1.20.1.1. AFR Plan Details

The Automatic Feeder Reconfiguration Plan Details display can be accessed from the AFR Plan display. This display shows detailed information about the selected AFR plan and is shown in three panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/AFRPlanDetail1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/AFRPlanDetail2_GridDisplay/<Substation Name>/`

Third Pane:

`netview://<Mode>/gridtabular/AFRPlanDetail3_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on AFRPlanDetail1_GridDisplay:

- **Station Name:** Substation name.
- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Pre-Plan Number of Customers De-Energized:** The number of de-energized customers.
- **Post-Plan Number of Customers De-Energized:** Number of customers not restored after the plan.
- **Pre-Plan Number of Transformers De-Energized:** Number of transformers de-energized.
- **Post-Plan Number of Transformers De-Energized:** Number of transformers not restored after

the plan.

- **Lost kW:** Total real power of all the energized loads before the plan minus the total real power of all the energized load after the plan, in kW.
- **Pre-Plan MAIFI:** Momentary Average Interruption Frequency Index (MAIFI) before plan.
- **Post-Plan MAIFI:** Momentary Average Interruption Frequency Index (MAIFI) after plan.
- **Pre-Plan SAIDI:** System Average Interruption Duration Index (SAIDI) before plan.
- **Post-Plan SAIDI:** System Average Interruption Duration Index (SAIDI) after plan.
- **Pre-Plan SAIFI:** System Average Interruption Frequency Index (SAIFI) before plan.
- **Post-Plan SAIFI:** System Average Interruption Frequency Index (SAIFI) after plan.
- **Number of Measurement Islands:** Number of measurement islands in the network selection.
- **Number of Non-Converged BLA Measurement Islands:** Number of BLA not converged measurement islands in the reconfiguration path of the plan.
- **Number of Scheduled Loading Measurement Islands:** Number of scheduled loading measurement islands in the reconfiguration path of the plan.
- **Number of Analyzed Tap Changing Transformers:** Number of tap changing transformers in the network selection that were analyzed in scope of the solution.

Navigation:

- "Number of Non-Converged BLA Measurement Islands" navigates to the Non-Converged BLA and Scheduled Loading Measurement Islands display in the new viewport.
- "Number of Scheduled Loading Measurement Islands" navigates to the Non-Converged BLA and Scheduled Loading Measurement Islands display in the new viewport.
- "Number of Analyzed Tap Changing Transformers" navigates to the Plan Estimated Tap Position display in the new viewport.

The following information is shown on AFRPlanDetail2_GridDisplay:

- **Pre-Plan Losses:** Losses before reconfiguration in kW.
- **Post-Plan Losses:** Losses after reconfiguration in kW.
- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against regular flow limit.
- **Highest Device Voltage (%):** Highest load voltages after reconfiguration in percentage.
- **Highest Voltage Device ID:** ID of the load with highest voltage.
- **Lowest Device Voltage (%):** Minimum load voltages after reconfiguration in percentage.
- **Lowest Voltage Device ID:** ID of the load with minimum voltage.
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration in percentage.

- **Highest Violated Voltage Limit:** Highest violated voltage limit after reconfiguration in percentage.
- **Highest Violated Voltage Device ID:** ID of the load with highest voltage violation.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration in percentage.
- **Lowest Violated Voltage Limit (%):** Lowest violated voltage limit after reconfiguration in percentage.
- **Lowest Violated Voltage Device ID:** ID of the load with lowest voltage violation.
- **Maximum Transformer Backfeed (kW/ph):** Maximum transformer backfeed, kW per phase.
- **Maximum Backfeed Power Transformer ID:** ID of the transformer with maximum backfeed.
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
- **Lost Revenue:** Lost revenue.
- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading:** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.
- **Type of Flow Limit Used:** Type of flow limit used by AFR to calculate the loading of the segment with maximum loading, either Regular or Derated.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Maximum Phase Trip Loading Device ID:** ID of the recloser/breaker with maximum loading against phase trip rating.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit.
- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the device with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the device with the minimum voltage.
- "Locate Highest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the highest voltage violation.
- "Locate Lowest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the lowest violated voltage.
- "Locate Maximum Backfeed Power Transformer" navigates to the Geographic Network view centered on the transformer with maximum backfeed voltage.
- "Protection Validation Results" opens the Protection Validation Results display.

- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.
- "Locate Maximum Phase Trip Loading Device" navigates to the Geographic Network view centered on the recloser/breaker with the maximum loading against phase trip rating.

The following information is shown on AFRPlanDetail3. The plans are ranked based on their total weighted PI, sorted from smallest to largest value. The total weighted PI is the sum of the weighted individual PIs (weight × PI). The weight of each individual PI is configured in the DNAF Configuration Editor (DCE). If the weight is 0, it means this PI is not considered. A PI with a larger weight has greater importance.

- **Total Weighted PI:** The total weighted PI is the sum of the weighted individual PIs.
- **Third-Party Owned Equipment Control Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of steps involving third-party owned equipment in the plan.
- **Number of Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of moves in the plan.
- **Cost Weighted PI:** This individual weighted PI indicates the cost of control moves. It is the sum of the cost of all control moves in the plan. The cost of the control is determined by the adjacency to the reconfigured area, whether it is an automatic or manual control, and the operation cost defined in the switch model.
- **Intermediate Cost Weighted PI:** This individual weighted PI indicates the cost of control moves during the intermediate switching state/solution.
- **Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations. It is calculated using the equation $\sum f_i^2$ to sum all branch violations, where f_i is zero if the branch i is not overloaded, and f_i is the difference between the flow value and the flow limit if branch i is overloaded.
- **Intermediate Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations during the intermediate switching state/solution.
- **Voltage violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations. It is calculated using the equation $\sum v_i^2$ to sum all load voltage violations, where vi is zero if the load voltage is within limit for load i, and v_i is the difference between the voltage value and the limit for load i if the load voltage upper or lower limit is violated.
- **Intermediate Voltage Violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations during the intermediate switching state/solution.
- **Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads.
- **Intermediate Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads during the intermediate switching state/solution.
- **Losses Weighted PI:** This individual weighted PI indicates the cost for the real power losses (in kW).

- **Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers.
- **Intermediate Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers during the intermediate switching state/solution.
- **Non-Converged or Scheduled Loading BLA Islands Weighted PI:** This individual weighted PI indicates the cost for the number of non-converged or scheduled loading BLA measurement islands involved in the plan.
- **Reverse Flow Through Non-Reversible Device Weighted PI:** This individual weighted PI indicates the cost for the number of reverse flow occurrences through non-reversible protective reclosers or breakers.
- **Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **Intermediate Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority during the intermediate switching state/solution.
- **CMI Weighted PI:** This individual weighted PI indicates the cost for the Customer Minutes Interrupted (CMI) for the plan. It is the sum of the interrupted minutes for all de-energized customers, which is calculated based on the automatic and manual switching device operation times that are specified in the problem formulation.
- **Priority CMI Weighted PI:** This individual weighted PI indicates the cost for the priority Customer Minutes Interrupted (CMI) for the plan using the equation $\sum(P_i \times t_i)$, where P_i is the priority of that customer, and t_i is the duration of the customer outage. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **LMI Weighted PI:** This individual weighted PI indicates the cost for the Load Minutes Interrupted (LMI) in kW for the plan. It is the sum of the de-energized load in kW multiplied by the duration of the load outage (in minutes).
- **Margin Weighted PI:** This individual weighted PI indicates the cost for the maximum flow margin of all branches after the final switching step (final network configuration). The margin is 0 if a branch is overloaded, and it is the difference between the flow value and the limit if a branch is not overloaded.
- **Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops.
- **Intermediate Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops during the intermediate switching state/solution.
- **Maximum Segment Margin Weighted PI**: This individual weighted PI indicates the cost for the maximum segment loading in the plan. This PI is useful for ranking plans that do not have flow violations, allowing a plan with the least heavily loaded branch device to be ranked higher.
- **MAIFI Weighted PI:** This individual weighted PI indicates the cost for Momentary Average Interruption Frequency Index (MAIFI), which tracks the average frequency of momentary interruptions per customer over a predefined area.

- **SAIDI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Duration Index (SAIDI), which tracks the average duration the customers are interrupted per customer over a predefined area.
- **SAIFI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Frequency Index (SAIFI), which tracks the average frequency of sustained interruptions per customer over a predefined area.
- **Feeder Voltage Drop Weighted PI:** This individual weighted PI indicates the cost for the largest voltage drop among all feeders in the plan.
- **TCUL Range Weighted PI:** This individual weighted PI indicates the cost for the deviation of the voltage from all feeders connected to a primary bus with a TCUL.
- **Protection Validation Weighted PI:** This individual weighted PI indicates the cost for the final plan step (final network configuration) execution of the Protection Validation function, which is the summation of the maximum interruption, maximum phase pickup, or minimum ground pickup violation using a pre-defined equation (found in *the Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*). This PI is only calculated for the top-ranked plans.
- **Protection Margin Weighted PI:** This individual weighted PI indicates the cost for the distance between the current/load flow (multiplied by a safety factor) to the flow limit.
- **Protection Reach Weighted PI:** This individual weighted PI indicates the cost for the protection reach. It checks whether a protection settings group can satisfy the minimum fault current.
- **Paralleling Circuits Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Paralleling Circuits with Reclosing Capable Protective Devices" switching check violation.
- **Ferroresonance Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Ferroresonance Switching Conditions" switching check violation.
- **3-Wire/4-Wire Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Feeding Single Phase-Ground Loads from 3 or 4 Wire Feeder Sections" switching check violation.

1.20.1.2. AFR Protection Validation Results for Plan

The Protection Validation Results for the Plan display can be accessed from FISR Plan, FISRCA Plan, PLOS Plan, and AFR Plan displays by clicking the Protection Validation Results button. Protection validation and short circuit results are shown for all of the breakers and reclosers in the plan.

Display call-up URL:

`netview://<Mode>/gridtabular/ReconfigFaultAnalysis_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

> **Note**

> This display is identical to the FISR results plan display. For detailed information, refer to the FISR Plan Protection Validation Results section.

1.20.2. AFR Plan Statistics

The Automatic Feeder Reconfiguration Plan Statistics display can be accessed from the

display.

The Plan Statistics display shows the detailed statistics associated with each plan. Each advanced application function has a different set of plan statistics, which are represented on this display. The statistics for FISR is shown in two panes.

Display call-up URL:

First pane:

`netview://<Mode>/gridtabular/AFRplanstat1_GridDisplay/`

Second pane:

`netview://<Mode>/gridtabular/AFRplanstat2_GridDisplay/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on AFRplanstat1_GridDisplay:

- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Pre-plan Number of Customers De-Energized:** Number of de-energized customers.
- **Post-plan Number of Customers De-Energized:** Number of customers not restored after the plan.
- **Pre-plan Number of Transformers De-Energized:** Number of transformers de-energized.
- **Post-plan Number of Transformers De-Energized:** Number of transformers not restored after the plan.
- **Pre-Plan MAIFI:** Momentary Average Interruption Frequency Index (MAIFI) before plan.
- **Post-Plan MAIFI:** Momentary Average Interruption Frequency Index (MAIFI) after plan.
- **Pre-Plan SAIDI:** System Average Interruption Duration Index (SAIDI) before plan.
- **Post-Plan SAIDI:** System Average Interruption Duration Index (SAIDI) after plan.
- **Pre-Plan SAIFI:** System Average Interruption Frequency Index (SAIFI) before plan.
- **Post-Plan SAIFI:** System Average Interruption Frequency Index (SAIFI) after plan.

The following information is shown on AFRplanstat2_GridDisplay:

- **Pre-Plan Losses:** The losses before the plan in kW.

- **Post-Plan Losses:** The losses after the plan in kW.
- **Lost kW:** Total real power of all the energized loads before the plan minus the total real power of all the energized load after the plan, in kW.
- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with the maximum loading against regular flow limit.
- **Highest Device Voltage:** Maximum load voltage after reconfiguration.
- **Highest Voltage Device ID:** The name of the load with the highest voltage (for the network selection/study area).
- **Lowest Device Voltage:** Minimum load voltage after reconfiguration.
- **Lowest Voltage Device ID:** The name of the load with the lowest voltage (for the network selection/study area).
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
- **Lost Revenue:** Estimated revenue-cost of production.
- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading:** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with the maximum loading against derated flow limit because of reserved capacity.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit.
- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the device with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the device with the minimum voltage.
- "Protection Validation Details" navigates to the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.

1.20.3. AFR Plan Weighted Performance Indexes

The Automatic Feeder Reconfiguration Plan Weighted Performance Indexes display shows the detailed Weighted Performance Indexes associated with each plan. This display can be accessed from the AFR Results Summary display by pressing the Plan Performance Indexes button.

Weighted Performance Indexes for AFR is shown in two panes.

First Pane:

`netview://<Mode>/gridtabular/ReconfigPerfIndex1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/ ReconfigPerfIndex2_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

> **Note**
>
> This display is identical to the FISR results plan display. For detailed information, refer to the FISR Plan Weighted Performance Indexes section.

1.21. Planned Outage Service

The Planned Outage Service display contains summaries of PLOS plans.

It can be accessed from the Station Output Summary and Advanced Application Status displays. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Planned Outage Switching option.

It allows the user to navigate to the detailed information related to PLOS plan by clicking the Get PLOS Plan button.

Display call-up URL:

`netview://<Mode>/gridtabular/plorsplansmain_GridDisplay/<Substation Name>`

Where, <Mode> is only ST (Study mode).

The following information is shown on this display:

- **Number of Plans:** Number of plans.
- **Solution Time:** Plan evaluation time.
- **Peak Solution Time:** Peak plan evaluation time.
- **Execution Time:** Plan execution time.

Navigation:

- "Get PLOS Plans" opens the PLOS Plan display.
- "Plan Statistics" opens the PLOS Plan Statistics display.
- "Plan Performance Indexes" opens the Plan Weighted Performance Indexes display.

1.21.1. PLOS Plan

The Planned Outage Switching Plan display can be accessed from Planned Outage Service display. Summaries of PLOS plans are shown in this display.

Display call-up URL:

`netview://<Mode>/gridtabular/plorsplans_GridDisplay/<Substation Name>/<Solution ID>`

Where, <Mode> is only ST (Study mode).

The following information is shown on plorsplans_GridDisplay:

- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Switching Validation Check State:** State of the switching validation check.
- **Number of Moves:** Number of moves in the plan.
- **Number of Customers Not Restored:** Number of customers not restored after the plan
- **Lost kW:** Total real power of all the energized loads before the plan minus the total real power of all the energized load after the plan, in kW.
- **Maximum Segment Loading:** Maximum segment loading after reconfiguration, in percentage.
  - It is against derated limit if adjusting limits based on reserved capacity is enabled.
  - It is against regular limit if adjusting limits based on reserved capacity is not enabled.
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration in percentage.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration in percentage.
- **Highest Device Voltage (%):** Maximum load voltage after reconfiguration in percentage.
- **Lowest Device Voltage (%):** Minimum load voltage after reconfiguration in percentage.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Total PI:** Total performance index.
- **Number of Analyzed Tap Changing Transformers:** Number of tap changing transformers in the network selection that were analyzed in scope of the solution.
- **Protection Validation:** The protection validation application solution status for the plan.

Navigation and controls:

- "Switching Validation Check State" opens the Switching Validation Check State display.
- "Show Steps" opens the PLOS Plan and Switching Plan Control Moves displays in one window in separate frames.
- "Number of Analyzed Tap Changing Transformers" navigates to the Plan Estimated Tap Position

display in the new viewport.

- "Protection Validation Results" navigates to the Protection Validation Results display.
- "Plan Details" navigates to the PLORSPlanDetail display.
- "Mark Devices" navigates to geographic in new viewport.
- "Implement Plan" performs steps in study mode (applies to study mode only).
- "Restore Plan" reverses (undoes) the plan steps in study mode (applies to study mode only).

1.21.1.1. PLOS Protection Validation Results for Plan

The Protection Validation Results for Plan display can be accessed from FISR Plan, FISRCA Plan, PLOS Plan, and AFR Plan displays by clicking the Protection Validation Results button. Protection validation and short circuit results are shown for all of the breakers and reclosers in the plan.

Display call-up URL:

`netview://<Mode>/gridtabular/ReconfigFaultAnalysis_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

> **Note**
>
> This display is identical to the FISR results plan display. For detailed information, refer to the FISR Plan Protection Validation Results section.

1.21.2. PLOS Plan Statistics

The Planned Outage Switching Plan Statistics display can be accessed from PLOS Plan display. Statistics of PLOS plans are shown in this display in two panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/POSplanstat1_GridDisplay/<Substation Name>`

Second Pane:

`netview://<Mode>/gridtabular/POSplanstat2_GridDisplay/<Substation Name>`

Where, <Mode> is only ST (Study mode).

The following information is shown on POSplanstat1_GridDisplay:

- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Pre-Plan Number of Customers De-Energized:** The number of de-energized customers.

- **Post-Plan Number of Customers De-Energized:** Number of customers not restored after the plan.
- **Number of Higher Priority Customers Not Restored:** Number of higher priority customers not restored after the plan.
- **Pre-Plan Number of Transformers De-Energized:** Number of transformers de-energized.
- **Post-Plan Number of Transformers De-Energized:** Number of transformers de-energized after the plan.
- **Lost kW:** Total real power of all the energized loads before the plan minus the total real power of all the energized load after the plan, in kW.

The following information is shown on POSplanstat2_GridDisplay:

- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration, in percentage.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with the maximum loading against regular flow limit.
- **Highest Device Voltage:** Maximum load voltage after reconfiguration.
- **Highest Voltage Device ID:** The name of the device with the highest voltage (for the network selection/study area).
- **Lowest Device Voltage:** Minimum device voltage after reconfiguration.
- **Lowest Voltage Device ID:** The name of the device with the lowest voltage (for the network selection/study area).
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
- **Lost Revenue:** Estimated revenue-cost of production.
- **Protection Validation:** The protection validation application solution status for the plan. Protection validation assesses two conditions: 1, that the protection settings are sufficient to operate for a short circuit at the end of the feeder and 2, that the circuit breaker rupture capacity is sufficient to interrupt a fault at close to the substation.
- **Derated Limit Maximum Segment Loading:** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit.
- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the device with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the device with the minimum voltage.

- "Protection Validation Details" navigates to the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.

1.21.3. PLOS Plan Weighted Performance Indexes

The Planned Outage Switching Plan Weighted Performance Indexes display shows the detailed Weighted Performance Indexes associated with each plan. This display can be accessed from the PLOS Results Summary display by pressing the Plan Performance Indexes button.

Weighted Performance Indexes for PLOS is shown in two panes.

First Pane:

`netview://<Mode>/gridtabular/ReconfigPerfIndex1_GridDisplay/<Substation Name>/`

Second Pane:

`netview://<Mode>/gridtabular/ ReconfigPerfIndex2_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

> **Note**
>
> This display is identical to the FISR results plan display. For detailed information, refer to the FISR Plan Weighted Performance Indexes section.

1.21.4. PLOS Plan Detail

The Planned Outage Switching Plan Detail can be accessed from PLOS Plan display. Details of PLOS plans are shown in this display in four panes.

Display call-up URL:

First Pane:

`netview://<Mode>/gridtabular/POSPlanDetail1_GridDisplay/`

Second Pane:

`netview://<Mode>/gridtabular/POSPlanDetail2_GridDisplay/`

Third Pane:

`netview://<Mode>/gridtabular/POSPlanDetail3_GridDisplay/`

Fourth Pane:

`netview://<Mode>/gridtabular/POSPlanDetail4_GridDisplay/`

Where, <Mode> is only ST (Study mode).

The following information is shown on POSPlanDetail1:

- **Station Name:** The name of substation.
- **Rank:** Rank of the plan.
- **State:** State of the plan.
- **Number of Moves:** Number of moves in the plan.
- **Pre-plan Number of Customers De-Energized:** Number of de-energized customers.
- **Post-Plan Number of Customers De-Energized:** Number of customers not restored after the plan.
- **Number of Higher Priority Customers Not Restored:** Number of higher priority customers not restored after the plan.
- **Pre-plan Number of Transformers De-Energized:** Number of transformers de-energized.
- **Post-plan Number of Transformers De-Energized:** Number of transformers not restored after the plan.
- **Lost kW:** Total real power of all the energized loads before the plan minus the total real power of all the energized load after the plan, in kW.
- **Number of Measurement Islands:** Number of measurement islands in the network selection.
- **Number of Non-Converged BLA Measurement Islands:** Number of BLA not converged measurement islands in the reconfiguration path of the plan.
- **Number of Scheduled Loading Measurement Islands:** Number of scheduled loading measurement islands in the reconfiguration path of the plan.
- **Number of Analyzed Tap Changing Transformers:** Number of tap changing transformers in the network selection that were analyzed in scope of the solution.

Navigation:

- "Number of Non-Converged BLA Measurement Islands" navigates to the Non-Converged BLA and Scheduled Loading Measurement Islands display in the new viewport.
- "Number of Scheduled Loading Measurement Islands" navigates to the Non-Converged BLA and Scheduled Loading Measurement Islands display in the new viewport.
- "Number of Analyzed Tap Changing Transformers" navigates to the Plan Estimated Tap Position display in the new viewport.

The following information is shown on POSPlanDetail2:

- **Regular Limit Maximum Segment Loading:** Maximum segment loading against regular flow limit after reconfiguration.
- **Regular Limit Maximum Loading Segment ID:** ID of the segment with the maximum loading against regular flow limit.
- **Highest Device Voltage:** Maximum load voltage after reconfiguration.
- **Highest Voltage Device ID:** The name of the load with the highest voltage (for the network

selection/study area).

- **Lowest Device Voltage:** Minimum load voltage after reconfiguration.
- **Lowest Voltage Device ID:** The name of the load with the lowest voltage (for the network selection/study area).
- **Highest Violated Voltage (%):** Highest load voltage violation after reconfiguration in percentage.
- **Highest Violated Voltage Limit (%):** Highest violated voltage limit after reconfiguration in percentage.
- **Highest Violated Voltage Device ID:** ID of the load with highest voltage violation.
- **Lowest Violated Voltage (%):** Lowest load voltage violation after reconfiguration in percentage.
- **Lowest Violated Voltage Limit (%):** Lowest violated voltage limit after reconfiguration in percentage.
- **Lowest Violated Voltage Device ID:** ID of the load with lowest voltage violation.
- **Customer Value of Uninterrupted Service:** Customer value of uninterrupted service.
- **Lost Revenue:** Estimated revenue-cost of production.
- **Protection Validation:** The protection validation application solution status for the plan.
- **Derated Limit Maximum Segment Loading:** Maximum segment loading against derated flow limit because of reserved capacity after reconfiguration, in percentage.
- **Derated Limit Maximum Loading Segment ID:** ID of the segment with maximum loading against derated flow limit because of reserved capacity.
- **Type of Flow Limit Used:** Type of flow limit used by PLOS to calculate the loading of the segment with maximum loading, either Regular or Derated.
- **Maximum Phase Trip Loading (%):** Maximum recloser/breaker loading against phase trip rating after reconfiguration, in percentage.
- **Maximum Phase Trip Loading Device ID:** ID of the recloser/breaker with maximum loading against phase trip rating.

Navigation:

- "Locate Maximum Loaded Branch Device" navigates to the Geographic Network view centered on the segment with the maximum loading against regular flow limit.
- "Locate Highest Voltage Device" navigates to the Geographic Network view centered on the load with the maximum voltage.
- "Locate Lowest Voltage Device" navigates to the Geographic Network view centered on the load with the minimum voltage.
- "Locate Highest Violated Voltage Device" navigates to the Geographic Network view centered on the device with the highest voltage violation.
- "Locate Lowest Violated Voltage Device" navigates to the Geographic Network view centered on

the device with the lowest violated voltage.

- "Protection Validation Results" opens the Protection Validation Results display.
- "Locate Derated Limit Maximum Loading Segment" navigates to the Geographic Network view centered on the segment with the maximum loading against derated flow limit because of reserved capacity.
- "Locate Maximum Phase Trip Loading Device" navigates to the Geographic Network view centered on the recloser/breaker with the maximum loading against phase trip rating.

The weighted performance index (PI) information is shown on the POSPlanDetail3_GridDisplay and the POSPlanDetail4_GridDisplay. The plans are ranked based on their total weighted PI, sorted from smallest to largest value. The total weighted PI is the sum of the weighted individual PIs (weight × PI). The weight of each individual PI is configured in the DNAF Configuration Editor (DCE). If the weight is 0, it means this PI is not considered. A PI with a larger weight has greater importance.

The following information is shown on POSPlanDetail3_GridDisplay:

- **Total Weighted PI:** The total weighted PI is the sum of the weighted individual PIs.
- **Third-Party Owned Equipment Control Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of steps involving third-party owned equipment in the plan.
- **Number of Moves Weighted PI:** This individual weighted PI indicates the cost for the total number of moves in the plan.
- **Cost Weighted PI:** This individual weighted PI indicates the cost of control moves. It is the sum of the cost of all control moves in the plan. The cost of the control is determined by the adjacency to the reconfigured area, whether it is an automatic or manual control, and the operation cost defined in the switch model.
- **Intermediate Cost Weighted PI:** This individual weighted PI indicates the cost of control moves during the intermediate switching state/solution.
- **Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations. It is calculated using the equation $\sum fi^2$ to sum all branch violations, where f_i is zero if the branch i is not overloaded, and f_i is the difference between the flow value and the flow limit if branch i is overloaded.
- **Intermediate Flow Violation Weighted PI:** This individual weighted PI indicates the cost for flow violations during the intermediate switching state/solution.
- **Voltage violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations. It is calculated using the equation $\sum v_i^2$ to sum all load voltage violations, where v_i is zero if the load voltage is within limit for load i, and v_i is the difference between the voltage value and the limit for load i if the load voltage upper or lower limit is violated.
- **Intermediate Voltage Violation Weighted PI:** This individual weighted PI indicates the cost for voltage violations during the intermediate switching state/solution.
- **Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads.

- **Intermediate Lost Load Weighted PI:** This individual weighted PI indicates the cost for the total real power of all de-energized loads during the intermediate switching state/solution.
- **Losses Weighted PI:** This individual weighted PI indicates the cost for the real power losses (in kW).
- **Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers.
- **Intermediate Dropped Customer Count Weighted PI:** This individual weighted PI indicates the cost for the number of de-energized customers during the intermediate switching state/solution.
- **Non-Converged or Scheduled Loading BLA Islands Weighted PI:** This individual weighted PI indicates the cost for the number of non-converged or scheduled loading BLA measurement islands involved in the plan.
- **Reverse Flow Through Non-Reversible Device Weighted PI:** This individual weighted PI indicates the cost for the number of reverse flow occurrences through non-reversible protective reclosers or breakers.

The following information is shown on POSPlanDetail4_GridDisplay:

- **Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **Intermediate Dropped Customer Priority Weighted PI:** This individual weighted PI indicates the cost for the dropped customer priority during the intermediate switching state/solution.
- **CMI Weighted PI:** This individual weighted PI indicates the cost for the Customer Minutes Interrupted (CMI) for the plan. It is the sum of the interrupted minutes for all de-energized customers, which is calculated based on the automatic and manual switching device operation times that are specified in the problem formulation.
- **Priority CMI Weighted PI:** This individual weighted PI indicates the cost for the priority Customer Minutes Interrupted (CMI) for the plan using the equation $\sum(P_i \times t_i)$, where P_i is the priority of that customer, and t_i is the duration of the customer outage. The customer priority is assigned based on the costs of uninterrupted service that are given for each load category.
- **LMI Weighted PI:** This individual weighted PI indicates the cost for the Load Minutes Interrupted (LMI) in kW for the plan. It is the sum of the de-energized load in kW multiplied by the duration of the load outage (in minutes).
- **Margin Weighted PI:** This individual weighted PI indicates the cost for the maximum flow margin of all branches after the final switching step (final network configuration). The margin is 0 if a branch is overloaded, and it is the difference between the flow value and the limit if a branch is not overloaded.
- **Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops.
- **Intermediate Radial Weighted PI:** This individual weighted PI indicates the cost for the total number of loops during the intermediate switching state/solution.

- **Maximum Segment Margin Weighted PI**: This individual weighted PI indicates the cost for the maximum segment loading in the plan. This PI is useful for ranking plans that do not have flow violations, allowing a plan with the least heavily loaded branch device to be ranked higher.
- **MAIFI Weighted PI:** This individual weighted PI indicates the cost for Momentary Average Interruption Frequency Index (MAIFI), which tracks the average frequency of momentary interruptions per customer over a predefined area.
- **SAIDI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Duration Index (SAIDI), which tracks the average duration the customers are interrupted per customer over a predefined area.
- **SAIFI Weighted PI:** This individual weighted PI indicates the cost for System Average Interruption Frequency Index (SAIFI), which tracks the average frequency of sustained interruptions per customer over a predefined area.
- **Feeder Voltage Drop Weighted PI:** This individual weighted PI indicates the cost for the largest voltage drop among all feeders in the plan.
- **TCUL Range Weighted PI:** This individual weighted PI indicates the cost for the deviation of the voltage from all feeders connected to a primary bus with a TCUL.
- **Protection Validation Weighted PI:** This individual weighted PI indicates the cost for the final plan step (final network configuration) execution of the Protection Validation function, which is the summation of the maximum interruption, maximum phase pickup, or minimum ground pickup violation using a pre-defined equation (found in the *Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide*). This PI is only calculated for the top-ranked plans.
- **Protection Margin Weighted PI:** This individual weighted PI indicates the cost for the distance between the current/load flow (multiplied by a safety factor) to the flow limit.
- **Protection Reach Weighted PI:** This individual weighted PI indicates the cost for the protection reach. It checks whether a protection settings group can satisfy the minimum fault current.
- **Paralleling Circuits Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Paralleling Circuits with Reclosing Capable Protective Devices" switching check violation.
- **Ferroresonance Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Ferroresonance Switching Conditions" switching check violation.
- **3-Wire/4-Wire Switching Validation Check Weighted PI:** This individual weighted PI indicates the cost for the number of switching steps with a "Feeding Single Phase-Ground Loads from 3 or 4 Wire Feeder Sections" switching check violation.

1.22. Reconfiguration Application Displays

1.22.1. Switching Validation Check State

Switching Validation Check State display is available for advanced DNAF functions (FISR, AFR, and PLOS). From this display the user can view the status of each switching validation check, including the paralleling circuits, ferroresonance, and 3-Wire/4-Wire switching validation checks.

To access the displays, click Switching Validation Check State on the AFR Plan, FISR Plan, and PLOS Plan displays.

Display call-up URL:

`netview://<Mode>/gridtabular/SwitchingValidationCheck_GridDisplay/<Substation Name>/<Solution ID>`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Paralleling Circuits:** The state of the paralleling circuits switching validation check.
- **Ferroresonance:** The state of the ferroresonance switching validation check.
- **3-Wire/4-Wire:** The state of the 3-Wire/4-Wire switching validation check.

1.22.2. Switching Plan Control Moves

Switching Plan Control Moves displays are available for advanced DNAF functions (FISR, AFR, FISRCA, and PLOS). From this display the user can view the characteristics of various switching solutions and their relative ranking. The plans are presented in order, with the highest ranked or "best" plans at the top. By selecting on the Plan Rank, the user can navigate to the Switching Steps display associated with the plan. For FISR plans the Restoration Switching Order number will be referenced on this display.

To access the displays, click Show Steps on the AFR Plan, FISR Plan, FISRCA Plan, and PLOS Plan displays.

Display call-up URL:

`netview://<Mode>/gridtabular/switchingplancontrolmoves_GridDisplay/<Substation Name>/`

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Step Number:** Step number of the move in the plan.
- **Device Type:** The device-type of the component (for example, SWITCH, JUMPER).
- **Device Name:** The name/number of the device as derived from the Asset Management System (the source of the Station Model Files).

- **Device ID:** Unique identifier of the device.
- **Subplan ID:** Sub-plan identifier. Applies to FISR plans when the parallel switching sequence functionality is enabled, and plans are split into sub-plans to be implemented in parallel.
- **Device Station And Feeder:** Dynamic substation name and feeder name.
- **Operation:** Name of the step operation (for example, status change).
- **Controllable:** Indicates if the device can be controlled (yes or no).
- **Initial State:** Initial position of the device.
- **Recommended State:** Plan recommended final position of the device.
- **Load Switching Effect:** Switching effect type for the switching control move (Dropped, Restored, Transfer, or Break Loop).
- **Amps Affected Phase A:** Affected load for phase A in Amps.
- **Amps Affected Phase B:** Affected load for phase B in Amps.
- **Amps Affected Phase C:** Affected load for phase C in Amps.
- **Number of State Change Devices:** Shows the number of switching devices in the solution area that changed their state during the solution evaluation and impact the associated plan.
- **Restoration Section Information:** Description of the involved restoration section information.

Navigation:

- "Switching Order" opens a Switching Order display with switching plan steps.
- "Daily Log" opens a Daily Log display with switching plan steps.
- "Mark Device" marks the device and shows the geographic view centered on the device in a new viewport.
- "Locate" locates the device on the geographic view in a new viewport.
- "Settings Group Template" shows the Settings Group Templates and Protection Settings Template Information displays with the devices' settings group template information.
- "Number of State Change Devices" shows the State Change Switching Devices display in the new viewport.

1.22.3. State Change Switching Devices

The State Change Switching Devices display shows switching devices in the solution area that changed their state during the solution evaluation and impact the associated plan. This display can be accessed from the Switching Plan Control Moves and Plan Control Moves displays.

Display call-up URL:

```
netview://<Mode>/gridtabular/StateChangeDevices_GridDisplay/<Station Name>/<Solution
ID>/<Control Move ID>
```

Where, <Mode> can be RT (Real-Time) or ST (Study).

The following information is shown on this display:

- **Device Id:** Unique identifier of the device.
- **Nominal Station:** Nominal substation.
- **Nominal Feeder:** Nominal feeder.

Navigation:

- "Locate" locates the device on the geographic view in a new viewport.

1.22.4. Protection Settings Group Updates

The Protection Settings Group Updates display shows all the information related to protection settings group updates such as previous and new settings group information and the name of the settings group template if it is available for the corresponding device. This tabular provides valuable information for users to determine why/how a settings group is changed for a device, and it allows the user to get this information without navigating to the FISR or AFR result file.

To access the displays, click Protection Settings Group Updates on the FISR or AFR Plan displays.

Display call-up URL:

```
netview://<Mode>/gridtabular/ProtectionSettingsGroupChanges_GridDisplay/<PlanID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Step Number:** Step number of the move in the plan.
- **Device Name:** The name of the device that updates its protection settings group.
- **Device ID:** The ID of the device that updates its protection settings group.
- **Previous Settings Group:** The name of the previously applied settings group.
- **New Settings Group:** The name of the new settings group.
- **Previous Phase Trip Current:** The previously applied phase trip current, in Amps.
- **New Phase Trip Current:** The new phase trip current, in Amps.
- **Previous Ground Trip Current:** The previously applied ground trip current, in Amps.
- **New Ground Trip Current:** The new ground trip current, in Amps.
- **Reach Safety Factor:** The protection reach safety factor.
- **New Phase Trip with Relay Safety Factor:** The new phase trip current multiplied by the protection reach safety factor, in Amps.
- **Minimum Phase Fault Type:** The minimum phase fault type that is simulated in the PRV application that is executed in the scope of a reconfiguration application to determine the minimum phase fault current. For example, a line-to-line fault between phases A and B is displayed as "AB".

- **Minimum Phase Fault Current:** The minimum phase fault current, in Amps.
- **Minimum Ground Fault Type:** The minimum ground fault type that is simulated in the PRV application that is executed in the scope of a reconfiguration application to determine the minimum ground fault current. For example, a single line-to-ground fault between phase A and ground is displayed as "AG".
- **Minimum Ground Fault Current:** The minimum ground fault current, in Amps.
- **Peak Load Safety Factor:** The peak load safety factor.
- **Maximum Load Current:** The maximum load current, in Amps.
- **Maximum Load Current with Peak Load Safety Factor:** The maximum load current multiplied by the peak load safety factor, in Amps.
- **Settings Group Template:** The name of the associated protection settings group template.

1.22.5. Plan Weighted Performance Indexes

The Plan Weighted Performance Indexes display is a common display for the applications FISR, AFR, and PLOS. It can be accessed from the FISR, AFR, and PLOS results summary displays.

This display has two panes and contains a list of plans and performance index values of the plans for FISR/AFR/PLOS respectively.

Display call-up URL:

First Pane

`netview://<Mode>/gridtabular/ReconfigPerfIndex1_GridDisplay/`

`netview://<Mode>/gridtabular/ReconfigPerfIndex2_GridDisplay/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on ReconfigPerfIndex1:

- **Total Weighted PI:** Total weighted performance index.
- **Third-Party Owned Equipment Control Moves Weighted PI:** Control moves with third-party equipment weighted performance index
- **Control Moves:** Number of moves in the plan performance index.
- **Total Cost:** Control cost performance index.
- **Total Intermediate Cost:** Control cost performance index during temporary moves.
- **Flow Violation:** Flow violation performance index.
- **Intermediate Flow Violation:** Flow violation performance index during temporary moves.
- **Bus Voltage Violation:** Bus voltage violation performance index.
- **Intermediate Bus Voltage Violation:** Bus voltage violation performance index during temporary moves.

- **Load Voltage Violation:** Load voltage violation performance index.
- **Intermediate Bus Voltage Violation:** Load voltage violation performance index during temporary moves.
- **Lost Load Weighted PI:** Lost load performance index.
- **Intermediate Lost Load Weighted PI:** Lost load performance index during temporary moves.
- **Losses Weighted PI:** Losses performance index.
- **Non-Converged or Scheduled Loading BLA Islands Weighted PI:** BLA not converged or scheduled loading measurement islands performance index.
- **Reverse Flow For Non-Reversible Recloser/Breaker Weighted PI:** Performance index for a new reverse flow through non-reversible protective reclosers or breakers.

The following information is shown on ReconfigPerfIndex2:

- **Dropped Customer Count Weighted PI:** Dropped customer count performance index.
- **Intermediate Dropped Customer Count Weighted PI:** Dropped customer count performance index during temporary moves.
- **Dropped Priority Customer Weighted PI:** Dropped priority load performance index.
- **Intermediate Dropped Priority Customer Weighted PI:** Dropped priority load performance index during temporary moves.
- **Customer Minutes Interrupted:** Customer Minutes Interrupted performance index.
- **Priority Customer Minutes Interrupted:** Priority Customer Minutes Interruption performance index.
- **Load Minutes Interrupted:** Load (kW) Minutes Interrupted performance index.
- **Maximum Margin:** Margin to limit performance index.
- **Radial Condition:** Radial condition performance index.
- **Intermediate Radial Condition:** Radial condition performance index during temporary moves.
- **Momentary Frequency – MAIFI:** Momentary average interruption frequency index.
- **Interruption Duration – SAIDI:** System average interruption duration index.
- **Interruption Frequency – SAIFI:** System average interruption frequency index.
- **Feeder Voltage Drop Weighted PI:** Voltage drop of the feeder.
- **Voltage Equalization:** Voltage equalization.
- **Protection Validation Weighted PI:** Protection performance index.
- **Protection Margin PI:** Protection margin performance index.
- **Protection Reach PI:** Protection reach performance index.
- **Paralleling Circuits Switching Validation Check Weighted PI:** Paralleling circuits switching validation check weighted performance index.
- **Ferroresonance Switching Validation Check Weighted PI:** Ferroresonance switching

validation check weighted performance index.

- **3-Wire/4-Wire Switching Validation Check Weighted PI:** 3-Wire/4-Wire switching validation check weighted performance index.
- **Maximum Number of Switching Steps:** Maximum number of switching steps performance index.
- **Maximum Number of Operations:** Maximum number of operations performance index.
- **Maximum Segment Margin:** Maximum segment margin performance index.

1.22.6. Non-Converged BLA and Scheduled Loading Measurement Islands

Non-Converged BLA and Scheduled Loading Measurement Islands displays are available for advanced DNAF functions (FISR, AFR, and PLOS). From this display the user can view the characteristics of non-converged and scheduled loading measurement islands in the plan reconfiguration path.

To access the displays, click Number of Non-Converged BLA Measurement Islands or Number of Scheduled Loading Measurement Islands on the AFR Plan Detail, FISR Plan Detail, and PLOS Plan Detail displays.

Display call-up URL:

```
netview://<Mode>/gridtabularnew/ReconfigNonConvergedBLAIsland_GridDisplay/<PlanStationName>/<
PlanID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Station:** Station name of the measurement island.
- **Island:** Name of the measurement island.
- **Island ID:** Identifier of the measurement island.
- **Branch:** Name of the measurement island's head branch.
- **Branch ID:** Identifier of the measurement island's head branch.
- **Allocation Type:** Measurement island load allocation type (kVA, or kW & kVAR).
- **Real Power Converged?:** Indicates whether real power was converged.
- **Reactive Power Converged?:** Indicates whether reactive power was converged.
- **Apparent Power Converged?:** Indicates whether apparent power was converged.

Navigation:

- "Locate" locates the measurement island head branch on the geographic view.

1.22.7. Plan Estimated Tap Position

The Plan Estimated Tap Position display provides details for tap changing transformers that were analyzed in scope of the solution. This display shows calculated tap positions for tap changing devices that self-regulate in reconfiguration applications. These internally calculated tap positions do not affect the existing tap positions in the real time or study mode environments. The tap regulation capability, either fixed or self-regulating, can impact the status of the reconfiguration application switching plans which can also impact closed-loop plan execution. This display can be accessed from FISR, AFR, and PLOS plan displays.

Display call-up URL:

`netview://<Mode>/gridtabular/PlanEstimatedTapPosition_GridDisplay/<Substation Name>/<Plan ID>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **ID:** Tap changer ID.
- **Transformer Name:** The associated transformer name.
- **Transformer Winding**: The transformer winding.
- **Device Type**: The transformer type.
- **Nominal Station**: Nominal substation.
- **Nominal Feeder**: Nominal feeder.
- **Control Type:** The control method used for the device (for example, Raise/Lower).
- **Applied Regulation Logic:** The tap changer regulation logic (Fixed or Regulating).
- **Effective Voltage Target:** Effective voltage target (per unit).
- **Effective Voltage Target Minimum**: Effective voltage target minimum (per unit).
- **Effective Voltage Target Maximum:** Effective voltage target maximum (per unit).
- **Initial Tap Position:** Initial tap position.
- **Post Plan Final Tap Position**: Post-plan final tap position.
- **Post Plan Phase A/B/C Voltage:** Post-plan phase A, B, and C voltage (per unit).

Navigation:

- "Locate" locates the power transformer on the network view.
- "Input Data" opens the Plan Estimated Tap Position Input Data display.

1.22.7.1. Plan Estimated Tap Position Input Data

This display shows tap changer input data for a reconfiguration plan. This display can be accessed from the Plan Estimated Tap Position display.

Display call-up URL:

```
netview://<Mode>/gridtabular/PlanEstimatedTapPositionInputData_GridDisplay_GridDisplay/<Subst
ation Name>/<Plan ID>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **ID:** Tap changer ID.
- **Transformer Name:** The associated transformer name.
- **R/L Measurement:** The Remote/Local measurement value.
- **A/M Measurement:** The Auto/Manual measurement value.
- **Blocked Measurement:** The Blocked measurement value.
- **PT Ratio:** Regulating transformer PT ratio.
- **CT Ratio:** Regulating tap CT ratio.
- **Nominal Voltage:** Nominal voltage.
- **Rated Voltage:** Voltage specified on the nameplate.
- **Regulation Node ID:** The ID of the network node that is used as the sensing point for voltage regulation by the controller.
- **Regulation Mode Name:** The name of the network node that is used as the sensing point for voltage regulation by the controller.
- **Regulation Direction:** Regulation direction.
- **Regulation Mode:** The name of the regulation mode.
- **Voltage Target:** Voltage target in Volts.
- **Voltage Bandwidth:** The regulation voltage bandwidth in Volts.
- **LDC Resistance:** The LDC resistance in Volts.
- **LDC Reactance:** The LDC reactance in Volts.
- **Voltage Reduction:** The voltage reduction percentage.

1.23. DNAF Application Diagnostic Information

The DNAF Application Diagnostic Information display can be accessed from the Station Output Summary display. The display can also be accessed from the substation toolbar by clicking Network Analysis Outputs and selecting the Diagnostic Summary option. The display provides diagnostic data about all DNAF applications executed for a substation.

Display call-up URL:

```
netview://<Mode>/gridtabular/DNAFDiagnosticInfo_GridDisplay/<Substation Name>
```

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the DNAFDiagnosticInfo display:

- **DNAF Application:** Name of the DNAF application.
- **Execution Time:** Time at which the latest solution was executed.
- **Diagnostic Details:** Application diagnostic details (for example, error, success, no new solution.

Navigation:

- "Result Details" navigates to the relevant results display. For example, the AFR Plan display.

1.24. Stations Summary for Protection Settings Group

The Stations Summary for Protection Settings Group display can be accessed from RT, ST, and SIM menus. It provides navigation to protection settings group data for all substations.

Display call-up URL:

netview://<Mode>/gridtabular/StationSummaryForProtectionSetting_GridDisplay

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following navigation option is shown on this display:

- "Protection Settings Group" navigates to the Protection Settings Group display.

1.24.1. Protection Settings Group

The Protection Settings Group display can be accessed from the Stations Summary for Protection Settings Group display. It provides protection settings data for a substation.

Display call-up URL:

`netview://<Mode>/gridtabular/ProtectionSettingGroup_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the ProtectionSettingGroup display:

- **Recloser ID:** ID of the recloser.
- **Recloser Name:** Name of the recloser.
- **Nominal Feeder:** Name of the nominal (owning) feeder.
- **Energizing Station:** Name of the dynamic (energizing) substation.
- **Energizing Feeder:** Name of the dynamic (energizing) feeder.
- **Switch Status:** Switch status (open or closed).
- **Notes:** Setting group notes.
- **Nominal Settings Group Template:** The nominal protection settings group template.

- **Active Settings Group Template:** The currently active protection settings group template.
- **Nominal Settings Group:** The name of the nominal protection settings group.
- **Active Settings Group:** The name of the currently active protection settings group.
- **Active Phase Pickup:** Phase trip current rating of the currently active protection settings group, in Amps.
- **Protection Setting Group 1-10:** Setting group sources.

Navigation:

- "Locate" navigates to the recloser on the geographic view.
- "Nominal Settings Group Template" shows the Protection Settings Template Information display for the selected template in the lower frame.
- "Active Settings Group Template" shows the Protection Settings Template Information display for the selected template in the lower frame.

1.25. Stations Summary for Settings Group Templates

The Stations Summary for Settings Group Templates display can be accessed from RT, ST, and SIM menus. It provides navigation to station protection settings group templates.

Display call-up URL:

`netview://<Mode>/gridtabular/StationSummaryForSGTemplates_GridDisplay`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationSummaryForSGTemplates display:

- **StationName:** Station name.

Navigation:

- "Settings Group Template" navigates to the Settings Group Templates display.

1.25.1. Settings Group Templates

The Settings Group Templates display can be accessed from Stations Summary for Settings Group Templates. It provides details for station protection settings group templates.

Display call-up URL:

`netview://<Mode>/gridtabular/SettingsGroupTemplates_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the SettingsGroupTemplates display:

- **ID:** Template id.

- **Name:** Template name.
- **Description:** Template description.
- **Number of Settings Groups:** Number of settings groups in the template.

Navigation:

- "Name" opens the Protection Settings Template Information display.

1.25.2. Protection Settings Template Information

The Protection Settings Template Information display can be accessed from the Settings Group Templates display. It provides details about protections settings groups and available settings group transitions for a template.

Display call-up URL:

`netview://<Mode>/gridtabular/SettingsGroups_GridDisplay/<Template Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the SettingsGroups display:

- **Settings Group Name:** The name of the settings group.
- **Value:** The value (index) of the settings group.
- **Phase Trip:** The phase trip current setting, in Amps.
- **Ground Trip:** The ground trip current setting, in Amps.
- **Mode:** The setting group mode, such as switch, recloser, sectionalizer, or custom.
- **SG1-SG8:** The settings group transition matrix, where (1 = allowed transition, and 0 = forbidden transition). For example, transition from settings group 3 to settings group 4 is allowed, and transition from settings group 3 to settings group 7 is forbidden.

1.26. Mobile Substation List

The Mobile Substation List display can be accessed from the main menu (real-time or study). The display contains a list of mobile stations and key information related to these stations.

Display call-up URL:

`netview://<Mode>/gridtabular/MobileSubStationsGridDisplay/`

Where, <Mode> can be RT (Real-Time) and ST (Study).

The following information is shown on this display:

- **Station Name:** Name of mobile substation.
- **EMS Name:** Name in the Energy Management System.

- **Connected:** Connection indicator; can be Yes or No.
- **Adjacent Station:** Neighboring station.
- **Description:** Description of the mobile substation.
- **X:** connection point of the mobile substation latitude.
- **Y:** connection point of the mobile substation longitude.
- **Geometry:** Name of geometry.
- **AOR:** Area of responsibility.

Navigation:

- "Locate" navigates to the element on the geographic view.

1.27. Tag Summary

The Tag Summary display contains a list of tags. This display can be accessed from the RT DMS menu.

The Tag Summary display can be helpful for troubleshooting but was not designed for the end user. During initial configuration of the tagging interface between SCADA and DMS, this display can show whether the ADMS server is successfully receiving tags from the tagging server.

Display callup URL:

`netview://<Mode>/gridtabular/TagSummary_GridDisplay/`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Tag Summary display:

- **ID:** Unique identifier of the tag.
- **Station:** The name of substation.
- **Name:** Distribution Management System unique identifier in the format <Substation>.<Feeder>.<Device_Type>.<Device_ID>.
- **TAG Type:** The tag type (for example, DO NOT OPERATE).
- **Comment:** The tag comment.
- **Placed By:** The user who placed the tag.

Navigation:

- "Locate" navigates to the tag on the geographic view

1.27.1. Tag Summary (for Plan Devices)

The Tag Summary display lists all tags placed on devices used in optimization and reconfiguration plans. The display can be accessed from the LVM Device Eligibility and Reconfiguration Device Eligibility displays.

It can be manually called up using the URL below:

`netview://<Mode>/gridtabular/OptDeviceTagSummary_GridDisplay/<Station Name./Optimization Device ID>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Tag Summary display:

- **Tag ID:** Unique identifier of the tag.
- **Station:** The name of substation.
- **Tag Type:** The name of tag type.
- **Placed By:** The user who placed the tag.

Navigation:

- "Locate" navigates to the tag on the geographic view.

1.27.2. Orphan Tag Summary

The Orphan Tag Summary display lists all tags placed on devices that no longer exist in the system. This display can be accessed from the RT DMS menu.

It can be manually called up using the URL below:

`netview://<Mode>/gridtabular/OrphanTagSummary_GridDisplay`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Tag Summary display:

- **Tag ID:** Unique identifier of the tag.
- **Station:** The name of substation.
- **Tag Type:** The name of tag type.
- **EMS ID:** ID in the Energy Management System.
- **DMS ID:** ID of in the Distribution Management System.
- **Tag Type:** Type of the tag.
- **Placed By:** The user who placed the tag.

Navigation:

- "Locate" navigates to the tag on the geographic view.

1.28. Annotation Summary

The Annotation Summary display provides the list of annotations available in the system for a specific annotation type. This display can be accessed from the ADMS menu bar > RT DMS menu > Annotation Summary > Specific type of annotation (instead of RT, this can be ST or SIM DMS).

Display call-up URL:

`netview://<Mode>/gridtabular/AnnoSummary_griddisplay/<Annotation_Type>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the AnnoSummary display:

- **Name:** Name of the Annotation.
- **ID:** Annotation unique identifier.
- **Status:** Status of the annotation.
- **Creation Time:** Time when annotation was created.
- **Update Time:** Time when the annotation was updated.
- **Station:** Name of the substation.
- **Service Center:** Name of the service center.
- **Device:** ID of the device.
- **Comment:** Comments specified when the annotation was created or updated.
- **Remove:** Removes the annotation.

Navigation:

- "Locate" locates to chosen annotation on the geographic display.

> **Important**
>
> The bitmap name assigned to an annotation state via the Viewer Configuration Editor serves as the state description. If the bitmap is renamed for an annotation index, the original state description for that index is removed, and the description of a "new/different" state is added.

1.28.1. Annotation Type Summary Pop-Up

This pop-up–like display provides end users with direct access to the displays containing the list of Annotation Type summaries. This display is accessible from the menu bar > RT DMS menu > Annotation Summary (instead of RT, this can be ST or SIM DMS). Each type of summary has a clickable icon that navigates to the related summary display.

Display call-up URLs:

- `netview://<Mode>/abnormaltempobject/AnnoSummary`

- `netview://<Mode>/annotempobject/AnnoSummary`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Annotation Summary Type display:

- **Annotation Type:** The name of the annotation type.

Navigation:

- Bitmap icon of an annotation type: Clicking an icon navigates to the related annotation type summary display, Annotation Summary.

1.28.2. Annotation Type Summary Display

The Annotation Type Summary display is a list of annotation types available in the form of a table with clickable cells. Clicking the name or the icon of an annotation type navigates to the related annotation summary.

Display call-up URLs:

- `netview://<Mode>/gridtabular/summaryAnnoTypeText_GridDisplay`
- `netview://<Mode>/gridtabular/AnnoSummaryTypeBitmap_griddisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on these displays:

- Columns are named by the name of the annotation type.

Navigation:

- Bitmap icon or name of an annotation type: Clicking an item opens the Annotation Summary display for the corresponding annotation type.

1.28.3. All Annotations Summary

The All Annotations Summary display can be accessed from the menu bar > RT DMS menu > Annotations Summary > All Annotations (instead of RT, this can be ST or SIMDMS).

The display contains a list of all annotations available in the form of a table and ability to navigate to the annotation on the geographic map.

Display call-up URLs:

- `netview://<Mode>/gridtabular/SummaryAllAnnotations_GridDisplayy`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the SummaryAllAnnotations display:

- **Name:** Name of the Annotation.
- **ID:** Annotation unique identifier.
- **Status:** Status of the annotation.
- **Creation Time:** Time when annotation was created.
- **Update Time:** Time when the annotation was updated.
- **Station:** Name of the substation.
- **Service Center:** Name of the service center.
- **Device:** ID of the device.
- **Comment:** Comments specified when the annotation was created or updated.
- **Remove:** Removes the annotation.
- **Area of Responsibility:** Name of the area of responsibility.
- **User:** User name.
- **Annotation Type:** The annotation type name.
- **Feeder:** Feeder name.

Navigation:

- "Locate" locates the chosen annotation on the geographic display.

1.29. Abnormal Summary

The Abnormal Summary display can be accessed from the menu bar > RT DMS > Abnormal Summary (instead of RT, this can be ST or SIM).

This display contains a list of abnormal summary displays, with a display name and a clickable icon that navigates to the related display.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryCutJumpGrd_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

Navigation:

- **Jumper:** Clicking this button navigates to the Abnormal Summary Jumper display.
- **Ground:** Clicking this button navigates to the Abnormal Summary Ground display.
- **Cut:** Clicking this button navigates to the Abnormal Summary Cut display.
- **Fault:** Clicking this button navigates to the Abnormal Summary Fault display.
- **Fault Indicators:** Clicking this button navigates to the Abnormal Summary Fault Indicator display.
-

Feeder Customer Count: Clicking this button navigates to the Abnormal Feeder Summary display.

- **All Abnormal:** Clicking this button navigates to the All Abnormal Summary display.

1.29.1. All Abnormal Summary

The All Abnormal Summary display can be accessed from the menu bar > RT DMS menu > Abnormal Summary > All Abnormal (instead of RT, this can be ST or SIM DMS).

This display contains a list of all network elements in abnormal state with short summary and ability to navigate to the devices on the geographic display.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryAllAbnormal_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the summaryAllAbnormal display:

- **Date Time:** Date and time when the change of state occurred. This is the time that the model was changed, for non-telemetered devices it does not necessarily match the time that the switching took place in the field.
- **Station:** Name of substation.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (for example, SWITCH, JUMPER).
- **Name:** The name of the network component.
- **ID:** The unique identifier of the network component.
- **Description:** Description.
- **Tag:** Associated tag (if exists).
- **Status:** The current status of the abnormal component, such as Open or Disconnected.
- **Operator:** The name of the operator who performed the last change of state.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Associated Transformer:** ID of the associated transformer if the component is an elbow switch.
- **Is Orphan:** Indicator (Yes/No) of whether the device is orphaned.
- **Op - Center:** The name of the service center.

Navigation:

- "Locate" locates the device in the current viewport.
- "Attributes" opens the Operator Attributes pop-up.
- "Tag" opens the Tagging Operations pop-up.

Context menu:

- "Locate" locates the device in the current viewport.

1.29.2. Abnormal Summary Cut

The Summary Cut display can be accessed from the menu bar > RT DMS > Abnormal Summary > Cut (instead of RT, this can be ST or SIM DMS).

This display contains a list of cuts with a short summary and the ability to navigate to the devices on the geographic display.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryCut_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the summaryCut display:

- **Date Time of Insertion:** Date and time when the cut was created.
- **Date Time of Last Status Update:** Date and time when the change of state occurred. This is the time that the model was changed.
- **Station:** Substation name.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (Cut).
- **Name:** Name of the Cut.
- **Description:** Description specified when creating or updating related cut.
- **Tag:** Related tag.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Associate TF:** Associated transformer.
- **State:** State of the cut switch, open or closed.
- **Operator:** The name of the operator who created the cut or performed the last change of state.

Navigation:

- "Locate" locates the device in the current viewport.

1.29.3. Abnormal Summary Fault

The Summary Fault display can be accessed from the menu bar > RT DMS > Abnormal Summary > Fault (instead of RT, this can be ST or SIM DMS).

This display contains a list of faults with a short summary and the ability to navigate to the devices on the geographic display.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryFault_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the summaryFault display:

- **Date Time:** Date and time when the change of state occurred. This is the time that the model was changed, for non-telemetered devices it does not necessarily match the time that the fault took place in the field
- **Station:** Substation name.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (fault).
- **Name:** Name of the fault.
- **Description:** Fault description.
- **Tag:** Related tag.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Associate TF:** Associated transformer.
- **State:** State of the fault.

Navigation:

- "Locate" locates the device in the current viewport

1.29.4. Abnormal Summary Jumper

The Abnormal Summary Jumper display can be accessed from the menu bar > RT DMS > Abnormal Summary > Jumper (instead of RT, this can be ST or SIM).

This display contains a list of jumpers with a short summary and the ability to navigate to the devices on the geographic display.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryJumper_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the summaryJumper display:

- **Date Time of Insertion:** Date and time when the jumper was created.
- **Date Time of Last Status Update:** Date and time when the change of state occurred. This is the time that the model was changed.

- **Station:** Substation name.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (Jumper).
- **Name:** Jumper name.
- **Description:** Jumper description.
- **Tag:** Related tag.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Associate TF:** Associated transformer
- **State:** State of the jumper (open or closed).
- **Operator:** The name of the operator who created the jumper or performed the last change of state.

Navigation:

- "Locate" locates the device in the current viewport.

1.29.5. Abnormal Summary Ground

The Ground Summary display lists the existing grounds in the distribution network. Ground or earth is a point in an electrical circuit from which other voltages are measured. Ground should have an appropriate current-carrying capability in order to serve as an adequate zero-voltage reference level.

This display can be accessed from the Abnormal Summary display by clicking the ground icon.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryGround_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the summaryGround display:

- **Date Time:** Date and time when the ground was inserted.
- **Station:** Substation name.
- **Feeder:** Feeder name.
- **Component Type:** The device-type of the component (Ground).
- **Name:** Name of the ground specified in the Temporary Ground pop-up when the ground was created.
- **Description:** Description of the temporary ground specified when the ground was created.
- **Tag:** Related tag.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.

- **Associate TF:** Associated transformer.
- **State:** State of the ground.

Navigation:

- "Locate" locates the device in the current viewport.

1.29.6. Abnormal Summary Fault Indicator

The Abnormal Summary Fault Indicator display can be accessed from the menu bar > RT DMS > Abnormal Summary > Fault Indicators (instead of RT, this can be ST or SIM).

This display contains a list of grounds with a short summary and the ability to navigate to the devices on the geographic display.

Display call-up URL:

- `netview://<Mode>/gridtabular/summaryfindabnormal_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the summaryfindabnormal display:

- **Station:** Substation name.
- **Feeder:** Feeder name.
- **ID:** ID of the fault indicator.
- **Status:** Status of the fault indicator (Set, Unset, or Unknown for a non-directional fault indicator; Unset, Both, Side 1, Side 2, or Unknown for a type 1 directional fault indicator; Unset, Forward Set, Reverse Set, or Unknown for a type 2 directional fault indicator).
- **Telemetered:** Indicated if the fault indicator can be controlled remotely (Yes or No).
- **Quality:** Fault indicator signal quality (good, bad, or suspect). Applies only to telemetered (SCADA) fault indicators.
- **Phases:** Phases set by the fault indicator.

Navigation:

- "Locate" locates the device in the current viewport.

1.29.7. Abnormal Feeder Summary

The Abnormal Feeder Summary display can be accessed from the menu bar > RT DMS > Abnormal Summary > Feeder Customer Count (instead of RT, this can be ST or SIM).

This display contains a list of feeders whose currently connected customer count is different from the nominal customer count.

Display call-up URL:

- `netview://<Mode>/gridtabular/abnormalfeederreport_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the abnormalfeederreport display:

- **Name:** Feeder name.
- **Feeder Area:** Feeder area.
- **Nominal Station:** Feeder nominal station.
- **Feeder Nominal Customer Count:** The nominal number of customers connected to the feeder.
- **Feeder Current Customer Count:** The number of customers currently connected to the feeder.

1.30. Stations Containing Third Party Equipment

The Stations Containing Third Party Equipment display lists substations that have devices that are owned by third-party companies. You can access this display from the RT DMS menu.

Display call-up URL:

- `netview://<Mode>/gridtabular/ThirdParty_Substations_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the substation.
- **Has Third Party Owned Equipment:** Specifies whether the substation has devices that are owned by third-party companies.

Navigation:

- "Has Third Party Owned Equipment" opens the Third Party Equipment Summary display for the selected substation.

1.30.1. Third Party Equipment Summary

The Third Party Equipment Summary display lists devices that are owned by third-party companies for a substation. This display can be called from the Stations Containing Third Party Equipment display.

Display call-up URL:

- `netview://<Mode>/gridtabular/ThirdParty_Equipment_GridDisplay/<Station_Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Station Name:** Name of the substation.
- **Feeder Name:** Name of the feeder.
- **ID:** Device ID.
- **Name:** Device name.
- **Device Type:** Type of the device.
- **External Company Name:** Name of the third-party company that owns the device. This field is displayed only if the device belongs to a third-party company.

Navigation:

- "Locate" navigates to the device on the network view.

1.31. Base Application Status

The Base Application Status display provides an interface where the dispatcher can view the status of base DNAF functions (DPF, FL, PRV, and SCC). This display can be accessed from RT, ST, and SIM menus.

Display call-up URL:

- `netview://<Mode>/gridtabular/BaseApplicationStatus_GridDisplay/`

  Where, <Mode> can be RT (Real-Time) or ST (Study).

Display call-up URL for SIM mode:

- `netview://SIM/gridtabular/BaseApplicationStatusSIM_GridDisplay/`

  The following information is shown on the BaseApplicationStatus display:

- **Station:** Name of the substation.
- **DPF Status:** Shows the status of DPF results.
- **BLA Consistency Check Status:** Indicates the number of devices which have a potential inconsistency issue for the station.
- **PRV Status:** Shows the status of the PRV results.
- **FL Status:** Shows the status of the FL results.
- **SCC Status:** Shows the status of the SCC results.
- **Solution Times:** Shows the solution time for the base applications.

Navigation:

- "DPF Status" opens the Distribution Power Flow summary containing the Distribution Power Flow Summary for Station and DPF Feeder Results displays.
-

"BLA Consistency Check Status" opens the Station Consistency Check Devices display.

- "PRV Status" navigates to the Protection Validation Results display.
- "FL Status" navigates to the Fault Location display.
- "SCC Status" navigates to the Short Circuit Calculation Summary display.
- "Solution Times" opens the Base Application Solution Times display.

1.31.1. Base Application Solution Times

The Base Application Solution Times display can be accessed from the Base Application Status display. It provides details about the solution and execution times of Distribution Power Flow and Protection Validation applications.

Display call-up URL:

- `netview://<Mode>/gridtabular/BaseApplicationSolutionTimes_GridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Base Application Solution Times display:

- **Station:** Name of the substation.
- **Last DPF Converged Solution Time:** Solution time of the last converged Distribution Power Flow.
- **Last DPF Execution Time:** Last DPF execution time.
- **Last PRV Solution Time:** The solution time of the last Protection Validation.
- **Last PRV Execution Time:** The execution time of the last Protection Validation.
- **Last FL Solution Time:** The solution time of the last Fault Location.
- **Last FL Execution Time:** The execution time of the last Fault Location.
- **Last SCC Solution Time:** The solution time of the last Short Circuit Calculation.
- **Last SCC Execution Time:** The execution time of the last Short Circuit Calculation.

1.31.2. Station Consistency Check Devices

The Station Consistency Check Devices display provides details about devices with voltage and/or flow inconsistencies during DPF execution. This display can be accessed by clicking the BLA Consistency Check Status button on the Base Application Status display.

Display call-up URL:

- `NETVIEW://<Mode>/GRIDTABULAR/SubstationBLAConsistencySwitch_GridDisplay/<Station Name >.`

  Where, <Mode> can be RT (Real-Time), ST (Study), or SIM (Simulator).

-

Name: Switch name.
- **ID:** Switch ID.
- **Phase:** The phase of the switch.
- **Explanation:** The inconsistency explanation, which can be one of the following.
  - Non-zero current (Amp) flow measurement detected on a deenergized device.
  - Zero current (Amp) flow measurement detected on an energized device.
  - Non-zero current (Amp) flow measurement detected on an isolating switching device.
  - Non-zero voltage measurement detected on a deenergized device.
  - Zero voltage measurement detected on an energized device.
  - Non-zero voltage measurement detected on an isolating switching device.
  - Non-zero real power (W) flow measurement detected on a deenergized device.
  - Non-zero real power (W) reverse flow measurement detected on a deenergized device.
  - Zero real power (W) flow measurement detected on an energized device.
  - Zero real power (W) reverse flow measurement detected on an energized device.
  - Non-zero real power (W) flow measurement detected on an isolating switching device.
  - Non-zero real power (W) reverse flow measurement detected on an isolating switching device.
  - Non-zero reactive power (VAR) flow measurement detected on a deenergized device.
  - Non-zero reactive power (VAR) reverse flow measurement detected on a deenergized device.
  - Zero reactive power (VAR) flow measurement detected on an energized device.
  - Zero reactive power (VAR) reverse flow measurement detected on an energized device.
  - Non-zero reactive power (VAR) flow measurement detected on an isolating switching device.
  - Non-zero reactive power (VAR) reverse flow measurement detected on an isolating switching device.
  - Non-zero Volt-Amp (VA) flow measurement detected on a deenergized device.
  - Zero Volt-Amp (VA) flow measurement detected on an energized device.
  - Non-zero Volt-Amp (VA) flow measurement detected on an isolating switching device.
  - Real power (W) flow direction mismatch.
  - Real power (W) reverse flow direction mismatch.
  - Reactive power (VAR) flow direction mismatch.
  - Reactive power (VAR) reverse flow direction mismatch.
  - Non-zero current (Amp) flow measurement detected on an open switching device.
  - Non-zero real power (W) flow measurement detected on an open switching device.
  -

Non-zero real power (W) reverse flow measurement detected on an open switching device.
   - Non-zero reactive power (VAR) flow measurement detected on an open switching device.
   - Non-zero reactive power (VAR) reverse flow measurement detected on an open switching device.
   - Non-zero Volt-Amp (VA) flow measurement detected on an open switching device.
   - Real power forward and reverse flow measurement inconsistent reading.
   - Reactive power forward and reverse flow measurement inconsistent reading.
   - Non-zero amps measurement detected on an energized device, while it is supposed to have zero flow.
   - Non-zero real power measurement detected on an energized device, while it is supposed to have zero flow.
   - Non-zero reactive power measurement detected on an energized device, while it is supposed to have zero flow.
   - Non-zero apparent power measurement detected on an energized device, while it is supposed to have zero flow.

Navigation:

- "Locate" navigates to the device on the network view.

1.32. Advanced Application Status

The Advanced Application Status (previously DSO Control) display shows the solution status of advanced functions (LVM, FISR, PLOS, AFR, and FISRCA). This display can be accessed from RT, ST, and SIM menus.

This display only shows stations where the dispatcher has been granted write access. The dispatcher can also disable advanced functions for a particular substation from the DNAF System Configuration display by setting the Disable flag. This disabling of functions is recorded in the System Activity Log.

A key piece of information on this display is the last solution status for each function:

- **Solved:** Successful solution.
- **Suspect:** A solution was obtained, but the results may be questionable.
- **Diverged:** Could not solve due to numerical difficulty.
- **Unsolved:** The solution not started or exited abnormally.

This display is automatically refreshed with data on a frequent basis to ensure that the dispatcher is kept up to date on current system conditions.

There are real-time, study, and simulator versions of this display that share similar capabilities.

Display call-up URLs:

- `netview://RT/gridtabular/dsocontrolRT_GridDisplay`
- `netview://ST/gridtabular/dsocontrol_GridDisplay`
- `netview://SIM/gridtabular/dsocontrolSIM_GridDisplay`

The following information is shown on the Advanced Application Status display:

- **Station:** Name of the substation.
- **LVM Status:** Status of last executed Load and Voltage/VAR Management.
- **FISR Status:** Status of last executed Fault Isolation and Service Restoration.
- **AFR Status:** Status of last executed AFR (Automatic Feeder Reconfiguration).
- **PLOS Status:** Status of last executed PLOS (Planned Outage Switching). Available in Study mode only.
- **FISRCA Status:** Status of last executed FISR (Fault Isolation and Service Restoration) Contingency Analysis. Available in Study mode only.

Navigation:

- "LVM Status" navigates to the Load and Volt/VAR Management display.
- "FISR Status" navigates to the FISR Results Summary display.
- "AFR Status" navigates to the AFR Results Summary display.
- "FISRCA Status" navigates to the FISRCA Solution Overview display. Available in Study mode only.
- "PLOS Status" navigates to the Planned Outage Service display. Available in Study mode only.
- "Advanced Application Solution Times" navigates to the Advanced Application Solution Times display.

1.32.1. Advanced Application Solution Times

The Advanced Application Solution Times display can be accessed from the Advanced Application Status display. It has details about the execution times of advanced DNAF applications.

Display call-up URL:

- `netview://RT/gridtabular/DSOTimesRT_GridDisplay/`
- `netview://ST/gridtabular/DSOTimes_GridDisplay/`
- `netview://SIM/gridtabular/DSOTimesSIM_GridDisplay/`

The following information is shown on the Advanced Application Solution Times display:

- **Substation:** Name of the substation.
- **Last LVM Solution Time:** Evaluation time of the last LVM (Load and Voltage/VAR Management) execution.
-

Last LVM Execution Time: Last LVM execution time.

- **Last FISR Solution Time:** Evaluation time of the last FISR (Fault Isolation and Service Restoration) execution.
- **Last FISR Execution Time:** Last FISR execution time.
- **Last AFR Solution Time:** Evaluation time of the Last AFR (Automatic Feeder Reconfiguration) execution.
- **Last AFR Execution Time:** Last AFR execution time.

For study mode, additional columns are displayed:

- **Last PLOS Solution Time:** Evaluation time of the Last PLOS (Planned Outage Switching) execution. Available in Study mode only.
- **Last PLOS Execution Time:** Last PLOS execution time. Available in Study mode only.
- **Last FISRCA Solution Time:** Evaluation time of the last FISR Contingency Analysis execution. Available in Study mode only.
- **Last FISRCA Execution Time:** Last FISR Contingency Analysis execution time. Available in Study mode only.

1.33. Application Input Summary

The Application Input Summary display provides the summary for inputs and measurements used for DNAF applications. The display can be accessed from RT, ST, and SIM menus by selecting the Application Input Summary menu.

Display call-up URL for RT mode:

- `netview://RT/gridtabular/ApplicationInputSummary_GridDisplay/`

Display call-up URL for ST mode:

- `netview://ST/gridtabular/ApplicationInputSummaryST_GridDisplay/`

Display call-up URL for SIM mode:

- `netview://SIM/gridtabular/ApplicationInputSummarySIM_GridDisplay/`

The following information is shown on the Application Input Summary display:

- **Station:** Name of the substation.
- **DNAF Capable:** DNAF (Distribution Network Analysis Functions) capability indicator. If the substation is DNAF capable that means that this Distribution Network Analysis Functions can be run on this substation.

Navigation:

- "Measurement Data" opens the Measurement Data display.
-

"Load Category Input" opens the Load to Voltage Coefficients display.

- "Station Feeder Input" opens the Station Feeder Input display.
- "Reference Bus Input" opens the Reference Bus Input for Station display.
- "LVM Device Eligibility" opens the LVM Device Eligibility display.
- "Reconfiguration Device Eligibility" opens the Reconfiguration Device Eligibility display.
- "Capacitor Input" opens the Capacitor Input display.
- "Tap Changer Input" opens the Tap Changer Input display.
- "Generator/ Distributed Energy Resource Input" opens the Generator/Distributed Energy Resource Input display.
- "Fault Location Input" opens the Fault Location Input display.
- "Regulation Mode Summary" opens the Regulation Mode Summary display.
- "Reserved Capacity Summary" opens the Reserved Capacity Summary display.

1.33.1. Load to Voltage Coefficients

The Load to Voltage Coefficients display allows you to enter the load to voltage parameters. The display can be accessed from the Application Input Summary display by clicking the Load Category button. You can also access this display from the substation toolbar by clicking Network Analysis Inputs and selecting the Load Categories option.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationManualLoadCategoryInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Station Load to Voltage Coefficients display:

- **Name:** Name of the schedule (for example, RES, SCI, MCI).
- **Manual Enabled:** Select this checkbox to enable manual coefficient settings.
- **Schedule AD:** Name of the AD schedule (for example, LoadVolt1).
- **Nominal A:** Nominal A value.
- **Manual A:** Manual A value.
- **Nominal D:** Nominal D value.
- **Manual D:** Manual D value.
- **Schedule BE:** Name of the BE schedule (for example, LoadVolt2).
- **Nominal B:** Nominal B value.
- **Manual B:** Manual B value.
- **Nominal E:** Nominal E value.

- **Manual E:** Manual E value.
- **Schedule CF:** Name of the CF schedule (for example, LoadVolt3).
- **Nominal C:** Nominal C value.
- **Manual C:** Manual C value.
- **Nominal F:** Nominal F value.
- **Manual F:** Manual F value.

1.33.2. Measurement Data

The Measurement Data display provides the list of measurements known by ADMS for the selected station. These measurements are used as inputs for the DPF solution. This display can be accessed from the Application Input Summary display, or by selecting the Go to Station Measurement Tabular option from the RT, ST, or SIM menu (note that substation name must be entered in the Browser toolbar's text box).

Display call-up URL:

- `netview://<Mode/gridtabular/measurements_GridDisplay/<Substation Name>/`

  Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the Measurements display:

- **ID:** SCADA composite identifier of the measurement.
- **Device Name:** Name of measurement component or node.
- **Device ID:** ID of measurement component or node.
- **Nominal Feeder:** Nominal feeder name.
- **Energizing Feeder:** Energizing feeder name.
- **Type:** Type of measurement.
- **Value:** Measurement value.
- **SCADA Quality:** SCADA quality, such as good, suspect, test mode or replaced.
- **Measurement Validation (DNAF):** Measurement validation status, such as valid or invalid.
- **Phasing:** Phase designation (for example, ABC for three phases, or A for only phase A).
- **Multi Phase:** This flag indicates whether the measurement is a multi-phase point type measurement.
- **Terminal Identifier:** Terminal identifier (from side, or to side).
- **Units:** Multiplier unit applicable for flow measurements (Base, Kilo, or Mega).
- **Measurement Utilized for Triggering Network Applications:** This flag indicates whether an analog voltage measurement is considered a trigger for advanced DNAF applications.
- **Measurement Utilized in Network Applications:** This flag indicates whether an analog

measurement value is considered for advanced DNAF applications.

- **Flip Analog:** Can be Yes or No.
- **Voltage Base:** Voltage base in volts.
- **Voltage Measurement Connection:** Voltage measurement connection (for example, line-to-line, line-to-ground).
- **SDIS:** SDIS value.
- **XDIS:** XDIS value.
- **MDIS:** MDIS value.
- **Time:** Measurement time.
- **Simulate Only:** Can be Yes or No.
- **Use Manual Quality in DNAF Applications:** If this checkbox is checked, and the corresponding checkbox "Manual Quality for DNAF Applications" is unchecked, then the measurement will not be used in DNAF applications. If this checkbox is checked, and the corresponding check-box "Manual Quality for DNAF Applications" is checked, then the measurement will be used in DNAF applications, regardless of the actual SCADA quality. If this checkbox is unchecked, then the actual SCADA quality will be used to decide usage in DNAF applications.
- **Manual Quality for DNAF Applications:** If the checkbox "Use Manual Quality in DNAF Applications" is checked, and this checkbox is unchecked, then the measurement will not be used in DNAF applications. If the checkbox "Use Manual Quality in DNAF Applications" is checked, and this checkbox is checked, then this measurement will be used in DNAF applications, regardless of the actual SCADA quality. If the checkbox "Use Manual Quality in DNAF Applications" is unchecked, then the actual SCADA quality will be used to decide usage in DNAF applications.

Navigation:

- "Locate" navigates to the measurement or the substation circle on the geographic view.

In Study mode, this display (measurements_ST_GridDisplay) provides two more columns for manual value input:

- **User Manual Value:** Option to allow manual value input.
- **Manual Value:** Manual entry of measurement value.

1.33.3. Station Feeder Input

The Station Feeder Input display allows modifying input used by the Network Analysis solution. This display shows the currently entered nominal values and allows overriding values to be entered related to the station feeders. This display can be accessed from the Application Input Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationFdrInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Station:** The name of substation
- **Nominal Load Scaling Factor:** The station load scaling factor (nominal value is 1.0). If a feeder head flow manual entry and flow measurement is not given for a load's feeder, the load's real and reactive power values are multiplied by this scaling factor.
- **Manual Load Scaling Setting:** Manual setting of the feeder's load scaling factor.
- **Manual LVM Maximum Power Factor Target Setting:** Maximum power factor target for all substation transformers (used in the LVM reactive area support problem formulation).
- **Manual LVM Minimum Power Factor Target Setting:** Minimum power factor target for all substation transformers (used in the LVM reactive area support problem formulation).
- **Current Temperature:** Manually entered temperature value (in °F).
- **Block All Regulating Transformers in the Station:** Check this box to block all regulators in the station. Tap positions do not change when the regulation is blocked
- **Enable Voltage Reduction for All Transformers in the Station:** Check this box to enable voltage reduction for all transformers that have voltage reduction capability

Navigation:

- "Locate" opens the substation network view in the same viewport.
- "Schematic" opens the substation schematic view in the same viewport.
- "Feeder Input" navigates to the Feeder Input for Station display.

1.33.4. Feeder Input for Station

The Feeder Input for Station display allows the user to view and enter feeder parameters used by the Distribution Network Analysis Functions (DNAF). It displays the list of feeders related to the substation.

The display can be accessed by clicking the Feeder Input button on the Station Feeder Input display. You can also access this display from the substation toolbar by clicking Network Analysis Inputs and selecting the Feeder option.

Display call-up URL:

- `netview://<Mode>/gridtabular/FeederInput_GridDisplay/<Substation Name>/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Feeder:** Feeder name.

- **Enable Manual Feeder Header Flow:** Click this field to enable the manual-input feeder head flow.
- **Manual Feeder Head Flow Phase A/B/C:** Manual-input fields for feeder head flow in phases A, B, and C, in Amps.
- **Manual Feeder Head Flow Power Factor:** Manual-input field of feeder head flow power factor.
- **Manual Load Scaling Setting:** Manual setting of the feeder's load scaling factor.
- **Manual Flow Limit Multiplier Setting:** The nominal value of the multiplier to change the flow limit.
- **Manual Feeder LVM Maximum Power Factor Target Setting:** The value of the feeder LVM maximum power factor target.
- **Manual Feeder LVM Minimum Power Factor Target Setting:** The value of the feeder LVM minimum power factor target.
- **Manual Bus High Voltage Violation Limit:** The value of the bus high voltage violation limit (expressed as a p.u. value).
- **Manual Bus Low Voltage Violation Limit:** The value of the bus low voltage violation limit (expressed as a p.u. value).
- **Manual Load High Voltage Violation Limit:** The value of the load high voltage violation limit (expressed as a p.u. value).
- **Manual Load Low Voltage Violation Limit:** The value of the load low voltage violation limit (expressed as a p.u. value).
- **Manual Bus High Voltage Optimization Limit:** The value of the bus high voltage optimization limit (expressed as a p.u. value).
- **Manual Bus Low Voltage Optimization Limit:** The value of the bus low voltage optimization limit (expressed as a p.u. value).

1.33.5. Reference Bus Input for Station

The Reference Bus Input for Station display can be accessed from the Application Input Summary display by clicking the Reference Bus Input for one of the stations. You can also access this display from the substation toolbar by clicking Network Analysis Inputs and selecting the Reference Bus Input option.

The Reference Bus object is used as the means of defining the reference voltage and phase angle for all distribution network analysis functions for the section of the network that is supplied by the bus. The reference bus values are normally set by the corresponding subtransmission voltage and phase angle calculated by the EMS State Estimator. It is also possible to manually set the Reference Bus values by checking the Enable check boxes and directly entering a voltage and phase angle.

Display call-up URL:

- `netview://<Mode>/gridtabular/RefBusInput_AN_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Name:** Name of the reference bus.
- **ID:** Unique ID of the reference bus.
- **Nominal Voltage Target:** The nominal voltage of the reference bus expressed as a percentage of the rated bus voltage.
- **Enable Manual Voltage Target:** Click to enable the override value for the reference bus voltage in place of the nominal value.
- **Manual Voltage Target (Phase A/B/C):** A manually editable field for the override value for the reference bus phase A, B, and C voltages in place of the nominal value in percentage.
- **Nominal Angle Target:** The nominal phase angle of the reference bus voltage, expressed in degrees.
- **Enable Angle Target:** Check this box to use the override value for the reference bus phase angle in place of the nominal value.
- **Manual Angle Target:** A manually editable field for the phase angle (expressed in degrees) that is used to override the nominal value when Enable is checked.
- **Connect Reference Bus Interface:** Check this box to connect the reference bus interface with the substation; Uncheck this box to disconnect the reference bus interface with the substation. This option is not applicable in RT.

1.33.6. LVM Device Eligibility

The LVM Device Eligibility (previously - Optimization Devices) display lists the devices with Distributed System Optimization capability and eligibility information. It shows the current operation mode as well as current status and allows set capacitor values and regulating transformer parameters to be set, including the device initial starting position. The information on this display can be used for setting up and testing LVM cases. This display can be accessed from the Application Input Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/OptDevice_griddisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the LVM Device Eligibility display:

- **Energizing Station:** The name of the substation that is currently energizing the device.
- **Energizing Feeder:** The name of the feeder that is currently energizing the device.
- **Name:** Name of the optimization device.
- **Type:** The type of device (Capacitor, LTC, or Regulator).
-

Energizing Feeder LVM Status:** LVM status for the energizing feeder (LVM Enabled / LVM Disabled).

- **Current Status Value:** Shows the current reactive power output of the device.
- **Eligibility Status:** Indicates whether the device is eligible for the Distribution System Optimizer control (Eligible/Not Eligible).
- **Reason Device is not Eligible:** The reason for the optimization control ineligibility.
- **Toggle Device Eligibility Mode:** Shows the device optimization control mode (SCADA Decides / User Disabled). Right-clicking this button shows a context menu where you can change the optimization control mode for the device; left-clicking refreshes the display to show changes. For generators and energy resources, the context menu includes options to enable or disable all devices of the selected type for the feeder or substation.
- **Tap Position Source:** The source of the optimization device tap position.
- **Current Status Value Measurement and Measurement Quality:** Shows the current status measurement and quality from SCADA.
- **Supervisory Cutout Status Measurement and Measurement Quality:** Supervisory cutout status and the quality of the measurement.
- **Auto/Manual Status Measurement and Measurement Quality:** Auto/manual status and the quality of the measurement.
- **Remote/Local Status Measurement and Measurement Quality:** Remote/local status and the quality of the measurement.
- **Control Available Status Measurement and Measurement Quality:** Control available status and the quality of the measurement.
- **Movement Cost:** Movement cost for the device (capacitor, LTC, regulator, DER or generator).
- **Prevent Operation Tags:** Shows the number of restricting tags.
- **Problem Formulation Name:** Problem formulation name.
- **ID:** Unique identifier of the optimization device.

Navigation:

- "Locate" navigates to the device location on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up for the device.
- "Show All Tags" opens the Tag Summary (for Plan Devices) display.

1.33.7. Reconfiguration Device Eligibility

The Reconfiguration Device Eligibility display lists devices (switches, composite switches, cuts, or temporary jumpers) that are eligible or potentially eligible for switching reconfiguration applications (AFR, FISR, and PLOS). This display can be accessed from the Application Input Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/Reconfig_eligibility_griddisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Reconfiguration Device Eligibility display:

- **Energizing Station:** The name of the substation that is currently energizing the device.
- **Energizing Feeder:** The name of the feeder that is currently energizing the device.
- **Name:** Name of the device.
- **ID:** Unique identifier of the device.
- **Type:** The type of device.
- **Status:** The status of the device.
- **FISR Eligibility:** Indicates whether the device is eligible for FISR (Eligible/Not Eligible). If the device is ineligible due to an upstream DA disabled device, this field allows the user to navigate to the upstream device.
- **AFR Eligibility:** Indicates whether the device is eligible for AFR (Eligible/Not Eligible). If the device is ineligible due to an upstream DA disabled device, this field allows the user to navigate to the upstream device.
- **PLOS Eligibility:** Indicates whether the device is eligible for PLOS (Eligible/Not Eligible).
- **Position Measurement Value:** Status measurement.
- **Switch Position Measurement Status:** Status measurement quality.
- **Normal Position:** Normal position for the device.
- **Lockout Status:** Flag indicating the lockout status of the device.
- **Hot Line Tag (Block Close):** Hot line tag status (Applied or Not Applied).
- **Device In Alarm Measurement:** In Alarm measurement.
- **Reclosing Enabled:** Flag indicating if the reclosing capability of the device is enabled.
- **Active Phase Trip Setting:** Phase trip setting.
- **Active Ground Trip Setting:** Ground trip setting.
- **Switch Position Measurement ID:** ID of the status measurement.

Navigation:

- "Locate" navigates to the device location on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up for the device.
- "Show All Tags" opens the Tag Summary (for Plan Devices) display.
- "Contained Switches" opens the contained switches, Switch display, of the composite switch.

1.33.8. Composite Switch Display

The Composite Switch display lists devices that are contained under a specific composite switch. This display can be accessed from the Reconfiguration Device Eligibility display.

Display call-up URL:

- `netview://<Mode>/gridtabular/Switch_griddisplay/<Substation Name>/<Composite Switch ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Composite Switch display:

- **Energizing Station:** The name of the substation that is currently energizing the device.
- **Energizing Feeder:** The name of the feeder that is currently energizing the device.
- **Name:** Name of the device.
- **ID:** Unique identifier of the device.
- **Type:** The type of device.
- **Feeder Main:** Indicates whether this is a feeder main device.
- **Status:** The status of the device.
- **FISR Eligibility:** Indicates whether the device is eligible for FISR (Eligible/Not Eligible).
- **AFR Eligibility:** Indicates whether the device is eligible for AFR (Eligible/Not Eligible).
- **PLOS Eligibility:** Indicates whether the device is eligible for PLOS (Eligible/Not Eligible).
- **Position Measurement Value:** Status measurement.
- **Switch Position Measurement Status:** Status measurement quality.
- **Normal Position:** Normal position for the device.
- **Lockout Status:** Flag indicating the lockout status of the device.
- **Hot Line Tag (Block Close):** Hot line tag status (Applied or Not Applied).
- **Device In Alarm Measurement:** In Alarm measurement.
- **Reclosing Enabled:** Flag indicating if the reclosing capability of the device is enabled.
- **Active Phase Trip Setting:** Phase trip setting.
- **Active Ground Trip Setting:** Ground trip setting.
- **Switch Position Measurement ID:** ID of the status measurement.

Navigation:

- "Locate" navigates to the device location on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up for the device.
- "Show All Tags" opens the Tag Summary (for Plan Devices) display.

1.33.9. Capacitor Input

The Capacitor Input Display can be accessed from the Application Input Summary display. It has details about all the capacitors present in that station and the details are displayed in three panes.

Display call-up URL:

First pane:

- `netview://<Mode>/gridtabular/StationCapInput1_GridDisplay/<Substation Name>`

Second pane:

- `netview://<Mode>/gridtabular/StationCapInput2_GridDisplay/<Substation Name>`

Third pane:

- `netview://<Mode>/gridtabular/StationCapInput3_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationCapInput1 display:

- **Name:** Name of the capacitor.
- **ID:** ID of the capacitor.
- **Nominal Feeder:** The nominal (owning) feeder for the capacitor.
- **Energizing Feeder:** The energizing feeder for the capacitor.
- **Control Capability:** The control capability of the capacitor.
- **Remote Enabled:** Denotes whether the capacitor can be remotely handled.
- **Local Control Type:** Denotes the local control type available for the capacitor.
- **Nominal On Level:** Nominal On level.
- **Nominal Off Level:** Nominal Off level.
- **Enable Manual On/Off Level:** Select this checkbox to enable manually entering the On and Off levels. This check box is conditional on the presence of a local controller.
- **Manual On Level:** Edit box where the value of the On level can be entered manually.
- **Manual Off Level:** Edit box where the value of the Off level can be entered manually.

Navigation:

- "Locate" locates the capacitor on the geographic view.

The following information is shown on the StationCapInput2 display:

- **Name:** Name of the capacitor.
- **ID:** ID of the capacitor.
- **Has Local Voltage Override Capability:** Denotes if the local voltage can be overridden.
-

Operational Voltage Override High Level: Value of overridden operational voltage high level.

- **Enable Manual Voltage Override High Level:** Override the voltage high level. This check box is conditional on the presence of a local controller.
- **Manual Voltage Override High Level:** Manual voltage override high level.
- **Operational Voltage Override Low Level:** Nominal voltage override low limit value.
- **Enable Manual Voltage Override Low Level:** Enable override of voltage low level. This check box is conditional on the presence of a local controller.
- **Manual Voltage Override Low Level:** Manual voltage override low level value.
- **Nominal Maximum Number of Daily Switching Operations:** Nominal maximum number of daily switching operations.
- **Enable Manual Maximum Number of Daily Switching Operations:** Enable manual override of number of daily switchings.
- **Manual Maximum Number of Switching Daily:** Manual value for the maximum number of daily switchings.

Navigation:

- "Locate" locates the capacitor on the geographic view.

The following information is shown on the StationCapInput3 display:

- **Name:** Name of the capacitor.
- **ID:** ID of the capacitor.
- **Nominal Hours Between Consecutive Switching:** Nominal Hours between consecutive switching operations.
- **Enable Manual Hours Between Consecutive Switching:** Enable manual hours between consecutive switching.
- **Manual Hours Between Consecutive Switching:** Manual hours between consecutive switching.
- **Optimization Eligibility Status:** Device optimization control mode (User Disabled or SCADA Decides).
- **Controller Operable:** Controller operable.
- **Steps:** Number of steps.
- **Enable Manual Nominal Size:** Enable manual input for nominal size of the capacitor. This option is available only in Study Mode.
- **Manual Nominal Size:** Manual nominal size value of the capacitor.

Navigation:

- "Locate" locates the capacitor on the geographic view.

1.33.10. Tap Changer Input

The Tap Changer Input display can be accessed from the Application Input Summary display. You can also access this display from the substation toolbar by clicking Network Analysis Inputs and selecting the Tap Changer option. The display shows the characteristics of the tap changers in the substation in two panes.

Display call-up URL:

First pane:

- `netview://<Mode>/gridtabular/StationTapInput1_GridDisplay/<Substation Name>`

Second pane:

- `netview://<Mode>/gridtabular/StationTapInput2_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationTapInput1 display:

- **Name:** ID of the tap changing device.
- **ID:** ID of the tap changing device.
- **Type:** Type of the device (power transformer or regulator).
- **Control Capability:** The control capability of the device.
- **Remote Enabled:** Shows the remote control capability. Can be Yes, No, or none (for Local Only and Fixed transformers).
- **Nominal Primary Tap Position:** Position of the regulating tap.
- **Enable Primary Tap Override:** Enables primary tap position override. This checkbox is available if the tap position is allowed to be changed by the DMS (tap exists and there is no associated tap position measurement).
- **Primary Tap Position Overridden Value:** Manually entered position of the regulating tap.
- **Primary Tap Position:** Current position of the regulating tap.
- **Nominal Secondary Tap Position:** Position of the regulating tap.
- **Enable Secondary Tap Override:** Enables secondary tap position override. This checkbox is available if the tap position is allowed to be changed by the DMS (tap exists and there is no associated tap position measurement).
- **Secondary Tap Position Overridden Value:** Manually entered position of the regulating tap.
- **Secondary Tap Position:** Current position of the regulating tap.
- **Nominal Tertiary Tap Position:** Position of the regulating tap.
- **Enable Tertiary Tap Override:** Enables tertiary tap position override. This checkbox is available if the tap position is allowed to be changed by the DMS (tap exists and there is no associated tap position measurement).
- **Tertiary Tap Position Overridden Value:** Manually entered position of the regulating tap.

- **Tertiary Tap Position:** Current position of the regulating tap.
- **Optimization Eligibility Status:** Device optimization control mode (User Disabled or SCADA Decides).

Navigation:

- "Locate" navigates to the device on the geographic view.

The following information is shown on the StationTapInput2 display:

- **Name:** ID of the tap changing device.
- **ID:** ID of the tap changing device.
- **Nominal Voltage Target:** Nominal Voltage Target.
- **Enable Manual Voltage Target:** Selecting this option enables manual voltage target.
- **Manual Voltage Target:** Manual voltage target value.
- **Nominal Voltage Bandwidth:** Nominal Voltage Bandwidth.
- **Enable Manual Voltage Bandwidth:** Selecting this option enables manual voltage bandwidth.
- **Manual Voltage Bandwidth:** Manual voltage bandwidth value.
- **Nominal Line Drop Compensation Resistance:** Nominal line drop compensation resistance in volts.
- **Enable Manual Nominal Line Drop Compensation Resistance:** Selecting this option enables the manual nominal line drop compensation resistance.
- **Manual Line Drop Compensation Resistance:** Manual line drop compensation resistance value in volts.
- **Nominal Line Drop Compensation Reactance:** Nominal Line drop compensation reactance in volts.
- **Enable Manual Nominal Line Drop Compensation Reactance:** Selecting this option enables the manual nominal line drop compensation reactance.
- **Manual Line Drop Compensation Reactance:** Manual line drop compensation reactance value in volts.
- **Nominal Voltage Reduction Value:** Nominal voltage reduction value.
- **Enable Manual Voltage Reduction Value:** Selecting this option enables the manual voltage reduction value. This check box is enabled if the device has voltage reduction capability and there is no SCADA measurement driving this status.
- **Manual Voltage Reduction Value:** Manual voltage reduction value.

Navigation:

- "Locate" navigates to the device on the geographic view.

1.33.11. Generator/Distributed Energy Resource Input

The Generator/Distributed Energy Resource Input display (formerly - Station Generator Input Display) can be accessed from the Application Input Summary display. It displays the characteristics of the generators in the station.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationGenInput_AN_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Generator/Distributed Energy Resource Input Display:

- **Name:** Name of the generator.
- **ID:** ID of the generator.
- **Nominal Voltage:** Nominal voltage of the generator.
- **Rated Voltage:** Voltage specified on the generator.
- **Type:** Generator type.
- **Connection Status:** Connection status (Connected or Disconnected).
- **Nominal Voltage Target:** Nominal voltage target in percentage.
- **Enable Manual Voltage Target:** Selecting this option enables the manual voltage target. The check box is displayed based on the type.
- **Manual Voltage Target:** Manual voltage target value in percentage.
- **Nominal Real Power Target:** Nominal real power target in kW.
- **Enable Manual Real Power Target:** Selecting this option enables the manual real power target.
- **Manual Real Power Target:** Manual real power target value in kW.
- **Nominal Reactive Power Target:** Nominal reactive power target in kVAR.
- **Enable Manual Reactive Power Target:** Selecting this option enables the manual reactive power target.
- **Manual Reactive Power Target:** Manual reactive power target value in kVAR.
- **Has Nominal Contributes To Fault Current Setting:** Can be Yes or No.
- **Enable Manual Contributes To Fault Current:** Selecting this option enables the manual Contributes To Fault Current setting.
- **Manual Contributes To Fault Current:** Manual Contributes To Fault Current setting. If enabled, the device can remain connected during a fault and contribute to fault current. This is considered during SSC or PRV calculations.

Navigation:

- "Locate" locates the device on the geographic view.

1.33.12. Fault Location Input

This display can be accessed from the Application Input Summary and Manual Entry Input Summary displays.

The Fault Location Input display allows the user to input and view the necessary data for running the Fault Location application on a given feeder. This display is typically used in study mode.

Display call-up URL:

- `netview://<Mode>/gridtabular/FaultlocationManualinput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** The name of the trigger device.
- **ID:** The unique identifier of the trigger device.
- **Device Type:** The device type of trigger device (line, breaker, and so on).
- **Phase A/B/C Fault:** Check box to enable/disable fault for phases A, B, and C.
- **Ground Fault:** Check box to enable/disable a ground fault.
- **Fault Type:** Indicates the type of fault set. For example, a Phase A and Ground fault is shown as "AG".
- **Phase A/B/C Fault Current:** Data entry field for setting the peak breaker fault current value for phases A, B, and C in amps.
- **Maximum Fault Current:** Data entry field for setting the peak breaker fault current value in amps.

Navigation:

- "Locate" locates the fault on the geographic view.

1.33.13. Regulation Mode Summary

The Regulation Mode Summary display lists the transformers utilizing advanced regulation. The available regulation modes are displayed for each transformer with the active mode highlighted. This display can be accessed from the Application Input Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/RegulationModeSummary_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Transformer ID:** The ID of the transformer.

- **Transformer Name:** The name of the transformer.
- **Phase:** Phases of the transformer.
- **Current Tap Position:** Current tap position.
- **Sensed Phase:** Transformer phase sensed.
- **Primary Voltage:** Primary side phase voltage and angle (line-to-ground).
- **Secondary Voltage:** Secondary side phase voltage and angle (line-to-ground).
- **Regulation Mode 1-4:** The available regulation modes sorted by index (displays "—" if not available).

Navigation:

- Clicking the name of a regulation mode opens the Regulation Mode Details display for that specific regulation mode.

1.33.14. Regulation Mode Details

The Regulation Mode Details display shows the defined parameters for a given regulation mode. This display can be accessed from the Regulation Mode Summary display by clicking a regulation mode name.

Display call-up URL:

- `netview://<Mode>/gridtabular/RegulationModeDetails_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Regulation Mode Name:** The name of the regulation mode.
- **Forward Regulation Mode:** The name of the forward regulation mode.
- **Forward Voltage Target (V):** The voltage target for forward regulation.
- **Forward Voltage Bandwidth (V):** The voltage bandwidth for forward regulation.
- **Forward LDC Resistance (V):** The LDC resistance for forward regulation defined in volts.
- **Forward LDC Reactance (V):** The LDC reactance for forward regulation defined in volts.
- **Forward Voltage Reduction Capable:** The status of forward voltage reduction capability.
- **Forward Voltage Reduction Level 1 (V):** The level 1 voltage reduction value for forward regulation.
- **Forward Voltage Reduction Level 2 (V):** The level 2 voltage reduction value for forward regulation.
- **Forward Voltage Reduction Level 3 (V):** The level 3 voltage reduction value for forward regulation.
- **Reverse Regulation Mode:** The name of the reverse regulation mode.

- **Reverse Voltage Target (V):** The voltage target for reverse regulation.
- **Reverse Voltage Bandwidth (V):** The voltage bandwidth for reverse regulation.
- **Reverse LDC Resistance (V):** The LDC resistance for reverse regulation defined in volts.
- **Reverse LDC Reactance (V):** The LDC reactance for reverse regulation defined in volts.
- **Reverse Voltage Reduction Capable:** The status of reverse voltage reduction capability.
- **Reverse Voltage Reduction Level 1 (V):** The level 1 voltage reduction value for reverse regulation.
- **Reverse Voltage Reduction Level 2 (V):** The level 2 voltage reduction value for reverse regulation.
- **Reverse Voltage Reduction Level 3 (V):** The level 3 voltage reduction value for reverse regulation.

1.33.15. Reserved Capacity Summary

The Regulation Mode Summary display lists reserved capacity objects for a station. This display can be accessed from the Application Input Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/ReservedCapacitySummary_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the reserved capacity.
- **ID:** ID of the reserved capacity.
- **Add Reserves on Common Path:** Specifies whether reserves are added on common paths.
- **Apply Full Reserve on Parallel Paths:** Specifies whether full reserve is applied on parallel paths.
- **Primary Feed Device ID:** ID of the feed device, primary feed.
- **Primary Feed Energizing Station and Feeder:** Energizing station and feeder of the primary feed.
- **Primary Feed Firm Commitment:** The firm commitment flag of the primary feed (can be Yes or No).
- **Primary Feed Station Reserved Capacity:** Reserved capacity station values of the primary feed.
- **Primary Feed Line Reserved Capacity:** Line reserved capacity of the primary feed.
- **Secondary Feed Device ID:** ID of the feed device, secondary feed.
- **Secondary Feed Energizing Station and Feeder:** Energizing station and feeder of the secondary feed.

- **Secondary Feed Firm Commitment:** The firm commitment flag of the secondary feed (can be Yes or No).
- **Secondary Feed Station Reserved Capacity:** Reserved capacity station values of the secondary feed.
- **Secondary Feed Line Reserved Capacity:** Line reserved capacity of the secondary feed.
- **Load ID 1/2:** The ID of associated loads 1 and 2.

Navigation:

- "Locate" navigates to the reserved capacity on the geographic view.
- "Analyst Attributes" opens the Analyst Attributes pop-up for the device.
- "Operator Attributes" opens the Operator Attributes pop-up for the device.
- "Loads" opens the Reserved Capacity Associated Loads display.

1.33.16. Station Input

The Station Input display allows the user to view and enter input parameters used by the Distribution Network Analysis Functions (DNAF).

Display call-up URL:

- `netview://<Mode>/gridtabular/subInput_GridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study) or SIM (Simulator).

The following information is shown on the subInput display:

- **Station Name:** Name of substation.
- **Station Loading Percentage:** Percent of station loading.
- **Block All Regulations:** Regulator blocking indicator; can be Yes or No.
- **Voltage Reduction:** Voltage reduction indicator; can be Yes or No.

1.33.17. Station Internals Input

The Station Internals Input display can be accessed from the substation toolbar by clicking Network Analysis Inputs and selecting the Station option. This display lists all station internal components that are eligible for manual data entry.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationInternalsInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **ID:** ID of the station internal device.
- **Name:** Name of the station internal device.
- **Component Type:** Type of the station internal device.
- **Component ID:** ID of the station internal device.

Navigation:

- "Manual Input" opens the data entry pop-up for the station internal device.

1.34. BLA Station Summary

The Bus Load Allocation Station Summary display provides Bus Load Allocation calculations by station. This display can be accessed from the RT and SIM menus.

Display call-up URL:

- `netview://<Mode>/gridtabular/BLAStationSummary_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the BLAStationSummary display:

- **Name:** Name of substation.
- **Area:** Area of substation.
- **DNAF Capable:** DNAF (Distribution Network Analysis Functions) capability indicator. If the substation is DNAF capable that means that this Distribution Network Analysis Functions can be run on this substation.
- **Maximum Number of BLA Iterations with DPF:** Maximum number of iterations for the BLA to solve.
- **BLA Converged – Real Power:** Indication that the BLA Analysis has converged for the real power value (Yes/No/Unsolved).
- **BLA Converged – Reactive Power:** Indication that the BLA Analysis has converged for the reactive power value (Yes/No/Unsolved).
- **BLA Converged – Apparent Power:** Indication that the BLA Analysis has converged for the apparent power value (Yes/No/Unsolved).
- **Number of BLA Solution Iterations with DPF:** Number of iterations performed by BLA to reach a solution.
- **BLA Check - Downstream Island Status:** The status of the BLA Less Accurate Downstream Method option. If a downstream island has a less complete analog measurement set than its upstream island (for example, an upstream measurement island has valid single-phase Watt and VAR measurements, and the downstream island only has valid single-phase Amp or VA measurements), BLA checks the percentage difference between the measured value and the DPF calculated value of these two island boundary measurement groups (including apparent power, real power, reactive power, current, power factor, and voltage, if available). If the

percentage difference between any measured value and DPF calculated value is larger than this measurement's standard deviation, the less accurate downstream measurement status is considered "Bad". This does not impact the solution, but it may point to areas where the measurements need to be reviewed.

- **Number of Measurement Islands:** Number of measurement islands.
- **Last Evaluation Time:** Time of last evaluation.
- **Highest Measurement Island Flow Mismatch:** Highest measurement island flow mismatch in percentage.
- **Measurement Island with Highest Flow Mismatch:** Name of the measurement island with the highest flow mismatch.
- **Measurement Error Count from Last Solution:** The number of measurement errors for the substation in the last solution.
- **Cumulative Measurement Error Count:** The total number of measurement errors for the substation.

Navigation:

- "Locate" navigates to the substation on the geographic display in the same viewport.
- "BLA Results Summary" opens the BLA Results Summary display.

1.34.1. BLA Results Summary

The Bus Load Allocation (BLA) Results Summary display provides BLA calculations by measurement island. The display can be accessed from the BLA Station Summary display or by clicking the BLA Results Summary button the Distribution Power Flow Summary for Station display.

Display call-up URL:

- `netview://<Mode>/gridtabular/MeasIslandSummary_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the MeasIslandSummary display:

- **Station:** Substation name.
- **Island:** Island name.
- **Island ID:** Measurement island internal index.
- **Island Type:** Type of the measurement island. For example, Feeder Head.
- **Branch:** Feeder branch ID.
- **Allocation Type:** Measurement island load allocation type (kVA, or kW & kVAR).
- **Mismatch Check Method:** Mismatch check method (for example, Single-Phase).
- **BLA Consistency Check Summary:** Indicates the number of devices which have a potential inconsistency issue.

- **Real Power Converged?:** Indicates whether real power was converged.
- **Reactive Power Converged?:** Indicates whether reactive power was converged.
- **Apparent Power Converged?:** Indicates whether apparent power was converged.
- **Phase A/B/C Computed Real Power:** Computed real power of phases A, B, and C in kW.
- **Three-Phase Computed Effective Real Power:** Computed three-phase effective real power in kW.
- **Phase A/B/C Real Power Target:** Net real power target for phases A, B, and C in kW.
- **Three-Phase Measured Effective Real Power:** Measured three-phase effective real power in kW.
- **Phase A/B/C Real Power Mismatch:** Real power mismatch in phases A, B, and C expressed in percentage.
- **Three-Phase Effective Real Power Mismatch:** Measured three-phase effective real power expressed in percentage.
- **Phase A/B/C Computed Reactive Power:** Computed reactive power of phases A, B, and C in kVAR.
- **Three-Phase Computed Effective Reactive Power:** Computed three-phase effective reactive power in kVAR.
- **Phase A/B/C Reactive Power Target:** Net reactive power target for phases A, B, and C in kVAR.
- **Three-Phase Measured Effective Reactive Power:** Measured three-phase effective reactive power in kVAR.
- **Phase A/B/C Reactive Power Mismatch:** Reactive power mismatch in phases A, B, and C expressed in percentage.
- **Three-Phase Effective Reactive Power Mismatch:** Measured three-phase effective reactive power expressed in percentage.
- **Phase A/B/C Computed Effective Apparent Power:** Computed apparent power of phases A, B, and C in kVA.
- **Three-Phase Computed Effective Apparent Power:** Computed three-phase effective apparent power in kVA.
- **Phase A/B/C Measured Effective Apparent Power:** Measured apparent power of phases A, B, and C in kVA.
- **Three-Phase Measured Effective Apparent Power:** Measured three-phase effective apparent power in kVA.
- **Phase A/B/C Effective Apparent Power Mismatch:** Apparent power mismatch in phases A, B, and C expressed in percentage.
- **Three-Phase Effective Apparent Power Mismatch:** Measured three-phase effective apparent power expressed in percentage.

Navigation:

- "Locate" locates the measurement island branch on the geographic view.
- "Mismatch by Iteration" opens the Convergence Summary for P, Q, Capacitor, and Generator.
- "BLA Consistency Check Summary" opens the Consistency Check Devices display.
- "Measurement Island Net Injection Summary" opens the Measurement Island Net Injection Summary display.
- "Island Load Information" opens the Island Load Information display.
- "Nominal Island Data Details" opens the Nominal Island Detail display.
- "Net Island Data Details" opens the Net Island Detail display.

1.34.2. Convergence Summary for P, Q, Capacitor, and Generator

The Convergence Summary for P, Q, Capacitor and Generator display can be accessed by clicking the Mismatch by Iteration button on the BLA Results Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/MeasilPQCapGenMismatch_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the MeasilPQCapGenMismatch display:

- **Name:** Name of the measurement island.
- **Iteration:** The iteration index.
- **Phase A/B/C Real Power Target kW:** Real power targets for phases A, B, and C expressed in kW.
- **Phase A/B/C Real Power Calculated kW:** Calculated real power for phases A, B, and C expressed in kW.
- **Phase A/B/C Real Power Mismatch kW:** The value of a difference between the target and calculated Real Powers in phases A, B, and C expressed in kW.
- **Phase A/B/C Reactive Power Target kVAR:** Reactive power targets for phases A, B, and C expressed in kVAR.
- **Phase A/B/C Reactive Power Calculated kVAR:** Calculated reactive power for phases A, B, and C expressed in kVAR.
- **Phase A/B/C Reactive Power Mismatch kVAR:** The value of a difference between the target and calculated Reactive Powers in phases A, B, and C expressed in kVAR.
- **Phase A/B/C Apparent Power Target kVA:** Apparent power targets for phases A, B, and C expressed in kVA.
- **Phase A/B/C Apparent Power Calculated kVA:** Calculated apparent power for phases A, B, and C expressed in kVA.

- **Phase A/B/C Apparent Power Mismatch kVA:** The value of a difference between the target and calculated Apparent Powers in phases A, B, and C expressed in kVA.
- **Phase A/B/C Capacitive Power kVAR:** Capacitive power for phases A, B, and C expressed in kVAR.
- **Phase A/B/C Generator/Distributed Energy Resource kW:** Generator's/distributed energy resource's power for phases A, B, and C in kW.
- **Phase A/B/C Generator/Distributed Energy Resource kVAR:** Generator's/distributed energy resource's power for phases A, B, and C in kVAR.

1.34.3. Measurement Island Net Injection Summary

The Measurement Island Net Injection Summary display can be accessed by clicking the Measurement Island Net Injection Summary button on the BLA Results Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/MeasilInjNoneSummary_GridDisplay/<Substation Name>/<Island ID>/<Feeder Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the MeasilInjNoneSummary display:

- **Feeder Name:** Name of the feeder.
- **Branch:** Name of the branch.
- **Allocation Type:** Measurement island load allocation type (kVA, or kW & kVAR).
- **Net Phase A/B/C Computed Real Power:** Computed real power for phases A, B, and C in kW.
- **Net Phase A/B/C Computed Reactive Power:** Computed reactive power for phases A, B, and C in kVAR.
- **Net Phase A/B/C Computed Apparent Power:** Computed reactive power for phases A, B, and C in kVA.
- **Number of Non Three-Phase Loads:** The number of non-three-phase loads in the measurement island that is supplied by the feeder head as the nominal energy source.
- **Number of Balanced Three-Phase Loads:** The number of balanced three-phase loads in the measurement island.
- **Total Real Power:** Total real power in kW.
- **Total Reactive Power:** Total reactive power in kVAR.
- **Three Phase Power Factor:** Three-phase power factor.
- **Real and Reactive Power Converged?:** Indicates whether real and reactive power were converged.
- **Apparent Power Converged?:** Indicates whether apparent power was converged.

Navigation:

- "Individual Injection Data" opens the Measurement Island Injection Summary display.
- "Nominal Island Data Details" opens the Nominal Island Detail display.
- "Net Island Data Details" opens the Net Island Detail display.

1.34.3.1. Measurement Island Injection Summary

The Measurement Island Injection Summary display can be accessed by clicking the Individual Injection Data button on the Measurement Island Net Injection Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/MeasIls_AnalogMeasGrPsMRY_GridDisplay/<Substation Name>/<Feeder Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Measurement Island Injection Summary display:

- **ID:** Analog measurement point ID.
- **Side 1 Measurement Island ID:** Side 1 measurement island ID.
- **Side 2 Measurement Island ID:** Side 2 measurement island ID.
- **Phase A/B/C kW:** Real power measured for phases A, B, and C in kW.
- **Phase A/B/C kVAR:** Reactive power measured for phases A, B, and C in kVAR.

1.34.3.2. Nominal Island Detail

The Nominal Island Detail display can be accessed by clicking the Nominal Island Data Details button on the Measurement Island Net Injection Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/NominalIslandDetail_GridDisplay/<Substation Name>/<Island Index>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the NominalIslandDetail display:

- **Island Validation:** Island validation status (for example, pass).
- **High Measured kW:** High measured real power in kW.
- **Low Measured kW:** Low measured real power in kW.
- **High Actual kW:** High actual real power in kW.
- **Low Actual kW:** Low actual real power in kW.

- **High Measured kVAR:** High measured reactive power in kVAR.
- **Low Measured kVAR:** Low measured reactive power in kVAR.
- **High Actual kVAR:** High actual reactive power in kVAR.
- **Low Actual kVAR:** Low actual reactive power in kVAR.
- **High Measured kVA:** High measured apparent power in kVA.
- **Low Measured kVA:** Low measured apparent power in kVA.
- **High Actual kVA:** High actual apparent power in kVA.
- **Low Actual kVA:** Low actual apparent power in kVA.

1.34.3.3. Net Island Detail

The Net Island Detail display can be accessed by clicking the Net Island Data Details button on the Measurement Island Net Injection Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/NetIslandDetail_GridDisplay/<Substation Name>/<Island Index>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the NominalIslandDetail display:

- **Island Validation:** Island validation status (for example, pass).
- **High Measured kW:** High measured real power in kW.
- **Low Measured kW:** Low measured real power in kW.
- **High Actual kW:** High actual real power in kW.
- **Low Actual kW:** Low actual real power in kW.
- **High Measured kVAR:** High measured reactive power in kVAR.
- **Low Measured kVAR:** Low measured reactive power in kVAR.
- **High Actual kVAR:** High actual reactive power in kVAR.
- **Low Actual kVAR:** Low actual reactive power in kVAR.
- **High Measured kVA:** High measured apparent power in kVA.
- **Low Measured kVA:** Low measured apparent power in kVA.
- **High Actual kVA:** High actual apparent power in kVA.
- **Low Actual kVA:** Low actual apparent power in kVA.

1.34.4. Island Load Information

The Island Load Information display can be accessed by clicking the Island Load Information button on the BLA Results Summary display.

Display call-up URL:

- `netview://<Mode>/gridtabular/IslandLoadInformation_GridDisplay/<Substation Name>/<Island Index>/<Feeder Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the IslandLoadInformation display:

- **Station:** Name of the substation.
- **Island:** Name of the measurement island.
- **Island Type:** Type of the measurement island (feeder head or downstream).
- **Three-Phase Balanced Loads:** The number of three-phase balanced loads in the measurement island that is supplied by the feeder head as the nominal energy source.
- **Open-Wye Open-Delta Transformer Bank Loads:** The number of open-wye open-delta transformer bank loads in the measurement island.
- **Three-Phase Unbalanced Loads:** The number of three-phase unbalanced loads in the measurement island.
- **Phase A/B/C Loads:** The number of phase A, B, and C loads in the measurement island.

1.34.5. Consistency Check Devices

The Consistency Check Devices display provides details about inconsistent devices identified during BLA execution. This display can be accessed by clicking the BLA Consistency Check Summary button on BLA Results Summary display.

Display call-up URL:

- `NETVIEW://<Mode>/GRIDTABULAR/MeasIslandBLAconsistSwt_GridDisplay/<Measurement Island Station Name>/<Measurement Island Index>/<Measurement Island Name>.`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

- **Name:** Switch name.
- **ID:** Switch ID.
- **Phase:** The phase of the switch.
- **Explanation:** The inconsistency explanation, which can be one of the following.
  - Non-zero current (Amp) flow measurement detected on a deenergized device.
  - Zero current (Amp) flow measurement detected on an energized device.
  - Non-zero current (Amp) flow measurement detected on an isolating switching device.
  - Non-zero voltage measurement detected on a deenergized device.
  - Zero voltage measurement detected on an energized device.

- Non-zero voltage measurement detected on an isolating switching device.
- Non-zero real power (W) flow measurement detected on a deenergized device.
- Non-zero real power (W) reverse flow measurement detected on a deenergized device.
- Zero real power (W) flow measurement detected on an energized device.
- Zero real power (W) reverse flow measurement detected on an energized device.
- Non-zero real power (W) flow measurement detected on an isolating switching device.
- Non-zero real power (W) reverse flow measurement detected on an isolating switching device.
- Non-zero reactive power (VAR) flow measurement detected on a deenergized device.
- Non-zero reactive power (VAR) reverse flow measurement detected on a deenergized device.
- Zero reactive power (VAR) flow measurement detected on an energized device.
- Zero reactive power (VAR) reverse flow measurement detected on an energized device.
- Non-zero reactive power (VAR) flow measurement detected on an isolating switching device.
- Non-zero reactive power (VAR) reverse flow measurement detected on an isolating switching device.
- Non-zero Volt-Amp (VA) flow measurement detected on a deenergized device.
- Zero Volt-Amp (VA) flow measurement detected on an energized device.
- Non-zero Volt-Amp (VA) flow measurement detected on an isolating switching device.
- Real power (W) flow direction mismatch.
- Real power (W) reverse flow direction mismatch.
- Reactive power (VAR) flow direction mismatch.
- Reactive power (VAR) reverse flow direction mismatch.
- Non-zero current (Amp) flow measurement detected on an open switching device.
- Non-zero real power (W) flow measurement detected on an open switching device.
- Non-zero real power (W) reverse flow measurement detected on an open switching device.
- Non-zero reactive power (VAR) flow measurement detected on an open switching device.
- Non-zero reactive power (VAR) reverse flow measurement detected on an open switching device.
- Non-zero Volt-Amp (VA) flow measurement detected on an open switching device.
- Real power forward and reverse flow measurement inconsistent reading.
- Reactive power forward and reverse flow measurement inconsistent reading.
- Non-zero amps measurement detected on an energized device, while it is supposed to have zero flow.
- Non-zero real power measurement detected on an energized device, while it is supposed to

have zero flow.

  - Non-zero reactive power measurement detected on an energized device, while it is supposed to have zero flow.
  - Non-zero apparent power measurement detected on an energized device, while it is supposed to have zero flow.

Navigation:

- "Locate" navigates to the device on the network view.

1.34.6. BLA Data

The Bus Load Allocation (BLA) Data display provides parameters and values that have been used for the bus load allocation calculations.

Display call-up URL:

- `netview://<Mode>/gridtabular/BLAdata_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the BLAdata display:

- **Maximum Number of BLA/DPF Iterations:** Maximum number of BLA DPF iterations.
- **Percent Flow Convergence (Real Power):** Measurement and real power flow calculation maximum mismatch in percentage.
- **Absolute Flow Convergence (Real Power):** Measurement and real power flow calculation maximum mismatch.
- **Percent Flow Convergence (Reactive Power):** Measurement and reactive power flow calculation maximum mismatch in percentage.
- **Absolute Flow Convergence (Reactive Power):** Measurement and reactive power flow calculation maximum mismatch.
- **DPF Evaluation Time:** Date and time DPF (Distribution Power Flow) last evaluated.
- **BLA/DPF Solution Iterations:** Number of BLA (Bus Load Allocation) DPF (Distribution Power Flow) iterations.
- **Converged – Real Power:** Flag indicating BLA DPF convergence.
- **Converged – Reactive Power:** Flag indicating BLA DPF convergence.

1.34.7. BLA Feeder Summary

The Bus Load Allocation Feeder Summary display provides Bus Load Allocation calculations by feeder.

Display call-up URL:

- `netview://<Mode>/gridtabular/BLAFeederSummary_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the BLAFeederSummary display:

- **Name:** Name of the feeder.
- **Feeder Head Island 3-Ph Balanced Load Count:** The number of three-phase balanced loads in the measurement island that is supplied by the feeder head as the nominal energy source.
- **Feeder Head Island Non 3-Ph Balanced Load Count:** Number of loads (other than balanced three-phase loads) in the measurement island that is supplied by the feeder head as the nominal energy source.
- **BLA Converged – Real Power:** The total real power supplied by the feeder as determined by the converged Bus Load Allocation process.
- **BLA Converged – Reactive Power:** The total reactive power supplied by the feeder as determined by the converged Bus Load Allocation process.
- **Measured Flow Phase A/B/C kW:** Flow measured at phase A, B, and C in kW.
- **Calculated Flow Phase A/B/C kW:** Flow calculated at phases A, B, and C in kW.
- **Real Power Mismatch Phase A/B/C:** Real power flow percentage mismatch for phases A, B, and C.
- **Measured Flow Phase A/B/C kVAR:** Flow measured at phase A, B, and C in kVAR.
- **Calculated Flow Phase A/B/C kVAR:** Flow calculated at phases A, B, and C in kVAR.
- **Reactive Power Mismatch Phase A/B/C:** Reactive power flow percentage mismatch for phases A, B, and C.
- **Three Phase Power Factor:** Three phase power factor.

1.34.8. BLA Schedule Data

The Bus Load Allocation Schedule Data display provides the list of Bus Load Allocation schedules.

Display call-up URL:

- `netview://<Mode>/gridtabular/BLAloadschedule_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Schedule Name:** Name of schedule

1.35. Manual Entry Input Summary

The Manual Entry Input Summary display can be accessed from the ADMS menu bar > RT DMS menu > Manual Input Summary (instead of RT, can be ST or SIM DMS). It provides the list of network elements with modified parameters.

Display call-up URL:

- `netview://<Mode>/gridtabular/ManualInputSummary_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Manual Entry Input Summary display:

- **Name:** Substation name.
- **System:** Shows the number of system-wide manually entered parameters. When clicked, navigates to the Manual Input Parameter Entry pop-up display.
- **Station:** Shows the number of station-wide manually entered parameters. When clicked, navigates to the Feeder Input display.
- **Power Transformer:** Shows the number of manually entered parameters for the power transformers. When clicked, navigates to the Power Transformer Input display.
- **Feeder:** Shows the number of manually entered parameters for feeders. When clicked, navigates to the Feeder Input display.
- **Reference Bus:** Shows the number of manually entered parameters for the reference bus. When clicked, navigates to the Reference Bus Input display.
- **Generator / Distributed Energy Resource:** Shows the number of manually entered parameters for generators and distributed energy resources. When clicked, navigates to the Generator/Distributed Energy Resource Input display.
- **Capacitor:** Shows the number of manually entered parameters for capacitors. When clicked, navigates to the Capacitor Input display.
- **Disconnected Transformer:** Shows the number of manually entered parameters for disconnected transformer. When clicked, navigates to the Disconnected Transformer Input display.
- **Regulator:** Shows the number of manually entered parameters for tap changers. When clicked, navigates to the Regulator Input display.
- **Series Reactor:** Shows the number of manually entered parameters for series reactors. When clicked, navigates to the Series Reactor Input display.
- **Line:** Shows the number of manually entered parameters for lines. When clicked, navigates to the Line Input display.
- **Switching Device:** Shows the number of manually entered parameters for switching devices. When clicked, navigates to the Switching Device Input display.
- **Load:** Shows the number of manually entered parameters for loads. When clicked, navigates to the Load Input display.
- **Bus:** Shows the number of manually entered parameters for buses. When clicked, navigates to the Bus Input display.
-

Load Category: Shows the number of manually entered voltage coefficients. When clicked, navigates to the Load to Voltage Coefficients (Filtered) display.

- **Fault Location:** Shows the number of manually entered fault location parameters. When clicked, navigates to the Fault Location Input display.

1.35.1. Power Transformer Input

The Power Transformer Input display can be accessed from the Manual Entry Input Summary display. It allows the user to input and review the transformer parameters, which were entered using the Transformer Parameters and Manual Input Parameter Entry data entry pop-up displays.

Display call-up URL:

- `netview://<Mode>/gridtabular/TransformerManualInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Power Transformer Input display:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.

Navigation:

- "Locate" locates the transformer on the geographic view.
- "Enter Device Data" opens the Transformer Parameters display.
- "Manual Input Parameter Entry" opens the Manual Input Parameter Entry display.

1.35.2. Feeder Input

The Feeder Input display can be accessed from the Manual Entry Input Summary display. It allows the user to input and review the station feeder parameters, which were entered using the Manual Input Parameter Entry data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/FeederManualInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Feeder Input display:

- **Feeder:** Feeder name.
- **Enable Manual Feeder Header Flow:** A flag to Enable/Disable manual header flow parameters.
- **Feeder Head Flow Phase A/B/C:** Data entry field for setting the feeder head flow rate for phases A, B, and C in Amps.
-

Feeder Head Flow Power Factor: Data entry field for setting the feeder head flow power factor.

Navigation:

- "Manual Input Parameter Entry" opens the Manual Input Parameter Entry display.

1.35.3. Reference Bus Input

The Reference Bus Input display can be accessed from the Manual Entry Input Summary display. It allows the user to input and review the reference bus parameters, which were entered on the Reference Bus Input for Station display and the Reference Bus Interface Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/RefBusManualInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Reference Bus Input display:

- **Name:** Name of the reference bus.
- **ID:** ID of the reference bus.

Navigation:

- "Locate" locates the reference bus on the geographic view.
- "Enter Device Data" opens the Reference Bus Interface Parameters pop-up.
- "Manual Input Parameter Entry" opens the Manual Input Parameter Entry display.

1.35.4. Generator/Distributed Energy Resource Input

The Generator/Distributed Energy Resource Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Generator/Distributed Energy Resource Input display and the Generator Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationGeneratorInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationGeneratorInput display:

- **Name:** Name of the generator/distributed energy resource.
- **ID:** ID of the generator/distributed energy resource.

Navigation:

- "Locate" locates the generator on the geographic view.
- "Enter Device Data" opens the Generator Parameters pop-up.

1.35.5. Capacitor Input

The Capacitor Input display can be accessed from the Manual Entry Input Summary display. This display shows the devices that were configured using the Capacitor Input display and the Capacitor Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationCapacitorInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the StationCapacitorInput display:

- **Name:** Name of the capacitor.
- **ID:** ID of the capacitor.

Navigation:

- "Locate" locates the capacitor on the geographic view.
- "Enter Device Data" opens the Capacitor Parameters pop-up.

1.35.6. Disconnected Transformer Input

The Disconnected Transformer Input display can be accessed from the Manual Entry Input Summary display. It provides the summary of the disconnected distribution transformers.

Display call-up URL:

- `netview://<Mode>/gridtabular/DisconnectedDistTF_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Station Disconnected Transformers display:

- **Name:** Name of the transformer.
- **ID:** ID of the transformer.
- **Energizing Feeder:** Feeder name.

Navigation:

- "Analyst Attributes" opens the Analyst Attributes pop-up.
- "Analyst Outputs" opens the Analyst Output pop-up.
- "Operator Attributes" opens the Operator Attributes pop-up.

- "Operator Outputs" opens the Operator Output pop-up.
- "Locate" locates the transformer on the geographic view.

1.35.7. Regulator Input

The Regulator Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Tap Changer Input display and Transformer Parameters data entry pop-up.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationRegulatorInput_GridDisplay/<Substation Name>`

  The following information is shown on the StationRegulatorInput display:

- **Name:** Name of the tap changing device.
- **ID:** ID of the tap changing device.

Navigation:

- "Locate" locates the regulator on the geographic view.
- "Enter Device Data" opens the Transformer Parameters pop-up.

1.35.8. Line Input

The Line Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Line Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/LineManualInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Line Input display:

- **ID:** ID of the line.
- **Name:** Name of the line.
- **Enable Manual Flow Limit:** Selecting this check box enables the manual flow limit capability.
- **Flow Limit of Phase A/B/C/N:** Nominal flow limit rate of phases A, B, and C, and the neutral.
- **Manual Flow Limit of Phase A/B/C/N:** Data entry field for setting the flow limit rate of phases A, B, and C, and the neutral.

Navigation:

- "Locate" navigates to the line on the geographic view.

1.35.9. Series Reactor Input

The Series Reactor Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Series Reactor Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/SeriesManualInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Series Reactor Input display:

- **ID:** ID of the line.
- **Name:** Name of the series reactor.

Navigation:

- "Locate" navigates to the line on the geographic view.
- "Enter Device Data" opens the Series Reactor Parameters pop-up.

1.35.10. Switching Device Input

The Switching Device Manual Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Switch Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/SwitchManualInput_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Switching Device Input display:

- **Name:** Name of the switching device.
- **ID:** ID of the switching device.
- **Nominal Flow Limit:** Nominal flow limit value.
- **Enable Manual Flow Limit:** Selecting this check box enables the manual flow limit capability.
- **Manual Flow Limit:** Data entry field for setting the flow limit rate.
- **Load Break Capable:** Flag indicating whether the switch can break load.
- **Enable Manual Load Break Capable:** Check this box to apply the manual Load Break Capable value.
- **Enable Manual Break Rating:** Check this box to apply the manual Load Break Rating value.
- **Manual Load Break Rating:** Data entry field for setting the load break rating.

1.35.11. Load Input

The Load Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Load Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationLoadInput_GridDisplay/<Substation Name>`

The following information is shown on the StationLoadInput display:

- **Name:** Name of the load.
- **ID:** ID of the load.

Navigation:

- "Locate" navigates to the associated transformer on the geographic view.
- "Enter Device Data" opens the Load Parameters pop-up display.

1.35.12. Bus Input

The Bus Input display can be accessed from the Manual Entry Input Summary display. It shows the devices that were configured using the Bus Parameters data entry pop-up display.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationBusInput_GridDisplay/<Substation Name>`

The following information is shown on the StationBusInput display:

- **Name:** Name of the bus.
- **ID:** ID of the bus.

Navigation:

- "Locate" navigates to the bus on the geographic view.
- "Enter Device Data" opens the Bus Parameters pop-up display.

1.35.13. Load to Voltage Coefficients (Filtered)

The Load to Voltage Coefficients (Filtered) display can be accessed from the Manual Entry Input Summary display by clicking the Load Category button. The display shows all manual entries made on the Load to Voltage Coefficients display.

Display call-up URL:

- `netview://<Mode>/gridtabular/StationManualFilteredLoadCategoryInput_GridDisplay/<Substation Name>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Station Load to Voltage Coefficients display:

- **Name:** Name of the schedule (for example, RES, SCI, MCI).
- **Manual Enabled:** Select this checkbox to enable manual coefficient settings.
- **Schedule AD:** Name of the AD schedule (for example, LoadVolt1).
- **Nominal A:** Nominal A value.
- **Manual A:** Manual A value.
- **Nominal D:** Nominal D value.
- **Manual D:** Manual D value.
- **Schedule BE:** Name of the BE schedule (for example, LoadVolt2).
- **Nominal B:** Nominal B value.
- **Manual B:** Manual B value.
- **Nominal E:** Nominal E value.
- **Manual E:** Manual E value.
- **Schedule CF:** Name of the CF schedule (for example, LoadVolt3).
- **Nominal C:** Nominal C value.
- **Manual C:** Manual C value.
- **Nominal F:** Nominal F value.
- **Manual F:** Manual F value.

1.36. Load Shed Overview

The Load Shed Overview display shows the load shed status for stations. It can be accessed from the ADMS menu bar > RT DMS menu > Load Shed Overview.

Display call-up URL:

- `netview://RT/gridtabular/LoadShedOverview_GridDisplay`

The following information is shown on the Load Shed Overview display:

- **Station:** Name of the substation.
- **Load Shed Devices:** Devices that are eligible for load shedding.
- **Devices Enabled for Shedding:** Devices that are enabled for the load shedding.
- **Devices Currently Shed:** Devices that are currently shed.
- **Rule Set:** Name of the rule set applied.

Navigation:

- "Station Details" opens the Load Shed Station Details display.

1.36.1. Load Shed Station Details

The Load Shed Station Details display shows the load shed details for the selected station. It can be accessed from the Load Shed Overview display by clicking the Station Details button.

Display call-up URL:

- `netview://<Mode>/gridtabular/LoadShedStationDetails_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Service Center:** Name of the service center.
- **Device Name:** Name of the device eligible for load shedding.
- **Feeder:** Name of the feeder the device belongs to.
- **Device Type:** Type of the device.
- **Status:** The status of the device.
- **KVA:** Device apparent power in kVA.
- **Load Shed Enabled:** Can be Yes or No.
- **Load Shed Active:** Can be Yes or No.
- **Default Device Classification:** The default classification assigned to a breaker or recloser in the model.
- **Default Feeder Classification:** The default classification assigned to a feeder in the model.
- **Final Load Shed Classification:** Current device classification.
- **Classification Reason:** Classification reason.
- **Lowest CLPU Grade:** The lowest cold load pick-up grade.

Navigation:

- "Locate" locates the device on the network view.

1.36.2. Load Shed Sort Values

The Load Shed Sort Values display shows the load shed sorting values for all devices that are eligible for load shed.

Display call-up URL:

- `netview://<Mode>/gridtabular/LoadShedSorting`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Area:** Name of the area.
- **Service Center:** Name of the service center.
- **Station:** Name of the station.
- **Energizing Feeder:** Name of the energizing feeder.
- **Device Type:** Type of the device.
- **Device Name:** Name of the device eligible for load shedding.
- **Status:** Device status (Open or Closed).
- **Enabled:** Load shed enabled status. Can be Yes or No.
- **Load Shed Class:** Load shed class.
- **CLPU Grade:** Cold load pick-up grade.
- **kVA:** The total power that can be shed by the device.
- **Sort Value:** Load shed sort value.

1.37. Automation Schemes List

The Automation Schemes List display lists automated local peer-to-peer switching schemes used to automate the fault isolation and service restoration functions (decentralized FISR). You can access this display from the RT DMS menu.

Display call-up URL:

- `netview://<Mode>/gridtabular/SchemeGridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **ID:** ID of the automation scheme. Clicking the ID field opens the Automation Scheme Teams List display filtered to show teams in the selected scheme.
- **Name:** Name of the automation scheme.
- **Description:** Description of the automation scheme.
- **Number of Teams in Scheme:** The number of teams in the scheme.
- **Teams in Scheme:** The list of teams in the scheme.
- **Number of Stations:** The number of stations associated with the scheme.
- **Stations:** List of stations associated with the scheme.

Context menu options:

- **Trace Scheme:** Traces the scheme on the network view.
-

List Scheme Teams: Opens the Automation Scheme Teams List display filtered to show teams in the selected scheme.

1.37.1. Automation Scheme Teams List

The Automation Scheme Teams List display lists teams in an automated local peer-to-peer switching scheme. You can access this display from the Automation Schemes List display.

Display call-up URL:

- `netview://<Mode>/gridtabular/TeamGridDisplay/<Scheme ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **ID:** ID of the team. Clicking the ID field opens the Automation Scheme Team Switches display filtered to show switching devices in the selected team.
- **Name:** Name of the team.
- **Description:** Description of the team.
- **Number of Switches:** The number of switching devices in the team.
- **Switches:** The list of team switch IDs.
- **Stations:** The list of stations that are associated with the team.

Context menu options:

- **Mark Team Members:** Marks team switching devices on the network view.
- **Trace Team:** Traces lines with switching devices that are associated with the team on the network view.
- **List Team Members:** Opens the Automation Scheme Team Switches display filtered to show switching devices in the selected team.

1.37.2. Automation Scheme Team Switches

The Automation Scheme Team Switches display lists switching devices that are part of an automated local peer-to-peer switching scheme team. You can access this display from the Automation Scheme Teams List display.

Display call-up URL:

- `netview://<Mode>/gridtabular/TeamSwitchGridDisplay/<Team ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

-

- **ID:** ID of the switching device.
- **Name:** Name of the switching device.
- **Team ID:** ID of the associated team.
- **Primary Station:** Name of the first station that is associated with the switching device.
- **Secondary Station:** Name of the second station that is associated with the switching device. (For boundary switches.)
- **Role:** Switching device role (primary or secondary).

Navigation:

- "Locate" locates the switching device on the geographic network view.

1.37.3. Automation Scheme Team Switches on Feeder

The Automation Scheme Team Switches on Feeder display lists switching devices that are part of automated local peer-to-peer switching schemes on a feeder associated with the selected breaker or recloser. You can access this display from the breaker, recloser, or switch right-click menu.

Display call-up URL:

- `netview://<Mode>/gridtabular/SchemeDevices_GridDisplay/<Substation Name>/<Feeder Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Nominal Feeder:** Name of the nominal feeder.
- **Energizing Feeder:** Name of the energizing feeder.
- **Energizing Station:** Name of the energizing station.
- **ID:** ID of the switching device.
- **Name:** Name of the switching device.
- **Team ID:** ID of the associated team.
- **Scheme ID:** ID of the associated scheme.
- **Primary Station:** Name of the first station that is associated with the switching device.
- **Secondary Station:** Name of the second station that is associated with the switching device. (For boundary switches.)
- **Role:** Switching device role (primary or secondary).
- **Control Time:** Time when the last device control was sent.
- **Type:** The control method used for the device (Arm/Disarm).
- **Measurement Type:** Control measurement type (for example, Control Prohibit Restoration).
- **Status:** Control measurement status (Enabled/Disabled).

- **Measurement ID:** Control measurement ID.
- **Auto Reclosing Enabled Status:** Shows the Auto Reclosing Enabled status.
- **Ground Trip Enabled Status:** Shows the Ground Trip Enabled status.
- **Prohibit Restoration Status:** Shows the Prohibit Restoration status.
- **Tag Placed:** Indicates whether a tag is placed on the device.
- **Switch Status:** Shows the current device status.
- **Switch Scada Station Name:** Shows the Scada name of the substation that is associated with the device.

Navigation:

- "Locate" locates the switching device on the geographic network view.

1.38. Switches Pending Geographic Information System Update

The Pending Switches display can be accessed from the ADMS menu bar > RT DMS menu > Pending Switches (instead of RT, can be ST or SIM DMS). The display lists all switches that currently have "Pending Permanent" status. This means that the normal state of the switch has been changed but the change has not yet been confirmed by an update of the geographic information system/asset management system data (GIS/AMS) data.

Display call-up URL:

- `netview://<Mode>/gridtabular/pendingSwitches_GridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Service Center:** The name of the service center.
- **Substation (Nominal):** The name of nominal substation.
- **Switch Device Name:** The name of the switch.
- **Feeder Name:** The name of the feeder.
- **Nominal State:** Nominal state of the switch.
- **Pending State Name:** The name of pending state of the switch.

Navigation:

- "Locate" navigates to the element on the geographic view.

1.39. Station Commissioning/Decommissioning

The Station Commissioning/Decommissioning display is accessible from the RT DMS menu. It shows a summary of whether each station currently contains equipment that is modeled in the distribution electrical network but is in the process of commissioning or decommissioning (that is, equipment that is planned to be put in service, planned to be removed, or was recently removed).

Display call-up URL:

- `netview://<Mode>/gridtabular/FutureStationCommissioningDecommissioning_GridDisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Substation name.
- **ID:** Substation ID.
- **Has Commission/Decommission Equipment:** This field can have two values: Yes or No. When clicked, opens the Equipment Commissioning/Decommissioning display.

1.39.1. Equipment Commissioning/Decommissioning

The Equipment Commissioning/Decommissioning display lists the equipment that is modeled in the distribution electrical network but is either planned to be put in service, planned to be removed, or was recently removed.

The Equipment Commissioning/Decommissioning display can be accessed from the Station Commissioning/Decommissioning display by clicking the **Has Commission/Decommission Equipment** cell. This display can also be called from the Substation device toolbar.

Display call-up URL:

- `netview://<Mode>/gridtabular/EquipmentCommissioningDecommissioning_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Equipment Commissioning and Decommissioning display:

- **Station Name:** Name of substation.
- **ID:** The unique identifier of the network component.
- **Name:** The name of the network component.
- **Device Type:** The network component device type (for example, Switch, Jumper).
- **Commission State:** Commissioning state of the device (ACTIVE, FUTURE, FUTURE REMOVED, or REMOVED).
- **Project ID:** ID of the commissioning project that is associated with the device.

- **Project Stage:** The device commissioning or decommissioning stage. You can use this field to manually set or update the project stage.
- **Constructed Phases:** The phases to which the future device will be connected.
- **Future Phases:** Individual phases that are constructed but are not energized.
- **Future Remove Phases:** Individual phases that are planned to be removed in future.
- **Removed Phases:** Individual phases that were recently removed.
- **Inside Station:** Indicator (Yes/No) of whether the device is a substation internals device.
- **Elbow Switch Associated Transformer :** If the future device is an elbow switch, this field shows the name of the associated distribution transformer.
- **Load Associated Transformer :** If the future device is a load, this field shows the name of the distribution transformer that will supply the load.

Navigation and controls:

- "Locate" navigates to the component on the geographic display.

Context menu options for the ID field:

- "Future to Active", "Active to Future", "Future Remove to Removed", or "Removed to Future Remove" updates the commissioning state to Active, Future, Future Removed, or Removed for the selected device. The context menu option depends on the current device commissioning state.
- "Commission/Decommission All Related Devices" toggles the commissioning state (Active to Future, Removed to Future Removed, and vice versa) for all devices associated with the same project ID and stage.

1.40. Distribution Transformer Load

The Distribution Transformer Load display can be accessed by selecting a distribution transformer on the geographic view and then clicking the Distribution XF Load button in the device-specific toolbar. The display is split into three panes.

Display call-up URLs:

First frame:

- `netview://<Mode>/gridtabular/disttfload1_GridDisplay/<Substation Name>/<Transformer ID>`

Second frame:

- `netview://<Mode>/gridtabular/disttfload2_GridDisplay/<Substation Name>/<Transformer ID>`

Third frame:

- `netview://<Mode>/gridtabular/disttfload3_GridDisplay/<Substation Name>/<Transformer ID>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the Disttfload1 display:

- **Name:** Name of the device. For loads that are added by the Converter tool, this value is populated with the CustomerName value from the customer file.
- **ID:** ID of the load. For loads that are added by the Converter tool, this value is populated with the LoadID value from the customer file.
- **Phasing:** Designation of transformer phases.
- **Connection:** Connection indicator.
- **Associated Transformer ID:** Unique identifier of the associated transformer.
- **Connection Node Voltage Phase A/B/C:** Voltage magnitude and angle for the A/B/C phase connection node.
- **Phase A/B/C:** Current magnitude and angle for A, B, and C phases in Amps.

Navigation:

- "Static Schedules" displays the Schedule grid display for the associated load.
- "Dynamic Schedules" displays the Schedule Points grid display for the associated load.
- "Transformer Analyst Attributes" displays the Analyst Attributes pop-up.
- "Transformer Analyst Outputs" displays the Analyst Output pop-up
- "Locate Transformer" locates the transformer on the geographic view.

The following information is shown on the Disttfload2 display:

- **ID:** ID of the load. For loads that are added by the Converter tool, this value is populated with the LoadID value from the customer file.
- **Real Power Phase A/B/C:** Real power values in phases A, B, and C in kW.
- **Reactive Power Phase A/B/C:** Reactive power in phases A, B, and C in KVAR.
- **Voltage Phase A/B/C:** Phase A, B, and C voltage in V.
- **Secondary Voltage Drop:** Voltage drop on the secondary side.
- **High Voltage Power Quality Index:** High power quality index in % kW.
- **Low Voltage Power Quality Index:** Low power quality index in % kW.
- **Imbalance Voltage Power Quality Index:** Imbalance power quality index in % kW.
- **Stability Voltage High Limit:** Stability voltage high limit in V.
- **Emergency Voltage High Limit:** Emergency voltage high limit in V.
- **Operational Voltage High Limit:** Operational voltage high limit in V.
- **Stability Voltage Low Limit:** Stability voltage low limit in V.
- **Emergency Voltage Low Limit:** Emergency voltage low limit in V.
- **Operational Voltage Low Limit:** Operational voltage low limit in V.
- **Optimization High Voltage Limit:** Optimization voltage high limit in V.

- **Optimization Low Voltage Limit:** Optimization voltage low limit in V.
- **Optimization Emergency High Voltage Limit:** Optimization emergency voltage high limit in V.
- **Optimization Emergency Low Voltage Limit:** Optimization emergency voltage low limit in V.

Navigation:

- "Load Manual Entry Parameters" shows the "Load Parameters" pop-up.
- "Static Schedule Summary" opens the Schedule display.

The following information is shown on the Disttfload3 display:

- **ID:** ID of the load.
- **Latest Outage Time:** Time of the latest outage.
- **Latest Restoration Time:** Latest time of restoration.
- **Last Outage Duration (h:min):** Duration of the latest outage.
- **Cold Load Multiplier:** Cold load multiplier.
- **Load Correction Factor 1:** Load correction Factor 1.
- **Load Correction Factor 2:** Load correction Factor 2.
- **Load Correction Factor 3:** Load correction Factor 3.
- **Load Correction Factor 4:** Load correction Factor 4.
- **Load Correction Factor 5:** Load correction Factor 5.
- **Load Correction Factor 6:** Load correction Factor 6.
- **Load Category Name:** Name of the load category.
- **Nominal Real Power Phase A/B/C:** Phase A, B, and C nominal real power in kW.
- **Nominal Reactive Power Phase A/B/C:** Phase A, B, and C nominal reactive power in kVAR.
- **Customer Count:** Number of customers supplied from the transformer.
- **Critical Customer Count:** Number of critical customers supplied from the transformer.
- **Value of Uninterrupted Service:** Value of uninterrupted service.

1.41. Traced Distribution Transformer Summary

The Traced Distribution Transformer Summary display is accessed from the geographic, line, or distribution transformer right-click menu. This display provides detailed information about all currently traced distribution transformers in the system.

Display call-up URL:

- `netview://<Mode>/gridtabular/traceddisttf_griddisplay`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

For the description of display fields, refer to Distribution Transformer Summary.

The Traced Distribution Transformer Summary display has the following unique fields:

- **Station Name:** Name of the substation that the distribution transformer belongs to.
- **Customer Count:** The number of customers that are associated with the transformer.
- **Critical Customer:** Specifies whether critical customers are associated with the transformer

1.42. Find Results Summary

The Find Results Summary display shows search result according to the search criteria specified by user on the Device Search tab of the Tabbed Find pop-up.

Display call-up URL:

- `netview://<Mode>/gridtabular/FindResultsGridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Device Type:** Bitmap indication of the device type.
- **Station Name:** Name of the substation.
- **Name:** Name of the network element.
- **ID:** Unique identifier of the network element.
- **Alternate Name:** Alternate name of the network element.
- **Address:** Address of the network element.
- **Geometry Name:** The name of the geometry.

Navigation:

- "Go to Geographic and Feeder Main" navigates to the geographic view with laterals hidden.
- "Go to Geographic with Laterals On" navigates to the geographic view with laterals displayed.
- "Go to non-geographic" navigates to the relevant view.
- "Navigate To Schematic View" navigates to the schematic view.
- "Navigate to URD loop" navigates to the URD loop view.
- "Navigate to station internals" navigates to the station internal view.
- "Navigate to SCADA" navigates to the SCADA view.

1.42.1. Global Find Summary

The Global Find Summary display shows search result according to the search criteria specified by user on the Analyst Search tab of the Tabbed Find pop-up.

Display call-up URL:

- `netview://<Mode>/gridtabular/GlobalFindGridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Device Type:** Bitmap indication of the device type.
- **Station:** Name of the associated substation.
- **Name:** Name of the network element.
- **ID:** unique identifier of the network element.
- **Alternate Name:** Alternate name of the network element.
- **Address:** Address of the network element.
- **Geometry Name:** The name of geometry.

Navigation:

- "Go to Geographic and Feeder Main" navigates to the geographic view with laterals hidden.
- "Go to Geographic with Laterals On" navigates to the geographic view with laterals displayed.
- "Go to non-geographic" navigates to the relevant view.
- "Navigate To Schematic View" navigates to the schematic view.
- "Navigate to URD loop" navigates to the URD loop view.
- "Navigate to station internals" navigates to the station internal view.
- "Navigate to SCADA" navigates to the SCADA view.

1.42.2. Customer Find Summary

The Customer Find Summary display shows search result according to the search criteria specified by user on the Customer Search tab of the Tabbed Find pop-up.

Display call-up URL:

- `netview://<Mode>/gridtabular/CustomerFindGridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on the CustomerFind display:

- **Account #:** Account number.
- **Name:** Customer name.
- **Phone:** Customer phone number.
-

Service Center: Service center.

- **Address:** Customer address.
- **Customer Key:** Customer key.
- **Critical Account:** Important account.
- **Inactive Account:** Registered by not active account.
- **City:** Name of the city the customer belongs to.

Navigation:

- "Locate" navigates to the customer on the geographic view.

1.43. LVM System Monitor Stations

The LVM System Monitor Stations display shows the LVM forecast and implementation summary for stations in the system.

To open this display, select the DNAF Display Plugin option from the RT DMS menu, and then select System Monitor Stations.

Display call-up URL:

- `netview://RT/gridtabular/LVMSystemMonitorStations`

The following information is shown on this display:

- **Station Name:** Name of the station.
- **Division:** Name of the division.
- **Forecast Problem Formulation:** Name of the forecast problem formulation.
- **Forecast State:** The forecast solution state, such as Pending, In Progress, or Solved.
- **Plan Solved Time:** Date and time of the last solution.
- **Plan Steps:** The number of steps in the plan.
- **Eligible Devices:** The number of eligible devices that are associated with the substation.
- **Ineligible Devices:** The number of ineligible devices that are associated with the substation.
- **Excluded Devices:** The number of excluded devices that are associated with the substation.
- **Implementation State:** The plan implementation state, such as None or Complete.
- **Implemented Problem Formulation:** Name of the implemented problem formulation.
- **Pre Implementation Problem Formulation:** Name of the problem formulation that was used before implementing the forecast problem formulation.
- **Measured Power Available:** Specifies whether power measurements are available for the substation.
- **Measured Voltage Available:** Specifies whether voltage measurements are available for the

substation.

1.44. Global Designated Switches

The Global Designated Switches display shows all switching devices with the Normal Open, Temporary Open, or Bad Open specific open state classification, as set by the operator.

Display call-up URL:

- `netview://<Mode>/gridtabular/DesignatedSwitchesGlobal_GridDisplay/`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Switch ID:** The unique identifier of the switching device.
- **Switch Name:** The name of the associated switching device.
- **Device Type:** The switch type of the associated switching device (for example, OH Single Blade Switch).
- **Station 1:** The name of the substation to which the switching device is connected.
- **Feeder 1:** The name of the feeder to which the switching device is connected.
- **Station 2:** The name of the second substation to which the switching device is connected.
- **Feeder 2:** The name of the second feeder to which the switching device is connected.
- **Current Position:** Switching device's current position (Open, Closed, or Deenergized).
- **Classification:** Switching device's specific open state classification (Normal Open, Temporary Open, or Bad Open).

Navigation:

- "Locate" locates the switching device on the geographic view.

1.44.1. Designated Switches

The Designated Switches display shows switching devices with the Normal Open, Temporary Open, or Bad Open specific open state classification for a specific substation.

Display call-up URL:

- `netview://<Mode>/gridtabular/DesignatedSwitches_GridDisplay/<Substation Name>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Switch ID:** The unique identifier of the switching device.
- **Switch Name:** The name of the associated switching device.

- **Device Type:** The switch type of the associated switching device (for example, OH Single Blade Switch).
- **Station 1:** The name of the substation to which the switching device is connected.
- **Feeder 1:** The name of the feeder to which the switching device is connected.
- **Station 2:** The name of the second substation to which the switching device is connected.
- **Feeder 2:** The name of the second feeder to which the switching device is connected.
- **Current Position:** Switching device's current position (Open, Closed, or Deenergized).
- **Classification:** Switching device's specific open state classification (Normal Open, Temporary Open, or Bad Open).

Navigation:

- "Locate" locates the switching device on the geographic view.

1.45. DNA Scheduler Monitor

The DNA Schedule Monitor tabular display lets the user monitor the DNAF server connections to the DNA Scheduler. It shows the scheduler summary, the latest requests the scheduler receives, the requests that are currently being solved, and the DNA server's status.

To open this display, select the DNAF Display Plugin option from the RT DMS menu, and then select the DNA Scheduler Monitor.

1.45.1. DNA Scheduler Summary

This DNA Scheduler Summary frame shows summary data, including the number of requests received, the number of requests dispatched, the number of requests completed, the number of requests dropped, and links to pending requests and latest solved requests.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerSummary_GridDisplay`

The following information is shown on this display:

- **Number of Requests Received:** The number of requests received by the scheduler.
- **Number of Requests Dispatched:** The number of requests dispatched by the scheduler.
- **Number of Requests Completed:** The number of requests completed by the DNA servers.
- **Number of Requests Dropped:** The number of requests dropped by the scheduler because the requests have timed out.

Navigation:

- "Pending Requests" navigates to the pending requests tabular display.

- "Latest Solved Requests" navigates to the latest solved requests tabular display.

1.45.2. Latest Request

The Latest Request frame shows the latest requests received by the scheduler.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerLatestRequest_GridDisplay`

The following information is shown on this display:

- **Request ID:** The identifier of this request.
- **DNAF Server Request Time:** The time when the request is issued to the DNAF server.
- **Scheduler Received Time:** The time when the scheduler receives the request.
- **Analysis Function:** The analysis function of the request, which can be DPF, PRV, SCC, FL, LVM, FISR, AFR, or PLOS.
- **Entry Substation:** The name of the request’s entry substation.

1.45.3. Solving Requests

The Solving Requests frame displays the requests that are currently being solved by the DNA servers.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerSolvingRequest_GridDisplay`

The following information is shown on this display:

- **Request ID:** The identifier of the request.
- **DNAF Server Request Time:** The time when the request is issued to the DNAF server.
- **Scheduler Received Time:** The time when the scheduler receives the request.
- **Analysis Function:** The analysis function of the request, which can be DPF, PRV, SCC, FL, LVM, FISR, AFR, or PLOS.
- **Entry Substation:** The name of the request’s entry substation.
- **DNA Start Time:** The time the request is sent to a DNA server.
- **Elapsed DNA Solution Time (Sec):** The elapsed time since the request was sent out.
- **DNA Server ID:** The identifier of the DNA server that takes the request.

1.45.4. DNA Server Status

The DNA Server Status frame shows the status data of each DNA server that is currently connected to scheduler.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerDnaServerSummary_GridDisplay`

The following information is shown on this display:

- **Host Name:** The name of the computer that hosts the DNA server.
- **Process ID:** The process ID of the DNA server.
- **Dna Server ID:** The identifier of the DNA server.
- **Status:** The status of the DNA server. It can be Ready or Busy.
- **Number of Requests Processed:** The number of requests already solved and/or being solved by the DNA server since the scheduler was started.

Navigation:

- "Request Details" navigates to the DNA requests tabular display.

1.45.5. Pending Requests

The Pending Requests display shows the requests held by the scheduler that are ready to be dispatched.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerPendingRequest_GridDisplay`

The following information is shown on this display:

- **Request ID:** The identifier of this request.
- **DNAF Server Request Time:** The time when the request is issued to the DNAF server.
- **Scheduler Received Time:** The time when the scheduler receives this request.
- **Analysis Function:** The function of the request, which can be DPF, PRV, SCC, FL, LVM, FISR, AFR, or PLOS.
- **Entry Substation:** The name of the request's entry substation.

1.45.6. Latest Solved Requests

The Latest Solved Requests display shows the latest solved requests by the currently connected DNA servers.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerLatestSolvedRequest_GridDisplay`

The following information is shown on this display:

- **Request ID:** The identifier of this request.
- **DNAF Server Request Time:** The time when the request is issued to the DNAF server.
- **Scheduler Received Time:** The time when the scheduler receives the request.
- **Analysis Function:** The analysis function of the request, which can be DPF, PRV, SCC, FL, LVM, FISR, AFR, or PLOS.
- **Entry Substation:** The name of the request's entry substation.
- **DNA Start Time:** The time the request is sent to a DNA server.
- **DNA End Time:** The time the scheduler is notified that the latest request is solved and the DNA server is ready.
- **DNA Solution Time (Sec):** The duration between the DNA Start Time and the DNA End Time.
- **DNA Server ID:** The identifier of the DNA server that takes the request.

1.45.7. Requests – DNA Server ID

The Requests display shows the list of the requests solved or being solved by the DNA server.

Display call-up URL:

- `netview://rt/gridtabular/DnaSchedulerDnaServerSolveRequest_GridDisplay`

The following information is shown on this display:

- **Request ID:** The identifier of this request.
- **DNAF Server Request Time:** The time when the request is issued to the DNAF server.
- **Scheduler Received Time:** The time when the scheduler receives the request.
- **Analysis Function:** The analysis function of the request, which can be DPF, PRV, SCC, FL, LVM, FISR, AFR, or PLOS.
- **Entry Substation:** The name of the request's entry substation.
- **Status:** The status of this request. It can be Solved or Solving.
- **DNA Start Time:** The time the request is sent to a DNA server.
- **DNA End Time:** The time the scheduler is notified that the latest request was solved and the DNA server is ready.

2. ADMS OMS Tabular Displays

For descriptions of the tabular displays associated with the Outage Management System (OMS) part of ADMS, refer to the *Outage Management System User Guide*.

3. ADMS Displays and Pop-Ups

This chapter describes ADMS displays, pop-ups, and dialogs where the user can review and enter the specific data.

3.1. Device Data Entry Displays

This section lists displays where the user can specify device parameters to be used in standard and advanced DNAF applications.

3.1.1. Manual Input Parameter Entry

The Manual Input Parameter Entry data entry pop-up display can be accessed from the ADMS menu bar > RT DMS menu > Manual Input Parameter Entry (can also be ST instead of RT). This display allows an operator to manually set parameters for the whole system, or system sub-elements (optimization groups, operating centers, stations, power transformers, and feeders) using the tree-like structure.

When entering the parameter values for an element that contains sub-elements, you can specify if they should inherit the manually entered values. This allows you to set values for the whole system, operating center, and so on. For bus and load voltage limits, the values will only inherit from substation (station internal buses and loads) or feeder level (station external buses and loads) though the tree-like UI has other sub-elements.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/MANUALPARAMS`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following types of parameters can be set:

- General parameters
  - **Temperature:** Temperature settings (for stations).
  - **Frequency:** Frequency settings (for stations).
  - **Load Scaling Factor:** All the loads active and reactive power are scaled (for feeders).
- Violation limits
  - **Trans. Flow Limit Scaling:** Flow limit scaling rate (for power transformers).
  - **Device Flow Limit Scaling:** Device flow limit scaling rate (for feeders).
  - **Bus High Voltage Limit:** Bus high voltage limit rate (for feeders), per unit.
  - **Bus Low Voltage Limit:** Bus low voltage limit rate (for feeders), per unit.
  - **Load High Voltage Limit:** Load high voltage limit rate (for feeders), per unit.

- **Load Low Voltage Limit:** Load low voltage limit rate (for feeders), per unit.
- Optimization limits
  - **Bus High Voltage Limit.** Bus high voltage limit rate (for feeders), per unit.
  - **Bus Low Voltage Limit.** Bus low voltage limit rate (for feeders), per unit.
  - **Load High Voltage Limit.** Load high voltage limit rate (for feeders), per unit.
  - **Load Low Voltage Limit.** Load low voltage limit rate (for feeders), per unit.
- Load and Volt/VAR Management (LVM):
  - **Feeder Max Power Factor Target:** Feeder maximum power factor target rate (for feeders).
  - **Feeder Min Power Factor Target:** Feeder minimum power factor target rate (for feeders).
  - **Station Max Power Factor Target:** Station and power transformer maximum power factor rate (for stations).
  - **Station Min Power Factor Target:** Station and power transformer minimum power factor rate (for stations).
- Real/Reactive Power Throttling
  - **Real Power Throttling Up:** The rate at which a real power increase is throttled during an LVM plan execution, in MW/min.
  - **Real Power Throttling Down:** The rate at which a real power decrease is throttled during an LVM plan execution, in MW/min.
  - **Reactive Power Throttling Up:** The rate at which a reactive power increase is throttled during an LVM plan execution, in MVAR/min.
  - **Reactive Power Throttling Down:** The rate at which a reactive power decrease is throttled during an LVM plan execution, in MVAR/min.

  The manual real/reactive power throttling values can be set for the entire system or for optimization groups.
- Fault Location (FL)
  - **Fault Impedance Resistance:** Fault impedance resistance in ohms.
  - **Fault Impedance Reactance:** Fault impedance reactance in ohms.
- Short Circuit Calculation (SCC)
  - **Fault Impedance Resistance:** Fault impedance resistance in ohms.
  - **Fault Impedance Reactance:** Fault impedance reactance in ohms.
- Protection Validation (PRV)
  - **Fault Impedance Resistance:** Fault impedance resistance in ohms.
  - **Fault Impedance Reactance:** Fault impedance reactance in ohms.

3.1.2. Bus Parameters

The Bus Parameters data entry pop-up display can be accessed from the Bus toolbar. The data entry pop-up allows the bus high voltage and low voltage limits for reconfiguration applications to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/BUSPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Voltage High Limit:** Voltage high limit.
  - **Operational Value:** Nominal operational voltage high limit.
  - **Override Enable (Operational):** A flag to enable/disable manual operational voltage high limit.
  - **Override Value (Operational):** Manual operational voltage high limit.
  - **Optimization Value:** Nominal optimization voltage high limit.
  - **Override Enable (Optimization):** A flag to enable/disable manual optimization voltage high limit.
  - **Override Value (Optimization):** Manual optimization voltage high limit.
- **Voltage Low Limit:** Voltage low limit.
  - **Operational Value:** Nominal operational voltage low limit.
  - **Override Enable (Operational):** A flag to enable/disable manual operational voltage low limit.
  - **Override Value (Operational):** Manual operational voltage low limit.
  - **Optimization Value:** Nominal optimization voltage low limit.
  - **Override Enable (Optimization):** A flag to enable/disable manual optimization voltage low limit.
  - **Override Value (Optimization):** Manual optimization voltage low limit.

> **Note**
>
> Bus voltage limits can be set to values be between 10% and 500% of the current base voltage.

- **Voltage Base:** Voltage base value in Volts. To change the voltage base (for example, from Volts to per unit), use the Primary Voltage Units options on the DPF tab of the DNAF Configuration Editor.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the bus values without closing the pop-up.

- **OK:** Click this button to apply the changes made to the bus values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the bus values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.3. Capacitor Parameters

The Capacitor Parameters data entry pop-up display can be accessed from the Capacitor and Capacitor Bank toolbars. This display-up allows a range of operating parameters for capacitors and capacitor banks to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/CAPACITORPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Control Capability:** Check this box to set the capacitor to remote control. Unchecking this box sets the capacitor to local control. Note that this check box is only displayed for devices that have local and remote control capability.
- **Local Control Type:** Displays whether the capacitor is local controllable.
- **Upper Control Limit:** The upper level at which the capacitor can be controlled. This only applies when the device is in local control mode.
    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.
- **Lower Control Limit:** The lower level at which the capacitor can be controlled. This only applies when the device is in local control mode.
    - **Nominal Value:** Nominal value.
    - **Override Value:** Manual value. To set a manual value, check the Override Enable box next to the Upper Control Limit option.
- **Voltage Override High Limit:** A voltage high limit value (expressed as a per-unit value) that can be used to override the nominal voltage high limit.
    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.
- **Voltage Override Low Limit:** A voltage low limit value (expressed as a per-unit value) that can be used to override the nominal voltage low limit.

    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.
- **Nominal Size per Phase (kVAR):** Capacitor nominal size per phase expressed in kVAR.
- **Hours Between Consecutive Switching:** The minimum elapsed time between consecutive switching operations on the capacitor, expressed in hours.
    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.
- **Maximum Number of Daily Switching Operations:** The maximum number of switching operations that can be performed on a capacitor within a 24-hour period.
    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.
- **Movement Cost:** The cost of the transformer switching operation.
    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the capacitor values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the capacitor values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the capacitor values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.4. Energy Resource Parameters

The Energy Resource Parameters data entry pop-up display can be accessed by clicking the Enter Data button on the Distribution Energy Resource toolbar. This data entry pop-up allows the real and reactive power targets to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/ENERGYRESPARAMS/<Substation Name>/<Device ID>`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this pop-up display:

- **Nominal Voltage:** Nominal voltage of the generator.
- **Generation Type:** Generation type (for example, PQ).
- **Voltage Target (%):** Nominal voltage target in percentage.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Override Output:** Check this box to apply manual output values.
  - **Effective Real Power Target (kW):** Effective real power output in kW.
  - **Effective Reactive Power Target (kVAR):** Effective reactive power output in kVAR.
- **Real Power Target (kW):** Scheduled real power target in kW.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Reactive Power Target (kVAR):** Scheduled reactive power target in kVAR.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Contributes To Fault Current:** Specifies whether the device can remain connected during a fault and contribute to fault current. This is considered during SSC or PRV calculations. Can be True or False.
  - **Nominal Value:** Nominal value. Can be True or False.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **State of Charge (%):** The amount of charge that is currently stored in the DER storage device. State of Charge is specified as a percentage of the device's capacity in kWh (for example, 60 = 60%). This field is only enabled for batteries.
- **Override Enable:** Check this box to apply the manual value.
- Override Value: Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the distribution energy resource values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the distribution energy resource values and to close the pop-up.

- **Cancel:** Click this button to close the pop-up without applying the changes to the distribution energy resource values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.5. Generator Parameters

The Generator Parameters data entry pop-up display can be accessed by clicking the Enter Data button on the Generator toolbar. This data entry pop-up allows the real and reactive power targets to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/GENERATORPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this pop-up display:

- **Nominal Voltage:** Nominal voltage of the generator.
- **Generation Type:** Generation type (for example, PQ).
- **Voltage Target (%):** Nominal voltage target in percentage.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Override Output:** Check this box to apply manual values.
  - **Effective Real Power Target (kW):** Effective real power target in kW.
  - **Effective Reactive Power Target (kVAR):** Effective reactive power target in kVAR.
- **Real Power Target (kW):** Scheduled real power target in kW.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Reactive Power Target (kVAR):** Scheduled reactive power target in kVAR.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the generator values without closing the pop-up.

- **OK:** Click this button to apply the changes made to the generator values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the generator values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.6. Inverter Parameters

The Inverter Parameters data entry pop-up display can be accessed by clicking the Enter Data button on the Inverter toolbar. This data entry pop-up allows the deliverable and absorbable power targets to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/INVERTERPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this pop-up display:

- **Maximum Real Power Delivered (kW):** The maximum real power the DER can deliver to the grid (in W).
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Maximum Reactive Power Delivered (kVAR):** The maximum reactive power the DER can deliver to the grid (in VAR).
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Maximum Apparent Power Delivered (kVA):** The maximum apparent power the DER can deliver to the grid (in VA).
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Maximum Real Power Absorbed (kVA):** The maximum real power the DER can absorb from the grid (in W).
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

- **Maximum Reactive Power Absorbed (kVAR):** The maximum reactive power the DER can absorb from the grid (in VAR).
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Maximum Apparent Power Absorbed (kVA):** The maximum apparent power the DER can absorb from the grid (in VA).
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the inverter values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the inverter values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the inverter values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.7. Line Parameters

The Line Parameters data entry pop-up display can be accessed from the Line toolbar. This data entry pop-up allows the flow limit and control eligibility for reconfiguration applications to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/LINEPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Flow Limit of Phase A/B/C:** Flow limit values for phases A, B, C, and neutral, in Amps.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the line values without closing the pop-up.

- **OK:** Click this button to apply the changes made to the line values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the line values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.8. Load Parameters

The Load Parameters data entry pop-up display can be accessed from the Load toolbar or from the Distribution Transformer Load tabular display (click the Load Manual Entry Parameters button).

This data entry pop-up allows setting the nominal voltage limit and nominal load values.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/LOADPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Operational Voltage High Limit (p.u):** The maximum operational voltage limit (expressed as a p.u. value) at the transformer load terminals. Voltages in excess of the limit are shown as a violation.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply a manual value.
  - **Override Value:** Manual value.
- **Operational Voltage Low Limit (p.u):** The minimum operational voltage limit (expressed as a p.u. value) at the transformer load terminals. Voltages less than the limit value are shown as a violation.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Select this box to apply a manual value.
  - **Override Value:** Manual value.
- **Optimization Voltage High Limit (p.u):** The maximum optimization voltage limit (expressed as a p.u. value) at the transformer load terminals. Voltages in excess of the limit are shown as a violation.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Select this box to apply a manual value.
  - **Override Value:** Manual value.
- **Optimization Voltage Low Limit (p.u):** The minimum optimization voltage limit (expressed as a p.u. value) at the transformer load terminals. Voltages less than the limit value are shown as a violation.
  - **Nominal Value:** Nominal value.

    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.
- **Nominal Load Overrides?:** Check this box to override all nominal load values together.
    - Phase A (or AB if delta connected) nominal load (kW)
        - **Nominal Value:** Nominal value.
        - **Override Value:** Manual value.
    - Phase A (or AB if delta connected) nominal load (kVAR).
        - **Nominal Value:** Nominal value.
        - **Override Value:** Manual value.
    - Phase B (or BC if delta connected) nominal load (kW)
        - **Nominal Value:** Nominal value.
        - **Override Value:** Manual value.
    - Phase B (or BC if delta connected) nominal load (kVAR)
        - **Nominal Value:** Nominal value.
        - **Override Value:** Manual value.
    - Phase C (or CA if delta connected) nominal load (kW)
        - **Nominal Value:** Nominal value.
        - **Override Value:** Manual value.
    - Phase C (or CA if delta connected) nominal load (kVAR)
        - **Nominal Value:** Nominal value.
        - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the load values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the load values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the load values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.9. Reference Bus Interface Parameters

The Reference Bus Interface Parameters data entry pop-up display can be accessed by clicking the Enter Data button on the Reference Bus Interface toolbar. This data entry pop-up allows the real and reactive power targets to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/RFBSIFPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this pop-up display:

- **Voltage Target (%):** The voltage target for phases A, B, and C, in percentage.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Angle Target (deg):** The angle target, in degrees.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Positive Sequence Resistance (Ohm):** The positive sequence resistance, in Ohms.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Positive Sequence Reactance (Ohm):** The positive sequence reactance, in Ohms.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Negative Sequence Resistance (Ohm):** The negative sequence resistance, in Ohms.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Negative Sequence Reactance (Ohm):** The negative sequence reactance, in Ohms.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Zero Sequence Resistance (Ohm):** The zero sequence resistance, in Ohms.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Zero Sequence Reactance (Ohm):** The zero sequence reactance, in Ohms.
  - **Nominal Value:** Nominal value.

    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the reference bus interface values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the reference bus interface values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the reference bus interface values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the reference bus interface.

3.1.10. Series Reactor Parameters

The Series Reactor Parameters data entry pop-up display can be accessed from the Series Reactor toolbar. This data entry pop-up allows the flow limit value to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/REACTORPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Flow Limit:** The maximum current flow to be allowed through a series reactor by the DNAF applications; expressed in amps.
    - **Nominal Value:** Nominal value.
    - **Override Enable:** Check this box to apply the manual value.
    - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the series reactor values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the series reactor values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the series reactor values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.11. Switch Parameters

The Switch Parameters data entry pop-up display can be accessed from device toolbars for all types of switching devices (distribution switches, fuses, SCADA controlled switches, and so on).

This data entry pop-up allows the switch flow limit and control eligibility for reconfiguration applications to be changed.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/SWITCHPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Load Break Capable:** Flag indicating whether the switch can break load.
- **Settings Group Template:** The settings group template. Applies to reclosers only.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Flow Limit:** The maximum current flow to be allowed through a switch by the DNAF applications expressed in amps.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Load Break Rating:** The maximum current flow to be allowed to break by the switch expressed in amps.
  - **Nominal Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the switching device values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the switching device values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the switching device values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.1.12. Transformer Parameters

The Transformer Parameters pop-up display can be accessed from device toolbars for regulators and LTC transformers. This data entry pop-up contains a range of parameters relating to the operation of the regulators and tap changing devices. Please be aware that manual entried parameters that related to voltage regulation, for example, voltage target, voltage bandwidth, only apply to regular regulation mode, not for advanced regulation mode.

Display call-up URL:

- `netview://<Mode>/DNAFPARAMS/TFPARAMS/<Substation Name>/<Device ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following inputs and parameters are shown on this display:

- **Control Capability:** Displays the Regulator/Tap Changer control capability (for example, None, Local Only, Remote).
- **Regulation Tap Side:** Displays the side of tap regulation (for example, None, Primary, Secondary).
- **Primary Tap Position:** The transformer primary side tap position. Must be an integer value.
  - **Operational Value:** Current tap position value (returns manual override value if set, otherwise returns measured/estimated tap position).
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

  Tap position can be changed for fixed tap changers but cannot be changed for locally controlled only taps.
- **Secondary Tap Position:** The transformer secondary side tap position. Must be an integer value.
  - **Operational Value:** Current tap position value (returns manual override value if set, otherwise returns measured/estimated tap position).
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

  Tap position can be changed for fixed tap changers but cannot be changed for locally controlled only taps.
- **Tertiary Tap Position:** The transformer tertiary side tap position. Must be an integer value.
  - **Operational Value:** Current tap position value (returns manual override value if set, otherwise returns measured/estimated tap position).
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

Tap position can be changed for fixed tap changers but cannot be changed for locally controlled only taps.

- **Advanced Regulation Mode:** The active advanced regulation mode.
  - **Operational Value:** Current value. Displays the SCADA measured regulation mode or the default mode if the measurement is invalid.
  - **Override Enable:** Check this box to apply the manual value. This box is disabled if the transformer is incapable of advanced regulation or if a valid regulation mode measurement exists.
  - **Override Value:** Manual value.
- **Voltage Target (Volts):** The regulation voltage target expressed as a per-unit value. This applies only to devices in local control mode.
  - **Override Enable:** Check this box to apply the manual value. This box is disabled if the transformer has an advanced regulation mode.
  - **Override Value:** Enter a value for the voltage target.
- **Voltage Bandwidth (Volts):** The regulation two-sided voltage bandwidth expressed as a per-unit value. This applies only to devices in local control mode.
  - **Override Enable:** Check this box to apply the manual value. This box is disabled if the transformer has an advanced regulation mode.
  - **Override Value:** Enter a value for the Voltage Bandwidth.
- **Line Drop Compensation Resistance (Volts):** The line drop compensation resistance setting expressed in volts. This applies only to devices in local control mode.
  - **Override Enable:** Check this box to apply the manual value. This box is disabled if the transformer has an advanced regulation mode.
  - **Override Value:** Enter a value for Line Drop Compensation Resistance.
- **Line Drop Compensation Reactance (Volts):** The line drop compensation reactance setting expressed in volts. This applies only to devices in local control mode.
  - **Override Enable:** Check this box to apply the manual value. This box is disabled if the transformer has an advanced regulation mode.
  - **Override Value:** Enter a value for Line Drop Compensation Reactance.
- **Voltage Reduction (Volts):** If the regulating device has voltage reduction capability, the amount of voltage reduction, expressed as a per-unit value. This applies only to devices in local control mode.
  - **Override Enable:** Check this box to apply the manual value. This box is disabled if the transformer has an advanced regulation mode.
  - **Override Value:** Enter a value for the Voltage Reduction.
- **Flow Limit (per Phase):** A flow limit for the transformer, in Amps.
  - **Operational Value:** The nominal flow limit of the transformer.
  -

  Override Enable: Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **Voltage Imbalance Limit:** The voltage imbalance limit for the transformer, in percentage.
  - **Operational Value:** The nominal voltage imbalance limit of the transformer.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.
- **DPF Back Calculation Priority:** Priority of LVM back calculation.
  - **Override Value:** The value of the LVM back calculation priority.
- **ASC Regulation On Status:** The Arc Suppression Coil (ASC) Regulation On status.
  - **Operational Value:** Nominal value (defined in the static model).
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value (True/False).
- **Movement Cost:** The cost of the transformer switching operation.
  - **Operational Value:** Nominal value.
  - **Override Enable:** Check this box to apply the manual value.
  - **Override Value:** Manual value.

The following buttons are shown on this display:

- **Apply:** Click this button to apply the changes made to the transformer values without closing the pop-up.
- **OK:** Click this button to apply the changes made to the transformer values and to close the pop-up.
- **Cancel:** Click this button to close the pop-up without applying the changes to the transformer values.
- **Navigate to Geographic (Globe Icon):** Click this button to navigate to a geographic view centered on the device.

3.2. OMS Displays

For descriptions of the pop-ups associated with the Outage Management System (OMS) part of ADMS, refer to the *Outage Management System User Guide*.

3.3. Advanced Metering Infrastructure (AMI) Displays

This section describes AMI-related displays.

3.3.1. CIS/AMI Customer Ping

The CIS/AMI Customer Ping display shows the customer information and metered voltage and allows pinging the selected customers. The display can be accessed by selecting a customer on a network view and clicking Call Ping Pop-Up Single Customer / Multiple Customers on the toolbar or the context menu.

Display call-up URL:

- `netview://RT/gridtabular/CustomerPingGridDisplay`

The following fields and controls are displayed on the CIS/AMI Customer Ping display:

- **Ping:** Selects a customer to ping.
- **Ping Status:** Ping status (pending, sent, successful, connection error, or failed).
- **AMI:** Indicates if the customer is equipped with an AMI meter (true or false).
- **Called:** Indicates if the customer called (true or false).
- **Name:** Full customer name.
- **Meter Number:** Meter number.
- **Address:** Customer address.
- **Volt A/B/C:** Voltage measured by the meter in Volts.
- **Select All:** Selects all customers in the list.
- **Deselect All:** Deselects all customers in the list.
- **Ping:** Pings all selected customers.
- **Cancel:** Closes the display.

3.3.2. Customer Analogs

The Customer Analogs display shows the customer meter measurements. The display can be accessed by selecting a customer on a network view and clicking Show Analogs Single Customer / Multiple Customers on the toolbar or the context menu.

> **Note**
>
> To show meter measurements on the Customer Analogs display, you must first request for the measurements using the Scan AMI Analogs Single Customer / Multiple Customers options.

Display call-up URL:

- `netview://RT/gridtabular/CustomerAnalogGridDisplay`

The following fields are displayed on the Customer Analogs display:

- **Name:** Full customer name.

- **Meter Number:** Meter number.
- **Communication Time:** Date and time of the measurement response. After a configurable time period, all measurements are expired and removed from the display.
- **Volt A/B/C:** Voltage measured by the meter in Volts.
- **Watt A/B/C:** Real power measured by the meter in Watts.
- **VAR A/B/C:** Reactive power measured by the meter in VARs.
- **Amp A/B/C:** Current measured by the meter in Amps.

3.4. Curtailment

The Curtailment dialog box allows you to disconnect/reconnect distributed energy resources (DERs) to the network to prevent backfeed conditions. This display can be called from the line context menu (available only for feeders with connected DERs).

Display call-up URL:

- `netview://RT/Popup/Curtailplan`
- `netview://RT/Popup/Curtailplan/<Substation Name>`

The following data and controls are displayed on the Curtailment frame:

- **Curtailable:** Total amount of kW that can be curtailed for the feeder.
- **Backfeed:** Total amount backfeed for the feeder.
- **Refresh:** Refreshes the data on the Curtailment dialog box.
- **Done:** Closes the Curtailment dialog box.

The following data and controls are displayed on the Action and Disconnected frames:

- **Disconnect Selected Devices:** Disconnects selected devices from the network.
- **Connect All:** Reconnects all devices to the network.
- **Connect Selected:** Reconnects selected devices to the network.
- **Select:** Selects the DER.
- **Locate:** Locates the DER on the network view.
- **Substation:** Substation name.
- **ID:** Unique identifier of the energy resource.
- **Output:** DER output in kW.

  The grid tabular displays on the Action and Disconnected frames are controlled by EnergySourceGridDisplay.xml.

3.5. DNAF Aggregation Summary Display

The DNAF Aggregation Summary display shows load and voltage aggregation values (real and reactive power demand, power factor, technical losses, and voltages at station and regulated buses) for the distribution system as a whole and for individual station optimization groups, divisions, substations, transformers, and feeders. The power flow and voltage values are obtained from SCADA measurements and DPF calculations. The display also shows the difference between the measured and calculated load values, and the weighted average for the measured and calculated voltage values.

For more information about value calculation for this display, refer to section "DNAF System Aggregation" in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

To open this display, select the DNAF Display Plugin option from the RT DMS drop-down menu, and then select DNAF Aggregation Summary.

The following information is shown on the DNAF Aggregation Summary display:

- **Calculated Primary/Secondary Real Power Demand:** Calculated real power at the source side (primary) and at the load side (secondary), if applicable, in MW. For power transformer level, the value is on the secondary side.
- **Measured Secondary Real Power Demand:** Measured secondary real power in MW.

  For transformer and feeder levels, the value is shown only if the measurement exists. For system and station levels, the value is aggregated from downstream transformers because there are no associated physical measurements. If any of the transformers do not have a measured value, the calculated value is used in the summary.
- **Secondary Real Power Demand Difference:** Difference between the calculated and measured secondary real power in percentage.
- **Calculated Native Real Power Demand:** Calculated native real power demand in MW.
- **Calculated DER Real Power Injection:** Calculated DER/DG real power injection in MW.
- **Real Power Loss:** Calculated real power loss in MW. The losses include losses in lines, transformers, regulators, series reactors, and other equipment.
- **Calculated Primary/Secondary Reactive Power Demand:** Calculated reactive power at the source side (primary) and at the load side (secondary), if applicable, in MVAR.
- **Measured Secondary Reactive Power Demand:** Measured secondary reactive power in MVAR.

  For transformer and feeder levels, the value is shown only if the measurement exists. For system and station levels, the value is aggregated from downstream transformers because there are no associated physical measurements. If any of the transformers do not have a measured value, the calculated value is used in the summary.
- **Secondary Reactive Power Demand Difference:** Difference between the calculated and measured secondary reactive power in percentage.

- **Calculated Native Reactive Power Demand:** Calculated native reactive power demand in MVAR.
- **Calculated DER Reactive Power Injection:** Calculated DER/DG reactive power injection in MVAR.
- **Reactive Power Loss:** Calculated reactive power loss in MVAR. The losses include losses in lines, transformers, regulators, series reactors, and other equipment.
- **Calculated Primary/Secondary Power Factor:** Calculated power factor at the source side (primary) and at the load side (secondary), if applicable.
- **Measured Secondary Power Factor:** Measured power factor at the load side, if the measurement exists.
- **Calculated Average Voltage, Station Internal Regulated Bus Only:** Calculated average voltage in per unit. Only station internal regulated MV buses are included.
- **Measured Average Voltage, Station Internal Regulated Bus Only:** Measured average voltage in per unit. Only station internal regulated MV buses are included.
- **Calculated Weighted Average Voltage, Station Internal Regulated Bus Only:** Weighted calculated average voltage in per unit. Only station internal regulated MV buses are included. The weighting factor is the calculated current value on the same bus.
- **Measured Weighted Average Voltage, Station Internal Regulated Bus Only:** Weighted measured average voltage in per unit. Only station internal regulated MV buses are included. The weighting factor is the measured current value on the same bus.
- **Calculated Average Voltage, All Regulated Bus:** Calculated average voltage in per unit, including both station MV buses and regulator buses on feeders.
- **Measured Average Voltage, All Regulated Bus:** Measured average voltage in per unit, including both station MV buses and regulator buses on feeders.
- **Calculated Weighted Average Voltage, All Regulated Bus:** Weighted calculated average voltage in per unit, including both station MV buses and regulator buses on feeders. The weighting factor is the calculated current value on the same bus.
- **Measured Weighted Average Voltage, All Regulated Bus:** Weighted measured average voltage in per unit, including both station MV buses and regulator buses on feeders. The weighting factor is the measured current value on the same bus.
- **Calculated Online Capacitor Reactive Power:** Calculated total reactive power injection of online capacitors in MVAR.
- **Calculated Offline Capacitor Reactive Power:** Calculated total reactive power injection of offline capacitors in MVAR.
- **Calculated Total Capacitor Reactive Power:** Calculated total reactive power injection of online and offline capacitors in MVAR.
- **Calculated Real Power Generation Capacity:** Calculated real power generation capacity in MW.
- **Calculated Reactive Power Generation Capacity:** Calculated reactive power generation

capacity in MVAR.

3.6. DNAF Violation Counts Summary Display

The DNAF Violation Counts Summary display presents the aggregated counts of various violations including flow imbalance, voltage imbalance, flow limits, and voltage limits across different components such as switches, lines, distribution transformers, and loads. These counts are provided for the entire distribution system as well as for individual station optimization groups, divisions, substations, transformers, and feeders. The violations are identified by the LIM function, which detects all limit violations following each DPF execution.

For more information about value calculation for this display, refer to section "DNAF Violation Counts" in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

To open this display, select the DNAF Display Plugin option from the RT DMS drop-down menu, and then select DNAF Violation Counts Summary.

The following information is shown on the DNAF Violation Counts Summary display:

- **Number of Switches with Flow Imbalance Violation:** Number of three phase switching devices with a flow imbalance limit violation.
- **Number of Loads with Voltage Imbalance violation:** Number of three phase loads with a voltage imbalance limit violation.
- **Number of Switches with Flow Limit Violation:** Number of switching devices with a flow limit violation.
- **Number of Lines with Flow Limit Violation:** Number of lines with a flow limit violation.
- **Number of Distribution Transformers with Flow Limit Violation:** Number of distribution transformers with a flow limit violation.
- **Number of Loads with Voltage Limit Violation:** Number of loads with a high or low voltage limit violation.

3.7. DER Manual Dispatch

The DER Manual Dispatch display shows measured and aggregated values for distributed energy resources (DERs) that are managed by the 2030.5 Gateway and SCADA and allows you to create Manual DER Dispatch plans. To open this display, select the DER Manual Dispatch option from the RT DMS drop-down menu. You can also open this display from the line context menu (available only for feeders with connected DERs).

Display call-up URL:

- `netview://RT/POPUP/DERMANUALDISPATCHUI`
- **Group Edit Section:** Controls for editing all selected rows. Start Time and Duration are per dispatch plan. To apply a group control, modify the required value (for example, Ctrl kW) and

then press Enter.

  - **Start Time:** The start date and time for a plan dispatch. The default value is the current date and time.
  - **Duration in Minutes:** Number of minutes the plan is dispatched.
  - **Control Connect:** Globally checks or unchecks all the selected Control Connect check boxes.
  - **Control kW Percentage of Maximum kW:** Global control for changing the kW setting (in percentage of the maximum kW).
  - **Control kVAR Percentage of Minimum kVAR:** Global control for changing the kVAR setting (in percentage of the Minimum kVAR).
  - **Control kVAR Percentage of Maximum kVAR:** Global control for changing the kVAR setting (in percentage of the Maximum kVAR).
  - **Control PF:** Global control for changing the Power Factor setting.

The DER Manual Dispatch grid includes the following fields and controls:

- **Element Tree:** Hierarchical element tree. The elements are listed in alphabetical order.
- **Select:** Selects an element in the tree.
- **Data Source:** DER measurement data source, (Scada or Gateway).
- **DER Device Type:** DER device type, such as Wind or Solar.
- **Aggregator:** Aggregator name.
- **Connected:** The DER connection status. For the system, divisions, stations, transformers, and feeders, a black square indicates different connection statuses of DERs under the element.
- **Control Connect:** Manually specified DER connection status. This value is used for sending as a control to an external interface.
- **Connection Status:** The connection control status.
- **kW:** The DER real power output in kW. Shows measured values for individual DERs and aggregated values for system, divisions, stations, transformers, and feeders.
- **Maximum kW:** The maximum DER real power output in kW.
- **Control kW:** Manually specified DER real power output value in kW. This value is used for sending as a control to an external interface.

  For energy storage devices, the Control kW entry upper and lower validation ranges depend on the device charging and discharging limits.

  - **Charging Limit (kW)**: The true real power charging limit computed with dynamic charge limit, current state of charge, maximum state of charge, and dispatch duration.

    Charging Limit = min \{Dynamic Charge kW Limit, (MaxSoC in kWh – SoC in kWh) / (Dispatch Duration in h)}

  - **Discharging Limit (kW)**: The true real power discharging limit computed with dynamic discharge limit, current state of charge, minimum state of charge, and dispatch duration.

Discharging kW Limit = min \{Dynamic Discharge kW Limit, (SoC in kWh – MinSoC in kWh) / (Dispatch Duration in h)}

- **kW Status:** The kW control status.
- **kVAR:** The DER reactive power output in kVAR. Shows measured values for individual DERs and aggregated values for system, divisions, stations, transformers, and feeders.
- **Minimum kVAR:** The minimum DER reactive power output in kVAR.
- **Maximum kVAR:** The maximum DER reactive power output in kVAR.
- **Control kVAR:** Manually specified DER reactive power output value in kVAR. This value is used for sending as a control to an external interface.
- **kVAR Status:** The kVAR control status.
- **PF:** The power factor for an element.
- **Control PF:** Manually specified DER power factor value. This value is used for sending as a control to an external interface.
- **PF Status:** The PF control status.
- **Curve:** DER curve. Applies only to DERs that can operate in autonomous mode.
- **Control Curve:** Manually specified DER curve. This value is used for sending as a control to an external interface.
- **Control Curve Status:** The curve control status.

3.8. SCADA/DER Devices Tree View

The SCADA/DER Devices Tree View display lists SCADA devices and distributed energy resource objects that exist in the system in the hierarchical element tree view. The devices are grouped by substations and feeders. To open this display, select the SCADA/DER Devices Tree View Display option from the RT DMS drop-down menu.

Display call-up URLs:

```
NETVIEW://rt/POPUP/DSLOAD?TABULAR=DeviceTreeView&;LEVEL_NAMES=<Hierarcy_Level_N>,<Hierarcy_Level_X>

NETVIEW://<Mode>/popup/DEVICETREEVIEW?TABULAR=DeviceTreeView&EXPANDTONODE=<Hierarcy_Level_N>,<Hierarcy_Level_X>

NETVIEW://<Mode>/popup/DEVICETREEVIEW?TABULAR=DeviceTreeView&EXPANDTONODE=<Device_ID>
```

Where:

+

```
<Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

<LEVEL_NAMES> defines option labels in the Set Expand Level To drop-down list on the display.
```

```
<EXPANDTONODE> specifies which hierarchy levels appear expanded when the display is called.
```

Display call-up examples:

```
NETVIEW://RT/popup/DEVICETREEVIEW?TABULAR=DeviceTreeView&LEVEL_NAMES=Station,Feeder,Device

NETVIEW://RT/popup/DEVICETREEVIEW?TABULAR=DeviceTreeView&EXPANDTONODE=RAVEN|F8861

NETVIEW://RT/popup/DEVICETREEVIEW?TABULAR=DeviceTreeView&expandtonode=020482E3-D10A-4697-A77A-EF7974B1
1736
```

The following information is shown on this display:

- **Element tree:** Hierarchical element tree, which includes the specified hierarchy levels, such as service centers, substations, feeders, and devices. The elements are listed in alphabetical order.
- **Name:** Device name.
- **Service Center:** Service center that the device belongs to.
- **Station:** Substation that is associated with the device.
- **Feeder:** Feeder that is associated with the device.
- **Is SCADA:** Specifies whether this is a SCADA device.
- **Is DER:** Specifies whether this energy resource is controlled by the 2030.5 Gateway.
- **Control ID:** Control measurement ID.
- **Control Type:** Control type.
- **Status:** Control status.
- **Available Status:** The list of available statuses.
- **Last Execution Status:** Status of last executed control.
- **Last Execution Time:** The control's last execution time.

Navigation:

- "Locate" navigates to the device on the network view.

The SCADA/DER Devices Tree View pop-up display can be customized to show multiple hierarchy levels. In addition, you can use the Device Tree View pop-up display logic to configure the hierarchy level tree view for other grid tabular displays. For more details, refer to the *Grid Tabular Builder User Guide*.

3.9. Faulted Section Determination

The Faulted Section Determination (FSD) display shows details for the FSD function. The display can be called from the power transformer context menu.

Display call-up URL:

- `NETVIEW://<Mode>/POPUP/FSD/<Substation_Name>/<Substation_Name>/POWERTRANSFORMER/<Device_ID>`

  Where: <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

3.9.1. Fault Section Determination Stage 1

The Faulted Section Determination (FSD) Stage 1 display shows details for the FSD Stage 1, which is run to identify and isolate a faulted feeder downstream of a faulted power transformer. The display appears on the Faulted Section Determination display, which can be called from the power transformer context menu.

Display call-up URL:

- `netview://RT/gridtabular/FaultedSectionDeterminationStage1_GridDisplay`

The following information is shown on this display:

- **Status**: FSD Stage 1 processing status for the feeder breaker (None, Ineligible, Pending, Executing, Not Faulted, or Faulted).
- **Feeder ID:** Feeder ID.
- **Feeder Name:** Feeder name.
- **Isolating CB:** Feeder head circuit breaker name.
- **Breaker ID:** Feeder head circuit breaker ID.
- **Ranking Weight:** Total feeder ranking that is calculated based on the feeder conditions and the configured weighting factors.
- **Fault Indication is Set:** Flag indicates whether a Fault Indicator is set.
- **Overhead Length:** Total feeder overhead conductor length, in miles.
- **Load:** Total load (in kW) connected to the feeder.
- **Number of Customers:** The total number of customers on the feeder.
- **Total Number of Priority Customers:** The total number of priority customers on the feeder.
- **Generation:** Total generation (in kW) connected to the feeder.

Navigation:

- "Locate" locates the feeder head circuit breaker on the geographic network view.

3.9.2. Fault Section Determination Stage 2

The FSD Stage 2 display shows details for the FSD Stage 2, which is run to identify the faulted feeder section. The display appears on the Faulted Section Determination display, which can be called from the power transformer context menu.

Display call-up URL:

- `netview://RT/gridtabular/FaultedSectionDeterminationStage2_GridDisplay`

The following information is shown in the Selected Switches tabular display:

- **Select**: This checkbox indicates the selected/unselected automated switching device to be used to identify the faulted feeder section. If checked, this switch will be used to generate the FSD Stage 2 plan.
- **Switch Name:** Displays the name of the automated switching device used in FSD Stage 2 for the faulted feeder.
- **Switch ID:** Displays the ID of the automated switching device used in FSD Stage 2 for the faulted feeder.

The following information is shown in the Feeder Sections tabular display:

- **Status**: FSD Stage 2 processing status for the feeder automated switch (None, Ineligible, Pending, Executing, Not Faulted, or Faulted).
- **Feeder Section:** Displays the feeder section ID.
- **Upstream Switch Name:** Displays the name of the upstream automated switching device for the feeder section.
- **Upstream Switch ID:** Displays the ID of the upstream automated switching device for the feeder section.
- **Downstream Switch Name(s):** Displays the name(s) of the downstream automated switching device(s) for the feeder section. Multiple devices are displayed as a comma-separated string.
- **Downstream Switch ID(s):** Displays the ID(s) of the downstream automated switching device(s) for the feeder section. Multiple devices are displayed as a comma-separated string.

Controls:

- **Mark Devices:** Marks the feeder section's upstream and downstream automated switching devices in the network view.

The following information is shown in the FSD Stage 2 Summary section of the FSD Stage 2 display.

- **Load (kW)**: The total native real power demand (load) before and after the FSD Stage 2 plan implementation.
- **Number of Customers:** The total number of the energized customers before and after the FSD Stage 2 plan implementation.
- **Number of Critical Customers:** The total number of the energized critical customers before and after the FSD Stage 2 plan implementation.
- **Generation (kW):** The total real power generation from generators and energy resources before and after the FSD Stage 2 plan implementation.

3.10. Traced Load and Generation

The Traced Load and Generation display shows total load and generation for the traced part of the network or downstream of the selected switching device or transformer. The display can be called using the Traced Load and Generation option from the geographic network view (empty space) context menu. This display can also be called using the Downstream Load and Generation option from the context menu for a switching device or transformer.

Display call-up URL:

- `netview://<Mode>/POPUP/DSLOAD/`

or:

- `netview://<Mode>/POPUP/DSLOAD/<Device DNOM Object Type>/<Substation Name>/<Device ID>/TRACEDOWN`

Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

For example:

- `netview://RT/POPUP/DSLOAD/DISTF/RAVEN/6-7412-1226-0-4/TRACEDOWN`

The following information is shown on this display:

- **Total Native Load:** Total native load for traced loads. The counter next to the Total Native Load field shows the number of traced loads. Click Total Native Load to open the Load display, which shows load data for traced loads.
  - **Total Native Load Real Power:** Total native load real power for phases A, B, C, and ABC in kW.
  - **Total Native Load Reactive Power:** Total native load reactive power for phases A, B, C, and ABC in kVAR.
  - **Current Phase A/B/C:** Total native load current on secondary side of Distribution Transformer for Phase A, B, and C in Amps.
- **Total Generation:** Total generation for traced generators, energy resources, and loads with embedded generation. The counter next to the Total Generation field shows the number of traced devices. Click Total Generation to open the Generation display, which shows generation for traced devices.
  - **Total Real Power:** Total generation real power for phases A, B, C, and ABC in kW.
  - **Total Reactive Power:** Total generation reactive power for phases A, B, C, and ABC in kVAR.
  - **Total State of Charge:** Total energy resource state of charge for phases A, B, C, and ABC in kWh.
  - **Current Phase A/B/C:** Total generation currents for Phase A, B, and C in Amps.

3.10.1. Load

The Load display shows load data for the specified loads. To access this display, click Total Native on the Traced Load and Generation display.

Display call-up URL:

- `netview://<Mode>/gridtabular/DownstreamLoad_GridDisplay/<Load_ID_1,Load_ID_N>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the load.
- **ID:** Unique identifier of the load.
- **Real Power Phase A/B/C:** Phase A, B, and C real power in kW.
- **Total Real Power:** Total real power in kW.
- **Reactive Power Phase A/B/C:** Phase A, B, and C reactive power in kVAR.
- **Total Reactive Power:** Total reactive power in kVAR.
- **Current Phase A/B/C:** Phase A, B, and C current in Amps.

Navigation:

- "Locate" locates the load on the geographic view.

3.10.2. Generation

The Generation display shows generation for the specified generators and energy resources. To access this display, click Total Generation on the Traced Load and Generation display.

Display call-up URL:

- `netview://<Mode>/gridtabular/DownstreamGeneration_GridDisplay/<EnergyResource_or_Generator_ID_1,EnergyResource_or_Generator_ID_N>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

The following information is shown on this display:

- **Name:** Name of the generator or energy resource.
- **ID:** Unique identifier of the generator or energy resource.
- **Real Power Phase A/B/C:** Phase A, B, and C real power in kW.
- **Total Real Power:** Total real power in kW.
- **Reactive Power Phase A/B/C:** Phase A, B, and C reactive power in kVAR.
- **Total Reactive Power:** Total reactive power in kVAR.
-

Current Phase A/B/C: Phase A, B, and C current in Amps.

- **Type:** Device type, such as generator or energy resource.
- **Energy Rating:** The useable capacity of the DER storage device, in kWh.
- **State of Charge:** The amount of charge that is stored in the DER storage device, in kWh.

Navigation:

- "Locate" locates the generator or energy resource on the geographic view.

3.11. Downstream DER for Distribution Transformer

The Downstream DER for Transformer display shows the details for distributed energy resources (DERs) that are located downstream of a distribution transformer.

To open this display, select the Downstream DER option from the distribution transformer context menu on the network view.

Display call-up URL:

- `netview://<Mode>/POPUP/ENERGYRES/<Substation Name>/<Substation Name>/<Distribution Transformer ID>/<Distribution Transformer ID>`

  Where, <Mode> can be RT (Real-Time), ST (Study), and SIM (Simulator).

For example:

- `netview://rt/POPUP/ENERGYRES/Neptune/Neptune/21222502/21222502`

The Downstream DER display is defined by the `DERDistTF_GridDisplay.xml` grid tabular display file.

The following information is shown on this display:

- **Real Time Power Transformer:** Energizing power transformer name.
- **Real Time Feeder:** Energizing feeder name.
- **Name:** Name of the energy resource.
- **Connected:** Connection status (Connected or Disconnected).
- **Aggregator:** Aggregator name.
- **Type:** DER device type, such as Wind or Solar.
- **Phasing:** DER device phasing.
- **Enrolled Program:** Enrolled programs of the DER.
- **Capacity (kVA):** Capacity in kVA.
- **Real Power Phase A/B/C:** The real power value at phases A, B, and C in kW.
- **Total Real Power (kW):** The total real power in kW.
- **Reactive Power Phase A/B/C:** The reactive power value at phases A, B, and C in kVAR.

- **Reactive Power Phase A/B/C:** The reactive power value at phases A, B, and C in kVAR.
- **Total Reactive Power (kVAR):** The total reactive power in kVAR.
- **Real Power Target (kW):** Real power target in kW.
- **Reactive Power Target (kVAR):** Reactive power target in kVAR.

Navigation:

- "Locate" navigates to the energy resource on the network view.
- "Analyst Attributes" displays the Analyst Attributes pop-up.
- "Analyst Outputs" displays the Analyst Output pop-up.
- "Operator Attributes" displays the Operator Attributes pop-up.
- "Operator Outputs" displays the Operator Output pop-up.

3.12. DPF CBO Results Pop-Up

The Check Before Operate Results popup shows the power flow and limit monitor results after the DPF check has been completed. This display automatically pops up after DPF CBO completes.

The Check Before Operate Results popup includes the following tabs:

- **Summary:** Shows the maximum loading, maximum voltage, and minimum voltage on the feeder(s) before and after operation. Loading and voltage violations are shown in red. For the operation of tie switches between two feeders, the information for the two feeders is summarized. For other device operations, only the information for the device’s associated feeder is summarized. The following information is shown on this tab:
    - **Max Loading % Before:** The highest loading that exists for a branch network element on the feeder(s) before operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
    - **Max Voltage % Before:** The highest voltage that exists for loads and buses on the feeder(s) before operation, expressed as a percentage of the nominal voltage. If there is a high voltage violation on the feeder(s), the cell background is colored red.
    - **Min Voltage % Before:** The lowest voltage that exists for loads and buses on the feeder(s) before operation, expressed as a percentage of the nominal voltage. If there is a low voltage violation on the feeder(s), the cell background is colored red.
    - **Max Loading % After:** The highest loading that exists for a branch network element on the feeder(s) after operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
    - **Max Voltage % After:** The highest voltage that exists for loads and buses on the feeder(s) after operation, expressed as a percentage of the nominal voltage. If there is a high voltage violation on the feeder(s), the cell background is colored red.

    - **Min Voltage % After:** The lowest voltage that exists for loads and buses on the feeder(s) after operation, expressed as a percentage of the nominal voltage. If there is a low voltage violation on the feeder(s), the cell background is colored red.
- **Switch:** Shows the maximum loading, current, real power, and reactive power through the switch and the voltages at the two sides of the switch before and after operation. This tab appears for switching device status changes. The following information is shown on this display:
    - **Max Loading % Before:** The highest loading that exists for the switch before operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
    - **Current Phase A/B/C (Amps) Before:** The current flowing through the switch before operation for phases A, B, and C in Amps.
    - **Real Power Phase A/B/C (kW) Before:** The real power flowing through the switch before operation for phases A, B, and C in kW.
    - **Reactive Power Phase A/B/C (kVAR) Before:** The reactive power flowing through the switch before operation for phases A, B, and C in kVAR.
    - **Voltage Phase A/B/C (%) Side 1 Before:** The voltage at the side 1 node of the switch before operation for phases A, B, and C in percentage of nominal voltage.
    - **Voltage Phase A/B/C (%) Side 2 Before:** The voltage at the side 2 node of the switch before operation for phases A, B, and C in percentage of nominal voltage.
    - **Max Loading % After:** The highest loading that exists for the switch after operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
    - **Current Phase A/B/C (Amps) After:** The current flowing through the switch after operation for phases A, B, and C in Amps.
    - **Real Power Phase A/B/C (kW) After:** The real power flowing through the switch after operation for phases A, B, and C in kW.
    - **Reactive Power Phase A/B/C (kVAR) After:** The reactive power flowing through the switch after operation for phases A, B, and C in kVAR.
    - **Voltage Phase A/B/C (%) Side 1 After:** The voltage at the side 1 node of the switch after operation for phases A, B, and C in percentage of nominal voltage.
    - **Voltage Phase A/B/C (%) Side 2 After:** The voltage at the side 2 node of the switch after operation for phases A, B, and C in percentage of nominal voltage.
- **Capacitor:** Shows the real power, reactive power, and current output from the capacitor and voltages at the connection node of the capacitor before and after operation. This tab appears for capacitor status changes. The following information is shown on this display:
    - **Current Phase A/B/C (Amps) Before:** The current injected by the capacitor before operation for phases A, B, and C in Amps.

  - **Real Power Phase A/B/C (kW) Before:** The real power consumed by the capacitor before operation for phases A, B, and C in kW.
  - **Reactive Power Phase A/B/C (kVAR) Before:** The reactive power injected by the capacitor before operation for phases A, B, and C in kVAR.
  - **Voltage Phase A/B/C (%) Side 1 Before:** The voltage at the connection node of the capacitor before operation for phases A, B, and C in percentage of nominal voltage.
  - **Current Phase A/B/C (Amps) After:** The current injected by the capacitor after operation for phases A, B, and C in Amps.
  - **Real Power Phase A/B/C (kW) After:** The real power injected by the capacitor after operation for phases A, B, and C in kW.
  - **Reactive Power Phase A/B/C (kVAR) After:** The reactive power injected by the capacitor after operation for phases A, B, and C in kVAR.
  - **Voltage Phase A/B/C (%) Side 1 After:** The voltage at the connection node of the capacitor after operation for phases A, B, and C in percentage of nominal voltage.
- **Regulation Tap:** Shows the maximum loading, real power, reactive power, and current through the tap changing transformer and the voltages at the two sides of the tap changing transformer for the regulation tap before and after operation. This tab appears for tap position operation of tap changing transformers. The following information is shown on this tab:
  - **Max Loading % Before:** The highest loading that exists for the tap changing transformer before operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
  - **Current Phase A/B/C (Amps) Before:** The current flowing through the tap changing transformer before operation for phases A, B, and C in Amps.
  - **Real Power Phase A/B/C (kW) Before:** The real power flowing through the tap changing transformer before operation for phases A, B, and C in kW.
  - **Reactive Power Phase A/B/C (kVAR) Before:** The reactive power flowing through the tap changing transformer before operation for phases A, B, and C in kVAR.
  - **Voltage Phase A/B/C (%) Side 1 Before:** The voltage at the side 1 node of the tap changing transformer before operation for phases A, B, and C in percentage of nominal voltage.
  - **Voltage Phase A/B/C (%) Side 2 Before:** The voltage at the side 2 node of the tap changing transformer before operation for phases A, B, and C in percentage of nominal voltage.
  - **Max Loading % After:** The highest loading that exists for the tap changing transformer after operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
- **Feeder 1:** Shows the current, real power, and reactive power through the feeder head branch and the maximum and minimum voltages on the first feeder before and after the device operation. The following information is shown on this display:
  - **Current Phase A/B/C (Amps) Before:** The current flow through the feeder head branch

before operation for phases A, B, and C in Amps.

- **Real Power Phase A/B/C (kW) Before:** The real power flow through the feeder head branch before operation for phases A, B, and C in kW.
- **Reactive Power Phase A/B/C (kVAR) Before:** The reactive power flow through the feeder head branch before operation for phases A, B, and C in kVAR.
- **Maximum Feeder Voltage Phase A/B/C (%) Before:** The maximum voltage on the first feeder before operation for phases A, B, and C in percentage of nominal voltage.
- **Minimum Feeder Voltage Phase A/B/C (%) Before:** The minimum voltage on the first feeder before operation for phases A, B, and C in percentage of nominal voltage.
- **Current Phase A/B/C (Amps) After:** The current flow through the feeder head branch after operation for phases A, B, and C in Amps.
- **Real Power Phase A/B/C (kW) After:** The real power flow through the feeder head branch after operation for phases A, B, and C in kW.
- **Reactive Power Phase A/B/C (kVAR) After:** The reactive power flow through the feeder head branch after operation for phases A, B, and C in kVAR.
- **Maximum Feeder Voltage Phase A/B/C (%) After:** The maximum voltage on the first feeder after operation for phases A, B, and C in percentage of nominal voltage.
- **Minimum Feeder Voltage Phase A/B/C (%) After:** The minimum voltage on the first feeder after operation for phases A, B, and C in percentage of nominal voltage.

- **Feeder 2:** Shows the current, real power, and reactive power through the feeder head branch and the maximum and minimum voltages on the second feeder before and after the device operation. This tab applies to open tie switches between two feeders. The following information is shown on this display:
  - **Current Phase A/B/C (Amps) Before:** The current flow through the feeder head branch before operation for phases A, B, and C in Amps.
  - **Real Power Phase A/B/C (kW) Before:** The real power flow through the feeder head branch before operation for phases A, B, and C in kW.
  - **Reactive Power Phase A/B/C (kVAR) Before:** The reactive power flow through the feeder head branch before operation for phases A, B, and C in kVAR.
  - **Maximum Feeder Voltage Phase A/B/C (%) Before:** The maximum voltage on the second feeder before operation for phases A, B, and C in percentage of nominal voltage.
  - **Minimum Feeder Voltage Phase A/B/C (%) Before:** The minimum voltage on the second feeder before operation for phases A, B, and C in percentage of nominal voltage.
  - **Current Phase A/B/C (Amps) After:** The current flow through the feeder head branch after operation for phases A, B, and C in Amps.
  - **Real Power Phase A/B/C (kW) After:** The real power flow through the feeder head branch after operation for phases A, B, and C in kW.
  - **Reactive Power Phase A/B/C (kVAR) After:** The reactive power flow through the feeder head branch after operation for phases A, B, and C in kVAR.

  - **Maximum Feeder Voltage Phase A/B/C (%) After:** The maximum voltage on the second feeder after operation for phases A, B, and C in percentage of nominal voltage.
  - **Minimum Feeder Voltage Phase A/B/C (%) After:** The minimum voltage on the second feeder after operation for phases A, B, and C in percentage of nominal voltage.
- **Affected Devices:** Shows details for a user-configured number of devices that are affected, including the maximum branch loading levels, high voltages, and low voltages before and after the device operation. The following information is shown on this display:
  - **Device Type and ID:** The type and ID of the affected device.
  - **Max Loading % Before:** The highest loading of the affected branch device before operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
  - **Max Voltage % Before:** The highest voltage of the affected loads and buses before operation, expressed as a percentage of the nominal voltage. If there is a high voltage violation on the feeder(s), the cell background is colored red.
  - **Min Voltage % Before:** The lowest voltage of the affected loads and buses before operation, expressed as a percentage of the nominal voltage. If there is a low voltage violation on the feeder(s), the cell background is colored red.
  - **Max Loading % After:** The highest loading of the affected branch device after operation, expressed as a percentage of the branch flow limits (flow/limit * 100). If the loading exceeds the limit, the cell background is colored red. The highest overload level is displayed for a violation.
  - **Max Voltage % After:** The highest voltage of the affected loads and buses after operation, expressed as a percentage of nominal voltage. If there is a high voltage violation on the feeder(s), the cell background is colored red.
  - **Min Voltage % After:** The lowest voltage of the affected loads and buses after operation, expressed as a percentage of nominal voltage. If there is a low voltage violation on the feeder(s), the cell background is colored red.

4. ADMS Attribute and Output Pop-Ups

This chapter contains descriptions of the standard attribute and output pop-ups provided by ADMS.

4.1. Analyst Attributes Pop-Ups

4.1.1. Autotransformer

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the transformer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the transformer is connected nominally.
- **Normally Looped:** Indicates if the transformer is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the transformer is normally de-energized (Yes/No).
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABCN for A, B, and C phases and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the transformer is connected.

- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Rating:** The total capacity of the autotransformer (MVA).
- **Type:** The transformer type.
- **Description:** Description of the transformer.
- **Model Name:** The name of the transformer model.
- **Nominal Core Loss (kW):** The nominal losses of the transformer core at rated load, expressed in kilowatts.
- **Total Impedance – R + jX (p.u. on Transformer Power Base):** The effective impedance of the transformer expressed in per-unit values, R+jX (for example, 0.0126 + j0.024).
- **Excitation Current (%):** The excitation current of the transformer expressed as a percentage of rated current.
- **Magnetizing Power Factor:** Magnetization power factor. This field is optional if the excitation current (%) value is provided.
- **Configuration Index:** Transformer model type and index. For example, StarStarX (1).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the transformer winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Winding Type:** The transformer winding type (Delta, Wye, Open).
- **Grounded:** Indicates if the transformer is grounded (Yes/No).
- **Grounding Impedance - R + jX (ohm):** The transformer grounding impedance expressed in ohms.
- **Winding Lags Primary Phase (Deg):** Winding lags primary phase in degrees.
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Minimum Tap Position:** Transformer primary/secondary side minimum tap position.
- **Nominal Tap Position:** Transformer primary/secondary side tap nominal position.
- **Maximum Tap Position:** Transformer primary/secondary side tap maximum position.
- **Tap Step Size (%):** The size of the tap step expressed in percentage.
- **Target Voltage:** Regulating transformer voltage target (per unit or configurable voltage base).

- **Tap Movement Cost:** Tap movement cost.
- **Phases Sensed:** The sensed phases (local regulation).
- **Two Sided Bandwidth:** The regulation two-sided voltage bandwidth expressed as a per-unit value. This refers only to devices in local control mode.
- **Line Drop Compensation Resistance (V):** Regulating transformer line drop compensation resistance (V).
- **Line Drop Compensation Reactance (V):** Regulating transformer line drop compensation reactance (V).
- **PT Ratio:** Regulating transformer PT ratio.
- **CT Ratio:** Regulating tap CT ratio.
- **CT Secondary Rated Amps:** The current rate in the secondary winding.
- **Voltage Reduction Capable:** Indicates if the transformer is capable of voltage reduction.
- **Voltage Reduction Value (%):** The voltage reduction value.

Measurement Data:

- **Tap Position (Regulating Tap):** Regulating tap position measurement.
- **Remote Control Status:** Flag (Yes/No) indicating remote control of the transformer. If Yes, the transformer is controlled remotely. If No, the transformer is controlled locally.
- **Regulation Status:** Flag indicating if the regulation capability of the device is enabled.
- **Voltage Reduction Status:** Flag indicating if the voltage reduction is available.

4.1.2. Breaker

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the breaker.
- **ID:** Unique identifier of the breaker.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the breaker.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the breaker belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the breaker is connected nominally.
- **Normally Looped:** Indicates if the breaker is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the breaker is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.

- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the breaker forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Station Boundary Device:** Indicates if the breaker forms a boundary between two stations. Can be Yes or No.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the breaker phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Team Names:** Names of automated scheme teams that the breaker is associated with.
- **Team IDs:** IDs of automated scheme teams that the breaker is associated with.
- **Scheme Name:** Name of the automated scheme the breaker is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the breaker is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the breaker is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Breaker’s current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the breaker.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.

- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the breaker is pending the change of normal position (Yes/No).
- **Dynamic Phases:** Dynamic phases information.
  - **Has Dynamic Phase Mismatch:** Indicates that the dynamic phases on either side of the breaker do not match.
  - **Has Phase Angle Mismatch:** Indicates that a phase angle difference (as calculated by DPF) is larger than the configured limit.
  - **Has Nominal Voltage Mismatch:** Indicates that the nominal voltage on either side of the breaker is different.

Engineering Data:

- **Model Name:** The name of the breaker model.
- **Size (Amps):** The maximum interruption rating of the breaker (amps).
- **Type:** The breaker type.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the breaker.
- **Switched Phase Independently:** Yes or No value.
- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the breaker is eligible for reconfiguration in FISR application.
- **Participating in Local Control Scheme:** Indicates if the breaker participates in the local control scheme.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by the Distribution Power Flow (DPF) and the flow direction is determined.
- **Ground Trip Capable:** Indicates whether the recloser can trip on a ground fault (Yes/No).
- **Ground Trip Status:** Normal status of ground trip (Enabled/Disabled).
- **Phase Trip Current (Amps):** Phase trip current setting in amps.
- **Ground Trip Current (Amps):** Ground trip current setting in amps.
- **Alternative Phase Trip Current (Amps):** Alternative phase trip current setting in amps.
- **Alternative Ground Trip Current (Amps):** Alternative ground trip current setting in amps.
- **Neutral Relay Setting (Amps):** The maximum allowed neutral trip setting in amps.
- **Interrupting Capacity (Amps):** Interrupting capacity in amps.
- **PRV Safety Factor:** Safety factor used to calculate minimum phase and ground pick-up limit.

- **Ground and Neutral Relay Safety Factor:** Safety factor used when comparing the phase imbalance flow (the vector sum of currents in the three phases) for breakers and reclosers to the minimum ground relay pick-up setting and neutral relay current setting.

Hold Fault Indication Information:

- **Hold Fault Indication Capable:** Specifies whether the breaker is capable of holding fault indication (defined in the static model).
- **Hold Fault Indication Active:** Specifies whether the associated fault indicator's SET status is currently being held. This field is hidden if the breaker is not capable of holding fault indication.

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag indicating the state of the breaker based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Hot Line Tag Status:** Flag indicating the hot line tag status.
- **Alarm Status:** Flag indicating the alarm status of the breaker.
- **Remote Control Status:** Flag indicating the remote control of the breaker. If Yes, the breaker is controlled remotely. If No, the breaker is controlled locally.
- **Reclosing Status:** Flag indicating if the reclosing capability of the device is enabled.
- **Lockout Status:** Flag indicating the lockout status of the breaker.
- **Prohibit Restoration Status:** Flag indicating the prohibit restoration status of the breaker.
- **Fault Indication Measurement:** Breaker fault indicator status measurements and measurement quality per phase.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.

4.1.3. Bus

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the bus.
- **ID:** Unique identifier of the bus.
- **Service Center:** The name of the service center.

- **Permission Area:** Permission area the bus belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the bus is connected at normal conditions.
- **Normally Looped:** Indicates if the bus is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the bus is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the bus phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the bus is connected.
- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the bus in KV.
- **Voltage Base:** Voltage base value in Volts.
- **Normal Operational Limit - High:** Normal operational high voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Operational Limit - Low:** Normal operational low voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Operational Limit - High:** Emergency operational high voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Operational Limit - Low:** Emergency operational low voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Optimization Limit - High:** Normal optimization high voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Optimization Limit - Low:** Normal optimization low voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Optimization Limit - High:** Emergency optimization high voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Optimization Limit - Low:** Emergency optimization low voltage limit, expressed in

per-unit or as configurable voltage base.

Measurement Data:

- **Associated Bus Voltage Magnitude Measurement Phase A/B/C:** Phase A, B, and C voltage magnitude at the bus.
- **Angle Measurement Phase A/B/C:** Phase A, B, and C phase angle at the bus.

4.1.4. Capacitor

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the capacitor.
- **ID:** Capacitor unique identifier.
- **Structure ID:** Unique identifier of a structure that is associated with the capacitor.
- **Address:** Address of the capacitor.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the capacitor belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the capacitor is connected at normal conditions.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the capacitor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the capacitor is connected.
- **Status:** Current (real-time) status of the capacitor (On or Off).
- **Current Operation Mode:** The current operation mode of the controller, local or remote.
-

Dynamic Feeder is Looped: Yes or No value.

- **Number of Daily Switching Operations:** The number of switching operations that were performed on a capacitor within a 24-hour period.

Engineering Data:

- **Total Size (kVAR):** The total capacity of the capacitor (kVAR).
- **Control Type:** Control type of the capacitor (for example, remote only).
- **Local Control Type:** The local control mode for the capacitor (None, Power Factor, Time, Temperature, VAR, Current, or Voltage).
- **Communication Control Mechanism:** Communication control mechanism of the capacitor.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the capacitor in kV.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages of the capacitor.
- **Nominal Real Power (W):** The nominal real power rate in W.
- **Nominal Reactive Power (kVAR):** The nominal reactive power rate in kVAR.
- **Model Name:** The shunt impedance model described in the Global file that is used to define the characteristics of this capacitor bank.
- **Connection Type**: The connection arrangement of the capacitors (Wye or Delta).
- **Grounded:** Indicates if the capacitor is grounded (Yes/No).
- **Number of Steps:** Number of steps it takes to change the capacitor. A fixed bank has one step. A switched bank that is either all on or all off also has one step.
- **Step Size (kVAR):** The size of the steps in which the capacitor bank can be switched, expressed in kVAR.
- **Cell Size (kVAR):** The size of the individual units that make up the capacitor bank, expressed in kVAR.
- **Time Between Consecutive Operations (Minutes):** The minimum elapsed time between consecutive switching operations on the capacitor, expressed in minutes.
- **Time of Last Switching Operation:** Time of the last switching operation.
- **Time Delay (Minutes):** Controller time delay.
- **Maximum Number of Daily Switching Operations:** The maximum number of daily switching operations for the capacitor.
- **Capacitor Movement Cost:** Cost of a capacitor switching operation.
- **Local Controller Type:** The type of local controller.
- **Phases Sensed:** The phases that are sensed for controlling the capacitor status (for example, A, B, C).
- **Time Regulation Schedule:** For time-controlled capacitor banks, the on/off time schedule as defined in the Global file.
- **Regulation Node ID:** The ID of the network node that is used as the sensing point for voltage

regulation by the controller.

- **Station Capacitor Regulates Transformer Node:** Indicates if the station capacitor can regulate the transformer node.
- **Station Capacitor Power Transformer:** The station capacitor power transformer.
- **Voltage Sensing (Line-to-Line):** For capacitors with local voltage control or voltage override, this field shows whether the regulation is based on the line-to-line or line-to-ground voltage. In cases where there is no local voltage control, this field is blank.
- **Regulation Branch ID:** The ID of the network branch that is used as the sensing point for VAR, Power Factor, and Current regulation by the controller.
- **Regulation Branch Type:** The type of network branch that is used as the sensing point for VAR, Power Factor, and Current regulation by the controller.
- **Regulation Branch End:** The regulation branch end.
- **Regulation Branch Direction:** The regulation branch direction.
- **On Level:** The level at which the capacitor is turned on, expressed in units of the local control type. This only applies when the device is in local control mode.
- **Off Level:** The level at which the capacitor is turned off, expressed in units of the local control type. This only applies when the device is in local control mode.
- **Voltage Override High Limit:** The voltage level at which the capacitor is turned off, expressed in per unit. This only applies when the device has voltage override.
- **Voltage Override Low Limit:** The voltage level at which the capacitor is turned on, expressed in per unit. This only applies when the device has voltage override.
- **Local Controller Override Type:** The override type of the local controller.
- **Override Schedule:** The override schedule.
- **Low Override:** The low override level.
- **High Override:** The high override level.
- **Local Controller Permissive Type:** The local controller permissive type.
- **Permissive Schedule:** The local controller permissive schedule.
- **Permissive Control Level 1:** The permissive control level 1 value.
- **Permissive Control Level 2:** The permissive control level 2 value.
- **Associated Switching Device:** The name (or ID if the name is not provided) of the associated switching device.
- **Associated Capacitor Bank:** The name (or ID if the name is not provided) of the capacitor bank to which this capacitor belongs.
- **Next On Capacitor of the Associated Capacitor Bank:** The name (or ID if the name is not provided) of the next on capacitor in the capacitor bank, based on the capacitor bank on sequence and the open/closed status of the capacitors in the capacitor bank.
- **Next Off Capacitor of the Associated Capacitor Bank:** The name (or ID if the name is not

provided) of the next off capacitor in the capacitor bank, based on the capacitor bank off sequence and the open/closed status of the capacitors in the capacitor bank.

Measurement Data:

- **Step Analog:** Step analog measurement.
- **Open/Closed Status:** Flag indicating the state of the capacitor based on measurement data.
- **Associated Step or Open/Closed Measurement:** The name of the associated step or the open/closed measurement.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Remote Control Status:** Flag indicating the remote control status of the capacitor. If Yes, the capacitor is controlled remotely. If No, the capacitor is controlled locally.
- **Auto Control Status:** Auto control status.
- **Control Available Status:** Indicator showing that the device is eligible for control based on an available status measurement (Application Eligibility - SCADA Decides). It works in real-time mode when SCADA measurements are being updated and the device has remote control capability.

4.1.5. Composite Switch

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the composite switch.
- **ID:** Unique identifier of the composite switch.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the composite switch.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the switch belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the switch is connected nominally.
- **Normally Looped:** Indicates if the switch is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the switch is normally de-energized (Yes/No).
- **Normal Position:** Normal position for the ganged device.
- **Feeder Main Boundary Device:** Indicates if the switch forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Station Boundary Device:** Indicates if the switch forms a boundary between two stations. Can be Yes or No.

- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the switch phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the switch is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the switch is connected.
- **Currently Open Phase(s):** Phases that are currently opened by the switch.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Current Position:** Switch's current position (Open/Closed).
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent State Name:** The name of the pending permanent state.

Engineering Data:

- **Model Name:** The name of the model defined in the Global file that describes the characteristics of the switch.
- **Size:** The maximum interruption rating of the switch (amps).
- **Type:** Name of the type of the composite switch.
- **Composite Switch Type:** Associated composite switch model as defined in the Global resources file.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **State Name:** The names of states.
- **Throwover Capability:** The modes of operation available for the throwover operation.
- **Automatic Operation:** Indicates if the switch can operate automatically.

- **Time Delay (s):** The time delay of the throwover operation (seconds).
- **Master Switch:** In a coordinated switching arrangement, this is the ID of the switch that senses the loss of supply and initiates the throwover. The master switch sends the throwover message to the other switches in the scheme.
- **Master Switch Address:** Address of the master switch in the coordinated switching arrangement.
- **State Name:** The names given to the preferred and alternate states of the switch (for example, Main/Preferred/Side 1 or Standby/Alternate/Side 2).
- **Phases Measured:** The list of phases that are measured on the preferred and alternate sides of the switch to sense the presence of supply (for example, A, B, C).

Measurement Data:

- **Associated Status Measurement ID:** Associated state measurement.
- **Status:** Flag (Open/Closed) indicating state of the switch based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Supervisory Cutout Status:** Supervisory cutout state of the switch, set by the switch control. The dispatcher cannot control the switch in the supervisory cutout state.
- **Alarm Status:** Flag (Yes/No) indicating alarm status of the switch.
- **Remote Control Status:** Flag (Yes/No) indicating remote control of the switch. If Yes, the switch is controlled remotely. If No, the switch is controlled locally.

4.1.6. Cut

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the cut.
- **ID:** Unique identifier of the cut.
- **Permission Area:** Permission area the cut belongs to.
- **Original Component ID:** ID of the original component where the cut is placed.
- **Original Component Type ID:** Type of the original component where the cut is placed.
- **Type:** Type of the cut.
- **Normal Station:** The name of the nominal substation the cut belongs to.
- **Normal Feeder:** The name of the feeder that is normally connected to the cut.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Distance from Node 1:** Distance from Node 1.

- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the cut is connected.
- **Feeder:** The name of the feeder to which the cut is connected.
- **Dynamic Phases:** Dynamic phases information.
  - **Has Dynamic Phase Mismatch:** Indicates that the dynamic phases on either side of the cut do not match.
  - **Has Phase Angle Mismatch:** Indicates that a phase angle difference (as calculated by DPF) is larger than the configured limit.

4.1.7. Distribution Transformer

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Structure ID:** Unique identifier of a structure that is associated with the transformer.
- **Address:** Address of the transformer.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the transformer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the transformer is connected nominally.
- **Normally Looped:** Indicates if the transformer is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the transformer is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Elbow Switches:** Elbow switches connected to the transformer.

- **Critical Customer Connected:** Indicates whether the transformer supplies critical customers (Yes or No).
- **Total Customers Connected:** The number of customers connected to phases A, B, C, and ABC.
- **Number of Critical Customers:** The number of critical customers on attached loads.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the transformer is connected.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Rating:** The total capacity of the transformer (kVA).
- **Type:** The type of transformer.
- **Description:** Description of the transformer.
- **Model Name:** The name of the transformer model.
- **Nominal Core Loss:** The nominal losses of the transformer core at rated load, expressed in kilowatts.
- **Total Impedance – R + jX (p.u. on Transformer Power Base):** The effective impedance of the transformer, expressed in per-unit values, R+jX (for example, 0.0126 + j0.024).
- **Excitation Current (%):** The excitation current of the transformer, expressed as a percentage of rated current.
- **Magnetizing Power Factor:** Magnetization power factor. This field is optional if the excitation current (%) value is provided.
- **Configuration Index:** Transformer model type and index. For example, Type6SimpleX (12).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the transformer winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Winding Type:** The transformer winding type (Delta, Wye, Open).
- **Grounded:** Indicates if the transformer is grounded (Yes/No).
-

Grounding Impedance - R + jX (ohm): The transformer grounding impedance expressed in ohms.

- **Winding Lags Primary Phase (Deg):** Winding lags primary phase in degrees.
- **Delta Secondary Grounding Phase:** Grounded phase winding for a transformer with a delta connected secondary.

4.1.8. DuctBank

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the selected duct bank.
- **ID:** The unique identifier of the selected duct bank.
- **Permission Area:** Permission area the duct bank belongs to.

4.1.9. Elbow

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the switch.
- **ID:** Switch unique identifier.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the switch.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the switch belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the switch is connected nominally.
- **Normally Looped:** Indicates if the switch is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the switch is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the switch forms a boundary between the external

interface and a feeder. Can be Yes or No.

- **Station Boundary Device:** Indicates if the switch forms a boundary between two stations. Can be Yes or No.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the commissioning project that is associated with the device.
- **Project Stage:** The device commissioning or decommissioning stage. You can use this field to manually set or update the project stage.
- **Constructed Phase(s):** Enumeration of the switch phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the switch is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the switch is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Switch’s current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the switch.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the switch is pending the change of normal position (Yes/No).

Engineering Data:

- **Model Name:** The name of the switch model.
- **Size (Amps):** The maximum interruption rating of the switch (amps).

- **Type:** The type of switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Switched Phase Independently:** Yes or No value.
- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the switch is eligible for reconfiguration in FISR application.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag indicating the state of the switch based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Supervisory Cutout Status:** Supervisory cutout state of the switch, set by the switch control. The dispatcher cannot control the switch in the supervisory cutout state.
- **Alarm Status:** Flag indicating the alarm status of the switch.
- **Remote Control Status:** Flag (Yes/No) indicating remote control of the switch. If Yes, the switch is controlled remotely. If No, the switch is controlled locally.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.

4.1.10. Energy Resource

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the energy resource.
- **ID:** Unique identifier of the energy resource.
- **Address:** Address of the energy resource.
- **Aggregator:** Aggregation group name.
- **Service Center:** The name of the service center.
-

Permission Area: Permission area the energy resource belongs to.

- **Normal Station and Feeder:** The name of the substation and feeder to which the energy resource is connected at normal conditions.
- **Normally De-Energized:** Indicates if the energy resource is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the energy resource phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.
- **Regulation Node:** ID of the regulation node.
- **Ownership:** Energy resource type ownership type. For distributed energy resources only.
- **Device Type:** Energy resource type (for example, Solar, Wind).
- **Enrolled Program:** Enrolled programs of the energy resource.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the energy resource is connected.
- **Last Disconnection Time:** The time when the last disconnection occurred (for disconnected energy resources).
- **Last Disconnect Reason:** The reason for the last disconnection (for disconnected energy resources).
- **Inverter ID:** The ID of the associated inverter.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the energy resource in kV.
- **Energy Resource Model Name:** Name of the energy resource model as defined in the Global file that is used to describe the characteristics of the energy resource.
- **KVA Size:** Size of the energy resource in kVA.
- **Rated Power Factor:** Rated power factor.
- **Voltage Schedule Name:** Name of the voltage schedule as defined in the Global file that is used to describe the voltage output characteristics of the energy resource.

- **Real Power Schedule Name:** Name of the real power schedule as defined in the Global file that is used to describe the real power output characteristics of the energy resource.
- **Reactive Power Schedule Name:** Name of the reactive power schedule as defined in the Global file that is used to describe the reactive power output characteristics of the energy resource.
- **Slack Bus:** Slack generating unit status.
- **Type:** The energy resource type.
- **Primary Source:** Can be Yes or No.
- **Wye Connected:** Can be Yes or No.
- **Grounded:** Can be Yes or No.
- **Contributes To Fault Current:** Specifies whether the device can remain connected during a fault and contribute to fault current. This is considered during SSC or PRV calculations.
- **Voltage Target (%):** Voltage target in percentage.
- **Voltage Regulation is Line to Line:** Can be Yes or No.
- **Phases Sensed:** The phases that are sensed for controlling the energy resource output (for example, A, B, C).
- **Control Capability:** Control capability of the energy resource.
- **Excitation Mode:** Excitation mode of the energy resource.
- **Connection Type:** Connection type of the energy resource.
- **Real Power Target (kW):** Real power target expressed in kW.
- **Reactive Power Target (kVAR):** Reactive power target expressed in kVAR.
- **High Disconnect Voltage Limit:** High voltage disconnect limit. Expressed as a 120 V base value or per-unit value depending on the energy resource model used.
- **Low Disconnect Voltage Limit:** Low voltage disconnect limit. Expressed as a 120 V base value or per-unit value depending on the energy resource model used.
- **LVM Control Type:** LVM control type, such as PF Control or VAR Control.
- **Advanced Applications Eligibility Override:** Can be Yes or No.
- **Maximum Voltage Before Separation From the Network:** Maximum voltage before separation from the network (per unit).
- **Minimum Voltage Before Separation From the Network:** Minimum voltage before separation from the network (per unit).
- **Maximum Frequency Before Separation From the Network:** Maximum frequency before separation from the network (Hz). By default, 63 Hz.
- **Minimum Frequency Before Separation From the Network:** Minimum frequency before separation from the network (Hz). By default, 56 Hz.
- **Planned Restoration Time (sec):** The planned restoration time for a DER/DG after re-energization of a feeder, in seconds.

- **Automatic Reconnect Capable:** Specifies whether the energy resource can automatically reconnect to the system when the system voltage is restored.
- **Minimum State of Charge:** The amount of charge below which the device should never be drawn, in kWh.
- **Maximum State of Charge:** The maximum amount of charge that can be stored in the DER storage device, in kWh.
- **Energy Rating:** The useable capacity of the DER storage device, in kWh.

DER Statuses:

- **Connection Status:** Connection status (Connected or Disconnected).
- **Communication Status:** Communication status (Online, Suspect, Offline, or Unknown).
- **Alarm Status:** Alarm status (for example, None or DER Fault Over Voltage).
- **Connection Source:** DER connection source (SCADA, 2030.5 Gateway, or None).

Dynamic Ratings:

- **Maximum Charge Rate:** Maximum charge rate in kW.
- **Maximum Discharge Rate:** Maximum discharge rate in kW.
- **Maximum Real Power:** Maximum real power in kW.
- **Maximum Reactive Power:** Maximum reactive power in kVAR.
- **Maximum Negative Reactive Power:** Maximum negative reactive power in kVAR.
- **Maximum Apparent Power:** Maximum apparent power in kVA.
- **Minimum Power Factor – Overexcited:** Minimum power factor when overexcited.
- **Minimum Power Factor – Underexcited:** Minimum power factor when underexcited.

Measured Values:

- **Total Real Power:** Three-phase real power in kW.
- **Real Power Phase A/B/C:** Real power for phases A, B, and C in kW.
- **Total Reactive Power:** Three-phase reactive power in kVAR.
- **Reactive Power Phase A/B/C:** Reactive power for phases A, B, and C in kVAR.
- **Voltage Phase A/B/C:** Voltage in kV LG.
- **Frequency:** Dynamic DER three-phase frequency value from a measurement.
- **State of Charge:** The amount of charge that is currently stored in the DER storage device, in kWh. This value is provided via the Supervisory Control and Data Acquisition (SCADA) system or the 2030.5 Gateway.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the energy resource (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the energy resource (kVAR).

4.1.11. Fault Indicator

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the fault indicator.
- **ID:** Fault indicator unique identifier.
- **Address:** Fault indicator address.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the fault indicator belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the fault indicator is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the fault indicator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the connection node.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the real-time substation and feeder the fault indicator belongs to.
- **Status:** Current (real-time) status of the fault indicator: Set, Unset, or Unknown for a non-directional fault indicator; Unset, Both, Side 1, Side 2, or Unknown for a type 1 directional fault indicator; Unset, Forward Set, Reverse Set, or Unknown for a type 2 directional fault indicator.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the fault indicator.
- **Rating (Amps):** The rated interruption capacity of the fault indicator (Amps).
- **Reset Mechanism:** The fault indicator’s reset mechanism (automatic or manual).
- **Directional:** Specifies whether the fault indicator is directional.
- **Disregard Phase Mismatch:** Specifies whether the phase-agnostic fault indication is applied. The fault indicator phase indication is not considered; the faulted phase is detected based on the device that triggers the fault. (Fault indicator phase indication may differ from the actual

SCADA faulted phase indication.)

Hold Fault Indication Information:

- **Hold Fault Indication Capable:** Specifies whether the fault indicator has an upstream breaker that is capable of holding fault indication (defined in the static model).
- **Hold Fault Indication Active:** Specifies whether the associated fault indicator's SET status is currently being held. This field is hidden if the upstream breaker is not capable of holding fault indication.

Measurement Data:

- **Fault Indicator Measurement:** Indicates phase where the fault was detected. The fault phase measurements are available for directional fault indicators; therefore, these fields are not applicable for the devices.

4.1.12. Feeder

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the feeder.
- **ID:** Unique identifier of the feeder.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **EMS NETMOM Load Name:** Load name.
- **Default Feeder Color:** Number of the default color used to display this feeder on the network view (expressed as an integer 1–16).
- **Feeder Head Node Name:** Name of the feeder head node.
- **Service Center:** The name of the service center associated with the feeder.
- **Feeder Evaluation Head Node Name:** The name of the evaluation node.

Engineering Data:

- **VOLTAGE:** Nominal voltage of the feeder in kV.
- **Number of Load Categories:** Number of load categories.

4.1.13. Fuse

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the fuse.
- **ID:** Fuse unique identifier.
- **Alternate ID:** Second unique identifier.
- **Structure ID:** Unique identifier of a structure that is associated with the fuse.
- **Address:** Address of the fuse.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the fuse belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the fuse is connected nominally.
- **Normally Looped:** Indicates if the fuse is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the fuse is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the fuse forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the fuse phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Loop Number:** Loop number of the fuse.
- **Solid Blade Fuse:** The Yes or No value.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the fuse is connected.

- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the fuse is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Fuse’s current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the fuse.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the fuse is pending the change of normal position (Yes/No).
- **Dynamic Phases:** Dynamic phases information.
    - **Has Dynamic Phase Mismatch:** Indicates that the dynamic phases on either side of the fuse do not match.
    - **Has Phase Angle Mismatch:** Indicates that a phase angle difference (as calculated by DPF) is larger than the configured limit.
    - **Has Nominal Voltage Mismatch:** Indicates that the nominal voltage on either side of the fuse is different.

Engineering Data:

- **Model Name:** The name of the fuse model.
- **Size (Amps):** The maximum interruption rating of the fuse (amps).
- **Type:** The type of fuse.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the fuse.
- **Switched Phase Independently:** Yes or No value.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the switch is eligible for reconfiguration in FISR application.
- **Incident Energy Category:** Displays the arc flash incident energy levels.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.
- **Protection Validation Boundary:** Specifies whether the fuse is a boundary in determining adaptive relaying Protection Validation (PRV) application zone.

4.1.14. Generator

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the generator.
- **ID:** Unique identifier of the generator.
- **Address:** Address of the generator.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the generator belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the generator is connected at normal conditions.
- **Normally De-Energized:** Indicates if the generator is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the generator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.
- **Regulation Node:** ID of the regulation node.
- **Device Type:** Generator or type (for example, Wind).
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the generator is connected.
- **Status:** Connection status (Connected/Disconnected).
- **Last Disconnection Time:** The time when the last disconnection occurred (for disconnected generators).
- **Last Disconnect Reason:** The reason for the last disconnection (for disconnected generators).
- **Inverter ID:** The ID of the associated inverter.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the generator in kV.
- **Generator Model Name:** Name of the generator model as defined in the Global file that is used

to describe the characteristics of the generator.

- **KVA Size:** Size of the generator in kVA.
- **Rated Power Factor:** Rated power factor.
- **Rated Amps:** The current rate.
- **Voltage Schedule Name:** Name of the voltage schedule as defined in the Global file that is used to describe the voltage output characteristics of the generator.
- **Real Power Schedule Name:** Name of the real power schedule as defined in the Global file that is used to describe the real power output characteristics of the generator.
- **Reactive Power Schedule Name:** Name of the reactive power schedule as defined in the Global file that is used to describe the reactive power output characteristics of the generator.
- **Slack Bus:** Slack generating unit status.
- **Type:** The generator type.
- **Primary Source:** Can be Yes or No.
- **Wye Connected:** Can be Yes or No.
- **Grounded:** Can be Yes or No.
- **Grounding Impedance (R+jX ohms):** The grounding impedance (in ohms).
- **Voltage Target (%):** Voltage target in percentage.
- **Voltage Regulation is Line to Line:** Can be Yes or No.
- **Phases Sensed:** The phases that are sensed for controlling the generator output (for example, A, B, C).
- **Real Power Target (kW):** Real power target expressed in kW.
- **Reactive Power Target (kVAR):** Reactive power target expressed in kVAR.
- **High Disconnect Voltage Limit:** High voltage disconnect limit. Expressed as a 120 V base value or per-unit value depending on the generator model used.
- **Low Disconnect Voltage Limit:** Low voltage disconnect limit. Expressed as a 120 V base value or per-unit value depending on the generator model used.
- **LVM Control Type:** LVM control type, such as PF Control or VAR Control.
- **Advanced Applications Eligibility Override:** Can be Yes or No.
- **Maximum Voltage Before Separation From the Network:** Maximum voltage before separation from the network (per unit).
- **Minimum Voltage Before Separation From the Network:** Minimum voltage before separation from the network (per unit).
- **Maximum Frequency Before Separation From the Network:** Maximum frequency before separation from the network (Hz). By default, 63 Hz.
- **Minimum Frequency Before Separation From the Network:** Minimum frequency before separation from the network (Hz). By default, 56 Hz.

- **Planned Restoration Time (sec):** The planned restoration time for a DER/DG after re-energization of a feeder, in seconds.
- **Automatic Reconnect Capable:** Specifies whether the generator can automatically reconnect to the system when the system voltage is restored.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the generator (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the generator (kVAR).

4.1.15. Ground

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the ground.
- **ID:** Unique identifier of the ground.
- **Address:** Address of the ground.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the ground belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the ground is connected at normal conditions.
- **Normally De-Energized:** Indicates if the ground is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the ground phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the ground is connected.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.

- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the ground.

4.1.16. Inverter

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **ID:** Inverter unique identifier.
- **Name:** Name of the inverter.

Attributes:

- **Inverter Size (kVA):** Inverter size (in kVA).
- **Inverter Phases:** The phasing of the inverter.
- **Inverter Node:** Inverter node ID.

Model Attributes:

- **Model Name:** The inverter model name.
- **Maximum Real Power Delivered (kW):** The maximum real power the DER can deliver to the grid (in kW).
- **Maximum Reactive Power Delivered (kVAR):** The maximum reactive power the DER can deliver to the grid (in kVAR).
- **Maximum Apparent Power Delivered (kVA):** The maximum apparent power the DER can deliver to the grid (in kVA).
- **Maximum Real Power Absorbed (kW):** The maximum real power the DER can absorb from the grid (in kW). For energy storage charging.
- **Maximum Reactive Power Absorbed (kVAR):** The maximum reactive power the DER can absorb from the grid (in kVAR). For energy storage charging.
- **Maximum Apparent Power Absorbed (kVA):** The maximum apparent power the DER can absorb from the grid (in kVA). For energy storage charging.
- **Peak Power Limit (kW):** The peak power target power level limit (in kW).
- **Rating (Amps):** The maximum nameplate AC current level of the DER (in Amps).
- **Short Circuit Rating Multiplier:** The short circuit multiplier of the nameplate AC rating.
- **Normal Operating Voltage (V):** The normal operating voltage for the DER site/service connection (in V).
- **Voltage Reference Offset (V):** An offset voltage adjustment for this DER, relative to the normal operating voltage (in V).

- **Disconnect Expiration Timeout (sec):** The time after which a command to disconnect expires and the device can reconnect (in seconds).
- **Setting Limit Time Window (sec):** The time after which a new setting takes place (in seconds).
- **Power Factor Reversion Timeout (sec):** The time after which a DER returns to its default power factor setting (in seconds).
- **Disconnect Capability:** The disconnect capability (Physical, Virtual, Physical-Virtual, or None).
- **Power Factor Type:** The power factor type (IEC Convention or IEEE Convention).

4.1.17. Line

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the line.
- **ID:** Line unique identifier.
- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.
- **Permission Area:** Permission area the line belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the line is connected nominally.
- **Normally Looped:** Indicates if the line is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the line is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the reactor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment.

This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the line is connected.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Downstream Customers:** Total Number of energized customers which are downstream of the line.

Engineering Data:

- **Length (ft):** Length of the line in feet.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the line.
- **Construction Type:** Construction type (for example, Overhead/Underground).
- **Construction Model Name:** Name of the construction model.
- **Conductor Model (Phases A, B, C, and N):** Name of the conductor model for phases and neutral.
- **Phase Impedance (R+jX Ohm):** The impedance in ohms, where R-resistance, jX-reactance ("impedance" is the total opposition [that is, resistance and reactance] that a circuit offers to the flow of alternating current at a given frequency).
- **Phase Admittance - G+jB (microsiemens):** Admittance (inverse of impedance) in microsiemens.
- **Sequence Impedance (R+jX Ohm):** The sequence impedance (in ohms).
- **Sequence Admittance - G+jB (microsiemens):** The Positive Sequence Admittance (inverse of impedance) in microsiemens.

Nominal Switchable Section Information:

- **Bounding Switching Devices:** Names of bounding switching devices.
- **Customers in Switchable Section:** The number of customers connected to phases A, B, C and ABC, and the number of critical customers connected for each switchable section.

4.1.18. Load

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the load.
- **ID:** Load unique identifier.
- **Address:** Address of the load.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the load belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the load is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the load phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Primary Meter:** Primary meter of the load.
- **Non-Conforming Load:** Indicates a non-conforming load. For a non-conforming load, the load values are not adjusted during the load allocation process (the initial schedule point look-up is applied).
- **Node:** ID of the node.
- **Number of Customers:** Number of customers associated with the load.
- **Number of Critical Customers:** Number of critical customers associated with the load.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the load is connected.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Latest Outage Time:** Last recorded outage time (for cold load pick-up calculations).
- **Latest Restoration Time:** Last recorded restoration time (for cold load pick-up calculations).

- **Last Outage Duration:** Last recorded load outage length, in hours and minutes (for cold load pick-up calculations).

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the load in kV.
- **Load Category:** Load category as defined in the Global file.
- **Connection**: The connection arrangement (Wye or Delta).
- **Nominal Real Power (kW):** The nominal real power rate in kW.
- **Nominal Reactive Power (kVAR):** The nominal reactive power rate in kVAR.
- **Nominal Power Factor:** The nominal power factor of the load.
- **Load has Embedded Generation:** Can be Yes or No.
- **Type of Load:** Type of customer load.
- **Generator Rating:** Generator rating in kVA.
- **Generator Type:** Generator type.
- **Number of Customers with Generation:** Number of customers with generation.
- **Number of Customers with Demand Response:** Number of customers with demand response.
- **Percentage of Load with Demand Response:** Percentage of customer loads with demand response.
- **Load Value Source:** Load value source (Dynamic Schedule, Telemetered, Manually Replaced, AMI Fixed, AMI Scaled, or Static Schedule).

4.1.19. Loop End

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the loop end.
- **ID:** Unique identifier of the loop end.
- **Permission Area:** Permission area the loop end belongs to.

4.1.20. Manhole

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the manhole.
- **ID:** Unique identifier of the manhole.

- **Reference Drawing:** Associated reference drawing.
- **Manhole Type:** Type of manhole as defined by the enumerated list.
- **Number of Objects in Manhole:** Number of objects (for example, switches) that are located within the manhole.
- **Permission Area:** Permission area the loop end belongs to.

4.1.21. Network Protector

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the network protector.
- **ID:** Unique identifier of the network protector.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the network protector.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the network protector belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the network protector is connected nominally.
- **Normally Looped:** Indicates if the network protector is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the network protector is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the network protector forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Station Boundary Device:** Indicates if the network protector forms a boundary between two stations. Can be Yes or No.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the network protector phases (ABC for A, B, and C phases).

- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Team Names:** Names of automated scheme teams that the network protector is associated with.
- **Team IDs:** IDs of automated scheme teams that the network protector is associated with.
- **Scheme Name:** Name of the automated scheme the network protector is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the network protector is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the network protector is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Network protector's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the network protector.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the network protector is pending the change of normal position (Yes/No).

Engineering Data:

- **Model Name:** The name of the network protector model.
- **Continuous Current Rating (Amps):** The rate of continuous current, in amperes.
- **Type:** The type of network protector.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the network protector.
- **Switched Phase Independently:** Yes or No value.

- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.
- **Operation Mode:** Operation mode.
- **Maximum Interruption Rating (Amps):** The maximum interruption rating of the network protector (amps).
- **Close and Latch Current (Amps):** The close and latch current (amps).
- **Master Relay Trip (Amps):** The master relay trip current (amps).
- **Phase Relay Trip (deg.):** The phase relay trip angle (deg.).
- **Master Relay Trip Reclose Pick-Up (Volts):** The master relay trip reclose pick-up voltage (volts).
- **Phase Relay Trip Reclose Pick-Up (deg):** The phase relay trip reclose pick-up angle (deg.).

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag (Open/Closed) indicating state of the network protector based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Alarm Status:** Flag indicating the alarm status of the network protector.
- **Remote Control Status:** Flag indicating the remote control of the network protector. If Yes, the network protector is controlled remotely. If No, the network protector is controlled locally.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.

4.1.22. Permanent Jumper

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the permanent jumper.
- **ID:** Unique identifier of the permanent jumper.
- **Alternate ID:** Second unique identifier.
-

Address: Address of the permanent jumper.

- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the permanent jumper belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the permanent jumper is connected nominally.
- **Normally Looped:** Indicates if the permanent jumper is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the permanent jumper is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the permanent jumper forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Station Boundary Device:** Indicates if the permanent jumper forms a boundary between two stations. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the permanent jumper phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the permanent jumper is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the permanent jumper is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Permanent jumper’s current position (Open/Closed).

- **Currently Open Phase(s):** Phases that are currently opened by the permanent jumper.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the permanent jumper is pending the change of normal position (Yes/No).

Engineering Data:

- **Model Name:** The name of the permanent jumper model.
- **Size (Amps):** The maximum interruption rating of the permanent jumper (amps).
- **Type:** The type of permanent jumper.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the permanent jumper.
- **Switched Phase Independently:** Yes or No value.
- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the permanent jumper is eligible for reconfiguration in FISR application.
- **Participating in Local Control Scheme:** Indicates if the permanent jumper participates in the local control scheme.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag indicating the state of the permanent jumper based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Alarm Status:** Flag indicating the alarm status of the permanent jumper.
- **Remote Control Status:** Flag indicating the remote control status of the permanent jumper. If Yes, the permanent jumper is controlled remotely. If No, the permanent jumper is controlled locally.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.
- **Load Break Rating:** The maximum current flow to be allowed to break by the permanent jumper, expressed in Amps.

4.1.23. Power Transformer

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the transformer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the transformer is connected nominally.
- **Normally Looped:** Indicates if the transformer is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the transformer is normally de-energized (Yes/No).
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the transformer is connected.

- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Number of Daily Switching Operations:** The number of switching operations performed in the current day.

Engineering Data:

- **Rating:** The total capacity of the transformer (MVA).
- **Type:** The type of transformer.
- **Description:** Description of the transformer.
- **Model Name:** The name of the transformer model.
- **Nominal Core Loss:** The nominal losses of the transformer core at rated load, expressed in kilowatts.
- **Total Impedance – R + jX (p.u. on Transformer Power Base):** The effective impedance of the transformer, expressed in per-unit values, R+jX (for example, 0.0126 + j0.024).
- **Excitation Current (%):** The excitation current of the transformer, expressed as a percentage of rated current.
- **Magnetizing Power Factor:** Magnetization power factor. This field is optional if the excitation current (%) value is provided.
- **Configuration Index:** Transformer model type and index. For example, DeltaStar1 (8).
- **Time Between Consecutive Operations (Minutes):** Time between consecutive operations in minutes.
- **Time of Last Operation:** Time of the last operation.
- **Maximum Number of Daily Operations:** The maximum number of operations in one day. This is set by the DNAF Configuration Editor.
- **Maximum Phase Voltage Imbalance:** The maximum phase voltage imbalance in percentage.
- **Advanced Regulation Control Definition ID:** The ID of a Regulator Control Definition object defined in the RegulatorControl.csv file.
- **Advanced Regulation Voltage Base:** Advanced regulation voltage base.
- **Voltage Bandwidth Reduction Compensation:** Specifies whether the transformer has voltage bandwidth reduction compensated.
- **Transformer Self-Protection:** Specifies the self-protection type of the transformer.
- **Arc Suppression Coil (ASC) is Connected (Secondary):** Specifies whether the Arc Suppression

Coil (ASC) fault indicator is connected to the secondary side of the power transformer.

- **ASC Regulation Default On Status (Secondary):** Specifies whether the ASC Regulation On status is enabled by default.
- **ASC Current Regulation Status (Secondary):** Shows the currently applied ASC Regulation On status.
- **ASC Default Coil Size (kVA) (Secondary):** The default ASC coil size (in kVA).
- **ASC Default Minimum Grounding Reactance (ohms) (Secondary):** The default ASC minimum grounding reactance (ohms).
- **ASC Default Maximum Grounding Reactance (ohms) (Secondary):** The default ASC maximum grounding reactance (ohms).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the transformer winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Winding Type:** The transformer winding type (Delta, Wye, Open).
- **Grounded:** Indicates if the transformer is grounded (Yes/No).
- **Grounding Impedance - R + jX (ohm):** The transformer grounding impedance, expressed in ohms.
- **Winding Lags Primary Phase (Deg):** Winding lags primary phase in degrees.
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Regulation Control Capability:** Tap changer regulation control capability (Fixed, Local Only, Remote Only, or Remote with Local).
- **Current Regulation Control Mode:** Tap changer current regulation control mode (None, Tap Position Setpoint, or Voltage Setpoint).
- **Remote Control Capability:** Tap changer remote control capability (None, Raise/Lower, Voltage Setpoint, Raise/Lower and Voltage Setpoint, or Voltage Reduction).
- **Current Remote Control Mode:** Tap changer current remote control mode (None, Raise/Lower, Voltage Setpoint, or Voltage Reduction).
- **Minimum Tap Position:** Transformer primary/secondary side minimum tap position.
- **Nominal Tap Position:** Transformer primary/secondary side tap nominal position.
- **Maximum Tap Position:** Transformer primary/secondary side tap maximum position.
- **Normal Tap Position:** Transformer primary/secondary side tap normal position. The field shows "--" (null) for "Regulating" or "None" (no tap exists for the winding) tap changer types, and an integer value between (or equal to) the minimum and maximum tap position for the "Fixed" tap changer type.
- **Tap Step Size (%):** The size of the tap step, expressed in percentage.
- **Tap Movement Cost:** Tap movement cost.
- **Phases Sensed:** The sensed phases (local regulation).

- **Target Voltage:** Regulating transformer voltage target (per unit or configurable voltage base). This row will be hidden if advanced regulation is enabled on the transformer.
- **Voltage Setpoint Deadband:** Voltage setpoint deadband. This row will be hidden if advanced regulation is enabled on the transformer.
- **Two-Sided Bandwidth:** The regulation two-sided voltage bandwidth, expressed as a per-unit value. This refers only to devices in local control mode. This row will be hidden if advanced regulation is enabled on the transformer.
- **Line Drop Compensation Resistance (V):** Regulating transformer line drop compensation resistance (V). This row will be hidden if advanced regulation is enabled on the transformer.
- **Line Drop Compensation Reactance (V):** Regulating transformer line drop compensation reactance (V). This row will be hidden if advanced regulation is enabled on the transformer.
- **PT Ratio:** Regulating transformer PT ratio.
- **CT Ratio:** Regulating tap CT ratio.
- **CT Secondary Rated Amps:** The current rate in the secondary winding.
- **Voltage Reduction Capable:** Indicates if the transformer is voltage reduction capable. This row will be hidden if advanced regulation is enabled on the transformer.
- **Voltage Reduction Value (%):** The voltage reduction value. This row will be hidden if advanced regulation is enabled on the transformer.

Measurement Data:

- **Tap Position (Regulating Tap):** Regulating tap position measurement.
- **Remote Control Status:** Flag (Yes/No) indicating remote control of the transformer. If Yes, the transformer is controlled remotely. If No, the transformer is controlled locally.
- **Regulation Status:** Flag indicating if the regulation capability of the device is enabled.
- **Voltage Reduction Status:** Flag indicating if the voltage reduction is available.
- **Blocked Status:** Blocked status. Can be Blocked or Unblocked.

Regulation Mode Data:

- **Active Regulation Mode:** The name of the active regulation mode. The following parameters are displayed only if an active regulation mode exists.
- **Regulation Mode:** The name of the regulation mode for the specified direction.
- **Regulation Node Side:** The regulation node side.
- **Voltage Target (V):** The voltage target for the specified direction.
- **Voltage Bandwidth (V):** The regulation voltage bandwidth for the specified direction.
- **Current Threshold (A):** The current threshold for the specified direction. The value is shown in the current base of the primary side of the CT (multiplied by CT ratio).
- **LDC Resistance (V):** The LDC resistance in volts for the specified direction.
- **LDC Reactance (V):** The LDC reactance in volts for the specified direction.

- **Voltage Reduction Capable:** The status of the voltage reduction capability for the specified direction.
- **Voltage Reduction Level 1:** The value of the level 1 voltage reduction for the specified direction.
- **Voltage Reduction Level 2:** The value of the level 2 voltage reduction for the specified direction.
- **Voltage Reduction Level 3:** The value of the level 3 voltage reduction for the specified direction.
- **Flow Direction Sensing Input:** Power quantity (active or reactive) used for flow direction sensing input.

Arc Suppression Coil Data:

- **Arc Suppression Coil Fault Status:** The arc suppression coil fault status.

4.1.24. Transformer Bank

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer bank.
- **ID:** Transformer bank identifier. This may not be unique since any of the transformers in the bank may have the same identifier.
- **Number of Transformers in Bank:** The number of transformers available in the bank.
- **Transformer1 ID**: Identifier of the first transformer in the list of transformers available in the bank.
- **Transformer2 ID**: Identifier of the second transformer in the list of transformers available in the bank.
- **Transformer3 ID**: Identifier of the third transformer in the list of transformers available in the bank.
- **Permission Area:** Permission area to which the transformer bank belongs.

4.1.25. Reactor

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reactor.
- **ID:** Unique identifier of the reactor.
- **Address:** Address of the reactor.

- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the reactor belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the reactor is connected nominally.
- **Normally Looped:** Indicates if the line is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the line is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the reactor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the line is connected.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Nominal Voltage:** Nominal voltage.
- **Model Name:** The name of the reactor model as defined in the Global file.
- **Impedance (R+jX ohms):** The reactor impedance (in ohms).

4.1.26. Recloser

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the recloser.
- **ID:** Unique identifier of the recloser.
- **Alternate ID:** Second unique identifier.
- **Structure ID:** Unique identifier of a structure that is associated with the recloser.
- **Address:** Address of the recloser.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the recloser belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the recloser is connected nominally.
- **Normally Looped:** Indicates if the recloser is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the recloser is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the recloser forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Station Boundary Device:** Indicates if the recloser forms a boundary between two stations. Can be Yes or No.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the recloser phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Normally Open Phase(s):** Normally open phases.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Team Names:** Names of automated scheme teams that the recloser is associated with.
- **Team IDs:** IDs of automated scheme teams that the recloser is associated with.
- **Scheme Name:** Name of the automated scheme the recloser is associated with.

- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the recloser is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the recloser is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Recloser’s current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the recloser.
- **Associated Block Close DER:** The name (ID) of the energy resource with size greater than the value specified in the Minimum DER (kW) field for the default FISR problem formulation in the DNAF Configuration Editor. FISR plans will include a step to prevent connection of this energy resource.

  When multiple energy resources are connected to a recloser, only the name of the first energy resource in the list is shown.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the recloser is pending the change of normal position (Yes/No).
- **Dynamic Phases:** Dynamic phases information.
  - **Has Dynamic Phase Mismatch:** Indicates that the dynamic phases on either side of the recloser do not match.
  - **Has Phase Angle Mismatch:** Indicates that a phase angle difference (as calculated by DPF) is larger than the configured limit.
  - **Has Nominal Voltage Mismatch:** Indicates that the nominal voltage on either side of the recloser is different.

Engineering Data:

- **Model Name:** The name of the recloser model.
- **Size (Amps):** The maximum interruption rating of the recloser (amps).
- **Type:** Switch type (recloser).

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the recloser.
- **Switched Phase Independently:** Yes or No value.
- **Bypass Switch:** Name of a bypass switch, if any.
- **Reversible Recloser:** Yes or No value.
- **Control Mechanism:** Recloser control type, such as electronic or hydraulic.
- **Nominal Operation Mode:** Nominal operation mode (for example, Recloser Mode).
- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the recloser is eligible for reconfiguration in FISR application.
- **Participating in Local Control Scheme:** Indicates if the recloser participates in the local control scheme.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.
- **Ground Trip Capable:** Indicates whether the recloser can trip on a ground fault (Yes/No).
- **Ground Trip Status:** Normal status of ground trip (Enabled/Disabled).
- **Phase Trip Current (Amps):** Phase trip current value in Amps.
- **Ground Trip Current (Amps):** Ground trip current value in Amps.
- **Neutral Relay Setting (Amps):** The maximum allowed neutral trip setting in Amps.
- **Interrupting Capacity (Amps):** Interrupting capacity in Amps.
- **PRV Safety Factor:** Safety factor used to calculate minimum phase and ground pick-up limit.
- **Ground and Neutral Relay Safety Factor:** Safety factor used when comparing the phase imbalance flow (the vector sum of currents in the three phases) for breakers and reclosers to the minimum ground relay pick-up setting and neutral relay current setting.
- **Settings Group Template Name:** The name of the nominal settings group template.
- **Initial Block Close:** Specifies whether the recloser connects an energy resource with size greater than the value specified in the Minimum DER (kW) field for the default FISR problem formulation in the DNAF Configuration Editor. FISR plans will include a step to prevent connection of this energy resource.

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag indicating the state of the recloser based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Hot Line Tag Status:** Hot line tag status (Applied or Not Applied).
- **Alarm Status:** Flag indicating the alarm status of the recloser.

- **Block Close Measurement Status:** Block Close measurement status (Applied or Not Applied).
- **Remote Control Status:** Flag (Yes/No) indicating remote control of the recloser. If Yes, the recloser is controlled remotely. If No, the recloser is controlled locally.
- **Operation Mode Status:** The operation mode status of the recloser.
- **Reclosing Status:** Flag indicating if the reclosing capability of the device is enabled.
- **Lockout Status:** Flag indicating the lockout status of the recloser.
- **Prohibit Restoration Status:** Flag indicating the prohibit restoration status of the recloser.
- **Active Settings Group Template:** The active settings group template index.
- **Active Settings Group Name:** The active settings group template name.
- **Active Settings Group ID:** The active settings group template ID.
- **Active Settings Group Description:** The active settings group template description.
- **Normal Settings Group:** The nominal settings group value (index).
- **Normal Settings Group Name:** The nominal settings group name.
- **Active Settings Group:** The active settings group value (index).
- **Active Settings Group Name:** The active settings group name.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.

4.1.27. Reference Bus Interface

This Analyst Attribute pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reference bus interface.
- **ID:** Unique identifier of the reference bus interface.
- **Address:** Address of the reference bus interface.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the reference bus interface belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the reference bus interface is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.

- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the reference bus interface is connected.
- **Status:** Reference bus interface connection status (Connected or Disconnected).
- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the reference bus interface.
- **Voltage Target Phase A (%):** The voltage target for phase A, in percentage.
- **Voltage Target Phase B (%):** The voltage target for phase B, in percentage.
- **Voltage Target Phase C (%):** The voltage target for phase C, in percentage.
- **Angle Target (deg):** The nominal angle target, in degrees.
- **Positive Sequence Resistance (Ohm):** The positive sequence resistance, in Ohms.
- **Positive Sequence Reactance (Ohm):** The positive sequence reactance, in Ohms.
- **Negative Sequence Resistance (Ohm):** The negative sequence resistance in, Ohms.
- **Negative Sequence Reactance (Ohm):** The negative sequence reactance, in Ohms.
- **Zero Sequence Resistance (Ohm):** The zero sequence resistance, in Ohms.
- **Zero Sequence Reactance (Ohm):** The zero sequence reactance, in Ohms.

4.1.28. Regulator

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the regulator.
- **ID:** Unique identifier of the regulator.
- **Address:** Address of the regulator.

- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the regulator belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the regulator is connected nominally.
- **Normally Looped:** Indicates if the regulator is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the regulator is normally de-energized (Yes/No).
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the regulator is connected.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Number of Daily Operations:** The number of operations performed in the current day.

Engineering Data:

- **Rating:** The nominal capacity of the regulator (kVA).
- **Effective Rating:** The actual capacity of the regulator (kVA).
- **Type:** The type of regulator.
- **Description:** Description of the regulator.
-

Model Name: The name of the regulator model.

- **Nominal Core Loss (kW):** The nominal losses of the regulator core at rated load, expressed in kilowatts.
- **Total Impedance – R + jX (p.u. on Transformer Power Base):** The effective impedance of the regulator expressed in per-unit values, R+jX (for example, 0.0126 + j0.024).
- **Excitation Current (%):** The excitation current of the regulator expressed as a percentage of rated current.
- **Magnetizing Power Factor:** Magnetization power factor. This field is optional if the excitation current (%) value is provided.
- **Configuration Index:** Transformer model type and index. For example, AutoBankX (10).
- **Time Between Consecutive Operations (Minutes):** Time between consecutive operations in minutes.
- **Time of Last Operation:** Time of the last operation.
- **Maximum Number of Daily Operations:** The maximum number of operations in one day. This is set by the DNAF Configuration Editor.
- **Maximum Phase Voltage Imbalance:** The maximum phase voltage imbalance in percentage.
- **Advanced Regulation Voltage Base:** Advanced regulation voltage base.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the regulator winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the regulator winding.
- **Winding Type:** The regulator winding type (Delta, Wye, Open).
- **Grounded:** Indicates if the regulator is grounded (Yes/No).
- **Grounding Impedance - R + jX (ohm):** The regulator grounding impedance expressed in ohms.
- **Winding Lags Primary Phase (Deg):** Winding lags primary phase in degrees.
- **Tap Changer Type:** Regulator primary/secondary side tap changer type.
- **Regulation Control Capability:** Tap changer regulation control capability (Fixed, Local Only, Remote Only, or Remote with Local).
- **Current Regulation Control Mode:** Tap changer current regulation control mode (None, Tap Position Setpoint, or Voltage Setpoint).
- **Remote Control Capability:** Tap changer remote control capability (None, Raise/Lower, Voltage Setpoint, Raise/Lower and Voltage Setpoint, or Voltage Reduction).
- **Current Remote Control Mode:** Tap changer current remote control mode (None, Raise/Lower, Voltage Setpoint, or Voltage Reduction).
- **Minimum Tap Position:** Regulator primary/secondary side minimum tap position.
- **Nominal Tap Position:** Regulator primary/secondary side tap nominal position.
- **Maximum Tap Position:** Regulator primary/secondary side tap maximum position.
- **Tap Step Size (%):** The size of the tap step expressed in percentage.

- **Tap Movement Cost:** Tap movement cost.
- **Phases Sensed:** The phases that are sensed for local control of the regulator voltage output (for example, A, B, C).
- **Target Voltage:** Regulating regulator voltage target (per unit or configurable voltage base). This row will be hidden if advanced regulation is enabled on the regulator.
- **Voltage Setpoint Deadband:** Voltage setpoint deadband. This row will be hidden if advanced regulation is enabled on the regulator.
- **Two-Sided Bandwidth:** The regulation two-sided voltage bandwidth expressed as a per-unit value. This refers only to devices in local control mode. This row will be hidden if advanced regulation is enabled on the regulator.
- **Line Drop Compensation Resistance (V):** Regulating transformer line drop compensation resistance (V). This row will be hidden if advanced regulation is enabled on the regulator.
- **Line Drop Compensation Reactance (V):** Regulating transformer line drop compensation reactance (V). This row will be hidden if advanced regulation is enabled on the regulator.
- **PT Ratio:** Regulating transformer PT ratio.
- **CT Ratio:** Regulating tap CT ratio.
- **CT Secondary Rated Amps:** The current rate in the secondary winding.
- **Voltage Reduction Capable:** Indicates if the regulator is voltage reduction capable. This row will be hidden if advanced regulation is enabled on the regulator.
- **Voltage Reduction Value (%):** The voltage reduction value. This row will be hidden if advanced regulation is enabled on the regulator.

Measurement Data:

- **Tap Position (Regulating Tap):** Regulating tap position measurement.
- **Remote Control Status:** Flag (Yes/No) indicating remote control of the regulator. If Yes, the regulator is controlled remotely. If No, the regulator is controlled locally.
- **Regulation Status:** Flag indicating if the regulation capability of the device is enabled.
- **Voltage Reduction Status:** Flag indicating if the voltage reduction is available.
- **Blocked Status:** Blocked status. Can be Blocked or Unblocked.

Regulation Mode Data:

- **Active Regulation Mode:** The name of the active regulation mode. The following parameters are displayed only if an active regulation mode exists.
- **Regulation Mode:** The name of the regulation mode for the specified direction.
- **Voltage Target (V):** The voltage target for the specified direction.
- **Voltage Bandwidth (V):** The regulation voltage bandwidth for the specified direction.
- **Current Threshold (A):** The current threshold for the specified direction. The value is shown in the current base of the primary side of the CT (multiplied by CT ratio).

- **LDC Resistance (V):** The LDC resistance in volts for the specified direction.
- **LDC Reactance (V):** The LDC reactance in volts for the specified direction.
- **Voltage Reduction Capable:** The status of the voltage reduction capability for the specified direction.
- **Voltage Reduction Level 1 (V):** The value of the level 1 voltage reduction for the specified direction.
- **Voltage Reduction Level 2 (V):** The value of the level 2 voltage reduction for the specified direction.
- **Voltage Reduction Level 3 (V):** The value of the level 3 voltage reduction for the specified direction.
- **Flow Direction Sensing Input:** Power quantity (active or reactive) used for flow direction sensing input.

4.1.29. Reserved Capacity

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reserved capacity.
- **ID:** Unique identifier of the reserved capacity.
- **Permission Area:** Permission area the substation belongs to.
- **Service Center:** The name of the service center.
- **Normal Station:** The name of the nominal substation the device belongs to.
- **Normal Feeder:** The name of the nominal substation the device belongs to.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the reserved capacity is connected.
- **Feeder:** The name of the feeder to which the reserved capacity is connected.

Engineering Data:

- **Relay Type:** The type of protection relay.
- **Phase Trip Current:** Phase trip current value in amps.
- **Ground Trip Current:** Ground trip current value in amps.

4.1.30. Sectionalizer

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the sectionalizer.
- **ID:** Unique identifier of the sectionalizer.
- **Alternate ID:** Second unique identifier.
- **Structure ID:** Unique identifier of a structure that is associated with the sectionalizer.
- **Address:** Address of the sectionalizer.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the sectionalizer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the sectionalizer is connected nominally.
- **Normally Looped:** Indicates if the sectionalizer is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the sectionalizer is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the sectionalizer forms a boundary between the external interface and a feeder. Can be Yes or No.
- **Station Boundary Device:** Indicates if the sectionalizer forms a boundary between two stations. Can be Yes or No.
- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sectionalizer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Team Names:** Names of automated scheme teams that the sectionalizer is associated with.
- **Team IDs:** IDs of automated scheme teams that the sectionalizer is associated with.
- **Scheme Name:** Name of the automated scheme the sectionalizer is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the sectionalizer is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the sectionalizer is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Sectionalizer's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the sectionalizer.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the sectionalizer is pending the change of normal position (Yes/No).
- **Dynamic Phases:** Dynamic phases information.
    - **Has Dynamic Phase Mismatch:** Indicates that the dynamic phases on either side of the sectionalizer do not match.
    - **Has Phase Angle Mismatch:** Indicates that a phase angle difference (as calculated by DPF) is larger than the configured limit.
    - **Has Nominal Voltage Mismatch:** Indicates that the nominal voltage on either side of the sectionalizer is different.

Engineering Data:

- **Model Name:** The name of the sectionalizer model.
- **Size (Amps):** The maximum interruption rating of the sectionalizer (amps).
- **Type:** The type of sectionalizer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the sectionalizer.
- **Switched Phase Independently:** Yes or No value.
- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the sectionalizer is eligible for reconfiguration in FISR application.
- **Participating in Local Control Scheme:** Indicates if the sectionalizer participates in the local

control scheme.

- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag indicating the state of the sectionalizer based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Alarm Status:** Flag indicating the alarm status of the sectionalizer.
- **Remote Control Status:** Flag indicating the remote control status of the sectionalizer. If Yes, the sectionalizer is controlled remotely. If No, the sectionalizer is controlled locally.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.

4.1.31. Sensor

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the sensor.
- **ID:** Unique identifier of the sensor.
- **Address:** Address of the sensor.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the sensor belongs to.
- **Normal Station and Feeder:** The name of the nominal substation and feeder the sensor belongs to.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sensor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
-

Removed Phase(s): Individual phases that were recently removed.

- **Node:** ID of the node.
- **Flow Branch ID:** ID of the flow branch.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the sensor is connected.
- **Dynamic Feeder is Looped:** Yes or No value.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the sensor.

Measurement Data:

- **Amp Measurement:** Current rate measurement.
- **Fault Indication Measurement:** Status of the associated fault indicator.

4.1.32. Splice Box

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the splice box.
- **ID:** Unique identifier of the splice box.
- **Reference Drawing:** Reference drawing.
- **Type:** Splice box type number.
- **Number of Objects:** Number of objects within the splice box.
- **Permission Area:** Permission area the splice box belongs to.

4.1.33. Structure

This Analyst Attributes pop-up can provide end users with any information that has been modeled in the structures data files. By default, the following information is shown:

Identifiers.

- **ID:** Structure unique identifier.
- **Name:** Structure name.
- **Owner:** Structure owner.

Engineering Data:

- **Type:** Structure subtype.
- **Composition:** Structure material, such as steel or wood (for poles).
- **Height:** Structure composition.
- **Arrangement:** Structure arrangement type.
- **Spacing:** Structure spacing type.
- **Class:** Structure class.
- **Use:** Structure use.
- **Is Metered:** Specifies whether the structure is metered.
- **Num Lights:** Specifies the number of lights (for streetlights).
- **Wattage 1:** Specifies the light 1 wattage.
- **Wattage 2:** Specifies the light 2 wattage.

Location Info:

- **X:** X coordinate of the structure.
- **Y:** Y coordinate of the structure.
- **Feeder Name:** Feeder name.
- **Feeder ID:** Feeder ID.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.

4.1.34. Substation

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Substation name.
- **ID:** Substation unique identifier.
- **Alternate Name:** Substation alternative name.
- **Address:** Address of the substation.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the substation belongs to.
- **Division:** The name of the division.

Connectivity Data:

- **Number of Abnormal Topology Conditions:** The number of abnormal topology conditions for the substation.
- **De-Energized Customer Count:** The number of de-energized customers for the substation.

Engineering Data:

- **Positive Sequence Equivalent Impedance (Ohms):** Positive sequence equivalent source resistance (for use in fault studies).
- **Negative Sequence Equivalent Impedance (Ohms):** Negative sequence equivalent source resistance (for use in fault studies).
- **Zero Sequence Equivalent Impedance (Ohms):** Zero sequence equivalent source resistance (for use in fault studies).

4.1.35. Switch

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the switch.
- **ID:** Switch unique identifier.
- **Alternate ID:** Second unique identifier.
- **Structure ID:** Unique identifier of a structure that is associated with the switch.
- **Address:** Address of the switch.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the switch belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the switch is connected nominally.
- **Normally Looped:** Indicates if the switch is normally looped (Yes/No).
- **Normally De-Energized:** Indicates if the switch is normally de-energized (Yes/No).
- **Normal Position (Ganged Device):** Normal position for the ganged device.
- **Normally Open Phase(s):** Normally open phases.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Feeder Main Boundary Device:** Indicates if the switch forms a boundary between the external interface and a feeder. Can be Yes or No.
-

Station Boundary Device: Indicates if the switch forms a boundary between two stations. Can be Yes or No.

- **Advanced Applications Eligibility Override:** Override the advanced applications eligibility.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the switch phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Team Names:** Names of automated scheme teams that the switch is associated with.
- **Team IDs:** IDs of automated scheme teams that the switch is associated with.
- **Scheme Name:** Name of the automated scheme the switch is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the switch is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the switch is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** The switch's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the switch.
- **Phase(s) Energized:** Phases that are currently energized.
- **Phase(s) De-Energized:** Phases that are currently de-energized.
- **Phase(s) Abnormally Energized:** Phases that are currently energized from the abnormal source.
- **Phase(s) Looped:** Phases that are currently looped.
- **Phase(s) Removed:** Phases that are removed.
- **Dynamic Feeder is Looped:** Yes or No value.
- **Pending Permanent Position Change:** Indicates if the switch is pending the change of normal position (Yes/No).
- **Dynamic Phases:** Dynamic phases information.

  - **Has Dynamic Phase Mismatch:** Indicates that the dynamic phases on either side of the switch do not match.
  - **Has Phase Angle Mismatch:** Indicates that a phase angle difference (as calculated by DPF) is larger than the configured limit.
  - **Has Nominal Voltage Mismatch:** Indicates that the nominal voltage on either side of the switch is different.

Engineering Data:

- **Model Name:** The name of the switch model.
- **Size (Amps):** The maximum interruption rating of the switch (amps).
- **Type:** The type of switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Switched Phase Independently:** Yes or No value.
- **Load Break Capable:** Yes or No value.
- **Load Break Rating:** The load break rate expressed in Amps.
- **Initial Reconfiguration Eligibility (FISR):** Indicates if the switch is eligible for reconfiguration in FISR application.
- **Participating in Local Control Scheme:** Indicates if the switch participates in the local control scheme.
- **Calculate Flow through the Switching Device:** Specifies whether the flow through the device is calculated by Distribution Power Flow (DPF) and whether the flow direction is determined.

Measurement Data:

- **Associated Open/Closed Measurement:** Associated state measurement.
- **Open/Closed Status:** Flag indicating the state of the switch based on measurement data.
- **Control Measurement Count:** The number of associated control measurements that are mapped between SCADA and DMS for the device.
- **Supervisory Cutout Status:** Supervisory cutout state of the switch, set by the switch control. The dispatcher cannot control the switch in the supervisory cutout state.
- **Alarm Status:** Flag indicating the alarm status of the switch.
- **Remote Control Status:** Flag indicating the remote control status of the switch. If Yes, the switch is controlled remotely. If No, the switch is controlled locally.

Boundary Switch Data (Debug):

- **Station 1 and Feeder 1:** Name of the first bounding substation and feeder.
- **Station 1 - Node 1/Node 2:** IDs of the node 1 and node 2 of the first bounding substation.
- **Station 2 and Feeder 2:** Name of the second bounding substation and feeder.
- **Station 2 - Node 1/Node 2:** IDs of the node 1 and node 2 of the second bounding substation.

4.1.36. Switch Cabinet

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the switch cabinet.
- **ID:** Unique identifier of the switch cabinet.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the switch cabinet.
- **Permission Area:** Permission area the substation belongs to.
- **Normal Station:** The name of the nominal substation the switch cabinet belongs to.
- **Feeder Main Connection:** The switch cabinet is connected to the feeder backbone (Yes/No).
- **Cabinet Type:** Switch cabinet type.
- **Switches:** The list of device names contained in switch cabinet.
- **Number of Objects inside Cabinet:** Number of devices that the switch cabinet contains.

4.1.37. Tag

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the tag.
- **ID:** Unique identifier of the tag.
- **Permission Area:** Permission area the tag belongs to.

4.1.38. Temporary Ground

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the temporary ground.
- **ID:** Unique identifier of the temporary ground.
- **Permission Area:** Permission area the temporary ground belongs to.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Engineering Data:

- **Ground Resistance:** Resistance to the ground expressed in per-unit values.

- **Ground Reactance:** Reactance to the ground expressed in per-unit values.
- **Neutral Phase Is Present:** Phases that are grounded (for example, ABC).

4.1.39. Temporary Switch

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the temporary switch.
- **ID:** Unique identifier of the temporary switch.
- **Permission Area:** Permission area the temporary switch belongs to.
- **Original Component ID:** ID of the original component where the temporary switch is placed.
- **Original Component Type ID:** Type of the original component where the temporary switch is placed.
- **Terminal:** Terminal.
- **Normal Station:** The name of the nominal substation the temporary switch belongs to.
- **Normal Feeder:** The name of the feeder that is normally connected to the temporary switch.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **Distance from Node 1:** Distance from node 1.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the temporary switch is connected.
- **Feeder:** The name of the feeder to which the temporary switch is connected.

4.1.40. Trouble Ticket

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the trouble ticket.
- **ID:** Unique identifier of the trouble ticket.
- **X Coordinate:** X coordinate for the placement of the ticket symbol.
- **Y Coordinate:** Y coordinate for the placement of the ticket symbol.
- **Ticket Type:** Type of the trouble ticket.
- **Status:** Status of the trouble ticket.
- **First Stage Device:** First switchable device upstream from the customer.

- **Permission Area:** Permission area the trouble ticket belongs to.

4.1.41. Transformer Bank

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer bank.
- **ID:** Unique identifier of the transformer bank.
- **Number of Objects in Bank:** Number of transformers in the transformer bank.
- **Permission Area:** Permission area the transformer bank belongs to.

4.1.42. UI Container

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the UI container.
- **ID:** Unique identifier of the UI container.
- **Address:** Address of the UI container.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the substation belongs to.
- **Normal Station:** The name of the nominal substation the UI container belongs to.
- **Container Type:** UI container type.
- **Number of Objects inside UI Container:** Number of devices that the UI container contains.

4.1.43. Vault

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the vault.
- **ID:** Unique identifier of the vault.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the vault.
- **Permission Area:** Permission area the substation belongs to.
- **Normal Station:** The name of the nominal substation the vault belongs to.

- **Feeder Main Connection:** The vault is connected to the feeder backbone (Yes/No).
- **Vault Type:** Vault type.
- **Number of Objects inside Vault:** Number of devices that the vault contains.

4.2. Analyst Output Pop-Ups

4.2.1. Autotransformer

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Power factor.
- **Maximum Phase Loading against Level 1 Limit (%):** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.

- **Loading Level Against Rating (%):** Real-time loading of the transformer as a percentage of rated load.
- **Grounding Current (Amps):** Grounding current rate and phase angle.
- **Loading Percentage against Level 1 Limit:** Real-time phase loading of the transformer as a percentage of the Level 1 load limit.
- **Current Side 1/Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Real Power Side 1/Side 2 (kW):** Real power flow in kW.
- **Reactive Power Side 1/Side 2 (kVAR):** Reactive power flow in kVAR.
- **Apparent Power Side 1/Side 2 (kVA):** Apparent power flow in kVA.
- **Real Power Copper Loss:** Real power winding (copper) loss of the transformer (kW).
- **Reactive Power Copper Loss:** Reactive power winding (copper) loss of the transformer (kVAR).
- **Real Power Core Loss:** Real power core (iron) loss of the transformer (kW).
- **Reactive Power Core Loss:** Reactive power core (iron) loss of the transformer (kVAR).

Engineering Data:

- **Rating:** The total capacity of the transformer (MVA).
- **Type:** The type of transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the transformer.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Tap Position:** Transformer primary/secondary side tap position.
- **Limits:** Data on limits.
- **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
- **Season:** The season of the level limit (for example, Entire Year).
- **Value (per Phase):** The level limit value.
- **Manually Entered Limit:** The manually entered limit value, in MVA.

4.2.2. Breaker

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** Name of the breaker.
- **ID:** Unique identifier of the breaker.
- **Alternate ID:** Second unique identifier of the breaker.
- **Address:** Address of the breaker.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the breaker phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages, in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Downstream DER/DG Injection Real Power (kW):** Calculated total downstream real power DER injection per phase.

- **Downstream DER/DG Injection Reactive Power (kVAR):** Calculated total downstream reactive power DER injection per phase.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the breaker (amps).
- **Type:** The type of breaker.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the breaker.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The nominal level limit value for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - Available Capacity Adjusted for Reserved Capacity (per Phase): Available capacity adjusted for reserved capacity.
    - **Derated Limit Loading:** Derated limit loading. Calculated as maximum flow divided by derated limit.
    - **Manually Entered Limit:** The manually entered limit value.

Protection Validation Results:

- **Maximum Fault (Maximum Interruption Violation):** Boolean value (Yes/No) that indicates if a fault placed at the load side of the breaker creates enough fault current to exceed the maximum interruption capability of the device (in Amps).
- **Maximum Fault (Maximum Interruption Current):** The maximum interruption capability of the breaker (in Amps).
- **Maximum Fault (Maximum Fault Current with Bolted Fault/Configured Fault Impedance):** The maximum short circuit current (in Amps) calculated for a location immediately downstream of the protective device evaluated with a zero-fault impedance or a user configured fault impedance or both.
- **Maximum Fault (Maximum Capacity Ratio with Bolted Fault/Configured Fault Impedance):** The ratio of the maximum calculated fault current evaluated with a zero-fault impedance or a user configured fault impedance or both to the maximum interruption current of the device (in %).
- **Minimum Phase Fault (Minimum Pick-Up Violation):** Boolean value (Yes/No) that indicates if a fault placed at the furthest impedance distance away from the breaker does not create enough fault current (pick-up current) to exceed the phase trip relay setting of the device.
-

Minimum Phase Fault (Minimum Pick-up Current with Safety Factor): The minimum phase pick-up current (multiplied by a safety factor) setting of the phase trip relay (in Amps).

- **Minimum Phase Fault (Minimum Fault Current with Bolted Fault/Configured Fault Impedance):** The minimum phase short circuit current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance or a user configured fault impedance or both.
- **Minimum Phase Fault (Minimum Pick-up Ratio with Bolted Fault/Configured Fault Impedance):** The ratio of the minimum phase pick-up current (multiplied by a safety factor) to the minimum phase fault current evaluated with a zero-fault impedance or a user configured fault impedance or both (in %).
- **Minimum Ground Fault (Minimum Pick-Up Violation):** Boolean value (Yes/No) that indicates if a fault placed at the furthest impedance distance away from the breaker does not create enough fault current to exceed the ground trip relay setting of the device (expressed in amps).
- **Minimum Ground Fault (Minimum Pick-up Current with Safety Factor):** The minimum ground pick-up current (multiplied by a safety factor) setting of the ground trip relay (in Amps).
- **Minimum Ground Fault (Minimum Fault Current with Bolted Fault/Configured Fault Impedance):** The minimum ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance, a user configured fault impedance, or both.
- **Minimum Ground Fault (Minimum Pick-up Ratio with Bolted Fault/Configured Fault Impedance):** The ratio of the minimum ground pick-up current (multiplied by a safety factor) to the minimum ground fault current evaluated with a zero-fault impedance or a user configured fault impedance or both (in %).

Study Mode Only Protection Validation Results:

- **Short Circuit Validation Status:** The short circuit analysis is valid (Yes/No). This field returns a value of Protected or Unprotected after checking the fault analysis protection validation calculation. This is an optional study mode only calculation performed within the PRV application.
- **Minimum Phase Fault (Protective Device Name):** The name of the device that protects the zone from the minimum phase fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.
- **Minimum Phase Fault (Protective Device ID):** The ID of the device that protects the zone from the minimum phase fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.
- **Minimum Phase Fault (Pick-Up Current Limit):** The limit setting for the minimum phase pick-up current (Amps). This field is only applicable for three-phase devices.
- **Minimum Phase Fault (Fault Current):** The actual minimum phase fault current as calculated by the short circuit analysis, in amperes.
- **Minimum Ground Fault (Protective Device Name):** The name of the device that protects the

zone from the minimum ground fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.

- **Minimum Ground Fault (Protective Device ID):** The ID of the device that protects the zone from the minimum ground fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.
- **Minimum Ground Fault (Pick-Up Current Limit):** The limit setting of the minimum ground protective device (Amps).
- **Minimum Ground Fault (Fault Current):** The actual minimum ground fault current as calculated by the short circuit analysis (Amps).
- **Short Circuit Coordination Status:** This field returns a value of Coordinated or Uncoordinated after checking the fault analysis protection coordination calculation. This is an optional study mode only calculation performed within the PRV application.
- **Upstream Protective Device Name:** The name of the coordinated upstream protection device. This device can be the further upstream protection device if the device immediately upstream is not coordinated.
- **Upstream Protective Device ID:** The ID of the coordinated upstream protection device.

4.2.3. Bus

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the bus.
- **ID:** Unique identifier of the bus.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the bus phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.

- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Calculated Voltage (LL):** The calculated line-to-line voltage for phases A, B, and C.
- **Calculated Voltage (LG):** The calculated line-to-ground voltage for phases A, B, and C.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the bus in KV.
- **Voltage Base:** Voltage base value in Volts.

4.2.4. Capacitor

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the capacitor.
- **ID:** Unique identifier of the capacitor.
- **Address:** Address of the capacitor.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the capacitor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Capacitor Position Validation:** Is the current step position of the capacitor bank valid (Yes/No).
- **Total Real Power (W):** Total real power in W.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Voltage (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.

- **Voltage (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Current (Amps):** Phase real-time current (amps).
- **Total Loss (W):** Total loss (watts).

Engineering Data:

- **Total Size (kVAR):** The total capacity of the capacitor (kVAR).
- **Control Type:** Control type of the capacitor (for example, remote only).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the capacitor in kV.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages of the capacitor.
- **Status:** Status of the capacitor (On/Off).

4.2.5. Composite Switch

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the switch.
- **ID:** Unique identifier of the composite switch.
- **Alternate ID:** Second unique identifier of the switch.
- **Address:** Address of the composite switch.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the switch phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Loading Percentage:** Loading percentage.
-

Current (Amps): Phase real-time current (amps).

- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Power factor.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value:** The level limit value.

4.2.6. Distribution Transformer

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Name:** The name of the distribution transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
-

Future Removed Phase(s): Individual phases that are planned to be removed.

- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Power factor.
- **Maximum Phase Loading against Level 1 Limit (%):** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the transformer as a percentage of rated load.
- **Grounding Current (Amps):** Grounding current rate and phase angle.
- **Loading Percentage against Level 1 Limit:** Real-time phase loading of the transformer as a percentage of the Level 1 load limit.
- **Current Side 1/Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Real Power Side 1/Side 2 (kW):** Real power flow in kW.
- **Reactive Power Side 1/Side 2 (kVAR):** Reactive power flow in kVAR.
- **Apparent Power Side 1/Side 2 (kVA):** Apparent power flow in kVA.
- **Service Drop Voltage Side 2:** Voltage at the service drop.
- **Real Power Copper Loss:** Real power winding (copper) loss of the transformer (kW).
- **Real Power Core Loss:** Real power core (iron) loss of the transformer (kW).
- **Reactive Power Copper Loss:** Reactive power winding (copper) loss of the transformer (kVAR).

Engineering Data:

- **Rating:** The total capacity of the transformer (kVA).
- **Type:** The type of transformer.

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the transformer winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value for phases A, B, and C.
  - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.

4.2.7. Elbow

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** The name of the switch.
- **ID:** The unique identifier of the switch.
- **Alternate ID:** Second identifier of the switch.
- **Address:** Address of the switch.
- **Project ID:** ID of the commissioning project that is associated with the device.
- **Project Stage:** The device commissioning or decommissioning stage. You can use this field to manually set or update the project stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.

- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages, in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value for phases A, B, and C.
  - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
  - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
  - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
  - **Manually Entered Limit:** The manually entered limit value.

4.2.8. Energy Resource

This Analyst Output pop-up provides end users with the following information:

Identifiers:

-

Name: Name of the energy resource.

- **ID:** Unique identifier of the energy resource.
- **Address:** Address of the energy resource.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the energy resource phases (ABC for A, B, and C phases).
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Branch power factor.
- **Voltage (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Current (Amps):** Real-time phase current (amps) and angle (deg).

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the energy resource in kV.
- **KVA Size:** Size of the energy resource in kVA.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the energy resource (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the energy resource (kVAR).

Energy Resource Injection Values:

- **Real Power Target:** The real power target type (name), X/Y values, and X/Y units.
- **Reactive Power Target:** The reactive power target type (name), X/Y values, and X/Y units.
-

- **Real Power Schedule Points:** The real power point 1 and point 2 X/Y values.
- **Reactive Power Schedule Points:** The reactive power point 1 and point 2 X/Y values.

4.2.9. Feeder

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the feeder.
- **ID:** Unique identifier of the feeder.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **DPF Solution Time:** Date and time of the most recent execution of the distribution power flow solution.

Engineering Outputs:

- **PQI High:** High power quality index (%kW).
- **PQI Low:** Low power quality index (%kW).
- **PQI Imbalance:** Imbalance of power quality index (%kW).

4.2.10. Fuse

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** Name of the fuse.
- **ID:** Unique identifier of the fuse.
- **Alternate ID:** Second identifier of the fuse.
- **Address:** Address of the fuse.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the fuse phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
-

Future Removed Phase(s): Individual phases that are planned to be removed.

- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Downstream DER/DG Injection Real Power (kW):** Calculated total downstream real power DER injection per phase.
- **Downstream DER/DG Injection Reactive Power (kVAR):** Calculated total downstream reactive power DER injection per phase.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the fuse (amps).
- **Type:** The type of fuse.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the fuse.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.

- **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
- **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
- **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
- **Manually Entered Limit:** Manually entered limit value.

4.2.11. Generator

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the generator.
- **ID:** Unique identifier of the generator.
- **Address:** Address of the generator.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the generator phases (ABC for A, B, and C phases).
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Branch power factor.
- **Voltage (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.

- **Current (Amps):** Real-time phase current (amps) and angle (deg).

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the generator in kV.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the generator (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the generator (kVAR).
- **Schedule Type:** Schedule type (Static or Dynamic).
- **Schedule Current Value:** Currently used output value in kW.

4.2.12. Line

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Name:** The name of the line.
- **ID:** Unique identifier of the line.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the reactor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current Side 1/ Side 2 (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.

- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Real Power Side 1/ Side 2 (kW):** Real power flow in kW.
- **Reactive Power Side 1/ Side 2 (kVAR):** Reactive power flow in kVAR.
- **Apparent Power Side 1/ Side 2 (kVA):** Apparent power flow in kVA.
- **Power Factor Side 1/ Side 2:** Power factor.
- **Total Real Power Loss (W):** Real power loss in W.
- **Total Reactive Power Loss (VAR):** Reactive power loss in VAR.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Nominal Switchable Section Flow A/B/C:** Data for nominal switchable section flow for phases A, B, and C.
    - Net Current (Amps): Current rate.
    - Net Real Power Flow (kW): Real power flow in kW.
    - Net Reactive Power Flow (kVAR): Reactive power flow in kVAR.
- **Fault Location Triggering Device Fault Current (Amps):** Current on fault indicator (Amps).

Engineering Data:

- **Length (ft):** Length of the line in feet.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the line.
- **Construction Type:** Construction type (for example, Overhead/Underground).
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The nominal level limit values for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
    - **Neutral Value:** Neutral wire limit value.
    - **Manually Entered Limit:** The manually entered limit values for phases A, B, C, and neutral, in Amps.

4.2.13. Load

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the load.
- **ID:** Unique identifier of the load.
- **Address:** Address of the load.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the load phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Embedded Generation Real Power (kW):** Total embedded real power generation in kW.
- **Embedded Generation Reactive Power (kVAR):** Total embedded reactive power generation in kVAR.
- **Total Native Load Real Power (kW):** Total native load real power in kW.
- **Total Native Load Reactive Power (kVAR):** Total native load reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Branch power factor.
- **Voltage (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Current (Amps):** Real-time phase current (amps) and angle (deg).

- **Cold Load Multiplier:** Cold load pickup multipliers for phases A, B, and C.
- **Voltage Imbalance Percent (%):** Voltage imbalance of the load expressed as a percentage.
- **High Power Quality Index (% kW):** Power quality index of the primary side.
- **Low Power Quality Index (% kW):** Power quality index of the secondary side.
- **Imbalance Power Quality Index (% kW):** Power quality index imbalance.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the load in kV.
- **Nominal Real Power (kW):** The nominal real power rate in kW.
- **Nominal Reactive Power (kVAR):** The nominal reactive power rate in kVAR.
- **Nominal Power Factor:** The nominal power factor of the load.
- **Normal Operational Limit - High:** Normal operational high voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Operational Limit - Low:** Normal operational low voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Operational Limit - High:** Emergency operational high voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Operational Limit - Low:** Emergency operational low voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Optimization Limit - High:** Normal optimization high voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Optimization Limit - Low:** Normal optimization low voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Optimization Limit - High:** Emergency optimization high voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Optimization Limit - Low:** Emergency optimization low voltage limit, expressed in per-unit or as configurable voltage base.
- **Voltage Imbalance Limit:** The voltage imbalance limit, expressed in per-unit or as configurable voltage base.
- **Schedule Type:** Schedule type (Static or Dynamic).
- **Schedule Current Value:** Currently used output value in kW.

4.2.14. Network Protector

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** The name of the network protector.
- **ID:** Unique identifier of the network protector.
- **Alternate ID:** Second identifier of the network protector.
- **Address:** Address of the network protector.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the network protector phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.

Engineering Data:

- **Continuous Current Rating (Amps):** The maximum interruption rating of the network protector (amps).

- **Type:** The type of network protector.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the network protector.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value:** The level limit value.

4.2.15. Permanent Jumper

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the permanent jumper.
- **ID:** Unique identifier of the permanent jumper.
- **Alternate ID:** Second identifier of the permanent jumper.
- **Address:** Address of the permanent jumper.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the permanent jumper phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.

- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the permanent jumper (amps).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the permanent jumper.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value:** The level limit value.
  - **Manually Entered Limit:** The manually entered limit value.

4.2.16. Power Transformer

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
-

External Company Name: The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Active Advanced Regulation Direction:** Direction in which advanced regulation is active. Can be Forward, Reverse, Idle (Below Flow Threshold), or None. "None" is the case when advanced regulation is not active.
- **DPF Sensed Flow Direction:** Flow direction estimated by DPF. Can be Forward, Reverse, or None.
- **Flow Direction Sensing Input:** Power quantity (active or reactive) used for flow direction sensing input.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Power factor.
- **Maximum Phase Loading against Level 1 Limit (%):** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the transformer as a percentage of rated load.
- **Grounding Current (Amps):** Grounding current rate and phase angle.
- **Loading Percentage against Level 1 Limit:** Real-time phase loading of the transformer as a percentage of the Level 1 load limit.
- **Current Side 1/Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Real Power Side 1/Side 2 (kW):** Real power flow in kW.
- **Reactive Power Side 1/Side 2 (kVAR):** Reactive power flow in kVAR.
- **Apparent Power Side 1/Side 2 (kVA):** Apparent power flow in kVA.
- **Real Power Copper Loss:** Real power winding (copper) loss of the transformer (kW).
- **Real Power Core Loss:** Real power core (iron) loss of the transformer (kW).

- **Reactive Power Copper Loss:** Reactive power winding (copper) loss of the transformer (kVAR).
- **Reactive Power Core Loss:** Reactive power core (iron) loss of the transformer (kVAR).

Engineering Data:

- **Rating:** The total capacity of the transformer (MVA).
- **Type:** The type of transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the transformer.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Tap Position:** Transformer primary/secondary side tap position.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value for phases A, B, and C.
  - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
  - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
  - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
  - **Derated Limit Loading:** Derated limit loading. Calculated as maximum flow divided by derated limit.
  - **Manually Entered Limit:** Manually entered limit value, in MVA.

4.2.17. Reactor

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the reactor.
- **ID:** Unique identifier of the reactor.
- **Address:** Address of the reactor.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.

- **Constructed Phase(s):** Enumeration of the reactor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current Side 1/Side 2 (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Real Power Side 1/Side 2 (kW):** Real power flow in kW.
- **Reactive Power Side 1/Side 2 (kVAR):** Reactive power flow in kVAR.
- **Apparent Power Side 1/Side 2 (kVA):** Apparent power flow in kVA.
- **Total Real Power Loss (W):** Real power loss in W.
- **Total Reactive Power Loss (VAR):** Reactive power loss in VAR.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Nominal Voltage:** The nominal reactor and phase voltages of the reactor.
- **Impedance (R+jX ohms):** The reactor impedance (in ohms).
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value:** The level limit value for phases A, B, and C.
    - **Derated Value** The level limit derated value for phases A, B, and C.
    - **Loading Adjusted for Reserved Capacity:** Loading adjusted for reserved capacity.
    - **Available Capacity Adjusted for Reserved Capacity:** Available capacity adjusted for

reserved capacity.

  - **Manually Entered Limit:** Manually entered limit value.

4.2.18. Recloser

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** The name of the recloser.
- **ID:** Unique identifier of the recloser.
- **Alternate ID:** Second identifier of the recloser.
- **Address:** Address of the recloser.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the recloser phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
-

- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Downstream DER/DG Injection Real Power (kW):** Calculated total downstream real power DER injection per phase.
- **Downstream DER/DG Injection Reactive Power (kVAR):** Calculated total downstream reactive power DER injection per phase.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the recloser (amps).
- **Type:** The type of recloser.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the recloser.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value for phases A, B, and C.
  - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
  - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
  - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
  - **Manually Entered Limit:** Manually entered limit value.

Protection Validation Results:

- **Maximum Fault (Maximum Interruption Violation):** Boolean value (Yes/No) that indicates if a fault placed at the load side of the recloser creates enough fault current to exceed the maximum interruption capability of the device (in Amps).
- **Maximum Fault (Maximum Interruption Current):** The maximum interruption capability of the recloser (in Amps).
- **Maximum Fault (Maximum Fault Current with Bolted Fault/Configured Fault Impedance):** The maximum short circuit current (in Amps) calculated for a location immediately downstream of the protective device evaluated with a zero-fault impedance, a user configured fault

impedance, or both.

- **Maximum Fault (Maximum Capacity Ratio with Bolted Fault/Configured Fault Impedance):** The ratio of the maximum calculated fault current evaluated with a zero-fault impedance, a user configured fault impedance, or both to the maximum interruption current of the device (in %).
- **Minimum Phase Fault (Minimum Pick-Up Violation):** Boolean value (Yes/No) that indicates if a fault placed at the furthest impedance distance away from the recloser does not create enough fault current (pick-up current) to exceed the phase trip relay setting of the device.
- **Minimum Phase Fault (Minimum Pick-up Current with Safety Factor):** The minimum phase pick-up current (multiplied by a safety factor) setting of the phase trip relay (in Amps).
- **Minimum Phase Fault (Minimum Fault Current with Bolted Fault/Configured Fault Impedance):** The minimum phase short circuit current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance, a user configured fault impedance, or both.
- **Minimum Phase Fault (Minimum Pick-up Ratio with Bolted Fault/Configured Fault Impedance):** The ratio of the minimum phase pick-up current (multiplied by a safety factor) to the minimum phase fault current evaluated with a zero-fault impedance, a user configured fault impedance, or both (in %).
- **Minimum Ground Fault (Minimum Pick-Up Violation):** Boolean value (Yes/No) that indicates if a fault placed at the furthest impedance distance away from the recloser does not create enough fault current to exceed the ground trip relay setting of the device (expressed in amps).
- **Minimum Ground Fault (Minimum Pick-up Current with Safety Factor):** The minimum ground pick-up current (multiplied by a safety factor) setting of the ground trip relay (in Amps).
- **Minimum Ground Fault (Minimum Fault Current with Bolted Fault/Configured Fault Impedance):** The minimum ground fault current (in Amps) calculated at the furthest downstream (highest impedance) location from the protective device in the protection area evaluated with a zero-fault impedance, a user configured fault impedance, or both.
- **Minimum Ground Fault (Minimum Pick-up Ratio with Bolted Fault/Configured Fault Impedance):** The ratio of the minimum ground pick-up current (multiplied by a safety factor) to the minimum ground fault current evaluated with a zero-fault impedance, a user configured fault impedance, or both (in %).

Study Mode Only Protection Validation Results:

- **Short Circuit Validation Status:** The short circuit analysis is valid (Yes/No). This field returns a value of Protected or Unprotected after checking the fault analysis protection validation calculation. This is an optional study mode only calculation performed within the PRV application.
- **Minimum Phase Fault (Protective Device Name):** The name of the device that protects the zone from the minimum phase fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.
-

Minimum Phase Fault (Protective Device ID): The ID of the device that protects the zone from the minimum phase fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.

- **Minimum Phase Fault (Pick-Up Current Limit):** The limit setting for the minimum phase pick-up current (Amps). This field is only applicable for three-phase devices.
- **Minimum Phase Fault (Fault Current):** The actual minimum phase fault current as calculated by the short circuit analysis, in amperes.
- **Minimum Ground Fault (Protective Device Name):** The name of the device that protects the zone from the minimum ground fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.
- **Minimum Ground Fault (Protective Device ID):** The ID of the device that protects the zone from the minimum ground fault current. If the device at the head of the zone is not able to protect the zone, devices progressively upstream are checked to see if they can protect the zone.
- **Minimum Ground Fault (Pick-Up Current Limit):** The limit setting of the minimum ground protective device (Amps).
- **Minimum Ground Fault (Fault Current):** The actual minimum ground fault current as calculated by the short circuit analysis (Amps).
- **Short Circuit Coordination Status:** This field returns a value of Coordinated or Uncoordinated after checking the fault analysis protection coordination calculation. This is an optional study mode only calculation performed within the PRV application.
- **Upstream Protective Device Name:** The name of the coordinated upstream protection device. This device can be the further upstream protection device if the device immediately upstream is not coordinated.
- **Upstream Protective Device ID:** The ID of the coordinated upstream protection device.

4.2.19. Regulator

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Name:** Name of the regulator.
- **ID:** Unique identifier of the regulator.
- **Address:** Address of the regulator.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.

- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Active Advanced Regulation Direction:** Direction in which advanced regulation is active. Can be Forward, Reverse, Idle (Below Flow Threshold), or None. "None" is the case when advanced regulation is not active.
- **DPF Sensed Flow Direction:** Flow direction estimated by DPF. Can be Forward, Reverse, or None.
- **Flow Direction Sensing Input:** Power quantity (active or reactive) used for flow direction sensing input.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Power Factor:** Power factor.
- **Maximum Phase Loading against Level 1 Limit (%):** Real-time maximum phase loading of the regulator as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the regulator as a percentage of rated load.
- **Loading Percentage against Level 1 Limit:** Real-time phase loading of the regulator as a percentage of the Level 1 load limit.
- **Current Side 1/Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Real Power Side 1/Side 2 (kW):** Real power flow in kW.
- **Reactive Power Side 1/Side 2 (kVAR):** Reactive power flow in kVAR.
- **Apparent Power Side 1/Side 2 (kVA):** Apparent power flow in kVA.

- **Real Power Copper Loss:** Real power winding (copper) loss of the regulator (kW).
- **Real Power Core Loss:** Real power core (iron) loss of the regulator (kW).
- **Reactive Power Copper Loss:** Reactive power winding (copper) loss of the regulator (kVAR).
- **Reactive Power Core Loss:** Reactive power core (iron) loss of the regulator (kVAR).

Engineering Data:

- **Rating:** The nominal capacity of the regulator (kVA).
- **Effective Rating:** The actual capacity of the regulator (kVA).
- **Type:** The type of regulator.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the regulator.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the regulator winding.
- **Tap Changer Type:** Regulator primary/secondary side tap changer type.
- **Tap Position:** Regulator primary/secondary side tap position.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value.
  - **Loading Adjusted for Reserved Capacity:** Loading adjusted for reserved capacity.
  - **Available Capacity Adjusted for Reserved Capacity:** Available capacity adjusted for reserved capacity.
  - **Derated Limit Loading:** Derated limit loading.
  - **Dynamic Limit - Regulation Step (%):** Dynamic limit for the regulation step in percentage. This field is empty if the dynamic limits are not available.
  - **Dynamic Limit - (Amps per phase):** Dynamic limit for the current per phase in Amps. This field is empty if the dynamic limits are not available.
  - **Manually Entered Limit:** The manually entered limit value.

4.2.20. Reference Bus Interface

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reference bus interface.
- **ID:** Unique identifier of the reference bus interface.
- **Address:** Address of the reference bus interface.
- **Service Center:** The name of the service center.

- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Voltage (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Current Side 1/ Side 2 (Amps):** Real-time phase current (amps) and angle (deg).

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the reference bus interface.

4.2.21. Sectionalizer

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** The name of the sectionalizer.
- **ID:** Unique identifier of the sectionalizer.
- **Alternate ID:** Second identifier of the sectionalizer.

- **Address:** Address of the sectionalizer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sectionalizer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the sectionalizer (amps).
- **Type:** The type of sectionalizer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the sectionalizer.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value:** The level limit value.

- **Manually Entered Limit:** The manually entered limit value.

4.2.22. Sensor

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the sensor.
- **ID:** Unique identifier of the sensor.
- **Address:** Address of the sensor.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the sensor belongs to.
- **Normal Station and Feeder:** The name of the nominal substation and feeder the sensor belongs to.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sensor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Node:** ID of the node.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Voltage (LG):** Phase voltage expressed as a 120 V base value, and phase angle in degrees.
- **Voltage (LL):** Linear voltage expressed as a 120 V base value, and phase angle in degrees.
- **Current (Amps):** Phase real-time current (amps).
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.

Engineering Data:

- **Nominal Voltage:** The nominal linear and phase voltages of the sensor.

4.2.23. Substation

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the substation.
- **ID:** Unique identifier of the substation.
- **Alternate Name:** Substation alternative name.
- **Address:** Address of the substation.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the substation belongs to.
- **Division:** The name of the division.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Protection Validation (PRV) Solution Time:** Date and time of the most recent protection validation solution.
- **Load and Volt/VAR Management (LVM) Solution Time:** Date and time of the most recent LVM solution.

4.2.24. Switch

This Analyst Output pop-up provides end users with the following information:

- **Power Flow Direction Arrows:** This pane visually indicates the direction that real (P) and reactive (Q) power flows through the device.

Identifiers:

- **Switch Number:** The name of the switch.
- **ID:** The unique identifier of the switch.
- **Alternate ID:** Second identifier of the switch.
- **Address:** Address of the switch.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).

- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Real Power (kW):** Real power flow in kW.
- **Reactive Power (kVAR):** Reactive power flow in kVAR.
- **Apparent Power (kVA):** Apparent power flow in kVA.
- **Power Factor:** Branch power factor.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Downstream DER/DG Injection Real Power (kW):** Calculated total downstream real power DER injection per phase. This field is applicable only for three phase switches.
- **Downstream DER/DG Injection Reactive Power (kVAR):** Calculated total downstream reactive power DER injection per phase. This field is applicable only for three phase switches.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).

- **Value (per Phase):** The level limit value for phases A, B, and C.
- **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
- **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
- **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
- **Derated Limit Loading:** Derated limit loading. Calculated as maximum flow divided by derated limit.
- **Manually Entered Limit:** The manually entered limit value.

4.2.25. Temporary Switch

This Analyst Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the temporary switch.
- **ID:** Unique identifier of the temporary switch.
- **Alternate ID:** Second identifier of the temporary switch.
- **Address:** Address of the temporary switch.
- **DPF Solution Time:** Date and time of the most recent distribution power flow solution.

Engineering Outputs:

- **Total kW:** Total power expressed in kilowatts.
- **Total kVAR:** Total kilovolt-amperes reactive.
- **Flow Limit (Amps):** Flow limit value in amps.
- **Loading Percentage:** Loading percentage.
- **Amps:** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **kW:** Power for phases A, B, and C expressed in kilowatts.
- **kVAR:** Kilovolt-amperes reactive for phases A, B, and C.
- **Angle Difference:** The difference of angle value between side 1 and side 2 voltages.

4.3. Operator Attributes Pop-Ups

4.3.1. Annotation

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the annotation.
- **Display:** Annotation display.
- **Comment:** Annotation comment.

4.3.2. Autotransformer

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the transformer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the transformer is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the transformer is connected.

Engineering Data:

- **Rating:** The total capacity of the autotransformer (MVA).
- **Type:** The type of transformer.
- **Description:** Description of the transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the transformer winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Winding Type:** The transformer winding type (Delta, Wye, Open).
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Minimum Tap Position:** Transformer primary/secondary side minimum tap position.
- **Nominal Tap Position:** Transformer primary/secondary side tap nominal position.
- **Maximum Tap Position:** Transformer primary/secondary side tap maximum position.
- **Tap Step Size (%):** The size of the tap step expressed as a percentage.

4.3.3. Breaker

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the breaker.
- **ID:** Unique identifier of the breaker.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the breaker.
- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.
- **Permission Area:** Permission area the breaker belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the breaker is connected nominally.
- **External Interface (OMS Interface):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (OMS Interface):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or

decommissioning.

- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the breaker phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Team Names:** Names of automated scheme teams that the breaker is associated with.
- **Team IDs:** IDs of automated scheme teams that the breaker is associated with.
- **Scheme Name:** Name of the automated scheme the breaker is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the breaker is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the breaker is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Breaker's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the breaker.
- **Bypass Switch:** Name of a bypass switch, if any.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the breaker (amps).
- **Type:** The type of breaker.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the breaker.
- **Ground Trip Capable:** Indicates whether the recloser can trip on a ground fault (Yes/No).
- **Ground Trip Status:** Normal status of ground trip (Enabled/Disabled).
- **Phase Trip Current (Amps):** Phase trip current value in amps.
- **Ground Trip Current (Amps):** Ground trip current value in amps.
- **Neutral Relay Setting (Amps):** The maximum allowed neutral trip setting in amps.
- **Interrupting Capacity (Amps):** Interrupting capacity in amps.
- **PRV Safety Factor:** Safety factor used to calculate minimum phase and ground pick-up limit.
- **Ground and Neutral Relay Safety Factor:** Safety factor used when comparing the phase imbalance flow (the vector sum of currents in the three phases) for breakers and reclosers to the minimum ground relay pick-up setting and neutral relay current setting.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the breaker based on measurement data.
- **Hot Line Tag Status:** Hot line tag status.
- **Alarm Status:** Flag indicating the alarm status of the breaker.
- **Remote Controlled Status:** Flag indicating the remote control status of the breaker. If Yes, the breaker is controlled remotely. If No, the breaker is controlled locally.
- **Reclosing Status:** Flag indicating the reclosing capability of the device is enabled.
- **Lockout Status:** Flag indicating the lockout status of the breaker.

4.3.4. Bus

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the bus.
- **ID:** Unique identifier of the bus.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the bus belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the bus is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the bus phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the bus is connected.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the bus in KV.

4.3.5. Capacitor

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the capacitor.
- **ID:** Capacitor unique identifier.
- **Address:** Address of the capacitor.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the capacitor belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the capacitor is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the capacitor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the capacitor is connected.
- **Status:** Current (real-time) status of the capacitor (On or Off).

Engineering Data:

- **Total Size (kVAR):** The total capacity of the capacitor (kVAR).
- **Control Type:** Control type of the capacitor (for example, remote only).
- **Local Control Type:** The local control mode for the capacitor (None, Power Factor, Time, Temperature, VAR, Current, or Voltage).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the capacitor, in kV.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages of the capacitor.
- **Time of Last Switching Operation:** Time of the last switching operation.

4.3.6. Composite Switch

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name/number of the switch.
- **ID:** Unique identifier of the composite switch.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the composite switch.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the switch belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the switch is connected nominally.
- **Normal Position:** The normal position for the ganged switch (for example, AB).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the switch is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the switch is connected.
- **Current Position:** Switch’s current position (Open/Closed).

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Model Name:** The name of the switch model.
- **Type:** Name of the type of the switch.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the switch.

- **Supervisory Cutout Status:** Supervisory cutout state of the switch, set by the switch control. The dispatcher cannot control the switch in the supervisory cutout state.
- **Alarm Status:** Flag indicating the alarm status of the switch.

4.3.7. Customer

This Analyst Output pop-up provides end users with the following information:

- **Account Number:** The account number of the customer.
- **Name:** Customer’s name.
- **Critical Account:** Indicates if this is a critical account.
- **Phone Number:** Customer’s phone number.
- **Address:** Customer’s address.
- **Meter Number:** Meter number of the customer.
- **Load ID:** Load ID of the customer.
- **Service Point:** Service point of the customer.
- **X/Y:** X and Y coordinates of the customer.
- **Service Center:** Service center the customer belongs to.
- **City:** City the customer belongs to.
- **Feeder:** Name of the feeder the customer is connected to.
- **AMI:** Number of customer’s meters.
- **Inactive Account:** Indicates if this is an inactive account.
- **Meter Brown Out:** Indicates the voltage reduction below a threshold level for a certain period of time. Can be Yes or No.
- **Suspended:** Indicates if the customer is suspended. Can be Yes or No.

4.3.8. Cut

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the cut.
- **ID:** Unique identifier of the cut.
- **Comment:** Comment for the cut.
- **Phase:** Phase designation.
- **Type:** Type of the cut.
- **Normal Station:** The name of the nominal substation the cut belongs to.

- **Normal Feeder:** The name of the feeder that is normally connected to the cut.
- **Node 1:** ID of the node 1.
- **Node 2:** ID of the node 2.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the cut is connected.
- **Feeder:** The name of the feeder to which the cut is connected.

4.3.9. Distribution Transformer

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **Permission Area:** Permission area the transformer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the transformer is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Elbow Switches:** Elbow switches connected to the transformer.
- **Critical Customer Connected:** Indicates whether the transformer supplies critical customers (Yes or No).
- **Total Customers Connected:** The number of customers connected to phases A, B, C, and ABC.
- **Number of Critical Customers:** The number of critical customers on attached loads.
- **External Company Name:** The name of the third-party company that owns the equipment.

This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the transformer is connected.

Engineering Data:

- **Rating:** The total capacity of the transformer (kVA).
- **Type:** The type of transformer.
- **Description:** Description of the transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the transformer.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Winding Type:** The transformer winding type (Delta, Wye, Open).

4.3.10. Duct Bank

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the duct bank.
- **ID:** Unique identifier of the duct bank.
- **Address:** Address of the duct bank.

4.3.11. Elbow

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the switch.
- **ID:** Switch unique identifier.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the switch.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the switch belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the switch is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.

- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the commissioning project that is associated with the device.
- **Project Stage:** The device commissioning or decommissioning stage. You can use this field to manually set or update the project stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the switch is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the switch is connected.
- **Customer Count:** Number of customers downstream of the switch.
- **Current Position:** Switch's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the switch.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the switch.
- **Supervisory Cutout Status:** Supervisory cutout state of the switch, set by the switch control. The dispatcher cannot control the switch in the supervisory cutout state.
- **Alarm Status:** Flag indicating the alarm status of the switch.
- **Remote Controlled Status:** Flag indicating the remote control status of the switch. If Yes, the switch is controlled remotely. If No, the switch is controlled locally.

4.3.12. Energy Resource

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the energy resource.
- **ID:** Unique identifier of the energy resource.
- **Address:** Address of the energy resource.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the energy resource belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the energy resource is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the energy resource phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the energy resource is connected.
- **Status:** Connection status (Connected/Disconnected).
- **Last Disconnection Time:** The time when the last disconnection occurred (for disconnected energy resources).
- **Last Disconnect Reason:** The reason for the last disconnection (for disconnected energy resources).
- **Inverter ID:** The ID of the associated inverter.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the energy resource, in kV.
- **Maximum Voltage Before Separation From the Network:** Maximum voltage before separation from the network (per unit).
- **Minimum Voltage Before Separation From the Network:** Minimum voltage before separation from the network (per unit).
- **Maximum Frequency Before Separation From the Network:** Maximum frequency before separation from the network (Hz). By default, 63 Hz.
- **Minimum Frequency Before Separation From the Network:** Minimum frequency before

separation from the network (Hz). By default, 56 Hz.

- **Minimum State of Charge:** The amount of charge below which the device should never be drawn, in kWh.
- **Maximum State of Charge:** The maximum amount of charge that can be stored in the DER storage device, in kWh.
- **Energy Rating:** The useable capacity of the DER storage device, in kWh.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the energy resource (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the energy resource (kVAR).
- **Planned Restoration Time (sec):** The planned restoration time for a DER/DG after re-energization of a feeder, in seconds.

4.3.13. Fault Indicator

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the fault indicator.
- **ID:** Fault indicator unique identifier.
- **Address:** Fault indicator address.
- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.
- **Permission Area:** Permission area the fault indicator belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the fault indicator is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the fault indicator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.

- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the real-time substation and feeder the fault indicator belongs to.
- **Status:** Current (real-time) status of the fault indicator: Set, Unset, or Unknown for a non-directional fault indicator; Unset, Both, Side 1, Side 2, or Unknown for a type 1 directional fault indicator; Unset, Forward Set, Reverse Set, or Unknown for a type 2 directional fault indicator.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the fault indicator.
- **Rating (Amps):** The rated interruption capacity of the fault indicator (Amps).

4.3.14. Feeder

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the feeder.
- **ID:** Unique identifier of the feeder.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **EMS NETMOM Load Name:** Load name.
- **Default Feeder Color:** Number of the default color used to display this feeder on the network view (expressed as an integer 1–16).
- **Feeder Head Node Name:** Name of the feeder head node.
- **Feeder Evaluation Head Node Name:** The name of the evaluation node.
- **Service Center:** The name of the service center associated with the feeder.

Engineering Data:

- **Nominal Voltage:** Nominal voltage of the feeder in kV.
- **Number of Load Categories:** Number of load categories.

4.3.15. Fuse

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the fuse.
- **ID:** Fuse unique identifier.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the fuse.
- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.
- **Permission Area:** Permission area the fuse belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the fuse is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the fuse phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Loop Number:** Loop number of the fuse.
- **Solid Blade Fuse:** The Yes or No value.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the fuse is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the fuse is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Fuse's current position (Open/Closed).

- **Currently Open Phase(s):** Phases that are currently opened by the fuse.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the fuse (amps).
- **Type:** The type of fuse.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the fuse.

4.3.16. Generator

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the generator.
- **ID:** Unique identifier of the generator.
- **Address:** Address of the generator.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the generator belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the generator is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the generator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the generator is connected.
- **Status:** Connection status (Connected/Disconnected).
- **Last Disconnection Time:** The time when the last disconnection occurred (for disconnected generators).
- **Last Disconnect Reason:** The reason for the last disconnection (for disconnected generators).
- **Inverter ID:** The ID of the associated inverter.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the generator, in kV.
- **Maximum Voltage Before Separation From the Network:** Maximum voltage before separation from the network (per unit).
- **Minimum Voltage Before Separation From the Network:** Minimum voltage before separation from the network (per unit).
- **Maximum Frequency Before Separation From the Network:** Maximum frequency before separation from the network (Hz). By default, 63 Hz.
- **Minimum Frequency Before Separation From the Network:** Minimum frequency before separation from the network (Hz). By default, 56 Hz.
- **Planned Restoration Time (sec):** The planned restoration time for a DER/DG after re-energization of a feeder, in seconds.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the generator (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the generator (kVAR).

4.3.17. Ground

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the ground.
- **ID:** Unique identifier of the ground.
- **Address:** Address of the ground.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the ground belongs to.
- **Normal Station and Feeder:** The name of the substation and feeder to which the ground is connected at normal conditions.
- **Normally De-Energized:** Indicates whether the ground is normally de-energized (Yes/No).
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the ground phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the ground is connected.
- **Feeder:** The name of the feeder to which the ground is connected.

Engineering Data:

- **Phase(s):** Enumeration of ground phases (A, B, and C phases, and neutral).

4.3.18. Inverter

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **ID:** Inverter unique identifier.
- **Name:** Name of the inverter.

Attributes:

- **Inverter Size (kVA):** Inverter size (in kVA).
- **Inverter Phases:** The phasing of the inverter.
- **Inverter Node:** Inverter node ID.

Model Attributes:

- **Model Name:** The inverter model name.
- **Maximum Real Power Delivered (kW):** The maximum real power the DER can deliver to the grid (in kW).
- **Maximum Reactive Power Delivered (kVAR):** The maximum reactive power the DER can deliver to the grid (in kVAR).
- **Maximum Apparent Power Delivered (kVA)**: The maximum apparent power the DER can deliver to the grid (in kVA).
- **Maximum Real Power Absorbed (kW):** The maximum real power the DER can absorb from the grid (in kW). For energy storage charging.
- **Maximum Reactive Power Absorbed (kVAR):** The maximum reactive power the DER can absorb from the grid (in kVAR). For energy storage charging.
- **Maximum Apparent Power Absorbed (kVA):** The maximum apparent power the DER can absorb from the grid (in kVA). For energy storage charging.

4.3.19. Line

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

-

Name: Name of the line.

- **ID:** Line unique identifier.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the line belongs to.
- **Normal Station and Feeder:** The name of the associated substation and Feeder to which the line is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Enumeration of phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the line is connected.

Engineering Data:

- **Length (ft):** Length of the line.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the line.
- **Construction Type:** Construction type (for example, Overhead/Underground).

4.3.20. Load

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the load.
- **ID:** Load unique identifier.
- **Address:** Address of the load.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area.
- **Normal Station and Feeder:** The name of the substation and feeder to which the load is connected at normal conditions.
- **Project ID:** ID of the project that is associated with the device commissioning or

decommissioning.

- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the load phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Primary Meter:** Primary meter of the load.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the load is connected.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the load in kV.
- **Nominal Real Power (kW):** The nominal real power rate in kW.
- **Nominal Reactive Power (kVAR):** The nominal reactive power rate in kVAR.
- **Nominal Power Factor:** The nominal power factor of the load.

4.3.21. Loop End

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the loop end.
- **ID:** Unique identifier of the loop end.

4.3.22. Manhole

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the manhole.
- **ID:** Unique identifier of the manhole.
- **Reference Drawing:** Associated reference drawing.
- **Manhole Type:** Type of manhole as defined by the enumerated list.
- **Number of Objects in Manhole:** Number of objects (for example, switches) that are located within the manhole.

4.3.23. Network Protector

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the network protector.
- **ID:** Unique identifier of the network protector.
- **Alternate ID:** Second identification number of the network protector.
- **Address:** Address of the network protector.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the network protector belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the network protector is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the network protector phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Team Names:** Names of automated scheme teams that the network protector is associated with.
- **Team IDs:** IDs of automated scheme teams that the network protector is associated with.
- **Scheme Name:** Name of the automated scheme the network protector is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the network protector is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the network protector is connected.
- **Customer Count:** The number of associated customers that can be de-energized.

- **Current Position:** Network protector's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the network protector.

Engineering Data:

- **Continuous Current Rating (Amps):** The maximum interruption rating of the network protector (amps).
- **Type:** The type of network protector.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the network protector.
- **Operation Mode:** Operation mode.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the network protector based on measurement data.
- **Alarm Status:** Flag indicating the alarm status of the network protector.
- **Remote Controlled Status:** Flag indicating the remote control status of the network protector. If Yes, the network protector is controlled remotely. If No, the network protector is controlled locally.

4.3.24. Permanent Jumper

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the permanent jumper.
- **ID:** Unique identifier of the permanent jumper.
- **Alternate ID:** Second identification number of the permanent jumper.
- **Address:** Address of the permanent jumper.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the permanent jumper belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the permanent jumper is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.

- **Constructed Phase(s):** Enumeration of the permanent jumper phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the permanent jumper is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the permanent jumper is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Permanent jumper's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the permanent jumper.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the permanent jumper (amps).
- **Type:** The type of permanent jumper.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the permanent jumper.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the permanent jumper based on measurement data.
- **Alarm Status:** Flag indicating the alarm status of the permanent jumper.
- **Remote Controlled Status:** Flag indicating the remote control status of the permanent jumper. If Yes, the permanent jumper is controlled remotely. If No, the permanent jumper is controlled locally.

4.3.25. Power Transformer

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.

- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.
- **Permission Area:** Permission area the transformer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the transformer is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the transformer is connected.
- **Number of Daily Switching Operations:** The number of switching operations performed in the current day.

Engineering Data:

- **Rating:** The total capacity of the autotransformer (MVA).
- **Type:** The type of transformer.
- **Description:** Description of the transformer.
- **Time of Last Operation:** Time of the last operation.
- **Maximum Number of Daily Operations:** The maximum number of operations in one day. This is set by the DNAF Configuration Editor.
- **Maximum Phase Voltage Imbalance:** The maximum phase voltage imbalance in percentage.
- **Arc Suppression Coil (ASC) is Connected (Secondary):** Specifies whether the Arc Suppression Coil (ASC) fault indicator is connected to the secondary side of the power transformer.
- **ASC Regulation Default On Status (Secondary):** Specifies whether the ASC default Regulation On status is True.
-

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the transformer winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Winding Type:** The transformer winding type (Delta, Wye, Open).
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Minimum Tap Position:** Transformer primary/secondary side minimum tap position.
- **Nominal Tap Position:** Transformer primary/secondary side tap nominal position.
- **Maximum Tap Position:** Transformer primary/secondary side tap maximum position.
- **Normal Tap Position:** Transformer primary/secondary side tap normal position. The field shows "--" (null) for "Regulating" or "None" (no tap exists for the winding) tap changer types, and an integer value between (or equal to) the minimum and maximum tap position for the "Fixed" tap changer type.
- **Tap Step Size (%):** The size of the tap step expressed in percentage.

4.3.26. Transformer Bank

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer bank.
- **ID:** Transformer bank identifier. This may not be unique since any of the transformers in the bank may have same identifier.
- **Number of Transformers in Bank:** The number of transformers available in the bank.
- **Transformer1 ID**: Identifier of the first transformer in the list of transformers available in the bank.
- **Transformer2 ID**: Identifier of the second transformer in the list of transformers available in the bank.
- **Transformer3 ID**: Identifier of the third transformer in the list of transformers available in the bank.
- **Permission Area:** Permission area to which the transformer bank belongs.

4.3.27. Reactor

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reactor.
- **ID:** Unique identifier of the reactor.

- **Address:** Address of the reactor.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the reactor belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the reactor is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the reactor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the reactor is connected.

Engineering Data:

- **Nominal Voltage:** Nominal voltage.
- **Impedance (R+jX ohms):** The reactor impedance (in ohms).

4.3.28. Recloser

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name (number) of the recloser.
- **ID:** Unique identifier of the recloser.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the recloser.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the recloser belongs to.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
-

County: The name of the associated county.

- **Municipality:** The name of the associated municipality.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the recloser is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the recloser phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Team Names:** Names of automated scheme teams that the recloser is associated with.
- **Team IDs:** IDs of automated scheme teams that the recloser is associated with.
- **Scheme Name:** Name of the automated scheme the recloser is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the recloser is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the recloser is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Recloser's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the recloser.
- **Bypass Switch:** Name of a bypass switch, if any.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the recloser (amps).
- **Type:** The type of recloser.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the recloser.
- **Reversible Recloser:** Indicates whether the recloser is reversible (Yes/No).
- **Ground Trip Capable:** Indicates Indication of whether the recloser can trip on a ground fault

(Yes/No).

- **Ground Trip Status:** Normal status of ground trip (Enabled/Disabled).
- **Phase Trip Current (Amps):** Phase trip current value in amps.
- **Ground Trip Current (Amps):** Ground trip current value in amps.
- **Neutral Relay Setting (Amps):** The maximum allowed neutral trip setting in amps.
- **Interrupting Capacity (Amps):** Interrupting capacity in amps.
- **PRV Safety Factor:** Safety factor used to calculate minimum phase and ground pick-up limit.
- **Ground and Neutral Relay Safety Factor:** Safety factor used when comparing the phase imbalance flow (the vector sum of currents in the three phases) for breakers and reclosers to the minimum ground relay pick-up setting and neutral relay current setting.
- **Initial Block Close:** Specifies whether the recloser connects an energy resource with size greater than the value specified in the Minimum DER (kW) field for the default FISR problem formulation in the DNAF Configuration Editor. FISR plans will include a step to prevent connection of this energy resource.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the recloser based on measurement data.
- **Hot Line Tag Status:** Flag indicating the hot line tag status.
- **Alarm Status:** Flag indicating the alarm status of the recloser.
- **Remote Controlled Status:** Flag indicating remote control of the recloser. If Yes, the recloser is controlled remotely. If No, the recloser is controlled locally.
- **Operation Mode Status:** Flag indicating the operation mode status.
- **Reclosing Status:** Flag indicating if the reclosing capability of the device is enabled.
- **Lockout Status:** Flag indicating the lockout status of the recloser.

4.3.29. Regulator

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the regulator.
- **ID:** Unique identifier of the regulator.
- **Address:** Address of the regulator.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the regulator belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the regulator is connected nominally.

- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the regulator is connected.
- **Number of Daily Operations:** The number of operations performed in the current day.

Engineering Data:

- **Rating:** The nominal capacity of the regulator (kVA).
- **Effective Rating:** The actual capacity of the regulator (kVA).
- **Type:** The type of regulator.
- **Description:** Description of the transformer.
- **Time of Last Operation:** Time of the last operation.
- **Maximum Number of Daily Operations:** The maximum number of operations in one day. This is set by the DNAF Configuration Editor.
- **Maximum Phase Voltage Imbalance:** The maximum phase voltage imbalance in percentage.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the regulator winding.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the regulator winding.
- **Winding Type:** The regulator winding type (Delta, Wye, Open).
- **Tap Changer Type:** Regulator primary/secondary side tap changer type.
- **Minimum Tap Position:** Regulator primary/secondary side minimum tap position.
- **Nominal Tap Position:** Regulator primary/secondary side tap nominal position.
- **Maximum Tap Position:** Regulator primary/secondary side tap maximum position.
- **Tap Step Size (%):** The size of the tap step expressed as a percentage.

4.3.30. Reference Bus Interface

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reference bus interface.
- **ID:** Unique identifier of the reference bus interface.
- **Address:** Address of the reference bus interface.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the reference bus interface belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the reference bus interface is connected nominally.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the reference bus interface is connected.
- **Status:** Reference bus interface connection status (Connected or Disconnected).

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the reference bus interface.

4.3.31. Reserved Capacity

This Analyst Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reserved capacity.
- **ID:** Unique identifier of the reserved capacity.
- **Description:** Description.
- **Service Center:** The name of the service center.
- **Normal Station:** The name of the nominal substation the device belongs to.

- **Normal Feeder:** The name of the nominal substation the device belongs to.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the reserved capacity is connected.
- **Feeder:** The name of the feeder to which the reserved capacity is connected.

4.3.32. Sectionalizer

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the sectionalizer.
- **ID:** Unique identifier of the sectionalizer.
- **Alternate ID:** Second identification number of the sectionalizer.
- **Address:** Address of the sectionalizer.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the sectionalizer belongs to.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the sectionalizer is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sectionalizer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Team Names:** Names of automated scheme teams that the sectionalizer is associated with.
- **Team IDs:** IDs of automated scheme teams that the sectionalizer is associated with.
- **Scheme Name:** Name of the automated scheme the sectionalizer is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

-

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the sectionalizer is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the sectionalizer is connected.
- **Customer Count:** The number of associated customers that can be de-energized.
- **Current Position:** Sectionalizer's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the sectionalizer.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the sectionalizer (amps).
- **Type:** The type of sectionalizer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the sectionalizer.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the sectionalizer based on measurement data.
- **Alarm Status:** Flag indicating the alarm status of the sectionalizer.
- **Remote Controlled Status:** Flag indicating the remote control status of the sectionalizer. If Yes, the sectionalizer is controlled remotely. If No, the sectionalizer is controlled locally.

4.3.33. Sensor

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the sensor.
- **ID:** Unique identifier of the sensor.
- **Address:** Address of the sensor.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the sensor belongs to.
- **Normal Station and Feeder:** The name of the nominal substation and feeder the sensor belongs to.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sensor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
-

Removed Phase(s): Individual phases that were recently removed.

- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station and Feeder:** The name of the substation and feeder to which the sensor is connected.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the sensor.

4.3.34. Splice Box

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the splice box.
- **ID:** Unique identifier of the splice box.
- **Reference Drawing:** Reference drawing.
- **Type:** Splice box type number.
- **Number of Objects:** Number of objects within the splice box.

4.3.35. Structure

This Operator Attributes pop-up provides end users with the following information:

Identifiers.

- **ID:** Structure unique identifier.
- **Name:** Structure name.
- **Feeder Name:** Feeder name.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.

4.3.36. Substation

This Operator Attributes pop-up provides end users with the following information:

Identifiers.

- **Name:** Substation name.
- **ID:** Substation unique identifier.
- **Alternate Name:** Substation alternative name.
- **Address:** Address of the substation.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the substation belongs to.
- **Division:** The name of the division.

4.3.37. Switch

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the switch.
- **ID:** Switch unique identifier.
- **Alternate ID:** Second unique identifier.
- **Address:** Address of the switch.
- **Service Center:** The name of the service center.
- **Location Description:** Information about the device location (for example, pole number or street number).
- **Permission Area:** Permission area the switch belongs to.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Normal Station and Feeder:** The name of the associated substation and feeder to which the switch is connected nominally.
- **External Interface (for OMS):** Indicates if the selected element belongs to the external interface. Can be Yes or No.
- **Externally Owned (for OMS):** Indicates if the selected element is externally owned. Can be Yes or No.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Team Names:** Names of automated scheme teams that the switch is associated with.

- **Team IDs:** IDs of automated scheme teams that the switch is associated with.
- **Scheme Name:** Name of the automated scheme the switch is associated with.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Connectivity Data (Real Time):

- **Station 1 and Feeder 1:** The name of the substation and feeder to which the switch is connected.
- **Station 2 and Feeder 2:** The name of the second substation and feeder to which the switch is connected.
- **Customer Count:** Number of customers downstream of the switch.
- **Current Position:** Switch's current position (Open/Closed).
- **Currently Open Phase(s):** Phases that are currently opened by the switch.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.

Measurement Data:

- **Open/Closed Status:** Flag indicating the state of the switch.
- **Supervisory Cutout Status:** Supervisory cutout state of the switch, set by the switch control. The dispatcher cannot control the switch in the supervisory cutout state.
- **Alarm Status:** Flag indicating the alarm status of the switch.
- **Remote Controlled Status:** Flag indicating the remote control status of the switch. If Yes, the switch is controlled remotely. If No, the switch is controlled locally.

4.3.38. Switch Cabinet

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the switch cabinet.
- **ID:** Unique identifier of the switch cabinet.
- **Alternate ID:** Second unique identifier.
- **Address:** Address where the switch cabinet is located.
- **Permission Area:** Permission area the substation belongs to.
- **Location Description:** Information about the device location (for example, pole number or

street number).

- **State:** The name of the associated state.
- **County:** The name of the associated county.
- **Municipality:** The name of the associated municipality.
- **Normal Station:** The name of the nominal substation the switch cabinet belongs to.
- **Feeder Main Connection:** The switch cabinet is connected to the feeder backbone (Yes/No).
- **Cabinet Type:** Switch cabinet type.
- **Switches:** The list of device names that the switch cabinet contains.

4.3.39. Tag

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the tag.
- **ID:** Unique identifier of the tag.

4.3.40. Temporary Ground

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the temporary ground.
- **ID:** Unique identifier of the temporary ground.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Engineering Data:

- **Neutral Phase is Present:** Phases that are grounded (for example, ABC).
- **Ground Resistance:** Resistance to the ground (p.u.).
- **Ground Reactance:** Reactance to the ground (p.u.).

4.3.41. Temporary Switch

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the temporary switch (jumper).

- **ID:** Unique identifier of the temporary switch.
- **Normal Station:** The name of the nominal substation the temporary switch belongs to.
- **Normal Feeder:** The name of the feeder that is normally connected to the temporary switch.
- **Component 1:** ID of the temporary switch's component 1.
- **Component 2:** ID of the temporary switch's component 2.
- **X1 Coordinate:** X coordinate of component 1.
- **Y1 Coordinate:** Y coordinate of component 1.
- **X2 Coordinate:** X coordinate of component 2.
- **Y2 Coordinate:** Y coordinate of component 2.

Model Data:

- **Phase(s):** Phase designation.
- **Size:** Size of the switch.

Connectivity Data (Real Time):

- **Station:** The name of the substation to which the temporary switch is connected.
- **Feeder:** The name of the feeder the temporary switch's component 1 belongs to.
- **Feeder2:** The name of the feeder the temporary switch's component 2 belongs to.
- **Current Position:** Current state, open or closed.

4.3.42. Transformer Bank

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer bank.
- **ID:** Unique identifier of the transformer bank.
- **Number of Objects in Bank:** Number of transformers in the transformer bank.

4.3.43. Trouble Ticket

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the trouble ticket.
- **ID:** Unique identifier of the trouble ticket.
- **X Coordinate:** The X coordinate of the trouble ticket on the geographic display.

- **Y Coordinate:** The Y coordinate of the trouble ticket on the geographic display.
- **Ticket Type:** Type of the trouble ticket.
- **Status:** Status of the trouble ticket.
- **First Stage Device:** First switchable device upstream from the customer.

4.3.44. UI Container

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the UI container.
- **ID:** Unique identifier of the UI container.
- **Address:** Address where the UI container is located.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the substation belongs to.
- **Normal Station:** The name of the nominal substation the UI container belongs to.
- **Container Type:** UI container type.

4.3.45. Vault

This Operator Attributes pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the vault.
- **ID:** Unique identifier of the vault.
- **Alternate ID:** Second unique identifier.
- **Address:** Address where the vault is located.
- **Permission Area:** Permission area the substation belongs to.
- **Normal Station:** The name of the nominal substation the vault belongs to.
- **Feeder Main Connection:** The vault is connected to the feeder backbone (Yes/No).
- **Vault Type:** Vault type.

4.4. Operator Output Pop-Ups

4.4.1. Autotransformer

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Loading Level Against Level 1 Limit (%):** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the transformer as a percentage of rated load.
- **Grounding Current (Amps):** Grounding current rate and phase angle.
- **Loading Percentage:** Loading percentage.
- **Current Side 1/ Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.

- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.

Engineering Data:

- **Rating:** The total capacity of the autotransformer (MVA).
- **Type:** The type of transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the transformer.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Tap Position:** Transformer primary/secondary side tap position.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value.

4.4.2. Breaker

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the breaker.
- **ID:** Unique identifier of the breaker.
- **Alternate ID:** Second unique identifier of the breaker.
- **Address:** Address of the breaker.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the breaker phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.

- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the breaker (amps).
- **Type:** The type of breaker.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the breaker.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
    - **Derated Limit Loading:** Derated limit loading. Calculated as maximum flow divided by derated limit.
    - **Manually Entered Limit:** The manually entered limit value.

4.4.3. Bus

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the bus.
- **ID:** Unique identifier of the bus.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the bus phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Calculated Voltage (LL):** The calculated line-to-line voltage for phases A, B, and C.
- **Calculated Voltage (LG):** The calculated line-to-ground voltage for phases A, B, and C.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages in the bus in KV.

4.4.4. Capacitor

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the capacitor.
- **ID:** Unique identifier of the capacitor.
- **Address:** Address of the capacitor.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the capacitor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
-

Removed Phase(s): Individual phases that were recently removed.

- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Capacitor Position Validation:** Is the current step position of the capacitor bank valid (Yes/No).
- **Total Real Power (W):** Total real power in W.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.

Engineering Data:

- **Total Size (kVAR):** The total capacity of the capacitor (kVAR).
- **Control Type:** Control type of the capacitor (for example, remote only).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the capacitor, in kV.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages of the capacitor.
- **Status:** Status of the capacitor (On/Off).

4.4.5. Composite Switch

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the switch.
- **ID:** Unique identifier of the switch.
- **Alternate ID:** Second unique identifier of the switch.
- **Address:** Address of the switch.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution

power flow solution.

- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results (for all associated switches):

- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.

Engineering Data (for all associated switches):

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value:** The level limit value.

4.4.6. Customer

This Operator Output pop-up provides end users with the following information:

- **Name:** Customer’s name.
- **Critical Customer:** Indicates if this is a critical customer.
- **Phone Number:** Customer’s phone number.
- **Address:** Customer’s address.
- **Priority Call:** If True, indicates a priority call.
- **Call?:** If Yes, True that this customer called.
- **Call Time:** Call time for this customer.
- **Comment:** Comment from this customer’s call.
-

AMI Off: If Yes, AMI reported this customer outage.

- **AMI On:** If Yes, AMI reported this customer ON.
- **AMR Time:** The time that AMI last reported for this customer.
- **Meter Brown Out:** Indicates the voltage reduction below a threshold level for a certain period of time. Can be Yes or No.

4.4.7. Distribution Transformer

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the distribution transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Loading Level Against Level 1 Limit (%):** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the transformer as a percentage of rated load.
- **Grounding Current (Amps):** Grounding current rate and phase angle.

- **Loading Percentage:** Loading percentage.
- **Current Side 1/ Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.

Engineering Data:

- **Rating:** The total capacity of the autotransformer (kVA).
- **Type:** The type of transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the transformer.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Limits:** Data on limits.
  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value for phases A, B, and C.
  - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.

4.4.8. Elbow

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the switch.
- **ID:** Unique identifier of the switch.
- **Alternate ID:** Second identifier of the switch.
- **Address:** Address of the switch.
- **Project ID:** ID of the commissioning project that is associated with the device.
- **Project Stage:** The device commissioning or decommissioning stage. You can use this field to manually set or update the project stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.

4.4.9. Energy Resource

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the energy resource.
- **ID:** Unique identifier of the energy resource.
- **Address:** Address of the energy resource.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the energy resource phases (ABC for A, B, and C phases).
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the energy resource in kV.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the energy resource (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the energy resource (kVAR).

4.4.10. Feeder

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the feeder.
- **ID:** Unique identifier of the feeder.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **DPF Solution Time:** Date and time of the most recent execution of the distribution power flow solution.

Engineering Outputs:

- **PQI High:** High power quality index (%kW).
- **PQI Low:** Low power quality index (%kW).
- **PQI Imbalance:** Imbalance of power quality index (%kW).

4.4.11. Fuse

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** Name of the fuse.
- **ID:** Unique identifier of the fuse.
- **Alternate ID:** Second identifier of the fuse.
- **Address:** Address of the fuse.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the fuse phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.

- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the fuse (amps).
- **Type:** The type of fuse.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the fuse.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.

4.4.12. Generator

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the generator.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **ID:** Unique identifier of the generator.
- **Address:** Address of the generator.
- **Constructed Phase(s):** Enumeration of the generator phases (ABC for A, B, and C phases).
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.

- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the generator in kV.
- **Real Power Output (kW):** Maximum and minimum real power outputs of the generator (kW).
- **Reactive Power Output (kVAR):** Maximum and minimum reactive power outputs of the generator (kVAR).

4.4.13. Incident

This Operator Output pop-up provides end users with the following information:

- **Order Number:** Incident ticket number.
- **Device:** Name of device.
- **Feeder:** Name of the feeder that is associated with the outage.
- **Start Time:** Incident start time.
- **End Time:** End time of the incident.
- **ITR/ETR:** The latest estimated time for restoration.
- **Affected Customers:** Number of customers affected.
- **Called Customers:** Number of customers that called.
- **AMI:** Number of AMI calls.
- **Crews:** Crew dispatched to the incident.
- **Clear:** Indicates in the incident is cleared (Yes or No).
- **First Call Time:** First time the customer called.
- **From Callback:** If Yes, indicates that the incident was created based on a customer callback.

4.4.14. Line

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the line.
- **ID:** Unique identifier of the line.
- **Project ID:** ID of the project that is associated with the device commissioning or

decommissioning.

- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the line phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current Side 1/Side 2 (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Length (ft):** Length of the line.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the line.
- **Construction Type:** Construction type (for example, Overhead/Underground).
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity

adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.

  - **Neutral Value:** Neutral wire limit value.
  - **Manually Entered Limit:** The manually entered limit value.

4.4.15. Load

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the load.
- **ID:** Unique identifier of the load.
- **Address:** Address of the load.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the load phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the load in kV.
- **Nominal Real Power (kW):** The nominal real power rate in kW.
- **Nominal Reactive Power (kVAR):** The nominal reactive power rate in kVAR.
- **Nominal Power Factor:** The nominal power factor of the load.
- **Normal Operational Limit - High:** Normal operational high voltage limit, expressed in per-unit or as configurable voltage base.

- **Normal Operational Limit - Low:** Normal operational low voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Operational Limit - High:** Emergency operational high voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Operational Limit - Low:** Emergency operational low voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Optimization Limit - High:** Normal optimization high voltage limit, expressed in per-unit or as configurable voltage base.
- **Normal Optimization Limit - Low:** Normal optimization low voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Optimization Limit - High:** Emergency optimization high voltage limit, expressed in per-unit or as configurable voltage base.
- **Emergency Optimization Limit - Low:** Emergency optimization low voltage limit, expressed in per-unit or as configurable voltage base.

4.4.16. Network Protector

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the network protector.
- **ID:** Unique identifier of the network protector.
- **Alternate ID:** Second identifier of the network protector.
- **Address:** Address of the network protector.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the network protector phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.

- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.

Engineering Data:

- **Continuous Current Rating (Amps):** The maximum interruption rating of the network protector (amps).
- **Type:** The type of network protector.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the network protector.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value:** The level limit value.

4.4.17. Order

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Order Name:** Order name.
- **Order Type:** Order type (for example, Restoration for outage incidents and Investigation or Maintenance for non-outage incidents).
- **Order Status:** Order status (for example, Field Complete).
- **Description:** Order description or order location description.
- **Owner ID:** The name of the user who owns the order.
- **Creation Time:** Date and time when the order was created.

4.4.18. Permanent Jumper

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the permanent jumper.
- **ID:** Unique identifier of the permanent jumper.
- **Alternate ID:** Second identifier of the permanent jumper.
- **Address:** Address of the permanent jumper.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the permanent jumper phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the permanent jumper (amps).
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the permanent jumper.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).

    - **Value:** The level limit value.

4.4.19. Planned Outage

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Outage:** The planned outage name. This field is populated with the station name and the affected device name. For devices without a Device Name (for example, fuses), the Device ID is used instead.
- **Device:** Name of the device affected by the planned outage.
- **Station:** Name of the substation related to the planned outage.
- **Created:** Date and time when the planned outage was created.
- **Start Time:** Date and time when the outage started.
- **End Time:** Date and time when the outage completed.
- **Primary Start Time:** Date and time when the outage is planned to start.
- **Primary End Time:** Date and time when the outage is planned to end.
- **Alternate Start Time:** Alternate date and time when the planned outage can start.
- **Alternate End Time:** Alternate date and time when the planned outage can end.
- **Use Alternate:** Use the alternate start and end date and time for the planned outage. This option can be enabled while editing the planned outage record.
- **Representative Name:** Company contact for the customers included in the outage notification.
- **Representative Number:** Phone number of the company contact.
- **Creator Comments:** Comment of the notification creator.

4.4.20. Power Transformer

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the transformer.
- **ID:** Unique identifier of the transformer.
- **Address:** Address of the transformer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the transformer phases (ABC for A, B, and C phases).

- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Loading Level Against Level 1 Limit (%):** Real-time maximum phase loading of the transformer as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the transformer as a percentage of rated load.
- **Grounding Current (Amps):** Grounding current rate and phase angle.
- **Loading Percentage:** Loading percentage.
- **Current Side 1/ Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.

Engineering Data:

- **Rating:** The total capacity of the autotransformer (MVA).
- **Type:** The type of transformer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the transformer.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the transformer winding.
- **Tap Changer Type:** Transformer primary/secondary side tap changer type.
- **Tap Position:** Transformer primary/secondary side tap position.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).

- **Season:** The season of the level limit (for example, Entire Year).
- **Value (per Phase):** The level limit value for phases A, B, and C.
- **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
- **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
- **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
- **Derated Limit Loading:** Derated limit loading. Calculated as maximum flow divided by derated limit.
- **Manually Entered Limit:** The manually entered limit value.

4.4.21. Reactor

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reactor.
- **ID:** Unique identifier of the reactor.
- **Address:** Address of the reactor.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the reactor phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.

- **Current Side 1/ Side 2 (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Nominal Voltage:** The nominal reactor and phase voltages of the reactor.
- **Impedance (R+jX ohms):** The reactor impedance (in ohms).
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value:** The level limit value for phases A, B, and C.
    - **Derated Value** The level limit derated value for phases A, B, and C.
    - **Loading Adjusted for Reserved Capacity:** Loading adjusted for reserved capacity.
    - **Available Capacity Adjusted for Reserved Capacity:** Available capacity adjusted for reserved capacity.
    - **Manually Entered Limit:** The manually entered limit value.

4.4.22. Recloser

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the recloser.
- **ID:** Unique identifier of the recloser.
- **Alternate ID:** Second identifier of the recloser.
- **Address:** Address of the recloser.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the recloser phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.

- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the recloser (amps).
- **Type:** The type of recloser.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the recloser.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.
    - **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
    - **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
    - **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
    - **Manually Entered Limit:** The manually entered limit value.

4.4.23. Reference Bus Interface

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the reference bus interface.
- **ID:** Unique identifier of the reference bus interface.
- **Address:** Address of the reference bus interface.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the reference bus interface phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.

Engineering Data:

- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the regulator.

4.4.24. Regulator

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the regulator.
- **ID:** Unique identifier of the regulator.
- **Address:** Address of the regulator.
- **Project ID:** ID of the project that is associated with the device commissioning or

decommissioning.

- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the regulator phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Total Real Power (kW):** Total real power in kW.
- **Total Reactive Power (kVAR):** Total reactive power in kVAR.
- **Total Apparent Power (kVA):** Total apparent power in kVA.
- **Loading Level Against Level 1 Limit (%):** Real-time maximum phase loading of the regulator as a percentage of the Level 1 load limit.
- **Loading Level Against Rating (%):** Real-time loading of the regulator as a percentage of rated load.
- **Loading Percentage:** Loading percentage.
- **Current Side 1/ Side 2 (Amps):** Real-time phase current (amps) and angle (deg).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.

Engineering Data:

- **Rating:** The nominal capacity of the regulator (kVA).
- **Effective Rating:** The actual capacity of the regulator (kVA).
- **Type:** The type of regulator.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the regulator.
- **Rated Voltage:** The rated line-to-line and line-to-ground voltages in the regulator winding.
- **Tap Changer Type:** Regulator primary/secondary side tap changer type.
- **Tap Position:** Regulator primary/secondary side tap position.
- **Limits:** Data on limits.

  - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
  - **Season:** The season of the level limit (for example, Entire Year).
  - **Value (per Phase):** The level limit value for phases A, B, and C.
  - **Dynamic Limit - Regulation Step (%):** Dynamic limit for the regulation step in percentage. This field is empty if the dynamic limits are not available.
  - **Dynamic Limit - (Amps per phase):** Dynamic limit for the current per phase in Amps. This field is empty if the dynamic limits are not available.

4.4.25. Sectionalizer

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the sectionalizer.
- **ID:** Unique identifier of the sectionalizer.
- **Alternate ID:** Second identifier of the sectionalizer.
- **Address:** Address of the sectionalizer.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of the sectionalizer phases (ABC for A, B, and C phases).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.

- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages in degrees.

Engineering Data:

- **Size (Amps):** The maximum interruption rating of the sectionalizer (amps).
- **Type:** The type of sectionalizer.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the sectionalizer.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).
    - **Season:** The season of the level limit (for example, Entire Year).
    - **Value (per Phase):** The level limit value for phases A, B, and C.
    - **Manually Entered Limit:** The manually entered limit value.

4.4.26. Substation

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** Name of the substation.
- **ID:** Unique identifier of the substation.
- **Alternate Name:** Substation alternative name.
- **Address:** Address of the substation.
- **Service Center:** The name of the service center.
- **Permission Area:** Permission area the substation belongs to.
- **Division:** The name of the division.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Protection Validation (PRV) Solution Time:** Date and time of the most recent protection validation solution.
- **Load and Volt/VAR Management (LVM) Solution Time:** Date and time of the most recent LVM solution.

4.4.27. Switch

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Switch Number:** The name of the switch.
- **ID:** Unique identifier of the switch.
- **Alternate ID:** Second identifier of the switch.
- **Address:** Address of the switch.
- **Project ID:** ID of the project that is associated with the device commissioning or decommissioning.
- **Project Stage:** The device commissioning or decommissioning stage.
- **Constructed Phase(s):** Enumeration of constructed phases (A, B, and C phases, and neutral).
- **Future Phase(s):** Individual phases that are constructed but are not energized.
- **Future Removed Phase(s):** Individual phases that are planned to be removed.
- **Removed Phase(s):** Individual phases that were recently removed.
- **External Company Name:** The name of the third-party company that owns the equipment. This field is displayed only if the device belongs to a third-party company.

Network Application Results:

- **Distribution Power Flow (DPF) Solution Time:** Date and time of the most recent distribution power flow solution.
- **Distribution Power Flow (DPF) Solution Status:** Status of the most recent distribution power flow solution.
- **Loading Percentage:** Loading percentage.
- **Current (Amps):** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Phase Angle Difference (deg):** The difference of angle value between side 1 and side 2 voltages, in degrees.
- **Reserved Capacity:** Reserved capacity.
- **Reserved Capacity Adjusted Flow:** Reserved capacity adjusted flow.

Engineering Data:

- **Size (Amps):** The rated interruption capacity of the switch (Amps).
- **Type:** Name of the type of the switch.
- **Nominal Voltage:** The nominal line-to-line and line-to-ground voltages of the switch.
- **Limits:** Data on limits.
    - **Name:** The name of the level limit (for example, Long Term, Medium Term, Short Term).

- **Season:** The season of the level limit (for example, Entire Year).
- **Value (per Phase):** The level limit value for phases A, B, and C.
- **Derated Value (per Phase):** The derated level limit value for phases A, B, and C. Calculated as the reserved capacity value subtracted from the level limit value.
- **Loading Adjusted for Reserved Capacity (per Phase):** Loading adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values divided by the actual limit value.
- **Available Capacity Adjusted for Reserved Capacity (per Phase):** Available capacity adjusted for reserved capacity. Calculated as a sum of maximum flow and reserved capacity values subtracted from the actual limit value.
- **Derated Limit Loading:** Derated limit loading. Calculated as maximum flow divided by derated limit.
- **Manually Entered Limit:** The manually entered limit value.

4.4.28. Temporary Switch

This Operator Output pop-up provides end users with the following information:

Identifiers:

- **Name:** The name of the temporary switch.
- **ID:** Unique identifier of the temporary switch.
- **Alternate ID:** Second identifier of the temporary switch.
- **Address:** Address of the temporary switch.
- **DPF Solution Time:** Date and time of the most recent execution of the distribution power flow solution.

Engineering Outputs:

- **Total kW:** Total power expressed in kilowatts.
- **Total kVAR:** Total kilovolt-amperes reactive.
- **Loading Percentage:** Loading percentage.
- **Amps:** Phase real-time current (amps).
- **Voltage Side 1/Side 2 (LL):** Line-to-line voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Voltage Side 1/Side 2 (LG):** Line-to-ground voltage expressed in the DPF base voltage value, and phase angle in degrees.
- **Angle Difference:** The difference of angle value between side 1 and side 2 voltages.

4.4.29. Unassociated Call

This Operator Output pop-up provides end users with the following information:

Identifiers:

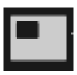
- **Critical:** Critical account type of the customer.
- **Call Time:** Date and time when the call was received.
- **Division:** Division the call belongs to.
- **Operating Center:** Operating center the call belongs to.
- **Account Number:** Account number of the customer associated with the call.
- **Customer Name:** Name of the customer associated with the call.
- **Address:** Address of the customer associated with the call.
- **Feeder:** Name of the feeder the call belongs to (based on the call location).
- **Substation:** Name of the substation the call belongs to (based on the call location).
- **Is Callback:** Indicates if this is a callback call.
- **Callback Number:** Customer callback number.
- **ID:** Unique ID of the call.
- **Call Type:** Shows the trouble type, such as Lights Out, Area Outage, or Blinking Lights.
- **Condition 1:** Describes the first trouble call condition, such as wires down or wires burning.
- **Condition 2:** Describes the second trouble call condition, such as pole on fire or pole leaning.
- **Description:** Displays the trouble call description, if any, such as pole/house or loud noise.
- **Comment:** Comment to the call.

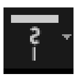
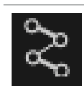
5. ADMS Toolbars

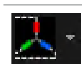
This chapter contains descriptions of the standard toolbars provided by ADMS.

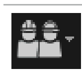
5.1. Geographic Toolbar

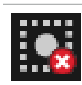
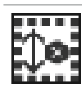
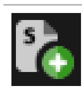
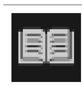
The following actions are available from this toolbar:

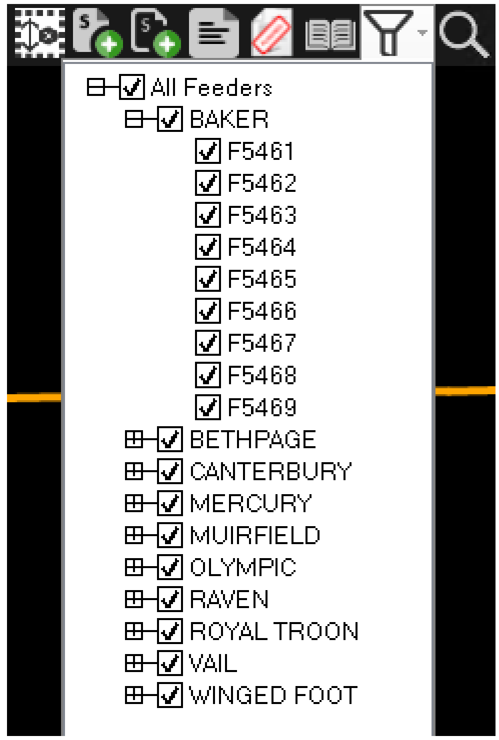
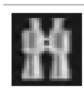
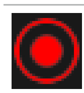
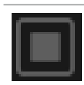
Icon	Function	Description
	**Load Surrounding Substations (Update Partial Model View)**	Re-centers the partial model view to show the currently centered station and its neighboring stations.
	**Picture in Picture**	Contains options to show the picture-in-picture window: • **Picture in Picture:** Shows the picture-in-picture window with geographic display. • **Station Overview:** Shows the picture-in-picture window with station overview. • **Child Window Navigation:** Shows the picture-in-picture window with navigation overview.
	**Background Layers**	Toggles different background layers ON/OFF. • **Roads:** Toggles display of roads and streets on and off. • **Water:** Toggles display of bodies of water on and off. • **Service Lines:** Toggles display of service lines on and off. • **Building Footprints:** Toggles display of buildings and structures on and off. • **Boundaries:** Toggles display of municipal boundaries on and off. • **Labels:** Toggles display of labels on and off. • **MicroGrid:** Toggles display of the micro grid on and off. • **Wildfire:** Toggles display of wildfires on and off. • **Fire Zones:** Toggles display of the fire zones grid on and off. • **Pole Labels:** Toggles display of pole labels on and off. • **Teams:** Toggles display of teams on and off. • **All On:** Toggles display of all background layers on and off.

Icon	Function	Description
	**Web Maps**	Contains options to show/hide web maps. (Internet connection is required.). The available options are: • Street • Aerial • Clouds • Radar • Off
	**Show/Hide Labels**	Contains options to toggle device labeling on and off, and specify label size: • **Zoom-Based Label:** Display labels according to labels zoom settings provided in the configuration file. • **Force Labels ON:** Toggles switch labeling ON regardless of labels zoom settings. • **Force Labels OFF:** Toggles switch labeling OFF. • **Size:** Sets label size to 80, 100, 120, and 200%.
	**Show/Hide Laterals**	Contains options to display laterals: • **Zoom-Based Lateral:** Display lateral according to lateral zoom settings provided in the configuration file. • **Force Laterals ON:** Toggles the display of laterals ON regardless of lateral zoom settings. • **Force Laterals OFF:** Toggles the display of laterals OFF.
	**Show/Hide Non-Reconfigurable Sections**	Shows only those elements that are capable of changing the configuration of the feeder backbone. All line sections, switches, and other devices that are not able to directly affect the configuration of the feeders are removed from the display.
	**Show/Hide Future**	Contains options to show equipment that is under commissioning: • **Future:** Toggles the display of future equipment. • **Future Removed:** Toggles the display of equipment to be removed. • **Removed:** Toggles the display of equipment that was recently removed. • **All On:** Toggles the display of all equipment under commissioning.
	**Show Halos**	Contains options to display violation and capacity margin halos on affected lines and/or devices: • **Voltage:** Toggles the display of voltage halos. • **Overload:** Toggles the display of overload halos. • **Capacity Margin:** Toggles the display of capacity margin halos. • **Potential Fault:** Toggles the display of potential fault halos. • **All:** Toggles the display of all halos.

Icon	Function	Description
	**Show/Hide Fault Indicator Halos**	Displays a menu to show/hide fault location halos based on fault indicator types: • **Hide All Halos:** Select this option to hide all fault indicator halos on the geographic view. • **Show Manual Halos Only:** Shows only manual fault indicator halos. Only the manual fault indicators are considered in the fault indicator halo analysis. Telemetered fault indicators (both set and unset) are ignored. • **Show Telemetered Halos Only:** Shows only telemetered fault indicator halos. Only the telemetered fault indicators are considered in the fault indicator halo analysis. Manual fault indicators (set, unset or unknown) are ignored. • **Show Combined Halos:** Both telemetered and manual fault indicator settings are used to determine the placement of the fault indicator halos.
	**Show/Hide Voltage Levels**	Contains a menu to show or hide network elements with different operating voltage levels. To show or hide the voltage levels, check the boxes in the drop-down menu. • **230 KV:** Show/hide 230kV network elements. • **138 KV:** Show/hide 138kV network elements. • **23 KV:** Show/hide 23kV network elements. • **13 KV:** Show/hide 13kV network elements. • **All On:** Show/hide network elements with all operating voltage levels.
	**Show/Hide Phase Colors**	Toggles between showing lines with feeder color and showing lines with phase color. • **Phase Colors:** The A phase is colored red, the B phase is colored green, the C phase is colored cyan, a two-phase line is colored yellow, and a three-phase line is colored light-orange. • **Feeder Colors:** Lines are colored with colors of the feeders they belong to (the feeder colors are defined in the Viewer Configuration Editor).
	**Show/Hide Phases (Nominal)**	Displays the menu to show/hide network elements by phase. The available options are: • A • B • C • All On
	**Show/Hide Overhead/Underground**	Displays the menu to show/hide overhead or underground devices: • **Overhead:** Select to show overhead components. • **Underground:** Select to show underground components. • **Both:** Select to show both overhead and underground components.

Icon	Function	Description
	**Show/Hide Annotations**	Clicking this button opens a drop-down list containing a set of check boxes that define annotations to be displayed. To specify the annotations to be displayed, check the boxes in the drop-down list. Check the All On box to display all annotations.
	**Show/Hide Structures**	Clicking this button opens a drop-down list containing a set of check boxes that define structure types to be displayed. The available check boxes options are: • Poles • Street Lights • Call Boxes • Restricted Areas • Work Boundaries • Hazard Areas • All On
	**Show/Hide Customers**	Displays a menu to show/hide customers on the geographic view: • **All:** Shows all customers. • **Critical:** Shows critical customers. • **Connected:** Shows only customers that are electrically connected to the network (both energized and de-energized). • **Disconnected:** Shows only customers that are not electrically connected to the network (invalid customers). • **With Activity:** Shows only customers with activity (de-energized customers or customers with calls or AMI Out messages). • **With Call:** Shows only customers that called. • **With Priority Call:** Shows only customers that issued low, high, or extreme priority calls. • **None:** Hides all customers.
	**Show/Hide Crews**	Displays a menu to show/hide crews on the geographic view: • **All:** Shows all crews. • **Dispatched:** Shows dispatched crews only. • **Available:** Shows available crews only. • **None:** Hides all customers.
	**Filter Crews by Type**	Displays a menu to show/hide specific crew types on the geographic view: • **First Responder:** Show/hides first responder crews. • **Foreman:** Show/hides foreman crews. • **All On:** Shows/hides all crew types. **Note** Crew types are configured by the CrewType coded translation item.

Icon	Function	Description
	**Show/Hide Vehicles**	Toggles showing crew vehicles with mobile data terminal (MDT) on the geographic view.
	**Filter Vehicles by Type**	Displays a menu to show/hide specific vehicle types on the geographic view: • **Pickup:** Show/hides pickups. • **Bucket Truck:** Show/hides bucket trucks. • **Derrick:** Show/hides derricks. • **Digger:** Show/hides diggers. • **Boom Truck:** Show/hides boom trucks. • **Service Truck:** Show/hides service trucks. • **Survey Truck:** Show/hides survey trucks. • **All On:** Shows/hides all vehicle types. **Note** Vehicle types are configured by the VehicleType coded translation item.
	**Clear Trace**	Clears all traces in all viewports.
	**Clear All Marks**	Clears the "mark" halo from all switches in all viewports.
	**Show Marked Devices**	Shows all marked devices on the geographic view.
	**Clear All Marks and Traces**	Clears the "mark" halo from all switches and also clears all traces in all viewports.
	**Add Marked Devices to SWO**	Includes the marked devices in a new switch order.
	**Add Marked Devices to Daily Log**	Includes the marked devices in the daily log.
	**Switch Order**	Opens the Switching Order application.
	**New Planned Outage Request**	Opens a new planned request in the Switching Order application.
	**Study Case Setup**	Displays the pop-up window for the creation and retrieval of savecases.

Icon	Function	Description
	**Feeder Filter**	Displays a menu to show/hide feeders on a geographic network view. 
	**Find Tool**	Opens the Find pop-up, which is used to locate network devices, customers, or land-based features on the current display or anywhere in the system.
	**Switching Order Search**	Opens the Switching Order Search pop-up, which is used to filter and open switching orders for edit or review.
	**Start SWO Recording**	Starts the Switching Order recording session to record the operator's actions as switching steps. This button is available in Study mode only.
	**Stop SWO Recording**	Stops the Switching Order recording session. This button is available in Study mode only.

5.2. Annotation Toolbar

The following actions are available from this toolbar:

- **Remove Annotation:** Removes the selected annotation.
- **Navigate to Related Display:** Opens the annotation pop-up display with the annotation details.

The following information is shown on the toolbar:

-

Bitmap designation of the annotation as a network element.

- Name of the annotation specified in the annotation pop-up.

5.3. Auto Transformer Toolbar

The following actions are available from the autotransformer toolbar:

- **Raise Step:** Steps up.
- **Lower Step:** Steps down.

  The Raise/Lower Step buttons are disabled if the Tap Position SCADA measurement exists or the transformer is blocked. If the measurement does not exist, the step regulation is enabled in the following cases:

  - The Remote and Manual statuses are determined by relevant SCADA measurements
  - The Remote status is determined by the SCADA measurement and the Auto/Manual SCADA measurement does not exist
  - The Remote/Local SCADA measurement does not exist, and the Manual status is determined by the SCADA measurement
  - The Remote/Local and Auto/Manual SCADA measurements do not exist, and the Remote status is determined statically or is set by the Toggle to Local/Remote toolbar button

- **Place Temporary Fault:** Opens a fault pop-up that allows creating a temporary fault with respect to the selected network element. The transformer must be de-energized.
- **Set Regulation On/Off:** Toggles the regulation to the On or Off state.

  If the Regulation (Auto/Manual) SCADA measurement exists, this button is disabled.

- **Attributes:**

  - **Operator:** Opens the Operator Attributes pop-up window for the transformer.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the transformer.
  - **Operator Output:** Opens the Operator Output pop-up window for the transformer.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the transformer.

- **Block/Unblock:** Blocks or unblocks the transformer.

  If the Blocked Status SCADA measurement exists, this button is disabled.

- **Disable/Enable Voltage Reduction:** Disables or enables voltage reduction. This button is enabled if the following conditions are met:

  - The regulator is in Local mode.
  - The voltage reduction capability is true for either forward or reverse direction (advanced

regulation mode).
    - The voltage reduction measurement does not exist.

- **Toggle to Local/Remote:** Toggles the transformer to Local or Remote mode.

  If the Local/Remote Status SCADA measurement exists, this button is disabled. If the measurement does not exist, the button is enabled if the Remote mode is determined statically.

- **Trace:** Tracing options:
    - **Trace Up:** Invokes a trace from the selected device to the power source.
    - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (switch, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
    - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
    - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked

  devices between the starting point and network end points, traces to all connected end points.
  - **Clear Trace:** Clears all traces in all viewports.

- **Show/Hide Other Feeders:** Toggles showing only the feeder the transformer belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, manage voltage, and so on.
- **Enter Parameters:** Opens the Transformer Parameters pop-up, which allows overriding the nominal values of the transformer.
- **Mark Auto Transformer:** Marks the selected auto transformer with a halo. Unmarks the device if it was already marked.
- **Add to Switching Order:** Adds the transformer to the Switching Order form.
- **Add to Daily Log:** Adds the transformer to the Daily Log.
- **Create Maintenance Order:** Creates a maintenance order for the transformer. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a transformer element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Transformer name. The tooltip shows the transformer's address.
- Designation of transformer phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Tap changer operation mode.
- Current step.

5.4. Breaker Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up allowing the placement or removal of tags for SCADA breakers.
- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up allowing the placement or removal of tags related to the given non-SCADA breaker.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Status:** Options for device status change:

    - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to open or close the breaker.
    - **Change Switch Status:** Changes the real-time status of the breaker to closed or open. Opening the breaker de-energizes the downstream network elements.
    - **Toggle Phases:** Opens the Toggle Switch Phases pop-up, which allows you to close and open individual phases of the breaker.
    - **Change Pending/Permanent Status:** Places the special mark at the breaker that informs about changing permanent status.
    - **Set Lockout Status:** Toggles the lockout status for the breaker.
- **Change Switch Classification:** Options to set the breaker's specific open state classification.
    - **Set Temporary Open:** Sets the breaker classification to Temporary Open.
    - **Set Bad Open:** Sets the breaker classification to Bad Open.
    - **Set Normal Open:** Sets the breaker classification to Normal Open.
    - **Reset:** Clears the specific open state classification.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode:
    - **User Enabled:** The device is available for control by the user. This option is available only in the Simulator and Study Mode.
    - **User Disabled:** The device is unavailable for control by the user.
    - **SCADA Decides:** The device is eligible for control based on an available status measurement. This option only works in real-time mode when SCADA measurements are being updated and the device has remote control capability.
- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the breaker.
    - **Operator Output:** Opens the Operator Output pop-up window for the breaker.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the breaker.
    - **Analyst Output:** Opens the Analyst Output pop-up window for the breaker.
- **Trace:** Tracing options:
    - **Trace Up:** Invokes a trace from the selected device to the power source.
    - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is

canceled.

  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.
  - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
  - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
  - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the breaker belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters pop-up window that allows entering breaker parameters.
- **Mark:**
  - **Mark Switch:** Marks the selected breaker with a halo. Unmarks the breaker if it was already marked.
  - **Mark All Potential Isolating Devices:** Marks all isolation switches (switches that can isolate the selected breaker from the network).
- **Add to Switching Order:** Adds a breaker to the Switching Order form.
- **Add to Daily Log:** Adds a breaker to the Daily Log.
- **Associate to Safety Document:** Adds a breaker record to the Safety Document.
- **Split Incident:** Splits a de-energized open device into a new incident. This can be used to split a

device from a predicted incident on an upstream device or break a large incident into multiple incidents.

- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected breaker.
- **Create Maintenance Order:** Creates a maintenance order for the breaker. (For the Enterprise OMS client types only.)
- **Block Outage Prediction:** Outage prediction options:
    - **Block Outage Prediction:** Blocks the device outage prediction.
    - **Resume Outage Prediction:** Resumes the device outage prediction.
    - **Toggle Block Outage Prediction:** Toggles blocking the prediction.

The following information is shown on the toolbar:

- Bitmap designation of a breaker network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Breaker name and address.
- Designation of breaker phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Current breaker state (Open, Partial, or Closed).

5.5. Bus Toolbar

The following actions are available from this toolbar:

- **Place Fault:** Opens the Fault pop-up window. This button is only enabled when the bus is de-energized.
- **Attributes:**
    - **Operator Attributes:** Opens the Operator Attributes pop-up window for the load.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the load.
    - **Operator Output:** Opens the Operator Output pop-up window for the load.
    - **Analyst Output:** Opens the Analyst Output pop-up window for the load.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, manage voltage,

and so on.

- **Enter Parameters:** Opens the Bus Parameters pop-up, which allows setting and overriding default parameters.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected bus.

The following information is shown on the toolbar:

- Bitmap designation of a bus element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Bus name and address.

5.6. Capacitor Toolbar

The following actions are available from this toolbar:

- **Place Tags:** Opens tag pop-up windows that allow placing or removing a tag on the capacitor.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Status:** Options for device status change:
  - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to toggle the capacitor status on or off. If a capacitor is part of a capacitor bank and the next capacitor to be switched on or off in the associated capacitor bank is not the current capacitor, this action is not allowed (the toolbar button is hidden).
  - **Change Capacitor Status:** Toggles the capacitor status on or off.
  - **Toggle Phases:** Opens the Toggle Switch Phases pop-up to toggle on or off individual phases of the capacitor.
- **Go to SCADA Control:** Opens the SCADA control pop-up for the capacitor.
- **Raise Step:** Click this button to increase the step of the capacitor.
- **Lower Step:** Click this button to decrease the step of the capacitor.
  - Currently the ADMS product does not support the Raise Step and Lower Step buttons. The Raise Step and Lower Step toolbar buttons are always disabled.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the capacitor.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the capacitor.
  - **Operator Output:** Opens the Operator Output pop-up window for the capacitor.
  -

Analyst Output: Opens the Analyst Output pop-up window for the capacitor.

- **Application Optimization Eligibility Status:** Toggles the device optimization control mode:
  - **User Enabled:** The device is available for control by the user. This option is available only in the Simulator and Study Mode.
  - **User Disabled:** The device is unavailable for control by the user.
  - **SCADA Decides:** The device is eligible for control based on an available status measurement. This option only works in real-time mode when SCADA measurements are being updated and the device has remote control capability.
  - **Not Controllable:** Indicates that the device is not eligible for control. This status is always read-only and cannot be changed. Applies to fixed capacitors only.
- **Toggle to Local/Remote:** If the Local/remote button is enabled, a user can toggle the Local/Remote value. The Local/Remote button is disabled if any one of the following conditions is met.
  - Capacitor control type is Fixed.
  - Capacitor control type is Local Only
  - Capacitor control type is Remote Only
  - A Local/Remote status SCADA measurement with good quality exists.
- **Set Inoperable/Operable:** Click to set the operation availability of the capacitor.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.
  - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
  - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end

points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the capacitor belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up display.
- **Enter Parameters:** Opens the Capacitor Parameters pop-up, which allows setting and overriding default parameters values.
- **Mark Capacitor:** Marks the selected capacitor with a halo. Unmarks the device if it was already marked.
- **Add to Switching Order:** Adds the capacitor to the Switching Order form.
- **Add to Daily Log:** Adds the capacitor to the Daily Log.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected capacitor.
- **Create Maintenance Order:** Creates a maintenance order for the temporary ground. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a capacitor element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Capacitor name and address.
- Designation of capacitor phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Current capacitor operation mode (for example, fixed).
- Current capacitor state (On or Off).
- Communication status.

5.7. Composite Switch Toolbar

The following actions are available from this toolbar:

- **Place DMS-Only Tags:** Opens a pop-up that allows adding a tag.
- **Place Tags from SCADA:** Opens the SCADA control display for the composite switch.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.

- **Status:** Controls to change the composite switch state. The states of the composite switch are defined in the static model.
    - **Scada switches**
        - **Go to SCADA Control (EMP):** Opens the SCADA control display for a SCADA controllable composite switch.
    - **Gang Switches**
        - **A Status:** Changes the state to A (feed from the A supply).
        - **B Status:** Changes the state to B (feed from the B supply).
        - **AB Status:** Changes the state to AB (feed from both the A and B supplies).
        - **Open Status:** Changes the state to Open.
    - **Throwover Switches**
        - **Preferred Status:** Changes the state to Preferred (feed from the preferred supply).
        - **Alternate Status:** Changes the state to Alternate (feed from the alternate supply).
        - **Off Status:** Changes the state to Off.
    - **RAC Switches**
        - **Close Status:** Changes the state to Close (the main switch is closed and the tie switch is open).
        - **Open Status:** Changes the state to Open (the main and tie switches are open).
        - **Tie Status:** Changes the state to Tie (the main switch is open and the tie switch is closed).
    - **RAG Switches**
        - **Close Status:** Changes the state to Close (the main switch is closed and the ground switch is open).
        - **Open Status:** Changes the state to Open (the main and ground switches are open).
        - **Ground Status:** Changes the state to Ground (the main switch is open and the ground switch is closed).
    - **Change Pending Permanent:** Places the special mark at the switch that informs about changing permanent status.
- **Change Switch Classification:** Options to set the composite switch's specific open state classification.
    - **Set Temporary Open:** Sets the composite switch classification to Temporary Open.
    - **Set Bad Open:** Sets the composite switch classification to Bad Open.
    - **Set Normal Open:** Sets the composite switch classification to Normal Open.
    - **Reset:** Clears the specific open state classification.
- **Operation Status:** Shows the current throwover operation status, Auto Disabled (Manual) or

Auto Enabled (Automatic). Clicking this button toggles the status. Applies to throwover switches.

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the composite switch.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the composite switch.
  - **Operator Output:** Opens the Operator Output pop-up window for the composite switch.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the composite switch.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (switch, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.

    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Auto-Enabled:** Toggles the status to status auto-enabled (normal mode).
- **Auto-Disabled:** Toggles the status to status auto-disabled (switches off the normal mode).
- **Navigate to Related Display:** Opens the new viewport displaying the internal view of the composite switch.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the switch belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up display.
- **Mark Composite Switch:** Marks the selected switch with a halo. Unmarks the switch if it was already marked.
- **Add to Switch Order:** Adds the composite switch to the Switching Order form.
- **Add to Daily Log:** Adds the composite switch to the Daily Log.
- **Create Maintenance Order:** Creates a maintenance order for the composite switch. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a switch element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Composite switch name and address.
- Designation of composite switch phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- State of the switch.

5.8. Crew Toolbar

The following actions are available from this toolbar:

- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the crew.

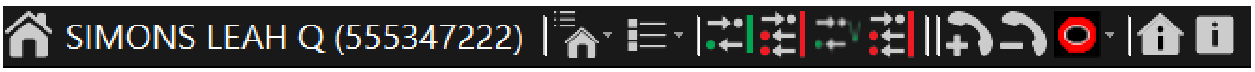
Operator Output: Opens the Operator Output pop-up window for the crew.

- **Update/Split Crew:** Opens the Crew Update Dialog to update or split the crew.

The following information is shown on the toolbar:

- Bitmap designation of a crew element.
- Crew ID.
- Crew Status.

5.9. Customer Toolbar



The following actions are available from this toolbar:

- **Customer List:**
  - **Add Customer:** Adds selected customers on the geographic view to the Call Customer List of the Planned Outage pop-up display, which allows notifying selected customers of outages and customizing notification messages.
  - **Remove Customer:** Removes customers from the Call Customer List of the Planned Outage pop-up display and deselects the customers on the geographic view.
  - **Customer Correct:** This option is used to associate a customer with another transformer. To do this, select a customer, select this option, then select a transformer, and select the Customer Correction option on the transformer toolbar.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the customer.
  - **Operator Output:** Opens the Operator Output pop-up window for the customer.
- **Call Ping Pop-Up – Single Customer:** Calls up the CIS/AMI Customer Ping display, which allows pinging of a selected customer. Changes the customer's icon coloring on the geographic view. A blue line indicates that a ping is in process, a green line indicates that a ping succeeded, and a red line indicates that a ping failed. The ping color is configurable in the Viewer Configuration Editor.
- **Call Ping Pop-Up – Service Point:** Calls up the CIS/AMI Customer Ping display, which allows pinging of all the customers associated with the same transformer as the selected customer. Changes the customers' icon coloring on the geographic view. A blue line indicates that a ping is in process, a green line indicates that a ping succeeded, and a red line indicates that a ping failed. The ping color is configurable in the Viewer Configuration Editor.
- **Ping AMI Customer:** Automatically pings the selected customer.
- **Ping All Customers on Transformer:** Automatically pings all the customers associated with the same transformer as the selected customer.

- **Trace:** Tracing options.
  - **Trace Up:** Invokes a trace from the selected customer to the power source.
  - **Clear Trace:** Clears all traces in all viewports.
- **Add/Update Customer Call:** Opens the Add Customer Call pop-up display to create a single customer incident record. The incident record is then showed on the Incident Detail OMS display.
- **Cancel Customer Call:** Cancels calls from the customer.
- **Create Single Customer Incident:** Creates a single customer incident for the customer. This option is only available for customers who are equipped with a smart meter, when the meter ping is Out.
- **Suspend Customer**
  - **Suspend Customer:** Suspends the customer that stays de-energized for a long period of time, and the service cannot be restored due to known cause (such as house on fire). This clears the associated single customer incident, if one exists, or excludes the customer from the upstream outage.
  - **Unsuspend Customer:** Unsuspends the previously suspended customer.
- **Associate Customer with Incident:** Opens the Associate Customer with Incident pop-up.
- **Split Out Customer into New Incident:** Creates a new incident for the de-energized customer.
- **Create Maintenance Order:** Creates a maintenance order for the customer. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a customer.
- Customer’s name and account number. The tooltip shows the address.

5.10. Cut Toolbar

The following actions are available from this toolbar:

- **Place Tags:** Opens the Tag pop-up, allowing a tag to be placed on or removed from the switch.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Switch:** Changes the status of the cut. The status can be changed to Closed or Open.
- **Edit Cut:** Opens a pop-up display to edit the temporary device.
- **Remove Cut:** Removes the cut (a "cut" is considered to be a temporary switch).
- **Attributes:**

    - **Operator:** Opens the Operator Attributes pop-up window for the cut.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the cut.
- **Label Position:**
    - **North:** Moves the label to the north (changes the side where the label is displayed) with respect to the selected network element.
    - **Northeast:** Moves the label to the northeast (changes the side where the label is displayed) with respect to the selected network element.
    - **East:** Moves the label to the east (changes the side where the label is displayed) with respect to the selected network element.
    - **Southeast:** Moves the label to the southeast (changes the side where the label is displayed) with respect to the selected network element.
    - **South:** Moves the label to the south (changes the side where the label is displayed) with respect to the selected network element.
    - **Southwest:** Moves the label to the southwest (changes the side where the label is displayed) with respect to the selected network element.
    - **West:** Moves the label to the west (changes the side where the label is displayed) with respect to the selected network element.
    - **Northwest:** Moves the label to the northwest (changes the side where the label is displayed) with respect to the selected network element.
- **Mark Cut:** Marks the cut on the geographic network view.
- **Add to Switching Order:** Adds the cut (switch) to the Switching Order form.
- **Add to Daily Log:** Adds the cut to the Daily Log.
- **Split Incident:** Splits the cut into a new incident. The cut must be open to enable this button.
- **Add Create Step to Switch Order:** Adds a step to create the cut to the Switching Order form.
- **Create Maintenance Order:** Creates a maintenance order for the cut. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a cut element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Cut name and address.
- Designation of cut phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- State of the cut (switch), open or closed.

5.11. Damage Assessment Location Toolbar

The following actions are available from this toolbar:

- **Move Damage Assessment Location:** Activates the move mode for the damage assessment location symbol.
- **Edit Damage Assessment Location:** Opens the Damage Assessment Status dialog box to edit damage assessment location properties.
- **Damage Assessment Location's Custom Fields:** Opens the Damage Assessment Location's Custom Fields display, which allows you to enter, view, or update custom damage assessment location properties.
- **Delete Damage Assessment Location:** Deletes the damage assessment location.

The following information is shown on the toolbar:

- Bitmap designation of a damage assessment location element.
- Associated incident name.

5.12. Distribution Transformer Toolbar

The following actions are available from this toolbar:

- **Disconnect/Connect Transformer:** Disconnects or connects the transformer from/to the network. Available only for transformers.
- **Confirm Disconnect/Connect Transformer:** Confirms disconnect or connect of the transformer from/to the network. Available only for transformers.
- **Disconnect/Connect Transformer Bank:** Disconnects or connects the transformer bank from/to the network. Available only for transformer banks.
- **Confirm Disconnect/Connect Transformer Bank:** Confirms disconnect or connect of the transformer bank from/to the network. Available only for transformer banks.
- **Place Temporary Fault:** Opens the Fault pop-up, which allows creating a fault in the selected distribution transformer. The transformer must be de-energized.
- **Distribution XF Load:** Opens the Distribution Transformer Load display.
- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the distribution transformer.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the distribution transformer.

Operator Output: Opens the Operator Output pop-up window for the distribution transformer.

  - **Analyst Output:** Opens the Analyst Output pop-up window for the distribution transformer.

- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protective Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.
  - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.

  - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
  - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the transformer belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Mark Distribution Transformer:** Marks the selected distribution transformer with a halo. Unmarks the device if it was already marked.
- **Add to Switching Order:** Adds the distribution transformer to the Switching Order form.
- **Add to Daily Log:** Adds the distribution transformer to the Daily Log.
- **Block Outage Prediction:** Blocks the device outage prediction.
- **Resume Outage Prediction:** Resumes the device outage prediction.
- **Toggle Block Outage Prediction:** Toggles blocking the prediction.
- **Customer Correction:** Used to associate a customer with another transformer. This option is enabled when the Customer Correct option on the customer toolbar is selected.
- **Dynamic Schedule:** Shows the Schedule Points grid display for the transformer.
- **Create Quick Order:** Creates a non-outage isolated incident for the transformer.
- **Create Maintenance Order:** Creates a maintenance order for the transformer. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a distribution transformer element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Transformer name. The tooltip shows the transformer's address.
- Designation of transformer phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.13. Elbow Switch Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up allowing the placement or removal of tags for SCADA elbows.

- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up allowing the placement or removal of tags related to the given elbow.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Switch:** Options for device status change:
    - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to open or close the elbow.
    - **Change Switch Status:** Changes the real-time status of the elbow, Closed or Open. Opening the switch de-energizes the downstream network elements.
    - **Toggle Phases:** Opens the Toggle Switch Phases pop-up, which allows you to close and open individual phases of the elbow.
    - **Change Pending/Permanent Status:** Places the special mark at the elbow that informs about changing permanent status.
- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the elbow.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the elbow.
    - **Operator Output:** Opens the Operator Output pop-up window for the elbow.
    - **Analyst Output:** Opens the Analyst Output pop-up window for the elbow.
- **Trace:** Tracing options:
    - **Trace Up:** Invokes a trace from the selected device to the power source.
    - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Up to Protective Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
    - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.

  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.
  - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
  - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
  - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the switch belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters pop-up window that allows entering elbow parameters.
- **Mark Switch:** Marks the selected elbow with a halo. Unmarks the elbow if it was already marked.
- **Add to Switching Order:** Adds an elbow to the Switching Order form.
- **Add to Daily Log:** Adds an elbow to the Daily Log.
- **Associate to Safety Document:** Adds an elbow record to the Safety Document.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Maintenance Order:** Creates a maintenance order for the elbow. (For the Enterprise OMS client types only.)
- **Create Quick Order:** Creates a non-outage isolated incident for the elbow.

The following information is shown on the toolbar:

- Bitmap designation of an elbow network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.

- Elbow name. The tooltip shows the elbow's address.
- Designation of elbow phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Current elbow state (Open or Closed).

5.14. Energy Resource Toolbar

The following actions are available from this toolbar:

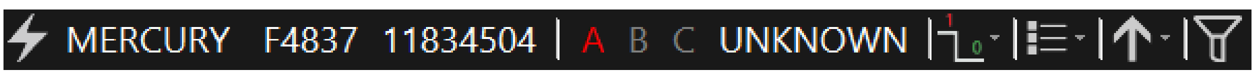
- **Place Tags:** Opens tag pop-up windows that allow the placement or removal of tags for the energy resource (DER).
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Connection Status:** Options for device status change:
  - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to connect or disconnect the energy resource. Applies to SCADA-controlled energy resources.
  - **Change Energy Resource Connection Status:** Connects or disconnects the DER. This button is disabled if the connection status SCADA measurement is available for the DER and the measurement quality is good.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the DER.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the DER.
  - **Operator Output:** Opens the Operator Output pop-up window for the DER.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the DER.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode (User Disabled / SCADA Decides).
- **Show/Hide Other Feeders:** Toggles showing only the feeder the DER belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Mark Energy Resource:** Marks the selected DER with a halo. Unmarks the device if it was already marked.
- **Enter Parameters:** Opens the Energy Resource Parameters display, which allows entering the voltage and power characteristics of the DER.
- **Schedules:**
  - **Static Schedules:** Shows the Schedule grid display for the DER.

- Dynamic Schedules: Shows the Schedule Points grid display for the DER.
  - **Autonomous Modes:** Opens the Autonomous Modes Details display for the DER.
- **Create Quick Order:** Creates a non-outage isolated incident for the DER.

The following information is shown on the toolbar:

- Bitmap designation of a DER network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- The name of the DER. The tooltip shows the address.
- Designation of DER phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.15. Fault Indicator Toolbar



The following actions are available from this toolbar:

- **Change Fault Status:** Toggles the fault indicator status to Set, Unset, or Unknown for a manual non-directional fault indicator. For non-directional SCADA fault indicators, the status can only be Set or Unset. For directional type 1 non-SCADA fault indicators, the status can be Both, Red (Side 1), Green (Side 2), Unset, or Unknown. For directional type 1 SCADA fault indicators, the status can be Both, Red, Green, or Unset. For directional type 2 non-SCADA fault indicators, the status can be Forward, Reverse, Unset, or Unknown. For directional type 2 SCADA fault indicators, the status can be Forward, Reverse, or Unset.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the fault indicator.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the fault indicator.
- **Trace:**
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the fault indicator belongs to.

The following information is shown on the toolbar:

- Bitmap designation of a fault indicator element.

- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Fault indicator unique identifier.
- Designation of fault indicator phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Fault indicator status: Set, Unset, or Unknown for a non-directional fault indicator; Unset, Both, Side 1, Side 2, or Unknown for a type 1 directional fault indicator; Unset, Forward Set, Reverse Set, or Unknown for a type 2 directional fault indicator.

5.16. Fuse Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up allowing the placement or removal of tags for SCADA fuses.
- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up allowing the placement or removal of tags related to the given fuse.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Status:** Options for device status change:
  - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to open or close the fuse.
  - **Change Switch Status:** Changes the real-time status of the fuse, Closed or Open. Opening the fuse de-energizes the downstream network elements.
  - **Confirm Change Switch Status:** Confirms the fuse status change and then changes the real-time status of the switch. When attempting to confirm open a fuse, the OMS engine invokes a pop-up display with a list of all downstream single customer and predicted incidents, including those in the predicted stable state. You can select incidents to be merged with the newly created upstream confirmed incident; the incidents that are not merged become predicted stable incidents.
  - **Toggle Phases:** Opens the Toggle Switch Phases pop-up, which allows you to close and open individual phases of the fuse.
  - **Change Pending/Permanent Status:** Places the special mark at the fuse that informs about changing permanent status.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the fuse.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the fuse.
  - **Operator Output:** Opens the Operator Output pop-up window for the fuse.

- Analyst Output: Opens the Analyst Output pop-up window for the fuse.

- **Application Optimization Eligibility Status:** Toggles the device optimization control mode:
  - **User Enabled:** The device is available for control by the user. This option is available only in the Simulator and Study Mode.
  - **User Disabled:** The device is unavailable for control by the user.
  - **SCADA Decides:** The device is eligible for control based on an available status measurement. This option only works in real-time mode when SCADA measurements are being updated and the device has remote control capability.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that

path and all line sections connected to it.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the fuse belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters pop-up window that allows entering fuse parameters.
- **Mark:**
    - **Mark Device:** Marks the selected fuse with a halo. Unmarks the fuse if it was already marked.
    - **Mark All Potential Isolating Devices:** Marks all isolation switches (switches that can isolate the selected fuse from the network).
- **Add to Switching Order:** Adds a fuse to the Switching Order form.
- **Add to Daily Log:** Adds a fuse to the Daily Log.
- **Associate to Safety Document:** Adds a fuse record to the Safety Document.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected fuse.
- **Create Maintenance Order:** Creates a maintenance order for the fuse. (For the Enterprise OMS client types only.)
- **Block Outage Prediction:** Blocks the device outage prediction.
- **Resume Outage Prediction:** Resumes the device outage prediction.
- **Toggle Block Outage Prediction:** Toggles blocking the prediction.

The following information is shown on the toolbar:

- Bitmap designation of a fuse network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Designation of fuse phases. Phases are colored according to the energization status (energized,

de-energized, or not constructed).

- Fuse name. The tooltip shows the address.
- Current fuse state (Open or Closed).

5.17. Generator Toolbar

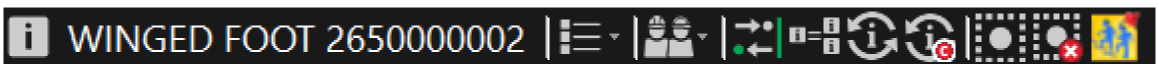
The following actions are available from this toolbar:

- **Place Tags:** Opens tag pop-up windows that allow the placement or removal of tags for the generator.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Status:** Options for device status change:
  - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to connect or disconnect the generator. Applies to SCADA-controlled generators.
  - **Change Generator Connection Status:** Connects or disconnects the generator. This button is disabled if a connection status SCADA measurement is available for the generator and the measurement quality is good.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the generator.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the generator.
  - **Operator Output:** Opens the Operator Output pop-up window for the generator.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the generator.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode: (User Disabled or SCADA Decides).
- **Show/Hide Other Feeders:** Toggles showing only the feeder the generator belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Generator Input display, which allows entering the voltage and power characteristics of the generator.
- **Mark Generator:** Marks the selected generator with a halo. Unmarks the device if it was already marked.
- **Schedules:**
  - **Static Schedules:** Shows the Schedule grid display for the generator.
  - **Dynamic Schedules:** Shows the Schedule Points grid display for the generator.

The following information is shown on the toolbar:

- Bitmap designation of a generator network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- The name of the generator. The tooltip shows the address.
- Designation of generator phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.18. Incident Toolbar



The following actions are available from this toolbar:

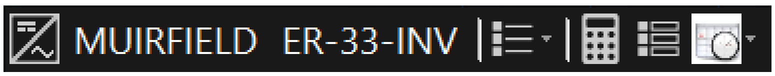
- **Attributes:**
  - **Operator Output:** Opens the Operator Output pop-up window for the incident.
- **Crew:**
  - **Dispatch Crew:** Opens the Dispatch Crew to Incident OMS tabular display.
  - **Dispatch Crew to Selected Incidents:** Opens the Dispatch Crew to Incident pop-up display.
- **Ping Customers:** Opens the Customers tab of the Update Incident pop-up.
- **Associate with Other Incidents:** Opens the Merge Incidents with Incident pop-up display, which allows you to associate selected incidents on the Geographic Network view with any incident from the incident list.
- **Merge Selected Incidents:** Merges all the incidents selected on the Geographic Network view by clicking the Select/Deselect Incident button.
- **Update Incident:** Opens the Update Incident display, which allows you to manage and clear the incident.
- **Incident's Custom Fields:** Opens the Incident's Custom Fields display, which allows you to enter, view, or update custom incident properties.
- **Select/Deselect Incident:** Allows you to select or deselect separate incidents for merging on the Geographic Network view. Click the Merge Selected Incident button to complete merging.
- **Clear All Incident Selections:** Deselects all the incidents on the Geographic Network view.
- **Create Follow-Up Order:** Opens the Follow Up Work Order pop-up display, which allows you to create a follow-up work order for the incident.

The following information is shown on the toolbar:

- Bitmap designation of an incident.

- Substation name.
- Associated device ID.
- Incident ID.

5.19. Inverter Toolbar



The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the inverter.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the inverter.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Inverter Parameters display, which allows entering the current characteristics of the inverter.
- **Create Maintenance Order:** Creates a maintenance order for the inverter. (For the Enterprise OMS client types only.)
- **Schedules:**
  - **Autonomous Modes:** Opens the Autonomous Modes Details display for the inverter.

The following information is shown on the toolbar:

- Bitmap designation of an inverter network element.
- Energizing substation name.
- Inverter name.

5.20. Line Toolbar

The following actions are available from this toolbar:

- **Place Cut:** Opens the Cut pop-up for a cut (temporary switch) to be inserted in the selected location and specifies its parameters.
- **Place Fault:** Opens the Fault pop-up, which allows creating a fault in the selected location and specifies its parameters. You can place a fault only on a de-energized line.

Place Temporary Ground: Opens the Temporary Ground pop-up, which allows creating a temporary ground and specifies its attributes.

- **Place Jumper:** Opens the Jumper pop-up, which allows creating a jumper. A jumper must be created between two points on lines. A jumper pop-up allows specifying phases to be connected as well as the initial state of the jumper (jumper switch).
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the line.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the line.
  - **Operator Output:** Opens the Operator Output pop-up window for the line.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the line.
- **Draw a Geo Schematic:** Shows the geo schematic view.
- **Draw a Multi-Loop Schematic:**
  - **Full:** Displays a URD loop schematic view. Available for underground laterals.
  - **Partial:** Displays a partial URD loop schematic view. Available for underground laterals.
- **Locate Dynamic Feeder Breaker on SCADA Oneline:** Opens a new viewport with a SCADA one-line display of the energizing substation and highlights the dynamic feeder breaker.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up Phase A/B/C:** Invokes a trace of the selected phase from the selected point to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down Phase A/B/C:** Invokes a trace of the selected phase from the point out to the end points, including laterals.

    - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it. This function only solves for points on the same feeder.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it. This function only solves only for points that have common phases.
    - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Study Case Setup:** Displays the pop-up window for the creation and retrieval of savecases.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the line belongs to. Applies to energized lines only.
- **Show/Hide Other Feeders (Line De-Energized):** Toggles showing only the feeder the line belongs to. Applies to de-energized lines only.
- **Network Analysis:** Network analysis options:
    - **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.

    - **Feeder Reconfiguration:** Opens a real-time DNAF solution pop-up. In the Application field, "Automatic Feeder Reconfiguration" is selected.
    - **Service Restoration:** Opens a real-time DNAF solution pop-up. In the Application field, "Fault Isolation and Service Restoration" is selected.
- **Enter Parameters:** Opens the Line Parameters display, which allows entering the current characteristics of the line.
- **Mark:**
    - **Mark All Potential Isolating Devices:** Marks all isolation switches (switches that can isolate the selected point on the line from the network).
    - **Mark and Trace All Potential Isolating Devices:** Marks all isolation switches and draws the trace between the switches.
    - **Mark Normal Open Devices:** Marks all normally open switches that are connected to the selected point on the line.
    - **Mark and Trace Normal Open Devices:** Marks all normally open switches that are connected to the selected point on the line and draws the trace between the switches.
    - **Mark Midpoint:** Marks the switch that is closest to the geographical midpoint of the feeder.
- **Clear All Marks:** Clear all marks (marked elements are highlighted with a blue dashed line halo).
- **Create Quick Order:** Creates a non-outage isolated incident for the line.
- **Create Maintenance Order:** Creates a maintenance order for the line. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a line network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- The name of the line. The tooltip shows the address.
- The size type of the line (for example, 1000 Al).
- Designation of line phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.21. Load Toolbar

The following actions are available from this toolbar:

- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the load.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the load.
    - **Operator Output:** Opens the Operator Output pop-up window for the load.
    - **Analyst Output:** Opens the Analyst Output pop-up window for the load.
- **Trace:**
    - **Trace Up:** Invokes a trace from the selected device to the power source.
    - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the load belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Load Parameters pop-up, which allows setting and overriding default parameters values.
- **Schedules:**
    - **Static Schedules:** Shows the Schedule grid display for the load.
    - **Dynamic Schedules:** Shows the Schedule Points grid display for the load.

The following information is shown on the toolbar:

- Bitmap designation of a load element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Load name.
- Designation of load phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.22. Loss of Voltage Cut Toolbar

The Loss of Voltage cut appears on a line upstream of a measurement device that detected the loss of voltage conditions. The Loss of Voltage cut works as a predicted cut that can be opened to reflect the current energization state or removed by the operator.

The following actions are available from the Loss of Voltage toolbar:

- **Switch:** Changes the status of the Loss of Voltage cut. The status can be changed to Closed or Open.
- **Edit Cut:** Opens a pop-up display to edit the Loss of Voltage cut.
- **Remove Cut:** Removes the Loss of Voltage cut.
- **Navigate to FISR Plans:** Opens the FISR Results Summary display for the associated substation.
- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the cut.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the cut.
- **Add Create Step to Switch Order:** Adds a step to create the cut to the Switching Order form.
- **Split Incident:** Splits the cut into a new incident. The cut must be open to enable this button.

The following information is shown on the toolbar:

- Bitmap designation of a Loss of Voltage component.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Loss of Voltage cut name and address. The name of the cut is automatically set as "Auto generated by FISR loss of voltage".
- Designation of Loss of Voltage cut phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.23. Network Protector Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up, which allows you to place or remove tags for SCADA network protectors.
- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up, which allows you to place or remove tags related to the given network protector.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Switch:** Options for device status change:
    - **Change Switch Status:** Changes the real-time status of the network protector to Closed or Open. Opening the network protector de-energizes the downstream network elements.
    - **Confirm Change Switch Status:** Confirms the network protector status change and then changes the real-time status of the switch. When attempting to confirm open a network

protector that is not predicted, the OMS engine opens a window with a list of all downstream single customer and predicted incidents, including those in the predicted stable state. You can select incidents to merge with the newly created upstream confirmed incident; the incidents that are not merged become predicted stable incidents.

  - **Toggle Phases:** Opens the Toggle Switch Phases window that allows you to close and open individual phases of the network protector.
  - **Change Pending/Permanent Status:** Places a special mark at the network protector that informs about changing permanent status.

- **Attributes:**
  - **Operator:** Opens the Operator Attributes window for the network protector.
  - **Analyst:** Opens the Analyst Attributes window for the network protector.
  - **Operator Output:** Opens the Operator Output window for the network protector.
  - **Analyst Output:** Opens the Analyst Output window for the network protector.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Traces upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an information message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Traces upstream from the starting point to the next upstream protective device (distribution transformer, fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an information message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Traces upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Traces downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an information message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Traces downstream from the starting point to the next downstream protective device. If there is no downstream protective device, an information message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Traces downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an information message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.

    - **Trace Section:** Traces the switchable section. Highlights all feeder backbones and laterals that are within the bounding switches.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the network protector belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters window, which allows you to enter network protector parameters.
- **Mark Switch:** Marks the selected network protector with a halo. Unmarks the network protector if it was already marked.
- **Add to Switching Order:** Adds a network protector to the Switching Order form.
- **Add to Daily Log:** Adds a network protector to the Daily Log.
- **Associate to Safety Document:** Adds a network protector record to the Safety Document.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected network protector.
- **Block Outage Prediction:** Blocks the device outage prediction.
- **Resume Outage Prediction:** Resumes the device outage prediction.
- **Create Maintenance Order:** Creates a maintenance order for the network protector. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a network protector network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Designation of network protector phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Network protector name. The tooltip shows the address.

- Current network protector state (Open or Closed).

5.24. Order Toolbar

The following actions are available from this toolbar:

- **Order Name:** Order name.
- **Order Type:** Order type (for example, Restoration or Investigation).
- **Order Status:** Order status (for example, New or Field Complete).
- **Attributes:**
  - **Operator Output:** Opens the Operator Output window for the order.
- **Order Worksheet:** Opens the Order Worksheet display for the order.

The following information is shown on this toolbar:

- Bitmap designation of an order.

5.25. Permanent Jumper Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up, which allows you to place or remove tags for SCADA permanent jumpers.
- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up, which allows you to place or remove tags related to the given permanent jumper.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Switch:** Options for device status change:
  - **Change Switch Status:** Changes the real-time status of the permanent jumper to Closed or Open. Opening the permanent jumper de-energizes the downstream network elements.
  - **Toggle Phases:** Opens the Toggle Switch Phases window that allows you to close and open individual phases of the permanent jumper.
  - **Change Pending/Permanent Status:** Places a special mark at the permanent jumper that informs about changing permanent status.
- **Place Cut:** Places a cut. This option is used to place a loss of voltage cut. Appears in study mode only.

- **Attributes:**
    - **Operator:** Opens the Operator Attributes window for the permanent jumper.
    - **Analyst:** Opens the Analyst Attributes window for the permanent jumper.
    - **Operator Output:** Opens the Operator Output window for the permanent jumper.
    - **Analyst Output:** Opens the Analyst Output window for the permanent jumper.
- **Trace:** Tracing options:
    - **Trace Up:** Invokes a trace from the selected device to the power source.
    - **Trace Up to SCADA Device:** Traces upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an information message appears, and the tracing is canceled.
    - **Trace Up to Protection Device:** Traces upstream from the starting point to the next upstream protective device (distribution transformer, fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an information message appears, and the tracing is canceled.
    - **Trace Up to Switching Device:** Traces upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
    - **Trace Down to SCADA Device:** Traces downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an information message appears, and the tracing is canceled.
    - **Trace Down to Protection Device:** Traces downstream from the starting point to the next downstream protective device. If there is no downstream protective device, an information message appears, and the tracing is canceled.
    - **Trace Down to Switching Device:** Traces downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an information message appears, and the tracing is canceled.
    - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
    - **Trace Section:** Traces the switchable section. Highlights all feeder backbones and laterals that are within the bounding switches.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.

    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder to which the permanent jumper belongs.
- **Run Network Analysis:** Opens a real-time DNAF solution window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters window, which allows you to enter permanent jumper parameters.
- **Add to Switching Order:** Adds the permanent jumper to the Switching Order form.
- **Associate to Safety Document:** Adds a permanent jumper record to the Safety Document.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.

The following information is shown on the toolbar:

- Bitmap designation of a permanent jumper network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Permanent jumper name. The tooltip shows the address.
- Designation of permanent jumper phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Current permanent jumper state (Open or Closed).

5.26. Planned Outage Toolbar

The following actions are available from this toolbar:

- **Attributes:**
    - **Operator Output:** Opens the Operator Output pop-up window for the planned outage.
- **Update Planned Outage:** Opens the Update Customer Notification pop-up display.

The following information is shown on the toolbar:

- Bitmap designation of a planned outage element.
- Selected outage component name.

5.27. Power Transformer Toolbar

The following actions are available from the regulator and power transformer toolbars:

- **Raise Step:** Steps up.
- **Lower Step:** Steps down.

  The Raise/Lower Step buttons are disabled for fixed tap changers.

  The following logic is applied for regulating transformers (LTCs and regulators) the Raise/Lower Step buttons are disabled if a Tap Position SCADA analog measurement with good quality exists, or the transformer is blocked. If a Tap Position analog measurement with good quality does not exist, the step regulation is enabled in the following cases:

  - Both Remote/Local and Auto/Manual SCADA status measurements with good quality exist and the values are Remote and Manual, respectively.
  - A good quality Remote/Local SCADA status measurement does not exist and an Auto/Manual SCADA measurement status with good quality exists and its value is Manual.
  - An Auto/Manual SCADA status measurement with good quality does not exist, and a Remote/Local SCADA status measurement with good quality exists and its value is Remote.
  - Good quality SCADA status measurements, whether Remote/Local or Auto/Manual, do not exist, with the Remote/Local value consistently taken from the static model being Remote.
- **Place Cut:** Places a cut. This option is used to place a loss of voltage cut. Appears in study mode only.
- **Place Temporary Fault:** Opens a fault pop-up that allows creating a temporary fault with respect to the selected network element. The transformer must be de-energized.
- **Set Regulation On/Off:** Toggles the regulation to the On or Off state.

  The Regulation On/Off button is disabled for fixed tap changers. For regulating transformer types, if a Regulation (Auto/Manual) SCADA status measurement with good quality exists, this button is disabled. Otherwise, the user can toggle the Auto/Manual value using the toolbar button.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the transformer.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the transformer.
  - **Operator Output:** Opens the Operator Output pop-up window for the transformer.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the transformer.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode:
  - **User Enabled:** The device is available for control by the user. This option is available only in

the Simulator and Study Mode.

  - **User Disabled:** The device is unavailable for control by the user.
  - **SCADA Decides:** The device is eligible for control based on an available status measurement. This option only works in real-time mode when SCADA measurements are being updated and the device has remote control capability.

- **Block/Unblock:** Blocks or unblocks the transformer.

  The Block/Unblock Step button is disabled for fixed tap changers. For regulating transformer types, if a Blocked Status SCADA status measurement with good quality exists, this button is disabled. Otherwise, the user can toggle the Block/Unblock value using the toolbar button.

- Disable/Enable Voltage Reduction**:** Click this button to Disables or Enables Voltage Reduction.
  - The Disable/Enable Voltage Reduction button is disabled for fixed tap changers. For regulating transformer types, this button is enabled if any one of the following conditions are met.
  - A good quality Voltage Reduction SCADA status measurement does not exist.
  - The transformer is not voltage reduction capable.
- The transformer is voltage reduction capable if any one of the following conditions is met:
  - A Voltage Reduction analog measurement with good quality exists.
  - A regulator has advanced regulation mode(s) and it is capable of regulating in the presently set flow direction.
  - If static transformer tap changer model defines at least one volt reduction step value (VOLTRED1, VOLTRED2, VOLTRED3), transformer is voltage reduction capable.
- **Toggle to Local/Remote:** Toggles the transformer to Local or Remote mode.

  The Toggle to Local/Remote button is disabled for fixed tap changers. For regulating transformer types, if Local/Remote button is enabled, the user can toggle the Local/Remote value using the toolbar button. The Local/Remote button is disabled in the following cases.

  - A Local/Remote SCADA status measurement with good quality exists.
  - The transformer has Local Only regulation control capability (there is no Remote regulation capability).
  - The transformer has Remote Only regulation control capability (there is no Local regulation capability).
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next

upstream protective device (switch, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.

  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.
  - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
  - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
  - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the transformer belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, manage voltage, and so on.
- **Enter Parameters:** Opens the Transformer Parameters pop-up, which allows overriding the nominal values of the transformer.
- **Mark Transformer:** Marks the selected transformer with a halo. Unmarks the device if it was already marked.
- **Add to Switching Order:** Adds the transformer to the Switching Order form.

- **Add to Daily Log:** Adds the transformer to the Daily Log.
- **Create Quick Order:** Creates a non-outage isolated incident for the transformer.

The following information is shown on the toolbar:

- Bitmap designation of a transformer element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Transformer name. The tooltip shows the transformer's address.
- Designation of transformer phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Tap changer operation mode.
- Current step.

5.28. Recloser Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up allowing the placement or removal of tags for SCADA reclosers.
- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up allowing the placement or removal of tags related to the given recloser.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Switch:** Options for device status change:
    - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to open or close the recloser.
    - **Change Switch Status:** Changes the real-time status of the recloser, Closed or Open. Opening the recloser de-energizes the downstream network elements.
    - **Toggle Phases:** Opens the Toggle Switch Phases pop-up, which allows you to close and open individual phases of the recloser.
    - **Change Pending/Permanent Status:** Places the special mark at the recloser that informs about changing permanent status.
    - **Set Lockout Status:** Toggles the lockout status for the recloser.
- **Change Switch Classification:** Options to set the recloser's specific open state classification.
    - **Set Temporary Open:** Sets the recloser classification to Temporary Open.
    - **Set Bad Open:** Sets the recloser classification to Bad Open.

    - **Set Normal Open:** Sets the recloser classification to Normal Open.
    - **Reset:** Clears the specific open state classification.
- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the recloser.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the recloser.
    - **Operator Output:** Opens the Operator Output pop-up window for the recloser.
    - **Analyst Output:** Opens the Analyst Output pop-up window for the recloser.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode:
    - **User Enabled:** The device is available for control by the user. This option is available only in the Simulator and Study Mode.
    - **User Disabled:** The device is unavailable for control by the user.
    - **SCADA Decides:** The device is eligible for control based on an available status measurement. This option only works in real-time mode when SCADA measurements are being updated and the device has remote control capability.
- **Trace:** Tracing options:
    - **Trace Up:** Invokes a trace from the selected device to the power source.
    - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled.
    - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
    - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.

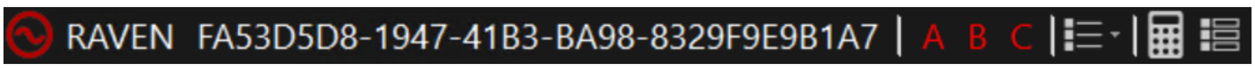
    - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
    - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the recloser belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters pop-up window that allows entering recloser parameters.
- **Mark:**
    - **Mark Switch:** Marks the selected recloser with a halo. Unmarks the recloser if it was already marked.
    - **Mark All Potential Isolating Devices:** Marks all isolation switches (switches that can isolate the selected recloser from the network).
- **Add to Switch Order:** Adds a recloser to the Switching Order form.
- **Add to Daily Log:** Adds a recloser to the Daily Log.
- **Associate to Safety Document:** Adds a recloser record to the Safety Document.
- **Block Outage Prediction:** Blocks the device outage prediction.
- **Resume Outage Prediction:** Resumes the device outage prediction.
- **Toggle Block Outage Prediction:** Toggles blocking the prediction.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected recloser.
- **Create Maintenance Order:** Creates a maintenance order for the recloser. (For the Enterprise

OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a recloser network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Recloser name. The tooltip shows the address.
- Designation of recloser phases. Phases are colored according to the energization status (energized, de-energized, or not constructed). Current recloser state (Open, Partial or Closed).

5.29. Reference Bus Interface Toolbar



The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the component.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the component.
  - **Operator Output:** Opens the Operator Output pop-up window for the component.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the component.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, manage voltage, and so on.
- **Enter Parameters:** Opens the Reference Bus Interface Parameters pop-up window that allows entering reference bus interface parameters.

The following information is shown on the toolbar:

- Bitmap designation of a reference bus interface network element.
- Energizing substation name.
- Reference bus interface name.
- Reference bus interface ID.
- Designation of reference bus interface phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.30. Sensor Toolbar

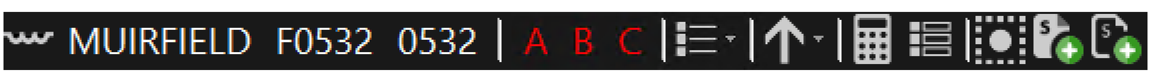
The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the sensor.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the sensor.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the sensor.
- **Tracing:**
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Clear Trace:** Clears all traces in all viewports.

The following information is shown on the toolbar:

- Bitmap designation of a sensor element.
- Sensor unique identifier.
- Sensor's fault indicator status (Set, Unset, or Unknown for a non-directional fault indicator; Unset, Both, Side 1, Side 2, or Unknown for a type 1 directional fault indicator; Unset, Forward Set, Reverse Set, or Unknown for a type 2 directional fault indicator).

5.31. Series Reactor Toolbar



The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the series reactor.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the series reactor.
  - **Operator Output:** Opens the Operator Output pop-up window for the series reactor.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the series reactor.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA:** Trace upstream from the starting point to the next upstream SCADA

controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.

  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down Phase A/B/C:** Invokes a trace of the selected phase from the point out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
  - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
  - **Trace Path:** Traces the shortest path between the selected points and highlights it.
  - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
  - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces continue to all connected end points.
  - **Clear Trace:** Clears all traces in all viewports.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up display.
- **Enter Parameters:** Opens the Series Reactor Parameters pop-up, which allows setting and overriding default parameter values.
- **Mark Series Reactor:** Marks the selected series reactor with a halo. Unmarks the device if it was already marked.
- **Add to Switching Order:** Adds the series reactor to the Switching Order form.

- **Add to Daily Log:** Adds the series reactor to the Daily Log.

The following information is shown on the toolbar:

- Bitmap designation of a series reactor element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Series reactor name and address.
- Designation of series reactor phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.32. Structure Toolbar

The following actions are available from this toolbar:

- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the structure.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the structure.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected structure.
- **Create Maintenance Order:** Creates a maintenance order for the structure. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a structure element.
- Type of a structure element.
- Name of a structure element.
- ID of a structure element.
- Owner/material of a structure element.

5.33. Substation Toolbar

The following actions are available from this toolbar:

- **Protection Settings:** Opens the Protection Settings Group grid display for the substation.

- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the substation.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the substation.
    - **Operator Output:** Opens the Operator Output pop-up window for the substation.
    - **Analyst Output:** Opens the Analyst Output pop-up window for the substation.
- **Station Model Conversion History:** Opens the Substation Conversion History display in a separate view port.
- **Navigate to Station Internals:** Displays an internal view of the substation.
- **Navigate to SCADA Display:** Opens a SCADA display in a separate viewport.
- **Future Equipment Summary:** Opens the Equipment Commissioning/Decommissioning display to show equipment that is not currently active.
- **Create Annotation:** Opens an annotation pop-up that allows creating a new annotation.
- **Clear Loss of Voltage Flag:** Removes a Loss of Voltage decoration from the substation circle.
- **Network Analysis Inputs:** Options to open manual data entry grid tabular and pop-up displays:
    - **Station:** Opens the Station Internals Input display.
    - **Feeder:** Opens the Feeder Input for Station display for the substation.
    - **Fault Location:** Opens the Fault Location display to show potential fault locations.
    - **LVM Eligibility:** Opens the LVM Device Eligibility display for the substation.
    - **Reference Bus:** Opens the Reference Bus Input for Station display to enter reference bus parameters.
    - **Capacitor:** Opens the Capacitor Control display to enter capacitor parameters.
    - **Tap Changer:** Opens the Tap Changer Input display to enter tap changer parameters.
    - **DER/DG:** Opens the Generator/Distributed Energy Resource Input display to enter generator and DER parameters.
    - **Load Categories:** Opens the Load to Voltage Coefficients display to enter load category parameters.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Network Analysis Outputs:**
    - **Diagnostic Summary:** Opens the DNAF Application Diagnostic Information display that provides diagnostic data about all DNAF applications executed for the substation.
    - **Distribution Power Flow:** Opens a new viewport containing the following DPF results tabular displays: DPF Feeder Results and Distribution Power Flow Summary for Station.
    - **Protection Validation:** Opens the Protection Validation Results tabular display in a

separate viewport.

  - **Load and Volt/VAR Management:** Opens the Load and Volt/VAR Management tabular display in a separate viewport.
  - **Fault Isolation and Service Restoration:** Opens the FISR Results Summary tabular display in a separate viewport.
  - **Automatic Feeder Reconfiguration Results:** Opens the AFR Results Summary tabular display in a separate viewport.
  - **Fault Location:** Opens the Fault Location Results tabular display in a separate viewport.
  - **Short Circuit Calculation:** Opens the Short Circuit Calculation Summary display in a separate viewport.
  - **Planned Outage Switching (ST Only):** Opens the Planned Outage Service display in a separate viewport.
  - **FISR Contingency Analysis (ST Only):** Opens the FISRCA Solution Overview in a separate viewport.
- **Add Create Step to Switch Order:** Adds a step to place a mobile substation to the Switching Order form. (Applies to mobile substations only.)

The following information is shown on the toolbar:

- Bitmap designation of a substation element.
- Name of the substation. The tooltip shows the address.

5.34. Switch Toolbar

The following actions are available from this toolbar:

- **Place Tags from EMS:** Opens the SCADA control pop-up allowing the placement or removal of tags for SCADA switches.
- **Place DMS-Only Tags:** Opens the Tagging Operations (Substations) pop-up allowing the placement or removal of tags related to the given switch.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Status:** Options for device status change:
  - **Go to SCADA Control (EMP):** Opens the SCADA control pop-up to open or close the switch.
  - **Change Switch Status:** Changes the real-time status of the switch, Closed or Open. Opening the switch de-energizes the downstream network elements.
  - **Toggle Phases:** Opens the Toggle Switch Phases pop-up, which allows you to close and open individual phases of the switch.
  - **Change Pending/Permanent Status:** Places the special mark at the switch that informs

about changing permanent status.

- **Change Switch Classification:** Options to set the switch's specific open state classification.
  - **Set Temporary Open:** Sets the switch classification to Temporary Open.
  - **Set Bad Open:** Sets the switch classification to Bad Open.
  - **Set Normal Open:** Sets the switch classification to Normal Open.
  - **Reset:** Clears the specific open state classification.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the switch.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the switch.
  - **Operator Output:** Opens the Operator Output pop-up window for the switch.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the switch.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode:
  - **User Enabled:** The device is available for control by the user. This option is available only in the Simulator and Study Mode.
  - **User Disabled:** The device is unavailable for control by the user.
  - **SCADA Decides:** The device is eligible for control based on an available status measurement. This option only works in real-time mode when SCADA measurements are being updated and the device has remote control capability.
- **Trace:** Tracing options:
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled.
  - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (switch, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled.
  - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled.
  - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next

    downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled.
    - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled.
    - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is canceled.
    - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions.
    - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points.
    - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the switch belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.
- **Enter Parameters:** Opens the Switch Parameters pop-up window that allows entering switch parameters.
- **Mark:**
    - **Mark Switch:** Marks the selected switch with a halo. Unmarks the switch if it was already marked.
    - **Mark All Potential Isolating Devices:** Marks all isolation switches (switches that can isolate the selected switch from the network).
- **Add to Switching Order:** Adds a switch to the Switching Order form.
- **Add to Daily Log:** Adds a switch to the Daily Log.
- **Associate to Safety Document:** Adds a switch record to the Safety Document.
- **Split Incident:** Splits a de-energized open device into a new incident. This can be used to split a device from a predicted incident on an upstream device or break a large incident into multiple incidents.

- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected switch.
- **Create Maintenance Order:** Creates a maintenance order for the switch. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a switch network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Switch name. The tooltip shows the address.
- Designation of switch phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Current switch state (Open, Partial or Closed).

5.35. Switch Cabinet Toolbar

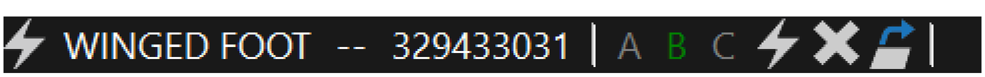
The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the switch cabinet.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the switch cabinet.
- **Navigate to Related Display:** Opens the new viewport displaying the internal view of the switch cabinet.

The following information is shown on the toolbar:

- Bitmap designation of the switch cabinet as a network element.
- The name of the substation and the name of the selected cabinet separated by a blank space.

5.36. Temporary Fault Toolbar



The following actions are available from this toolbar:

- **Edit Fault:** Opens the fault pop-up to edit the selected fault.

- **Remove Fault:** Removes the selected fault.
- **Navigate to FISR Plans:** Opens the FISR Results Summary display.

The following information is shown on the toolbar:

- Bitmap designation of a fault network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Fault name. The tooltip shows the address.
- Designation of fault phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.37. Temporary Ground Toolbar

The following actions are available from this toolbar:

- **Edit Temporary Ground:** Opens the Temporary Ground pop-up to edit the selected temporary ground.
- **Remove Temporary Ground:** Removes the selected temporary ground.
- **Add Create Step to Switch Order:** Adds a step to create the temporary ground to the Switching Order form.
- **Create Maintenance Order:** Creates a maintenance order for the temporary ground. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a temporary ground network element.
- Substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Designation of temporary ground phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Temporary ground name. The tooltip shows the address.

5.38. Temporary Switch (Jumper) Toolbar

The following actions are available from this toolbar:

- **Place Tags:** Opens the Tagging Operations (Substation) pop-up, which allows the placement or removal of tags for the switch.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Edit Jumper:** Opens the jumper pop-up window.
- **Switch:** Changes the status of the temporary switch. There can be two statuses: Open or Closed. When the status is Open, it allows performing downstream and upstream traces.
- **Remove Jumper:** Removes the selected temporary switch.
- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the switch.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the switch.
  - **Operator Output:** Opens the Operator Output pop-up window for the switch.
  - **Analyst Output:** Opens the Analyst Output pop-up window for the switch.
- **Move Label:**
  - **North:** Moves the label to the north (changes the side where the label is displayed) with respect to the selected network element.
  - **Northeast:** Moves the label to the northeast (changes the side where the label is displayed) with respect to the selected network element.
  - **East:** Moves the label to the east (changes the side where the label is displayed) with respect to the selected network element.
  - **Southeast:** Moves the label to the southeast (changes the side where the label is displayed) with respect to the selected network element.
  - **South:** Moves the label to the south (changes the side where the label is displayed) with respect to the selected network element.
  - **Southwest:** Moves the label to the southwest (changes the side where the label is displayed) with respect to the selected network element.
  - **West:** Moves the label to the west (changes the side where the label is displayed) with respect to the selected network element.
  - **Northwest:** Moves the label to the northwest (changes the side where the label is displayed) with respect to the selected network element.
- **Tracing:**
  - **Trace Up:** Invokes a trace from the selected device to the power source.
  - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals.
  - **Clear Trace:** Clears all traces in all viewports.
- **Show/Hide Other Feeders:** Toggles showing only the feeder the switch belongs to.
- **Run Network Analysis:** Opens a real-time DNAF solution pop-up window that allows you to

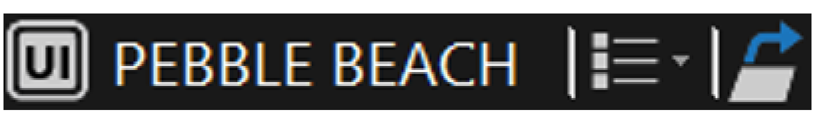
perform network analysis as well as manage power flow, transfer capacity, voltage management, and so on.

- **Mark:**
    - **Mark Switch:** Marks the selected switch with a halo. Unmarks the switch if it was already marked.
    - **Mark All Potential Isolating Devices:** Marks all isolation switches (switches that can isolate the selected switch from the network).
- **Add to Switching Order:** Adds the temporary switch to the Switching Order form.
- **Add to Daily Log:** Adds the temporary switch record to the Daily Log.
- **Add Create Step to Switch Order:** Adds a step to create the temporary switch to the Switching Order form.
- **Create Maintenance Order:** Creates a maintenance order for the switch. (For the Enterprise OMS client types only.)

The following information is shown on the toolbar:

- Bitmap designation of a switch network element.
- Energizing substation name. Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Switch name. The tooltip shows the address.
- Designation of switch phases. Phases are colored according to the energization status (energized, de-energized, or not constructed).
- Current switch state (Open, Partial or Closed).

5.39. UI Container Toolbar



The following actions are available from this toolbar:

- **Attributes:**
    - **Operator:** Opens the Operator Attributes pop-up window for the UI container.
    - **Analyst:** Opens the Analyst Attributes pop-up window for the UI container.
- **Navigate to Related Display:** Displays the UI container content (internal devices) in a new viewport.

The following information is shown on the toolbar:

- Bitmap designation of the UI container element.
- Substation name.
- UI container name. The tooltip shows the address.

5.40. Transformer Bank Toolbar

The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the transformer bank.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the transformer bank.

The following information is shown on the toolbar:

- Energizing substation name.

  Clicking the substation name navigates to the geographic view centered on the substation.
- Energizing feeder name.
- Transformer bank name.
- Designation of phases.

  Phases are colored according to the energization status (energized, de-energized, or not constructed).

5.41. Vault Toolbar

The following actions are available from this toolbar:

- **Attributes:**
  - **Operator:** Opens the Operator Attributes pop-up window for the vault.
  - **Analyst:** Opens the Analyst Attributes pop-up window for the vault.
- **Navigate to Related Display:** Displays the vault content (internal devices) in a new viewport.

The following information is shown on the toolbar:

- Bitmap designation of the vault.
- Substation name.

- Vault name. The tooltip shows the address.

5.42. Vehicle Toolbar

The following information is shown on the toolbar:

- Bitmap designation of the vehicle.
- Vehicle ID.
- Vehicle Type.

6. ADMS Contextual Menus

This chapter contains descriptions of the standard contextual menus provided by ADMS.

6.1. Geographic Network View

The following actions are available from this menu:

- **Help:** Opens the *Outage Management System User Guide*.
- **Place Annotation:** Opens an annotation pop-up that allows creating a new annotation and places it at the specified geographical coordinates.
- **Manage Mobile Energy Resources:** Opens the Mobile Substation pop-up, which allows placing and removing mobile substations.
- **Create Safety Doc:** Creates a Safety Document.
- **Create Clearance:** Opens the Create Clearance pop-up to create a tag clearance.
- **Add Call:** Opens the Add/Update Unassociated Call data input pop-up window to create a new location call. Depending on the Call Action Matrix configuration, creates an unassociated call or non-outage incident.
- **Jumper End at X, Y:** Specifies the end position of a jumper-to-nowhere to the current position of the mouse cursor. Available only if a jumper start position is already marked on a line.
- **Real Time**: Navigates to the current view in real-time mode.
- **Study**: Navigates to the current view in study mode.
- **Simulation**: Navigates to the current view in simulator mode.
- **Zoom-Based Lateral:** Displays the lateral according to lateral zoom settings provided in the configuration file.
- **Turn Laterals On/Off:** Toggles the display of laterals On/Off regardless of lateral zoom settings.
- **Zoom-Based Label:** Display labels according to the zoom level settings provided in the configuration file.
- **Turn Labels On/Off:** Toggles device labeling On/Off regardless of label zoom settings.
- **Turn Line Labels On/Off:** Toggles line labeling On/Off regardless of label zoom settings. This option is only available on loop schematic views.
- **Show All Customers:** Shows all customers.
- **Customers with Activity:** Shows customers with activity (de-energized customers or customers with calls or AMI Out messages).
- **Hide All Customers:** Hides all customers.
- **Traced Distribution Transformer List:** Opens the Traced Distribution Transformer Summary display with the list of all currently traced transformers in the system.

- **Traced Customer List:** Opens the Traced Customer List (Create Planned Outage) window for currently traced customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Clear Customer List:** Removes customers from the Customer List pop-up window and deselects customers on the geographic view.
- **Traced Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation for the currently traced part of the network.
- **Start SWO Recording:** Starts the Switching Order recording session to record the operator’s actions as switching steps. This option is available in Study mode only.
- **Stop SWO Recording:** Stops the Switching Order recording session. This option is available in Study mode only.
- **Add Marked Devices to SWO:** Adds all marked devices to the Switching Order form.
- **Add Marked Devices to Daily Log:** Adds all marked devices to the Daily Log.
- **Clear All Marks:** Clear all marks (marked elements are highlighted with a blue dashed line halo).
- **Clear Trace:** Clears all traces in all viewports.
- **Clear All Marks and Traces:** Clear all marks and traces in all viewports.
- **Power Flow Animation:** Power flow animation options.
  - **PA:** Enables the real power flow animation for phase A.
  - **PB:** Enables the real power flow animation for phase B.
  - **PC:** Enables the real power flow animation for phase C.
  - **PABC:** Enables the real power flow animation for phases A, B, and C.
  - **QA:** Enables the reactive power flow animation for phase A.
  - **QB:** Enables the reactive power flow animation for phase B.
  - **QC:** Enables the reactive power flow animation for phase C.
  - **QABC:** Enables the reactive power flow animation for phases A, B, and C.
  - **Off:** Disables the power flow animation.
- **Station Measurement Tabular:** Opens the Measurement Data display for the station that is associated with the selected location.
- **Add Damage Assessment Location:** Opens the Add Damage Assessment Location dialog box to add a damage assessment location report.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected location.
- **Create Planned Outage:** Opens the Traced Customer List (Create Planned Outage) window.
- **Create Maintenance Order:** Creates a maintenance order for the selected location. (For the Enterprise OMS client types only.)

6.2. Annotation

The following actions are available from this menu:

- **Edit Annotation:** Opens the annotation pop-up display with the annotation details.
- **Remove Annotation:** Removes the selected annotation.
- **Move Annotation:** Activates the move mode for the annotation.

6.3. Breaker

The following actions are available from this menu:

- **Label:** Indicates the substation name that the breaker belongs to, the breaker name, and the breaker state (open or closed).
- **Go to SCADA Control:** Opens the SCADA control pop-up to open or close the breaker.
- **Change State:** Toggles the real-time status of the breaker to closed or open.
- **Open Breaker:** Changes the real-time status of the breaker to open.
- **Close Breaker:** Changes the real-time status of the breaker to closed.
- **Add to Daily Log:** Adds the breaker to the Daily Log.
- **Add to Switch Order:** Adds a breaker to the Switching Order form.
- **Create Hot Line Hold:** Opens the Switching Order application and creates a hot line hold document for the breaker.
- **Create Incident:** Creates a separate incident, adds a record to the Incident Detail tabular display, and marks the affected device on the geographic view. Allows splitting a big incident into multiple smaller incidents. This command is visible for open breakers only.
- **Create Predicted Incident:** Creates a predicted stable incident for the selected breaker.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected breaker.
- **Create Confirmed Outage:** Creates a confirmed incident for the selected breaker. (For the Enterprise OMS client types only.)
- **Create Task:** Options to create an order task for the selected breaker. (For the Enterprise OMS client types only.)
    - **Open:** Opens a dialog to create an Open task for the selected breaker.
    - **Close:** Opens a dialog to create a Close task for the selected breaker.
    - **Other:** Opens a dialog to create a Maintenance, Investigation, Make Safe, or Assessment task for the selected breaker.
- **Create Maintenance Order:** Creates a maintenance order for the selected breaker. (For the

Enterprise OMS client types only.)

- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag for the selected breaker.
- **Place Do Not Operate Tag (SCADA):** Automatically places the Do Not Operate tag for the selected SCADA breaker.
- **Place Hold Open Tag:** Automatically places the Hold Open tag for the selected breaker.
- **Place Hold Open Tag (SCADA):** Automatically places the Hold Open tag for the selected SCADA breaker.
- **Place DMS Tag:** Opens the Tagging Operations window to place a tag for the selected breaker.
- **Associate To Safety Doc:** Adds a breaker record to the Safety Document.
- **Create Hot Line Hold Safety Doc Template:** Creates a hot line hold safety document and adds the associated devices to the safety doc.
- **Create Transfer of Control Safety Doc Template:** Creates a transfer of control safety document and adds the associated devices to the safety doc.
- **Mark/Unmark (1) Switch:** Marks/unmarks the breaker with mark type 1. The marked breaker is highlighted by a halo with a blue dashed line.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Downstream:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Downstream Customers:** Opens the Traced Customer List (Create Planned Outage) window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.
- **Find Nearest Structure:** Selects a structure that is closest to the breaker.
- **Mark Team(s) Switches:** Marks/unmarks members of the automated scheme teams that are associated with the breaker.
    - **Mark/Unmark (1):** Marks/unmarks team members with mark type 1.
    - **Mark/Unmark (2):** Marks/unmarks team members with mark type 2.
    - **Mark/Unmark (3):** Marks/unmarks team members with mark type 3.
- **Trace Scheme Switches:** Draws traces between members of the automated scheme teams that are associated with the breaker.
- **Scheme Switches on Feeder:** Opens the Automation Scheme Team Switches on Feeder display, which lists switching devices that are part of automated local peer-to-peer switching schemes on a feeder associated with the breaker.

- **Faulted Section Determination:** Shows the Faulted Section Determination display where you can run the FSD application. Appears only for breakers downstream of power transformers with ASC.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that breaker state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that breaker state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the breaker is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the breaker is enabled for controls. This option is only available in the simulator mode.

6.4. Bus

The following actions are available from this menu:

- **Future To Active:** Changes the device commissioning status from Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Create Quick Order:** Creates a non-outage isolated incident for the selected bus.
- **Find Nearest Structure:** Selects a structure that is closest to the bus.

6.5. Capacitor

The following actions are available from this menu:

- **Label:** Indicates the substation name that the capacitor belongs to, the capacitor name, and the capacitor state (open or closed).
- **Change State:** Toggles the real-time status of the capacitor to on or off.
- **SCADA Control Popup:** Opens the SCADA control pop-up to enable or disable the capacitor.
- **Manual Only:** Changes the capacitor status to Manual Only.
- **Add to Exclusions List:** Adds the capacitor to the LVM device exclusions list.
- **Remove from Exclusions List:** Removes the capacitor from the LVM device exclusions list.
- **Add to Daily Log:** Adds the capacitor to the Daily Log.

- **Add to Switch Order:** Adds the capacitor to the Switching Order form.
- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag for the selected capacitor.
- **Place Do Not Operate Tag (SCADA):** Automatically places the Do Not Operate tag for the selected SCADA capacitor.
- **Place Hold Open Tag:** Automatically places the Hold Open tag for the selected capacitor.
- **Place Hold Open Tag (SCADA):** Automatically places the Hold Open tag for the selected SCADA capacitor.
- **Place DMS Tag:** Open the Tagging Operations pop-up, which allows the placement or removal of tags related to the selected capacitor.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected capacitor.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Find Nearest Structure:** Selects a structure that is closest to the capacitor.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Create Maintenance Order:** Creates a maintenance order for the capacitor. (For the Enterprise OMS client types only.)
- **Block State Control:** Sets the Block State Control status to Block. This specifies that capacitor state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that capacitor state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the capacitor is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the capacitor is enabled for controls. This option is only available in the simulator mode.

6.6. Crew

The following actions are available from this menu:

- **Label:** Indicates the crew name and radio Assigned.
- **Assigned:** Changes the crew status to Dispatched.
- **Dispatched:** Changes the crew status to Dispatched.
- **EnRoute:** Changes the crew status to EnRoute.

- **OnSite:** Changes the crew status to OnSite.
- **Update/Split Crew:** Opens the Crew Update pop-up display, which allows you to add an employee to the crew, split one team into two different crews, or create a new crew.
- **Move Crew:** Activates the move mode for the crew.
- **Find Nearest Structure:** Selects a structure that is closest to the crew.

6.7. Customer

The following actions are available from this menu:

- **Add/Update Customer Call:** Opens the Add Customer Call pop-up display to create a single customer incident record. The incident record is shown on the Incident Detail OMS display.
- **Cancel Customer Call:** Cancels calls from the customer.
- **Show Customer Calls History:** Opens the Customer Calls display, which shows call history for a customer.
- **Show Customer Outage History:** Opens the Incident Details display, which shows incidents that are associated with the customer. (Future functionality.)
- **Associate Customer with Incident:** Opens the Associate Customer with Incident pop-up.
- **Split Out Customer into New Incident:** Creates a new incident for the de-energized customer.
- **Ping and Request Meter Voltage:** Pings the selected customer and requests the meter voltage value.

  A blue circle around the customer symbol indicates that a ping is in process, a green circle indicates that a ping succeeded, a yellow circle indicates AMI connection failure (the meter may be temporarily out), and a red circle indicates that a ping failed. The ping color is configurable in the Viewer Configuration Editor.

- **Ping All Customers on Transformer:** Automatically pings all the customers associated with the same transformer and requests the meter voltage values.
- **Call Ping Pop-Up – Single Customer:** Calls up the CIS/AMI Customer Ping display, which shows customer meter data and allows you to ping the selected customer.
- **Call Ping Pop-Up – Service Point:** Calls up the CIS/AMI Customer Ping display, which shows customer meter data and allows you to ping the customers connected to the same transformer.
- **Disconnect Meter:** Disconnects the customer’s meter.
- **Reconnect Meter:** Reconnects the customer’s meter.
- **Move Customer:** Activates the move mode for the customer.
- **Suspend Customer:** Suspends the customer that stay de-energized for a long period of time, and the service cannot be restored due to known cause (such as house on fire). This clears the associated single customer incident, if exists, or excludes the customer from the upstream outage.

- **Unsuspend Customer:** Unsuspends the previously suspended customer.
- **Scan AMI Analogs:** Sends the command for detecting load side voltage for the selected customer.
- **Scan AMI Analogs – Multiple Customers:** Sends the command for detecting load side voltage for the customers connected to the same transformer.
- **Show Analogs – Single Customer:** Calls up the Customer Analogs display, which shows the analog meter data for the selected customer. You need to scan AMI analogs before showing them.
- **Show Analogs – Multiple Customers:** Calls up the Customer Analogs display, which shows the analog meter data for the customers connected to the same transformer. You need to scan AMI analogs before showing them.
- **Add Customer to List:** Adds the selected customer to the Create Planned Outage display to notify the selected customers of a planned outage.
- **Remove Customer From List:** Removes the selected customer from the Create Planned Outage display and deselects the customer on the geographic view.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Up:** Invokes a trace from the selected customer to the power source. You can also specify a trace color.
- **Find Nearest Structure:** Selects a structure that is closest to the customer.
- **Create Maintenance Order:** Creates a maintenance order for the customer. (For the Enterprise OMS client types only.)

6.8. Cut

The following actions are available from this menu:

- **Label:** Indicates the substation name that the cut belongs to, the cut name, and the cut state (open or closed).
- **Open Cut:** Changes the real-time status of the cut to open.
- **Close Cut:** Changes the real-time status of the cut to closed.
- **Remove:** Removes the cut.
- **Add to Daily Log:** Adds a cut to the Daily Log.
- **Add to Switch Order:** Adds a cut to the Switching Order form.
- **Add Create Step to Switch Order:** Adds a step to create the cut to the Switching Order form.
- **Create Incident:** Creates a separate incident, adds a record to the Incident Detail tabular display, and marks the affected device on the geographic view. Allows splitting a big incident into multiple smaller incidents. This command is visible for open cuts only.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you

to create a customer notification about the planned outage.

- **Create Quick Order:** Creates a non-outage isolated incident for the selected cut.
- **Create Task:** Options to create an order task for the selected cut. (For the Enterprise OMS client types only.)
  - **Open:** Opens a dialog to create an Open task for the selected cut.
  - **Close:** Opens a dialog to create a Close task for the selected cut.
  - **Other:** Opens a dialog to create a Maintenance, Investigation, Make Safe, or Assessment task for the selected cut.
- **Create Maintenance Order:** Creates a maintenance order for the cut. (For the Enterprise OMS client types only.)
- **Associate to Safety Document:** Adds a cut record to the Safety Document.
- **Mark/Unmark Cut:** Marks/unmarks the cut with mark type 1. The marked cut is highlighted by a halo with a blue dashed line.
- **Find Nearest Structure:** Selects a structure that is closest to the cut.

6.9. Damage Assessment Location

The following actions are available from this menu:

- **Move Damage Assessment Location:** Activates the move mode for the damage assessment location symbol.
- **Edit Damage Assessment Location:** Opens the Damage Assessment Status dialog box to edit damage assessment location properties.
- **Delete Damage Assessment Location:** Deletes the damage assessment location.

6.10. Distribution Transformer

The following actions are available from this menu:

- **Label:** Indicates the substation name that the transformer belongs to, the transformer name, and the transformer state (connected or disconnected).
- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Connect Transformer:** Connects the transformer to the network.

- **Disconnect Transformer:** Disconnects the transformer from the network.
- **Found Open:** Disconnects the transformer from the network. Creates a confirmed incident with the FoundOpen flag set.
- **Add to Switch Order:** Adds the distribution transformer to the Switching Order form.
- **Add to Daily Log:** Adds the distribution transformer to the Daily Log.
- **Create Historical Incident:** Creates a historical incident for the distribution transformer and adds a record to the History Correction tabular display.
- **Create Incident:** Creates a separate incident and adds a record to the Incident list. Allows splitting a big incident into multiple smaller incidents. This command is visible for disconnected transformers.
- **Create Predicted Incident:** Creates a predicted stable incident for the selected distribution transformer.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected transformer.
- **Create Confirmed Outage:** Creates a confirmed incident for the selected transformer. (For the Enterprise OMS client types only.)
- **Create Task:** Options to create an order task for the selected transformer. (For the Enterprise OMS client types only.)
  - **Open:** Opens a dialog to create an Open task for the selected transformer.
  - **Close:** Opens a dialog to create a Close task for the selected transformer.
  - **Other:** Opens a dialog to create a Maintenance, Investigation, Make Safe, or Assessment task for the selected transformer.
- **Create Maintenance Order:** Creates a maintenance order for the transformer. (For the Enterprise OMS client types only.)
- **Distribution XF Load:** Opens the Distribution Transformer Load display.
- **Traced Distribution Transformer List:** Opens the Traced Distribution Transformer Summary display with the list of all currently traced transformers in the system.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Upstream:** Invokes a trace from the transformer to the power source. You can also specify a trace color.
- **Trace Downstream:** Traces the transformer and the connected customers. You can also specify a trace color.
- **Downstream Customers:** Opens the Traced Customer List (Create Planned Outage) window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.

- **Downstream DER:** Opens the Downstream DER display, which shows details for energy resources that are connected to the secondary side of the transformer.
- **Ping All Customers on Transformer:** Automatically pings all customers associated with the transformer.
- **Show Schedule Chart:** Shows the load chart based on dynamic schedules.
- **Find Nearest Structure:** Selects a structure that is closest to the transformer.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Mark as Failed Equipment:** Marks the transformer with mark type 3 to allow adding this transformer to the incident's failed equipment list.

6.11. Energy Resource

The following actions are available from this menu:

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **SCADA Control Popup:** Opens the SCADA control pop-up to connect or disconnect the energy resource. Applies to SCADA-controlled energy resources.
- **Connect:** Changes the DER connection status to connected.
- **Disconnect:** Changes the DER connection status to disconnected.
- **Add to Exclusions List:** Adds the energy resource to the LVM device exclusions list.
- **Remove from Exclusions List:** Removes the energy resource from the LVM device exclusions list.
- **Static Schedules:** Shows the Schedule grid display.
- **Dynamic Schedules:** Shows the Schedule Points grid display.
- **Show Schedule Chart:** Shows the generation chart based on dynamic schedules.
- **Create Quick Order:** Creates a non-outage isolated incident for the energy resource.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that the energy resource state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that the energy

resource state controls are allowed. This option is only available in the simulator mode.

- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the energy resource is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the energy resource is enabled for controls. This option is only available in the simulator mode.

6.12. Fault Indicator

The following actions are available from this menu:

- **Change State:** Options to change fault indicator state.
- **Change Settings:** Options to change fault indicator settings (for directional telemetered fault indicators).
- **Find Nearest Structure:** Selects a structure that is closest to the fault indicator.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected fault indicator.

6.13. Feeder

The following actions are available from this menu:

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Find Nearest Structure:** Selects a structure that is closest to the feeder.

6.14. Fuse

The following actions are available from this menu:

- **Label:** Indicates the substation name that the fuse belongs to, the fuse switch name, and the fuse state (open or closed).
- **Open Fuse:** Changes the real-time status of the fuse to open.
- **Close Fuse:** Changes the real-time status of the fuse to closed.
- **Found Open:** Opens the fuse. Creates a confirmed incident with the FoundOpen flag set.
- **Confirm Outage:** Confirms the fuse outage and then changes the real-time status of the fuse

to open. When attempting to confirm open a fuse, the OMS engine invokes a pop-up display with a list of all downstream single customer and predicted incidents, including those in the predicted stable state. You can select incidents to be merged with the newly created upstream confirmed incident; the incidents that are not merged become predicted stable incidents.

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Add to Switch Order:** Adds the fuse to the Switching Order form.
- **Add to Daily Log:** Adds the fuse to the Daily Log.
- **Create Historical Incident:** Creates a historical incident for the fuse and adds a record to the History Correction tabular display.
- **Create Incident:** Creates a separate incident and adds a record to the Incident Detail OMS display. Allows splitting a big incident into multiple smaller incidents. This command is visible for open fuses only.
- **Create Predicted Incident:** Creates a predicted stable incident for the selected fuse.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected fuse.
- **Create Confirmed Outage:** Creates a confirmed incident for the selected fuse. (For the Enterprise OMS client types only.)
- **Create Task:** Options to create an order task for the selected fuse. (For the Enterprise OMS client types only.)
    - **Open:** Opens a dialog to create an Open task for the selected fuse.
    - **Close:** Opens a dialog to create a Close task for the selected fuse.
    - **Other:** Opens a dialog to create a Maintenance, Investigation, Make Safe, or Assessment task for the selected fuse.
- **Create Maintenance Order:** Creates a maintenance order for the fuse. (For the Enterprise OMS client types only.)
- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag for the fuse.
- **Place Hold Open Tag:** Automatically places the Hold Open tag for the fuse.
- **Place DMS Tag:** Opens the Tagging Operations window to place a tag for the fuse.
- **Associate to Safety Document:** Adds a fuse record to the Safety Document.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.

- **Mark/Unmark Switch:** Marks the selected fuse with a halo. Unmarks the fuse if it was already marked.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Downstream:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Downstream Customers:** Opens the Traced Customer List (Create Planned Outage) window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.
- **Find Nearest Structure:** Selects a structure that is closest to the fuse.
- **Mark as Failed Equipment:** Marks the fuse with mark type 3 to allow adding this fuse to the incident’s failed equipment list.

6.15. Generator

The following actions are available from this menu:

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **SCADA Control Popup:** Opens the SCADA control pop-up to connect or disconnect the generator. Applies to SCADA-controlled generators.
- **Add to Exclusions List:** Adds the generator to the LVM device exclusions list.
- **Remove from Exclusions List:** Removes the generator from the LVM device exclusions list.
- **Find Nearest Structure:** Selects a structure that is closest to the generator.
- **Create Maintenance Order:** Creates a maintenance order for the generator. (For the Enterprise OMS client types only.)
- **Create Quick Order:** Creates a non-outage isolated incident for the selected generator.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that the generator state controls are blocked. This option is used to simulate device malfunction. This

option is only available in the simulator mode.

- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that the generator state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the generator is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the generator is enabled for controls. This option is only available in the simulator mode.

6.16. Incident

The following actions are available from this menu:

- **Update Incident:** Opens the Update Incident display, which allows you to manage and clear the incident.
- **Incident's Custom Fields:** Opens the Incident's Custom Fields display, which allows you to enter, view, or update custom incident properties.
- **Confirm/Restore Incident:** Confirms the device outage (for devices that are predicted out) or confirms power restoration (for devices that are de-energized) and then changes the real-time status of the device.
- **Dispatch Crew:** Opens the Crews tab of the Update Incident display.
- **Navigate to OMS:** Shows the Incident Details display with the corresponding incident record selected. If the incident record was not present in the previously open Incident Details display (for example, due to applied filters), another Oms Display Plugin tab with the Incident Details display appears with the selected incident.
- **Merge Downstream Incidents:** Opens the Merge Downstream Incidents pop-up display to merge downstream confirmed, manually confirmed, predicted, and single customer incidents to a confirmed incident. Applies to downstream incidents on all phases.
- **Merge Downstream Predicted Incidents:** Merges downstream predicted and single customer incidents to a confirmed incident. Applies to downstream incidents on the same phase.
- **Merge Downstream Predicted Incidents (All Phases):** Merges downstream predicted and single customer incidents to a confirmed incident. Applies to downstream incidents on all phases.
- **Reset Predicted Stable:** Resets the Predicted Stable state for the incident. This option appears only for single customer and predicted incidents in the Predicted Stable state.
- **Set Predicted Stable:** Sets the Predicted Stable state for the incident. This option appears only for single customer and predicted incidents that are not in the Predicted Stable state.
- **Undo Confirmation:** This option is available for confirmed incidents.
  - For a confirmed incident that originated as a result of an operator's action, such as opening a device, this option closes the device and removes the incident.

    - For a confirmed incident that originated by confirming a predicted incident, this option closes the device, removes the incident, and re-creates a predicted incident with the same incident number, properties, and crew assignments based on downstream customer calls and AMI-out signals.

- **Move Up:** Moves the incident to the next upstream protective device and predicts the device out. This option is available for single customer and predicted incidents.
- **Move Down:** Clears the prediction on a protective device and creates downstream incidents in the Predicted Stable state. The device does not become blocked for prediction and can be re-predicted when more customer calls or AMI-out events are received.
- **Merge Selected Incidents:** Merges all the incidents selected on the Geographic Network view by clicking the Select Incident button.
- **Convert Incident:** Opens the Convert Incident pop-up display to convert a non-outage incident into an outage incident or vice versa. Applies to non-outage and single customer incidents only (except for incidents generated by AMI Power Out messages).
- **Associate with Other Incident:** Opens the Merge Incidents with the Incident pop-up display, which allows you to associate the selected incidents on the Geographic Network view with any incident from the incident list.
- **Dispatch Crew to Selected Incidents:** Opens the Dispatch Crew to Incident pop-up display.
- **Dispatch Specific Crew by WFM:** Dispatches a specific crew type to an incident via WFM.
- **Clear All Incident Selections:** Deselects all the incidents on the geographic network view.
- **Move Incident:** Activates the move mode for the incident.
- **Select/Deselect Incident:** Allows you to select or deselect separate incidents for merging on the Geographic Network view. Click the Merge Selected Incident button to complete merging.
- **Find Nearest Structure:** Selects a structure that is closest to the incident.
- **Confirm:** Updates the associated Check Prediction task status to Complete - Confirmed Open. (For the Enterprise OMS client types only.)
- **Move Up:** Moves the incident and the associated order to the next upstream protective device.
- **Move Down:** Cancels prediction for the device of record and creates incidents and orders downstream based on the received customer calls and AMI Out signals.
- **No Outage:** Updates the associated Check Prediction or Repair task status to Complete - No Outage. (For the Enterprise OMS client types only.)
- **Repair:** Updates the associated Check Prediction task status to Complete - Repaired. (For the Enterprise OMS client types only.)
- **Problem Confirmed:** Updates the associated Investigate task status to Complete - Found Problem. (For the Enterprise OMS client types only.)
- **Problem Resolved:** Updates the associated Fix Problem task status to Complete - Fixed Problem. (For the Enterprise OMS client types only.)
- **No Problem:** Updates the associated Investigate task status to Complete - Failed to Find Problem. (For the Enterprise OMS client types only.)

- **Problem Not Resolved:** Updates the associated Fix Problem task status to Complete - Failed to Fix Problem. (For the Enterprise OMS client types only.)
- **Order Worksheet:** Opens the Order Worksheet display in a separate tab, which allows you to make work-order-related changes and track the progress of the order. (For the Enterprise OMS client types only.)
- **Change Lead Crew Status:** Shows the following options to change the lead crew status:
  - Assigned
  - Dispatched
  - En Route
  - Arrived
  - Released
- **Take Ownership:** Associates the order with the current operator.
- **Assign Crews:** Opens the Crew Selection dialog, where you can assign a crew to the order.
- **Request Crew Assignment:** Opens the Request from the Workforce Management dialog, where you can request automatic crew assignment to the order by the Workforce Management System.

6.17. Inverter

The following actions are available from this menu:

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Find Nearest Structure:** Selects a structure that is closest to the inverter.

6.18. Line

The following actions are available from this menu:

- **Manage Curtailment:** Opens the Curtailment dialog box, for controlling the output from distributed energy resources (DERs).

  > **Note**
  >
  > This option is available only for feeders with connected DERs.

- **DER Manual Dispatch UI:** Opens the DER Manual Dispatch display, for monitoring and controlling DERs.

> **Note**
>
> This option is available only for feeders with connected DERs.

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Place Cut:** Opens the Cut pop-up. You can insert a cut (temporary switch) in the selected location and specify its parameters.
- **Place Jumper:** Opens the Jumper pop-up, for creating a jumper. A jumper must be created between two points on lines. The jumper pop-up allows you to specify phases to be connected as well as the initial state of the jumper (jumper switch).
- **Place Fault:** Opens the Fault pop-up. You can insert a fault in the selected location and specify its parameters.
- **Place SIM Fault:** Opens the Fault pop-up for the line in simulator mode. You can insert a fault in the selected location and specify its parameters.
- **Add to SWO as Temporary Ground:** Adds the selected line to the list of temporary grounds in the Equipment section in the currently open switching order (or creates a new switching order).
- **Check SWO Conflicts:** Opens the Check Conflicts popup to show conflicting switching orders for the station and feeders that are associated with the line.
- **Place Temporary Ground:** Opens the Temporary Ground pop-up. You can insert a temporary ground in the selected location and specify its parameters.
- **Navigate to Source:** Navigates to a geographic view centered on the substation that is associated with the line.
- **Schematic View:**
  - **URD Schematic:** Shows the URD schematic view.
  - **Schematic:** Shows the geo schematic view.
  - **Schematic Filter:** Shows the geo schematic view filtered to show the associated feeder.
  - **Orthogonal:** Shows the orthogonal geo schematic view.
  - **Orthogonal Filter:** Shows the orthogonal geo schematic view filtered to show the associated feeder.
  - **Split View:** Opens a new panel at the right of the display and shows the geo schematic view.
  - **Automatic Feeder Schematic:** Shows the schematic view for the feeder main (medium

    voltage) part for the feeder that is associated with the selected line.
    - **Automatic Feeder Schematic (Adjacent):** Shows the schematic view for the feeder main (medium voltage) part for the feeder that is associated with the selected line and the adjacent feeders.
- **Traced Distribution Transformer List:** Opens the Traced Distribution Transformer Summary display with the list of all currently traced transformers in the system.
- **Tracing:** Tracing options:
    - **Trace Up:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
    - **Trace Up Phase A/B/C:** Invokes a trace of the selected phase from the selected point to the power source. You can also specify a trace color.
    - **Trace Up to SCADA Device:** Trace upstream from the starting point to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears, and the tracing is canceled. You can also specify a trace color.
    - **Trace Up to Protection Device:** Trace upstream from the starting point to the next upstream protective device (fuse, recloser, sectionalizer, and so on). If there is no upstream protective device, an info message appears, and the tracing is canceled. You can also specify a trace color.
    - **Trace Up to Switching Device:** Trace upstream from the starting point to the next upstream switching device, including temporary switches, such as cuts and jumpers. If there is no upstream switching device, an info message appears, and the tracing is canceled. You can also specify a trace color.
    - **Trace Up to Recloser/Breaker:** Trace upstream from the starting point to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears, and the tracing is canceled. You can also specify a trace color.
    - **Trace Down:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
    - **Trace Down Phase A/B/C:** Invokes a trace of the selected phase from the point out to the end points, including laterals. You can also specify a trace color.
    - **Trace Down to SCADA Device:** Trace downstream from the starting point to the next downstream SCADA controllable (not only telemetered) device. If there is no downstream SCADA device, an info message appears, and the tracing is canceled. You can also specify a trace color.
    - **Trace Down to Protection Device:** Trace downstream from the starting point to the next downstream protective device (fuse, recloser, sectionalizer, and so on). If there is no downstream protective device, an info message appears, and the tracing is canceled. You can also specify a trace color.
    - **Trace Down to Switching Device:** Trace downstream from the starting point to the next downstream switching device, including temporary switches, such as cuts and jumpers. If there is no downstream switching device, an info message appears, and the tracing is

canceled. You can also specify a trace color.

    - **Trace Up and Down:** Performs both the Trace Up and Trace Down functions. You can also specify a trace color.
    - **Trace Section:** Traces the switchable section. Highlights all the feeder backbones and laterals that are within the bounding switches. You can also specify a trace color.
    - **Trace Between:** Traces the shortest path between the selected points and highlights that path and all line sections connected to it. You can also specify a trace color.
    - **Trace Path:** Traces the shortest path between the selected points and highlights it. You can also specify a trace color.
    - **Trace Extent:** Traces all connected end points of the network that are not energy sources, including branches and laterals. You can also specify a trace color.
    - **Trace to Marks:** Traces from the starting point to the marked device. If there are no marked devices between the starting point and network end points, traces to all connected end points. You can also specify a trace color.
    - **Clear Trace:** Clears all traces in all viewports.
- **Downstream Customers:** Opens the Traced Customer List (Create Planned Outage) window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Network Analysis:** Options to run a DNAF application for the selected line. (Visibility depends on the DNAF Configuration Editor settings that enable applications for RT and ST.) A study mode Distribution Power Flow (DPF) execution which is triggered from the line right-click menu in the geographic display will always utilize the default Generator/DER Output setting, Use Existing Generator/DER Output.
    - Distribution Power Flow
    - Protection Validation
    - Load and Volt/VAR Management
        - Minimize Demand
        - Loss Minimization
        - Reactive Area Support
        - Reduce Violations
        - Voltage Reduction
        - Voltage Reduction 1
        - Voltage Reduction 2
        - Power Factor Control
        - No Optimization
    - Fault Isolation and Service Restoration
        - Minimum Unserved kW (Auto+Manual)

    - Minimum Unserved kW (Auto Only)
    - Minimum Customer Minutes Interrupted
    - Minimum Loss of Customer Values of Uninterrupted Service
    - Minimum Number of Switching Operations
    - Minimum Voltage Drop Along Reconfigured Feeders
  - Automatic Feeder Reconfiguration
    - Reduce Overloads and Violations
    - Load Shedding
    - Return to Normal
  - Fault Location
  - Short Circuit Calculation
  - Planned Outage Study (Study mode only)
    - Minimum Unserved kW (Auto+Manual)
    - Minimum Unserved kW (Auto Only)
    - Minimum Customer Minutes Interrupted
    - Minimum Loss of Customer Values of Uninterrupted Service
    - Minimum Number of Switching Operations
    - Minimum Voltage Drop Along Reconfigured Feeders
  - FISR Contingency Analysis (Study mode only)
    - Minimum Unserved kW (Auto+Manual)
    - Minimum Unserved kW (Auto Only)
    - Minimum Customer Minutes Interrupted
    - Minimum Loss of Customer Values of Uninterrupted Service
    - Minimum Number of Switching Operations
    - Minimum Voltage Drop Along Reconfigured Feeders
- **Network Analysis Results:** Options to show DNAF application results. (Visibility depends on the DNAF Configuration Editor settings that enable applications for RT and ST.)
  - Distribution Power Flow
  - Protection Validation
  - Load and Volt/VAR Management
  - Fault Isolation and Service Restoration
  - Automatic Feeder Reconfiguration
  - Fault Location
  - Short Circuit Calculation

  - Planned Outage Study (Study mode only)
  - FISR Contingency Analysis (Study mode only)

- **Distance Calculation:** Distance calculation options:
  - **Distance to Source:** Calculates the network distance from the selected point on the line to the power source. This function is not available if the selected line is de-energized.
  - **Distance Between Points:** Calculates the network distance between the selected point on the line and another device or selected point on the same network. (If the two points have no path between them, no distance is calculated.) In addition, calculates the straight-line distance between the selected point on the line and another device or selected point.
- **Switching Advisor:** (For study and simulator modes only.)
  - **Distribution Power Flow:** Runs DPF for the substation that is associated with the line.
  - **Isolate Section:** Runs PLOS for the line.
  - **Get Plan:** Shows the Planned Outage Service display.
- **Power Flow Result:** Opens an attribute pop-up with DPF results for the line.
- **Show Voltage Profile:** Shows the voltage profile. (Future functionality).
- **Create Quick Order:** Creates a non-outage isolated incident for the selected line.
- **Find Nearest Structure:** Selects a structure that is closest to the line.
- **Create Planned Outage:** Opens the Traced Customer List (Create Planned Outage) window.
- **Create Maintenance Order:** Creates a maintenance order for the line. (For the Enterprise OMS client types only.)
- **Create Hot Line Hold Safety Doc Template:** Creates a hot line hold safety document and adds the associated devices to the safety doc.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.

6.19. Load

The following actions are available from this menu:

- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **Show Schedule Chart:** Shows the load chart based on dynamic schedules.
-

Create Quick Order: Creates a non-outage isolated incident for the selected load.

6.20. Order

The following actions are available from this menu:

- **Update Incident:** Opens the Update Incident display, which allows you to manage and clear the incident.
- **Incident's Custom Fields:** Opens the Incident's Custom Fields display, which allows you to enter, view, or update custom incident properties.
- **Confirm/Restore Incident:** Confirms the device outage (for devices that are predicted out) or confirms power restoration (for devices that are de-energized) and then changes the real-time status of the device.
- **Dispatch Crew:** Opens the Crews tab of the Update Incident display.
- **Navigate to OMS:** Shows the Incident Details display with the corresponding incident record selected. If the incident record was not present in the previously open Incident Details display (for example, due to applied filters), another Oms Display Plugin tab with the Incident Details display appears with the selected incident.
- **Merge Downstream Incidents:** Opens the Merge Downstream Incidents pop-up display to merge downstream confirmed, manually confirmed, predicted, and single customer incidents to a confirmed incident. Applies to downstream incidents on all phases.
- **Merge Downstream Predicted Incidents:** Merges downstream predicted and single customer incidents to a confirmed incident. Applies to downstream incidents on the same phase.
- **Merge Downstream Predicted Incidents (All Phases):** Merges downstream predicted and single customer incidents to a confirmed incident. Applies to downstream incidents on all phases.
- **Set Predicted Stable:** Sets the Predicted Stable state for the incident. This option appears only for orders that are associated with single customer or predicted incidents not in the Predicted Stable state.
- **Set Predicted:** Resets the Predicted Stable state for the incident. This option appears only for orders that are associated with single customer or predicted incidents in the Predicted Stable state.
- **Undo Confirmation:** This option is available for restoration orders that are associated with confirmed incidents.
  - For a confirmed incident that originated as a result of an operator's action, such as opening a device, this option closes the device and removes the incident.
  - For a confirmed incident that originated by confirming a predicted incident, this option closes the device, removes the incident, and re-creates a predicted incident with the same incident number, properties, and crew assignments based on downstream customer calls and AMI-out signals.

- **Merge Downstream Outages:** Merges downstream restoration orders to the order. This option appears only for orders that are associated with predicted incidents.
- **Confirm Outage:** Updates the associated Check Prediction task status to Confirmed Open. Updates the device state to Open. The incident becomes confirmed. This option is available for restoration orders with a Check Prediction task.
- **Move Up:** Moves the incident and the associated order to the next upstream protective device.
- **Move Down:** Cancels prediction for the device of record and creates incidents and orders downstream based on the received customer calls and AMI Out signals.
- **No Outage:** Sets the Check Prediction task result to No Outage (the incident becomes cancelled, and the order becomes closed). This option is available for restoration orders with a Check Prediction task.
- **Repair:** Sets the Repair task result to Repaired (the device becomes closed, and the incident becomes Pending Clear if all customers are restored). This option is available for restoration orders with a Repair task.
- **Problem Confirmed:** Sets the Investigate task result to Problem Confirmed. Creates a Fix Problem task. This option is available for investigation orders with an Investigate task.
- **Problem Resolved:** Sets the Fix Problem task result to Problem Resolved. This option is available for investigation orders with a Fix Problem task.
- **No Problem:** Sets the Investigate task result to Failed to Find Problem. This option is available for investigation orders with an Investigate task.
- **Problem Not Resolved:** Sets the Fix Problem task result to Problem Not Resolved. This option is available for investigation orders with a Fix Problem task.
- **Order Worksheet:** Opens the Order Worksheet display, which allows you to make order-related changes and track the progress of the order.
- **Change Lead Crew Status:** Shows the following options to change the lead crew status:
    - Assigned
    - Dispatched
    - En Route
    - Arrived
    - Released
- **Take Ownership:** Associates the order with the current operator.
- **Assign Crews:** Opens the Crew Selection dialog, where you can assign a crew to the order.
- **Request Crew Assignment:** Opens the Request from the Workforce Management dialog, where you can request automatic crew assignment to the order by the Workforce Management System.
- **Move Order:** Moves the order to a new position on the network view.

6.21. Permanent Jumper

The following actions are available from this menu:

- **Find Nearest Structure:** Selects a structure that is closest to the permanent jumper. This option only appears for devices outside of the station internals.
- **Add to Switch Order:** Adds the permanent jumper to the Switching Order form.

6.22. Planned Outage

The following actions are available from this menu:

- **Label:** Indicates the device associated with the planned outage.
- **Update Planned Outage Request:** Opens the Update Customer Notification pop-up display.
- **Move Planned Outage Request:** Activates the move mode for the planned outage.
- **Cancel Planned Outage Request:** Cancels the planned outage.
- **Delete Planned Outage Request:** Deletes the planned outage.
- **Convert to Planned Incident:** Creates an incident from the planned outage.
- **Open Switching Order:** Opens the Switching Order display. This field only appears for planned outages that are associated with a switching order.

6.23. Power Transformer

The following actions are available from this menu:

- **Label:** Indicates the substation name that the transformer belongs to and the transformer name.
- **Add to Daily Log:** Adds the transformer to the Daily Log.
- **Add to Switch Order:** Adds the transformer to the Switching Order form.
- **Toggle to Local:** Toggles the transformer to local mode.
- **Run Network Analysis:** Options to run a DNAF application for the selected power transformer. (Visibility depends on the DNAF Configuration Editor settings that enable applications for RT and ST.)
    - Distribution Power Flow
    - Protection Validation
    - Load and Volt/VAR Management
    - Fault Isolation and Service Restoration
    - Automatic Feeder Reconfiguration

    - Fault Location
    - Short Circuit Calculation
    - Planned Outage Study (Study mode only)
    - FISR Contingency Analysis (Study mode only)
- **Enter Parameters:** Opens the Transformer Parameters pop-up window, which allows overriding the default values of the transformer parameters. Please be aware that manual entried parameters that related to voltage regulation, for example, voltage target, voltage bandwidth, only apply to regular regulation mode, not for advanced regulation mode.
- **Application Optimization Eligibility Status:** Toggles the device optimization control mode (SCADA Decides or User Disabled).
- **Unblock:** Unblocks the power transformer.
- **Block:** Blocks the power transformer.
- **Add to Exclusions List:** Adds the power transformer to the LVM device exclusions list.
- **Remove from Exclusions List:** Removes the power transformer from the LVM device exclusions list.
- **Disable Voltage Reduction:** Disables voltage reduction.
- **Enable Voltage Reduction:** Enables voltage reduction.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Downstream:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Downstream Customers:** Opens the Traced Customer List (Create Planned Outage) window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.
- **Find Nearest Structure:** Selects a structure that is closest to the transformer.
- **Create Quick Order:** Creates a non-outage isolated incident for the transformer.
- **Faulted Section Determination:** Shows the Faulted Section Determination display where you can run the FSD application. This option is only available for power transformers with ASC.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that the transformer state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that the transformer state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the

transformer is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.

- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the transformer is enabled for controls. This option is only available in the simulator mode.

6.24. Reactor

The following actions are available from this menu:

- **Find Nearest Structure:** Selects a structure that is closest to the reactor.

6.25. Recloser

The following actions are available from this menu:

- **Label:** Indicates the substation name, the recloser name, and the recloser status (open or closed).
- **SCADA Control Popup:** Opens the SCADA control pop-up to enable or disable the recloser.
- **SCADA ARE Control Popup:** Opens the SCADA ARE control pop-up to change the automatic reclosing (ARE) status for the recloser.
- **Change State:** Toggles the real-time status of the recloser to open or closed.
- **Open Recloser:** Opens the recloser.
- **Close Recloser:** Closes the recloser.
- **Toggle Phases:** Opens the Toggle Switch Phases pop-up, which allows you to close and open individual phases of the reclosers.
- **Found Open:** Opens the recloser. Creates a confirmed incident with the FoundOpen flag set.
- **Confirm Outage:** Confirms the recloser outage and then changes the real-time status of the recloser to open. When attempting to confirm open a recloser, the OMS engine invokes a pop-up display with a list of all downstream single customer and predicted incidents, including those in the predicted stable state. You can select incidents to be merged with the newly created upstream confirmed incident; the incidents that are not merged become predicted stable incidents.
- **Recloser Details:** Shows a SCADA display with recloser details.
- **Add to Daily Log:** Adds the recloser to the Daily Log.
- **Add to Switch Order:** Adds the recloser to the Switching Order form.
- **Create Hot Line Hold:** Opens the Switching Order application and creates a hot line hold document for the recloser.
- **Create Historical Incident:** Creates a historical incident for the recloser and adds a record to the History Correction tabular display.

- **Create Predicted Incident:** Creates a predicted stable incident for the selected recloser.
- **Create Incident:** Creates a separate incident and adds a record to the Incident list. Allows splitting a big incident into multiple smaller incidents. This command is visible for open reclosers.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected recloser.
- **Create Confirmed Outage:** Creates a confirmed incident for the selected recloser. (For the Enterprise OMS client types only.)
- **Create Task:** Options to create an order task for the selected recloser. (For the Enterprise OMS client types only.)
  - **Open:** Opens a dialogue to create an Open task for the selected recloser.
  - **Close:** Opens a dialogue to create a Close task for the selected recloser.
  - **Other:** Opens a dialogue to create a Maintenance, Investigation, Make Safe, or Assessment task for the selected recloser.
- **Create Maintenance Order:** Creates a maintenance order for the selected recloser. (For the Enterprise OMS client types only.)
- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag for the selected recloser.
- **Place Do Not Operate Tag (SCADA):** Automatically places the Do Not Operate tag for the selected SCADA recloser.
- **Place Hold Open Tag:** Automatically places the Hold Open tag for the selected recloser.
- **Place Hold Open Tag (SCADA):** Automatically places the Hold Open tag for the selected SCADA recloser.
- **Place DMS Tags:** Opens the Tagging Operations window to place a tag for the selected recloser.
- **Associate to Safety Document:** Adds a recloser record to the Safety Document.
- **Create Hot Line Hold Safety Doc Template:** Creates a hot line hold safety document and adds the associated devices to the safety doc.
- **Mark/Unmark (1) Switch:** Marks/unmarks the recloser with mark type 1. The marked recloser is highlighted by a halo with a blue dashed line.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Downstream:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Downstream Customers:** Opens the Create Planned Outage window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.

- **Distance Calculation:** Distance calculation options:
  - **Distance to Source:** Calculates the network distance from the device to the power source. (If the selected device is de-energized, no distance is calculated.)
  - **Distance Between Points:** Calculates the network distance between the device and another device or selected point on the same network. (If the two points have no path between them, no distance is calculated.) In addition, calculates the straight-line distance between the switch and another device or selected point.
- **Find Nearest Structure:** Selects a structure that is closest to the recloser.
- **Mark Team(s) Switches:** Marks/unmarks members of the automated scheme teams that are associated with the recloser.
  - **Mark/Unmark (1):** Marks/unmarks team members with mark type 1.
  - **Mark/Unmark (2):** Marks/unmarks team members with mark type 2.
  - **Mark/Unmark (3):** Marks/unmarks team members with mark type 3.
- **Trace Scheme Switches:** Draws traces between members of the automated scheme teams that are associated with the recloser.
- **Scheme Switches on Feeder:** Opens the Automation Scheme Team Switches on Feeder display, which lists switching devices that are part of automated local peer-to-peer switching schemes on a feeder associated with the recloser.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that the recloser state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that the recloser state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the recloser is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the recloser is enabled for controls. This option is only available in the simulator mode.

6.26. Regulator

The following actions are available from this menu:

- **Add to Exclusions List:** Adds the regulator to the LVM device exclusions list.
- **Remove from Exclusions List:** Removes the regulator from the LVM device exclusions list.
- **Find Nearest Structure:** Selects a structure that is closest to the regulator.

- **Create Maintenance Order:** Creates a maintenance order for the regulator. (For the Enterprise OMS client types only.)
- **Create Quick Order:** Creates a non-outage isolated incident for the selected regulator.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that the regulator state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that the regulator state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the regulator is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the regulator is enabled for controls. This option is only available in the simulator mode.

6.27. Sectionalizer

The following actions are available from this menu:

- **Label:** Indicates the substation name, the sectionalizer name, and the sectionalizer status (open or closed).
- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **SCADA Control Popup:** Opens the SCADA control pop-up to enable or disable the sectionalizer.
- **Change State:** Toggles the real-time status of the sectionalizer to open or closed.
- **Open Switch:** Opens the sectionalizer.
- **Close Switch:** Closes the sectionalizer.
- **Found Open:** Opens the sectionalizer. Creates a confirmed incident with the FoundOpen flag set.
- **Add to Daily Log:** Adds the sectionalizer to the Daily Log.
- **Add to Switch Order:** Adds the sectionalizer to the Switching Order form.
- **Add as Isolation Switch to Switch Order:** Adds the selected sectionalizer to the list of isolating switches in the Equipment section in the currently open switching order (or creates a new switching order).
- **Create Historical Incident:** Creates a historical incident for the sectionalizer and adds a record

to the History Correction tabular display.

- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected sectionalizer.
- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag for the selected sectionalizer.
- **Place Do Not Operate Tag (SCADA):** Automatically places the Do Not Operate tag for the selected SCADA sectionalizer.
- **Place Hold Open Tag:** Automatically places the Hold Open tag for the selected sectionalizer.
- **Place Hold Open Tag (SCADA):** Automatically places the Hold Open tag for the selected SCADA sectionalizer.
- **Place DMS Tags:** Opens the Tagging Operations window to place a tag for the selected sectionalizer.
- **Associate to Safety Document:** Adds a sectionalizer record to the Safety Document.
- **Create Hot Line Hold Safety Doc Template:** Creates a hot line hold safety document and adds the associated devices to the safety doc.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Mark/Unmark (1) Switch:** Marks/unmarks the sectionalizer with mark type 1. The marked sectionalizer is highlighted by a halo with a blue dashed line.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Downstream:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Downstream Customers:** Opens the Create Planned Outage window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.
- **Filter Feeders:** Filters the display to show only the feeder the switch belongs to.
- **Undo Feeder Filter:** Undoes the display filtering by the switch feeder.
- **Distance Calculation:** Distance calculation options:
  - **Distance to Source:** Calculates the network distance from the device to the power source. (If the selected device is de-energized, no distance is calculated.)
  - **Distance Between Points:** Calculates the network distance between the device and another device or selected point on the same network. (If the two points have no path between them, no distance is calculated.) In addition, calculates the straight-line distance

between the switch and another device or selected point.

- **Mark Team(s) Switches:** Marks/unmarks members of the automated scheme teams that are associated with the sectionalizer.
  - **Mark/Unmark (1):** Marks/unmarks team members with mark type 1.
  - **Mark/Unmark (2):** Marks/unmarks team members with mark type 2.
  - **Mark/Unmark (3):** Marks/unmarks team members with mark type 3.
- **Trace Scheme Switches:** Draws traces between members of the automated scheme teams that are associated with the sectionalizer.
- **Scheme Switches on Feeder:** Opens the Automation Scheme Team Switches on Feeder display, which lists switching devices that are part of automated local peer-to-peer switching schemes on a feeder associated with the sectionalizer.
- **Find Nearest Structure:** Selects a structure that is closest to the sectionalizer.
- **Mark as Failed Equipment:** Marks the sectionalizer with mark type 3 to allow adding this sectionalizer to the incident’s failed equipment list.

6.28. Sensor

The following actions are available from this menu:

- **Distance Calculation:** Distance calculation options:
  - **Distance to Source:** Calculates the network distance from the sensor to the power source. (If the selected device is de-energized, no distance is calculated.)
  - **Distance Between Points:** Calculates the network distance between the sensor and another device or selected point on the same network. (If the two points have no path between them, no distance is calculated.) In addition, calculates the straight-line distance between the sensor and another device or selected point.
- **Find Nearest Structure:** Selects a structure that is closest to the sensor.

6.29. Safety Document

The following actions are available from this menu:

- **Label:** Indicates the safety document ID.
- **Edit Document:** Edits the safety document.
- **Move Document:** Moves the safety document.

6.30. Structure

The following actions are available from this menu:

- **Create Non-Outage Incident:** Creates a non-outage isolated incident for the structure.
- **Mark as Failed Equipment:** Marks the structure with mark type 3 to allow adding this structure to the incident's failed equipment list.

6.31. Substation

The following actions are available from this menu:

- **Label:** Indicates the substation name.
- **Navigate to SCADA:** Opens a station SCADA display in a separate viewport.
- **Navigate to Station Internals:** Opens a station internal display in the same viewport. Station internal shows substation schema containing all nested objects of the selected substation.
- **Navigate to Schematic:** Opens the geo schematic display in the same viewport.
- **Find Nearest Structure:** Selects a structure that is closest to the substation.
- **DNAF System Configuration Display:** Opens the DNAF System Configuration display.
- **Station Feeder Input Display:** Opens the Station Feeder Input display.
- **Add Create Step to Switch Order:** Adds a step to place a mobile substation to the Switching Order form. (Applies to mobile substations only.)
- **Create Task:** Creates an order task for the substation. (For the Enterprise OMS client types only.)
- **Distribution Power Flow Results:** Opens a new viewport containing the following DPF results tabular displays: DPF Feeder Results and Distribution Power Flow Summary for Station.
- **Protection Validation Results:** Opens the Protection Validation Results tabular display in a separate viewport.
- **Load and Volt/VAR Management Results:** Opens the Load and Volt/VAR Management tabular display in a separate viewport.
- **Fault Isolation and Service Restoration Results:** Opens the FISRCA Results Summary display in a separate viewport.
- **Automatic Feeder Reconfiguration Results:** Opens the AFR Results Summary tabular display in a separate viewport.
- **Fault Location Results:** Opens the Fault Location Results tabular display in a separate viewport.
- **Short Circuit Calculation Results:** Opens the Short Circuit Calculation Summary tabular display in a separate viewport.
- **Move Mobile Station:** Moves the mobile substation on the network view.

6.32. Switch

The following actions are available from this menu:

- **Label:** Indicates the substation name, the switch name (number), and the switch status (open or closed).
- **Future To Active:** Changes the device commissioning status from "Future" to "Active".
- **Future Removed To Removed:** Changes the device commissioning status from "Future Removed" to "Removed".
- **Removed To Future Removed:** Changes the device commissioning status from "Removed" to "Future Removed".
- **Active To Future:** Changes the device commissioning status from "Active" to "Future".
- **SCADA Control Popup:** Opens the SCADA control pop-up to enable or disable the switch.
- **Change State:** Toggles the real-time status of the switch to open or closed.
  - **Open Switch:** Opens the switch.
  - **Close Switch:** Closes the switch.
- **Found Open:** Opens the switch. Creates a confirmed incident with the FoundOpen flag set.
- **Add to Daily Log:** Adds the switch to the Daily Log.
- **Add to Switch Order:** Adds the switch to the Switching Order form.
- **Add as Isolation Switch to Switch Order:** Adds the selected switch to the list of isolating switches in the Equipment section in the currently open switching order (or creates a new switching order).
- **Create Historical Incident:** Creates a historical incident for the switch and adds a record to the History Correction tabular display.
- **Create Incident:** Creates a separate incident, adds a record to the Incident Detail tabular display, and marks the affected device on the geographic view. Allows splitting a big incident into multiple smaller incidents. This command is visible for open switches only.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected switch.
- **Create Confirmed Outage:** Creates a confirmed incident for the selected switch. (For the Enterprise OMS client types only.)
- **Create Maintenance Order:** Creates a maintenance order for the switch. (For the Enterprise OMS client types only.)
- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag for the selected switch.
- **Place Do Not Operate Tag (SCADA):** Automatically places the Do Not Operate tag for the selected SCADA switch.

- **Place Hold Open Tag:** Automatically places the Hold Open tag for the selected switch.
- **Place Hold Open Tag (SCADA):** Automatically places the Hold Open tag for the selected SCADA switch.
- **Place DMS Tags:** Opens the Tagging Operations window to place a tag for the selected switch.
- **Associate to Safety Document:** Adds a switch record to the Safety Document.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Mark/Unmark (1) Switch:** Marks/unmarks the switch with mark type 1. The marked switch is highlighted by a halo with a blue dashed line.
- **Clear Trace:** Clears all traces in all viewports.
- **Trace Downstream:** Invokes a trace from the selected device out to the end points, including laterals. You can also specify a trace color.
- **Trace Upstream:** Invokes a trace from the selected device to the power source. You can also specify a trace color.
- **Downstream Customers:** Opens the Create Planned Outage window for the downstream customers. You can create a planned outage, customize the notification message, and notify selected customers of the outage.
- **Downstream Load and Generation:** Opens the Traced Load and Generation display, which shows the total load, generation, and embedded generation downstream of the device.
- **Filter Feeders:** Filters the display to show only the feeder the switch belongs to.
- **Undo Feeder Filter:** Undoes the display filtering by the switch feeder.
- **Distance Calculation:** Distance calculation options:
    - **Distance to Source:** Calculates the network distance from the device to the power source. (If the selected device is de-energized, no distance is calculated.)
    - **Distance Between Points:** Calculates the network distance between the switch and another device or selected point on the same network. (If the two points have no path between them, no distance is calculated.) In addition, calculates the straight-line distance between the switch and another device or selected point.
- **Mark Team(s) Switches:** Marks/unmarks members of the automated scheme teams that are associated with the switch.
    - **Mark/Unmark (1):** Marks/unmarks team members with mark type 1.
    - **Mark/Unmark (2):** Marks/unmarks team members with mark type 2.
    - **Mark/Unmark (3):** Marks/unmarks team members with mark type 3.
- **Trace Scheme Switches:** Draws traces between members of the automated scheme teams that are associated with the switch.
- **Scheme Switches on Feeder:** Opens the Automation Scheme Team Switches on Feeder display, which lists switching devices that are part of automated local peer-to-peer switching schemes

on a feeder associated with the switch.

- **Find Nearest Structure:** Selects a structure that is closest to the switch.
- **Mark as Failed Equipment:** Marks the switch with mark type 3 to allow adding this switch to the incident's failed equipment list.
- **Block State Control:** Sets the Block State Control status to Block. This specifies that the switch state controls are blocked. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Allow State Control:** Sets the Block State Control status to Allow. This specifies that the switch state controls are allowed. This option is only available in the simulator mode.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that the switch is disabled for any controls except Enable Device. This option is used to simulate device malfunction. This option is only available in the simulator mode.
- **Enable Device:** Changes the Device Disabled status to Enable. This specifies that the switch is enabled for controls. This option is only available in the simulator mode.

6.33. Switch Cabinet

- **Open Picture in Picture for Container:** Opens a separate window in one corner of the main screen for the switch cabinet. This window can be zoomed separately to the main network view, to provide an alternate view of the section of the network currently being worked on.
- **Navigate to Related Display:** Opens the new viewport displaying the internal view of the switch cabinet.
- **Find Nearest Structure:** Selects a structure that is closest to the switch cabinet.

6.34. Tag

The following actions are available from this menu:

- **Move Tag:** Activates the move mode for the tag.

6.35. Temporary Switch (Jumper)

The following actions are available from this menu:

- **Label:** Indicates the substation name that the temporary switch belongs to, the temporary switch name, and the temporary switch state (open or closed).
- **Open Temporary Switch:** Changes the real-time status of the temporary switch to open.
- **Close Temporary Switch:** Changes the real-time status of the temporary switch to closed.
- **Remove Temporary Switch:** Removes the temporary switch.

- **Add to Daily Log:** Adds a temporary switch to the Daily Log.
- **Add to Switch Order:** Adds a temporary switch to the Switching Order form.
- **Add Create Step to Switch Order:** Adds a step to create the temporary switch to the Switching Order form.
- **Create Incident:** Creates a separate incident, adds a record to the Incident Detail tabular display, and marks the affected device on the geographic view. Allows splitting a big incident into multiple smaller incidents. This command is visible with open fuses only.
- **Create Planned Outage:** Opens the Create Planned Outage pop-up display, which allows you to create a customer notification about the planned outage.
- **Create Quick Order:** Creates a non-outage isolated incident for the selected switch.
- **Create Task:** Options to create an order task for the selected switch. (For the Enterprise OMS client types only.)
  - **Open:** Opens a dialogue to create an Open task for the selected switch.
  - **Close:** Opens a dialogue to create a Close task for the selected switch.
  - **Other:** Opens a dialogue to create a Maintenance, Investigation, Make Safe, or Assessment task for the selected switch.
- **Create Maintenance Order:** Creates a maintenance order for the switch. (For the Enterprise OMS client types only.)
- **Place Do Not Operate Tag:** Automatically places the Do Not Operate tag on the selected switch.
- **Place Hold Open Tag:** Automatically places the Hold Open tag on the selected switch.
- **Place DMS Tags:** Opens the Tagging Operations window to place a tag for the temporary switch.
- **Associate to Safety Document:** Adds a temporary switch record to the Safety Document.
- **Create Clearance Safety Doc Template:** Creates a clearance safety document and adds the associated devices to the safety doc.
- **Mark/Unmark Switch:** Marks the selected cut with a halo. Unmarks the temporary switch if it was already marked.
- **Move Jumper:** Activates the move mode for the jumper.
- **Distance Calculation:** Distance calculation options:
  - **Distance to Source:** Calculates the network distance from the device to the power source. (If the selected device is de-energized, no distance is calculated.)
  - **Distance Between Points:** Calculates the network distance between the device and another device or selected point on the same network. (If the two points have no path between them, no distance is calculated.) In addition, calculates the straight-line distance between the switch and another device or selected point.

6.36. UI Container

The following action is available from this menu:

- **Open Picture in Picture for Container:** Opens a separate window in one corner of the main screen for the UI container. This window can be zoomed separately to the main network view, to provide an alternate view of the section of the network currently being worked on.

6.37. Unassociated Call

The following actions are available from this menu:

- **Update Unassociated Call:** Opens the Add/Update Unassociated Call pop-up display to update information about the unassociated call.
- **Delete Unassociated Call:** Deletes the unassociated call record and removes the icon from the geographic view.
- **Associate Call with Incident:** Opens the Associate Call with Incident pop-up tabular display with a list of incidents. Adds the unassociated call to the selected incident and removes the unassociated call record from the Unassociated Calls OMS display and the call icon from the geographic view.
- **Associate Call with Selected Incident:** Adds the unassociated call to the selected incident and removes the unassociated call record from the Unassociated Calls OMS display and the call icon from the geographic view.
- **Create Incident from Call:** Creates a separate incident record from the unassociated call and adds it to the Incident Detail OMS display, removes the unassociated call record from the Unassociated Calls OMS display, and changes the call icon to the incident icon on the geographic view.
- **Move Call:** Activates the move mode for the unassociated call.

6.38. Vehicle

- **Move Vehicle:** Activates the move mode for the vehicle.
- **Find Nearest Structure:** Selects a structure that is closest to the vehicle.